MATH2121: Linear Algebra

WU Jinjun

November 13, 2026

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1 Linear Systems and Matrices

1.1 Notation

- $\mathbb{R}$: real numbers.
- $\mathbb{Q}$: rational numbers $\frac{p}{q}$.
- $\mathbb{Z}$: integers $\{\ldots, -2, -1, 0, 1, 2, \ldots\}$.
- $\mathbb{N}$: nonnegative integers $\{0, 1, 2, \ldots\}$.

1.2 Systems of Linear Equations

Definition. A *linear equation* in variables $x_1, x_2, \ldots, x_n$ is an equation of the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where $a_1, \ldots, a_n, b \in \mathbb{R}$. Equivalently, $\sum_{i=1}^{n} a_ix_i = b$.

Example. The following are linear equations in x_1, x_2, x_3:

$$3x_1 = 2x_2, \quad 3x_1 + 4x_2 - \sqrt{2}x_3 = 7, \quad 0 = 0, \quad 0 = 1.$$

The following are **not** linear equations:

$$3x_1^2 + 4x_2 = 7, \quad x_1x_2 = x_3, \quad \sqrt{x_1} = x_2.$$

Definition. A *system of linear equations* (or *linear system*) is a list of linear equations in the same set of variables. A *solution* is a list $(s_1, \ldots, s_n)$ such that substituting $x_i = s_i$ makes all equations true. A solution is *nonzero* if at least one $s_i \neq 0$.

Definition. Two linear systems are *equivalent* if they have the same set of solutions.

Number of solutions

Theorem. A linear system (in any number of variables) has either 0, 1, or infinitely many solutions.

Proof (algebraic). Suppose the linear system has two distinct solutions $\vec{s} = (s_1, \ldots, s_n)$ and $\vec{r} = (r_1, \ldots, r_n)$, so $\vec{s} \neq \vec{r}$. Consider the associated *homogeneous system*, obtained by replacing each b_i with 0. For each equation $\sum_{j=1}^{n} a_{ij}x_j = b_i$, since $\vec{s}$ and $\vec{r}$ are both solutions we have $\sum_{j=1}^{n} a_{ij}s_j = b_i$ and $\sum_{j=1}^{n} a_{ij}r_j = b_i$. Subtracting gives $\sum_{j=1}^{n} a_{ij}(s_j - r_j) = 0$, so $\vec{s} - \vec{r} = (s_1 - r_1, \ldots, s_n - r_n)$ is a nonzero solution of the homogeneous system. Let $\vec{t} = \vec{s} - \vec{r} \neq \vec{0}$. For any scalar $c \in \mathbb{R}$, the vector $\vec{r} + c\vec{t}$ satisfies $\sum_{j=1}^{n} a_{ij}(r_j + ct_j) = b_i + c \cdot 0 = b_i$, so $\vec{r} + c\vec{t}$ is a solution. As c ranges over $\mathbb{R}$, the vectors $\vec{r}, \vec{r} + \vec{t}, \vec{r} + 2\vec{t}, \vec{r} + 3\vec{t}, \ldots$ are all distinct (since $\vec{t} \neq \vec{0}$), giving infinitely many solutions. Hence a system with at least two solutions has infinitely many. Therefore the only possibilities are 0, 1, or infinitely many solutions. ■

Proof (geometric). Each equation $\sum_{j=1}^{n} a_{ij}x_j = b_i$ defines a hyperplane in $\mathbb{R}^n$. The

solution set is the intersection of all these hyperplanes. If the intersection is empty, there are 0 solutions. If the intersection is a single point, there is 1 solution. If the intersection contains two distinct points $\vec{s}$ and $\vec{r}$, then the entire line $\{\vec{r} + c(\vec{s} - \vec{r}) : c \in \mathbb{R}\}$ lies in the intersection (each hyperplane, containing two points of the line, contains the whole line), giving infinitely many solutions. ■

Definition. A linear system is *consistent* if it has at least one solution, and *inconsistent* if it has zero solutions.

1.3 Matrices

Definition. A *matrix* is a rectangular array of numbers. A matrix is $m \times n$ (or m-by-n) if it has m rows and n columns. We write A_{ij} for the entry in the ith row and jth column.

Definition. Given a linear system

$$\begin{aligned} a_{11}x_1 + \cdots + a_{1n}x_n &= b_1 \\ &\vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

the *coefficient matrix* is $\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}$ and the *augmented matrix* is

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} & b_1 \\ \vdots & & \vdots & \vdots \\ a_{m1} & \cdots & a_{mn} & b_m \end{pmatrix}.$$

The coefficient matrix is $m \times n$; the augmented matrix is $m \times (n+1)$.

1.4 Elementary Row Operations

Definition. The *elementary row operations* on a matrix are:

1. **Replacement:** Replace a row by the sum of itself and a multiple of another row.
2. **Scaling:** Multiply a row by a nonzero number.
3. **Interchange:** Swap two rows.

Definition. Two matrices are *row equivalent* if one can be transformed to the other by a sequence of zero or more row operations.

Each row operation is reversible

- Scaling by $c \leftrightarrow$ scaling by $\frac{1}{c}$.
- Swapping rows i and $j \leftrightarrow$ swapping rows i and j.
- Replacing row i by (row i + $c\cdot$ row j) $\leftrightarrow$ replacing row i by (row i − $c\cdot$ row j).

Row equivalence preserves solutions

Theorem. If the augmented matrices of two linear systems are row equivalent, then the systems are equivalent (i.e., have the same solutions).

Proof. It suffices to show that a single elementary row operation on the augmented matrix produces a new system with the same solution set. There are three cases. Let the original augmented matrix correspond to the system $\sum_{j=1}^{n} a_{ij}x_j = b_i$ for $i = 1, \ldots, m$.

1. **Replacement:** Replace row i by (row i + $c\cdot$ row k) with $k \neq i$ and $c \in \mathbb{R}$. The ith equation becomes $\sum_{j=1}^{n}(a_{ij}+c\,a_{kj})x_j = b_i + c\,b_k$. If $\vec{s}$ solves the original system, then $\sum a_{ij}s_j = b_i$ and $\sum a_{kj}s_j = b_k$, so $\sum(a_{ij}+c\,a_{kj})s_j = b_i + c\,b_k$, meaning $\vec{s}$ solves the new system. Conversely, if $\vec{s}$ solves the new system, then $\sum(a_{ij}+c\,a_{kj})s_j = b_i + c\,b_k$, and using $\sum a_{kj}s_j = b_k$ (the kth equation is unchanged) we recover $\sum a_{ij}s_j = b_i$. So the solution sets are equal.
2. **Scaling:** Multiply row i by $c \neq 0$. The ith equation becomes $\sum_{j=1}^{n}(c\,a_{ij})x_j = c\,b_i$, i.e., $c\sum a_{ij}x_j = cb_i$. Since $c \neq 0$, dividing by c recovers the original equation, so the solution sets are equal.
3. **Interchange:** Swap rows i and k. This merely reorders the equations, which does not change the solution set (a list $(s_1, \ldots, s_n)$ satisfies all equations regardless of their order).

Since each elementary row operation preserves the solution set, any sequence of them (i.e., row equivalence) preserves the solution set. ■

1.5 Echelon Form and RREF

Definition. A row in a matrix is *nonzero* if not every entry is zero. The *leading entry* in a nonzero row is the first nonzero entry (from left to right).

Definition. A matrix with m rows and n columns is in *echelon form* if:

(E1) When a row is nonzero, every row above it is also nonzero.

(E2) The leading entry in a nonzero row is strictly to the right of the leading entry of any earlier row.

These imply:

(E3) If a row is nonzero, then every entry below its leading entry in the same column is zero.

Definition. A matrix in echelon form is *reduced* if:

(R1) Each nonzero row has leading entry 1.

(R2) The leading 1 in each nonzero row is the only nonzero number in its column.

Such a matrix is said to be in *reduced echelon form*.

Uniqueness of RREF

Theorem. Each matrix A is row equivalent to exactly one matrix RREF(A) in reduced echelon form, called the *reduced echelon form* of A.

Proof. Existence follows from the row reduction algorithm, which produces a reduced echelon form. We prove uniqueness. Suppose A is row equivalent to two reduced echelon matrices R and S. Since R and S are both row equivalent to A, they are row equivalent to each other. We show $R = S$ by induction on the number of columns. The key idea: the pivot columns of any echelon form of A are in the same positions (this can be shown by induction on the number of pivot columns, since the leftmost nonzero column must contain a pivot in both R and S, and reducing the submatrix below that pivot column gives the same structure recursively). So R and S have pivots in the same positions. In a reduced echelon form, each pivot column is a standard basis vector $\vec{e}_i$ (with 1 in the pivot row and 0 elsewhere). Since R and S have the same pivot positions, each pivot column of R equals the corresponding pivot column of S. For a non-pivot column, its entries are determined: the entry in a

pivot row equals the coefficient needed to express that column as a linear combination of the pivot columns to its left, which is uniquely determined by the column itself and the pivot columns. Since both R and S are obtained from A by row operations (which preserve linear relations among columns), the non-pivot columns of R and S coincide. Therefore $R = S$. ■

1.6 Solving a Linear System from RREF

Definition. A *pivot position* in a matrix A is the location containing a leading 1 in $\mathrm{RREF}(A)$. A *pivot column* is a column containing a pivot position. An entry in a pivot position is called a *pivot*.

Definition. For a linear system in variables $x_1, \ldots, x_n$ with augmented matrix A, the variable x_i is *basic* if i is a pivot column of A, and *free* otherwise.

Solutions from RREF

Theorem. Consider a linear system whose augmented matrix is A.

- The system has **0 solutions** if the last column of A contains a pivot (i.e., $\mathrm{RREF}(A)$ has a row $(0 \ \cdots \ 0 \ \ 1)$, giving the false equation $0 = 1$).
- The system has **1 solution** if there are no free variables and the last column is not a pivot.
- The system has **infinitely many solutions** otherwise (i.e., the last column is not a pivot but there is at least one free variable).

Proof. Row-reduce A to $\mathrm{RREF}(A) = R$. By the previous theorem, R represents an equivalent system with the same solution set.

1. If the last column of R is a pivot column, then some row of R is $(0 \ \cdots \ 0 \ \ 1)$, corresponding to the equation $0 = 1$, which has no solution. So the system is inconsistent: 0 solutions.
2. If the last column is not a pivot and there are no free variables, then every variable $x_1, \ldots, x_n$ is basic (each corresponds to a pivot column). Each basic variable is uniquely determined (e.g., $x_i = b_i$ for some determined value b_i), so there is exactly one solution.
3. If the last column is not a pivot but there is at least one free variable, then the system is consistent (no row $(0 \ \cdots \ 0 \ \ 1)$) and each free variable can take any real value. For each assignment of the free variables, the basic variables are uniquely determined. Since there is at least one free variable ranging over $\mathbb{R}$, there are infinitely many solutions.

These three cases are exhaustive and mutually exclusive. ■

General solution procedure. After computing $\mathrm{RREF}(A)$:

1. Identify basic and free variables.
2. Write each basic variable in terms of the free variables.
3. Choose arbitrary values for free variables; solve for basic variables.

Corollary. If

$$\boxed{\mathrm{RREF}(A) = \begin{pmatrix} 1 & 0 & \cdots & 0 & b_1 \\ 0 & 1 & \cdots & 0 & b_2 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1 & b_m \end{pmatrix},}$$

then the unique solution is $(x_1, \ldots, x_m) = (b_1, \ldots, b_m)$.

Proof. The RREF shows each variable is a basic variable (every column $1, \ldots, m$ is a pivot column) and the last column is not a pivot. By the previous theorem, there is exactly one solution. Reading off the RREF, the ith equation is $x_i = b_i$ (the leading

1 in column i gives x_i plus zeros from other entries). So $x_i = b_i$ for all i, giving the unique solution $(x_1, \ldots, x_m) = (b_1, \ldots, b_m)$. ■

1.7 Row Reduction Algorithm

Algorithm (Row reduction to echelon form). Input: an $m \times n$ matrix A.

1. Begin with the leftmost nonzero column. This is a pivot column; the pivot position is the top position.
2. Select a nonzero entry in the current pivot column. If needed, swap its row to the top.
3. Use row operations to create zeros below the pivot position.
4. Cover the row containing the current pivot; apply steps 1–3 to the remaining $(m-1)\times n$ submatrix. Repeat until the entire matrix is in echelon form.
5. Starting from the rightmost pivot, rescale to leading entry 1, then create zeros *above* each pivot. Repeat going right to left.

Output: RREF(A).

Pivot positions are invariant

In any echelon form E of a matrix A, the locations of the leading entries are the same. Thus pivot positions can be found from *any* echelon form—one need not fully reduce to find pivots.

Practice Problem #1.14 (RREF classification of ± 1 matrices)

Problem. Let X be the set of all 3×3 matrices whose entries are $+1$ or -1 (there are $2^9 = 512$ such matrices). What forms can $E = \text{RREF}(A)$ take as A ranges over X? For each such E, exactly how many matrices $A \in X$ have $\text{RREF}(A) = E$?
Solution. There are 11 possible reduced echelon forms. The key trick is *row-first normalization*: multiply every row whose first entry is -1 by -1 (this is an elementary row operation and does not change RREF). The normalized rows all begin with 1 and can only be

$$v_1 = (1 \quad 1 \quad 1)^\top,\ v_2 = (1 \quad 1 \quad -1)^\top,\ v_3 = (1 \quad -1 \quad 1)^\top,\ v_4 = (1 \quad -1 \quad -1)^\top.$$

Each ordered triple of normalized rows comes from exactly $2^3 = 8$ original matrices, because each row can independently have either sign. We count by cases:

- *One distinct normalized row.* All three normalized rows are equal. The RREF is $\begin{pmatrix} 1 & \epsilon & \eta \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ with $\epsilon, \eta \in \{\pm 1\}$. There are 4 such reduced forms, each with $4 \cdot 8 = 32$ original matrices.
- *Two distinct normalized rows.* The non-zero part of RREF belongs to one of $\begin{pmatrix} 1 & \epsilon & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1 & 0 & \epsilon \\ 0 & 1 & 0 \end{pmatrix}$, $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \epsilon \end{pmatrix}$. This gives 6 more reduced forms, each with 48 original matrices ($\binom{4}{2} = 6$ pairs, minus the 2 constant triples per pair, times 8).
- *Three distinct normalized rows.* Among any three distinct types, two have the same second coordinate and the opposite third. After one row swap and row reduction, all three pivots are nonzero, giving RREF I_3. There are $4 \cdot 3 \cdot 2 \cdot 8 = 192$ such original matrices.

Check: $32 + 6 \cdot 48 + 192 = 512$, consistent.

Clever trick: row-first normalization

When matrix entries have sign symmetry (like ± 1 rows), multiply rows to force a uniform sign in the first non-zero entry. This is an elementary row operation that

does *not* change RREF, but dramatically reduces case work. Each normalized matrix corresponds to 2^m originals (independent sign choices per row), so counts are recovered post-hoc.

Practice Problem #1.15 (RREF of $M_{ij} = i + j$)

Problem. Let M be the $n \times n$ matrix with $M_{ij} = i + j$. Find $\mathrm{RREF}(M)$.

Solution. Let $\vec{u} = (1 \quad 1 \quad \cdots \quad 1)^\top$. The original ith row is $\vec{r}_i = (i+1, i+2, \ldots, i+n) = \vec{r}_1 + (i-1)\vec{u}$. Row-reduce:

1. $R_2 \leftarrow R_2 - R_1$, turning R_2 into $\vec{u}$;
2. for $i \geq 3$, $R_i \leftarrow R_i - R_1 - (i-1)R_2$; all these rows become zero.

The first two rows are now $\vec{r}_1 = (2, 3, \ldots, n+1)$ and $\vec{u} = (1, 1, \ldots, 1)$. Then $R_1 \leftarrow R_1 - 2R_2$ gives $(0, 1, \ldots, n-1)$. Exchange R_1 and R_2, eliminate the leading entries in column 1, and the final RREF is

$$\mathrm{RREF}(M) = \begin{pmatrix} 1 & 0 & -1 & \cdots & 2-n \\ 0 & 1 & 2 & \cdots & n-1 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix},$$

with exactly $n-2$ zero rows. Special cases: $n = 1$ gives (1), $n = 2$ gives I_2.

Clever trick: eliminate rows first, then columns

When rows are simple linear combinations of a few fixed vector patterns, first row-reduce all high rows to zero and only then fully reduce the survivors. This avoids chasing pivots column by column.

2 Vectors and Linear Combinations

2.1 Vectors

Definition. A *(column) vector* of size n is an $n \times 1$ matrix: $\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$. We write $\mathbb{R}^n$ for the set of all vectors with exactly n rows. Each v_i is a real number.

Vector operations:

- *Addition* (same size only): $\begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} + \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} = \begin{pmatrix} v_1 + u_1 \\ \vdots \\ v_n + u_n \end{pmatrix}$.
- *Scalar multiplication* ($c \in \mathbb{R}$): $c \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} cv_1 \\ \vdots \\ cv_n \end{pmatrix}$.
- *Subtraction*: $\vec{u} - \vec{v} = \vec{u} + (-1)\vec{v}$.

Definition. The *zero vector* in $\mathbb{R}^n$ is $\vec{0} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$. We always have $\vec{0} + \vec{v} = \vec{v} + \vec{0} = \vec{v}$.

Geometric intuition in $\mathbb{R}^2$

A vector $\begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ is drawn as an arrow from the origin to (a_1, a_2). The sum $\vec{a} + \vec{b}$ is the diagonal of the parallelogram with sides $\vec{a}$ and $\vec{b}$.

2.2 Linear Combinations and Span

Definition. If $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p \in \mathbb{R}^n$ and $c_1, \ldots, c_p \in \mathbb{R}$ (scalars), then

$$\vec{y} = c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_p\vec{v}_p$$

is a *linear combination* of $\vec{v}_1, \ldots, \vec{v}_p$ with coefficients $c_1, \ldots, c_p$.

Definition. The *span* of vectors $\vec{v}_1, \ldots, \vec{v}_p \in \mathbb{R}^n$ is the set of all their linear combinations:

$$\operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_p\} = \{c_1\vec{v}_1 + \cdots + c_p\vec{v}_p : c_1, \ldots, c_p \in \mathbb{R}\} \subseteq \mathbb{R}^n.$$

Span $\Leftrightarrow$ consistent system

Proposition. A vector equation $x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n = \vec{b}$ (where $\vec{a}_1, \ldots, \vec{a}_n, \vec{b} \in \mathbb{R}^m$) has the same solutions as the linear system with augmented matrix $\begin{pmatrix} \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n & \vec{b} \end{pmatrix}$.

Proof. Write each vector $\vec{a}_j = \begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$. The vector equation $x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n = \vec{b}$ means, in the ith coordinate, $\sum_{j=1}^{n} a_{ij}x_j = b_i$ for $i = 1, \ldots, m$. But these are exactly the m equations of the linear system with coefficient matrix (a_{ij}) and augmented matrix $\begin{pmatrix} \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n & \vec{b} \end{pmatrix}$. Therefore a list $(s_1, \ldots, s_n)$ satisfies the vector equation if and only if it satisfies each of the m scalar equations, i.e., it is a solution of the linear system. The two have identical solution sets. ■

Corollary. $\vec{b} \in \operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_p\}$ if and only if the matrix $\begin{pmatrix} \vec{v}_1 & \vec{v}_2 & \cdots & \vec{v}_p & \vec{y} \end{pmatrix}$ is the augmented matrix of a consistent linear system (i.e., the last column is not a pivot column).

Proof. By definition, $\vec{b} \in \operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_p\}$ if and only if there exist scalars $x_1, \ldots, x_p$ with $\vec{b} = x_1\vec{v}_1 + \cdots + x_p\vec{v}_p$, i.e., the vector equation has a solution. By the proposition, this is equivalent to the linear system with augmented matrix $\begin{pmatrix} \vec{v}_1 & \cdots & \vec{v}_p & \vec{b} \end{pmatrix}$ having a solution, i.e., being consistent. By the RREF criterion, a system is consistent if and only if the last column of the augmented matrix is not a pivot column (no row $\begin{pmatrix} 0 & \cdots & 0 & 1 \end{pmatrix}$). ■

Span of a single vector

$\operatorname{span}\{\vec{0}\} = \{\vec{0}\}$ (a single point). For $\vec{v} \neq \vec{0}$, $\operatorname{span}\{\vec{v}\} = \{c\vec{v} : c \in \mathbb{R}\}$ is the line through the origin in the direction of $\vec{v}$. This is infinite unless $\vec{v} = \vec{0}$.

Geometric picture of span.

- In $\mathbb{R}^2$: span is a point, a line through origin, or all of $\mathbb{R}^2$.

- In $\mathbb{R}^3$: span is a point, a line through origin, a plane through origin, or all of $\mathbb{R}^3$.

Practice Problem #2.14 (Span is idempotent)

Problem. Interpret the statement "for any $S \subseteq \mathbb{R}^n$, $\operatorname{span}(\operatorname{span} S) = \operatorname{span} S$" and prove it formally.
Solution. Taking linear combinations twice produces nothing new beyond taking them once. Let $W = \operatorname{span} S$.
$W \subseteq \operatorname{span} W$: every $\vec{w} \in W$ is a linear combination of vectors in W, namely $\vec{w} = 1 \cdot \vec{w}$, so this is trivial.

$\operatorname{span} W \subseteq W$: let $\vec{u} \in \operatorname{span} W$. By definition, $\vec{u} = \sum_{j=1}^{m} a_j \vec{w}_j$ with $\vec{w}_j \in W$. Each $\vec{w}_j = \sum_{i=1}^{r_j} b_{ji} \vec{s}_{ji}$ with $\vec{s}_{ji} \in S$. Substitute:

$$\vec{u} = \sum_{j=1}^{m} a_j \sum_{i=1}^{r_j} b_{ji} \vec{s}_{ji} = \sum_{\text{all } (j,i)} (a_j b_{ji}) \vec{s}_{ji},$$

a finite linear combination of vectors from S, hence $\vec{u} \in W$. If $S = \emptyset$, both sides equal $\{\vec{0}\}$.

Key intuition

$\operatorname{span} S$ is by definition a subspace, and subspaces are exactly the sets already closed under linear combinations. So applying span a second time does nothing. This trivial-sounding property underlies many results: the basis theorem, rank-nullity, etc.

3 Matrix Equations and Matrix Operations

3.1 Matrix-Vector Product

Definition. If A is an $m \times n$ matrix with columns $\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_n \in \mathbb{R}^m$ and $\vec{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{R}^n$, then the *matrix-vector product* $A\vec{v}$ is the vector in $\mathbb{R}^m$ given by:

$$A\vec{v} = \begin{pmatrix} \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = v_1\vec{a}_1 + v_2\vec{a}_2 + \cdots + v_n\vec{a}_n \in \mathbb{R}^m.$$

Thus $A\vec{v}$ is the linear combination of the columns of A with coefficients from $\vec{v}$.

Dimensions and linearity

- $A\vec{v}$ is defined only when #columns of A = #rows of $\vec{v}$. If A is $m \times n$ and $\vec{v} \in \mathbb{R}^n$, then $A\vec{v} \in \mathbb{R}^m$.
- A transforms $\mathbb{R}^n \to \mathbb{R}^m$. This transformation is **linear**:
  1. $A(\vec{u} + \vec{v}) = A\vec{u} + A\vec{v}$ for all $\vec{u}, \vec{v} \in \mathbb{R}^n$.
  2. $A(c\vec{v}) = c(A\vec{v})$ for all $\vec{v} \in \mathbb{R}^n$, $c \in \mathbb{R}$.

Entry-wise formula. The ith entry of $A\vec{v}$ is the dot product of row i of A with $\vec{v}$:

$$(A\vec{v})_i = a_{i1}v_1 + a_{i2}v_2 + \cdots + a_{in}v_n.$$

3.2 Matrix Equations

Definition. If A is $m \times n$, $\vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ (variables), and $\vec{b} \in \mathbb{R}^m$, then $A\vec{x} = \vec{b}$ is a *matrix equation.*

Three equivalent views

Proposition. The matrix equation $A\vec{x} = \vec{b}$ has the same solutions as:

1. The vector equation $x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n = \vec{b}$.
2. The linear system with augmented matrix $\begin{pmatrix} \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n & \vec{b} \end{pmatrix}$.

Proof. By definition of the matrix-vector product, $A\vec{x} = \begin{pmatrix} \vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n \end{pmatrix}\vec{x} = x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n$. So $A\vec{x} = \vec{b}$ is precisely the vector equation $x_1\vec{a}_1 + \cdots + x_n\vec{a}_n = \vec{b}$, and they have identical solutions. By the proposition in Section 2, the vector equation has the same solutions as the linear system with augmented matrix $\begin{pmatrix} \vec{a}_1 & \cdots & \vec{a}_n & \vec{b} \end{pmatrix}$. Therefore all three formulations share the same solution set. ■

Proposition. $A\vec{x} = \vec{b}$ has a solution $\Leftrightarrow$ $\vec{b}$ is a linear combination of the columns of A $\Leftrightarrow$ $\vec{b} \in \text{span}\{\vec{a}_1, \ldots, \vec{a}_n\}$.

Proof. $A\vec{x} = \vec{b}$ has a solution $\Leftrightarrow$ there exist $x_1, \ldots, x_n$ with $A\vec{x} = \vec{b}$ $\Leftrightarrow$ there exist $x_1, \ldots, x_n$ with $x_1\vec{a}_1 + \cdots + x_n\vec{a}_n = \vec{b}$ (by the definition of matrix-vector product) $\Leftrightarrow$ $\vec{b}$ is a linear combination of $\vec{a}_1, \ldots, \vec{a}_n$ $\Leftrightarrow$ $\vec{b} \in \text{span}\{\vec{a}_1, \ldots, \vec{a}_n\}$ (by the definition of span). ■

Four equivalent conditions

Theorem. Let A be an $m \times n$ matrix. The following are equivalent:

1. For each $\vec{b} \in \mathbb{R}^m$, the equation $A\vec{x} = \vec{b}$ has a solution.
2. Each $\vec{b} \in \mathbb{R}^m$ is a linear combination of the columns of A.
3. The span of the columns of A is $\mathbb{R}^m$ ("the columns of A span $\mathbb{R}^m$").
4. A has a pivot position in every row.

Proof. We show (a) $\Leftrightarrow$ (b) $\Leftrightarrow$ (c) $\Leftrightarrow$ (d).

(a) $\Leftrightarrow$ (b): By the previous proposition, $A\vec{x} = \vec{b}$ has a solution if and only if $\vec{b}$ is a linear combination of the columns of A. So (a) (for each $\vec{b}$, $A\vec{x} = \vec{b}$ has a solution) is equivalent to (b) (each $\vec{b}$ is a linear combination of the columns of A).

(b) $\Leftrightarrow$ (c): Saying each $\vec{b} \in \mathbb{R}^m$ is a linear combination of the columns of A is exactly saying $\mathbb{R}^m \subseteq \text{span}\{\text{columns of } A\}$. Since the columns of A are in $\mathbb{R}^m$, the span is a subset of $\mathbb{R}^m$. Hence $\text{span}\{\text{columns}\} = \mathbb{R}^m$, which is (c).

(c) $\Leftrightarrow$ (d): Suppose (c): the columns of A span $\mathbb{R}^m$. Then for each $\vec{b} \in \mathbb{R}^m$, the system with augmented matrix $\begin{pmatrix} \vec{a}_1 & \cdots & \vec{a}_n & \vec{b} \end{pmatrix}$ is consistent, so the last column is never a pivot column. This means no row of RREF(A) is all zeros (otherwise appending $\vec{b} = \vec{e}_i$ for the corresponding row would make the last column a pivot), so every row of RREF(A) has a pivot: (d). Conversely, suppose (d): every row of A has a pivot. Then RREF(A) has no zero rows, so for any $\vec{b}$, the augmented matrix $\begin{pmatrix} A & \vec{b} \end{pmatrix}$ cannot have a pivot in the last column (a pivot in the last column would require a zero row in the coefficient part, contradicting every row having a pivot). So the system is always consistent, giving (c).

■

3.3 Sums and Scalar Multiples

Definition. If $T, U : \mathbb{R}^n \to \mathbb{R}^m$ are linear with the same domain and codomain:

- *Sum:* $(T+U)(\vec{v}) = T(\vec{v}) + U(\vec{v})$.
- *Scalar multiple:* $(cT)(\vec{v}) = c\,T(\vec{v})$ for $c \in \mathbb{R}$.

Both $T+U$ and cT are linear transformations.

Definition. If $A = (a_{ij})$ and $B = (b_{ij})$ are $m \times n$ matrices:

- $A + B = (a_{ij} + b_{ij})$ (entry-wise sum, same size required).
- $cA = (c\,a_{ij})$ (scalar multiplication).

Proposition. The standard matrix of $T+U$ is $A+B$; the standard matrix of cT is cA.

Proof. Let A and B be the standard matrices of T and U respectively, so $T(\vec{v}) = A\vec{v}$ and $U(\vec{v}) = B\vec{v}$ for all $\vec{v}$. Then for every $\vec{v} \in \mathbb{R}^n$, $(T+U)(\vec{v}) = T(\vec{v}) + U(\vec{v}) = A\vec{v} + B\vec{v} = (A+B)\vec{v}$ (using entry-wise definition of matrix addition and linearity of matrix-vector product). So $A+B$ is a matrix representing $T+U$. By uniqueness of the standard matrix, $A+B$ is the standard matrix of $T+U$. Similarly, $(cT)(\vec{v}) = c\,T(\vec{v}) = c(A\vec{v}) = (cA)\vec{v}$ (using entry-wise scalar multiplication), so cA is the standard matrix of cT. ■

Properties of matrix addition. For matrices of the same size and $a, b \in \mathbb{R}$:

1. $A + B = B + A$.
2. $(A+B) + C = A + (B+C)$.
3. $A + 0 = A$ (zero matrix).
4. $a(A+B) = aA + aB$.
5. $(a+b)A = aA + bA$.
6. $a(bA) = (ab)A$.

Proof: Let $A = (a_{ij})$, $B = (b_{ij})$, $C = (c_{ij})$ be $m \times n$ matrices, and let 0 denote the $m \times n$ zero matrix. We prove each property entry-wise.

1. *Commutativity* $A + B = B + A$: The (i,j) entry of $A+B$ is $a_{ij} + b_{ij}$, and the (i,j) entry of $B+A$ is $b_{ij} + a_{ij}$. Since addition of real numbers is commutative, $a_{ij} + b_{ij} = b_{ij} + a_{ij}$ for all i, j. Hence $A + B = B + A$.
2. *Associativity* $(A+B)+C = A+(B+C)$: The (i,j) entry of $(A+B)+C$ is $(a_{ij}+b_{ij})+c_{ij}$, and the (i,j) entry of $A+(B+C)$ is $a_{ij}+(b_{ij}+c_{ij})$. Since addition of real numbers is associative, $(a_{ij} + b_{ij}) + c_{ij} = a_{ij} + (b_{ij} + c_{ij})$ for all i, j. Hence $(A+B)+C = A+(B+C)$.
3. *Additive identity* $A + 0 = A$: The (i,j) entry of $A+0$ is $a_{ij} + 0 = a_{ij}$, which is the (i,j) entry of A. Hence $A + 0 = A$.
4. *Scalar distributivity over matrix sum* $a(A+B) = aA + aB$: The (i,j) entry of $a(A+B)$ is $a(a_{ij} + b_{ij})$, and the (i,j) entry of $aA + aB$ is $aa_{ij} + ab_{ij}$. Since real multiplication distributes over real addition, $a(a_{ij} + b_{ij}) = aa_{ij} + ab_{ij}$ for all i, j. Hence $a(A+B) = aA + aB$.
5. *Scalar distributivity over scalar sum* $(a+b)A = aA + bA$: The (i,j) entry of $(a+b)A$ is $(a+b)a_{ij}$, and the (i,j) entry of $aA + bA$ is $aa_{ij} + ba_{ij}$. Since real multiplication distributes over real addition, $(a+b)a_{ij} = aa_{ij} + ba_{ij}$ for all i, j. Hence $(a+b)A = aA + bA$.
6. *Scalar associativity* $a(bA) = (ab)A$: The (i,j) entry of $a(bA)$ is $a(ba_{ij})$, and the (i,j) entry of $(ab)A$ is $(ab)a_{ij}$. By associativity of real multiplication, $a(ba_{ij}) = (ab)a_{ij}$ for all i, j. Hence $a(bA) = (ab)A$.

■

3.4 Composition and Matrix Multiplication

Definition. If $U : \mathbb{R}^n \to \mathbb{R}^m$ and $T : \mathbb{R}^m \to \mathbb{R}^k$, the *composition* $T \circ U : \mathbb{R}^n \to \mathbb{R}^k$ is $(T \circ U)(\vec{v}) = T(U(\vec{v}))$. If T and U are linear, then $T \circ U$ is linear.

Order matters

$U \circ T$ is not defined unless $k = n$. Even when both $T \circ U$ and $U \circ T$ are defined, generally $T \circ U \neq U \circ T$ (composition is not commutative).

Definition. If A is $k \times m$ (standard matrix of T) and B is $m \times n$ (standard matrix of U), then the *matrix product* AB is the $k \times n$ standard matrix of $T \circ U$:

$$AB = C \quad \text{where} \quad C\vec{v} = A(B\vec{v}) \text{ for all } \vec{v} \in \mathbb{R}^n.$$

Column-wise formula for AB

Theorem. If $B = \begin{pmatrix} \vec{b}_1 & \vec{b}_2 & \cdots & \vec{b}_n \end{pmatrix}$ then

$$\boxed{AB = \begin{pmatrix} A\vec{b}_1 & A\vec{b}_2 & \cdots & A\vec{b}_n \end{pmatrix}.}$$

Each column of AB is A times the corresponding column of B.

Proof. Let A be $m \times p$ with columns $\vec{a}_1, \ldots, \vec{a}_p$, and $B = \begin{pmatrix} \vec{b}_1 & \cdots & \vec{b}_n \end{pmatrix}$ be $p \times n$. By definition, AB is the matrix of the composite $T \circ U$ where $T(\vec{v}) = A\vec{v}$ and $U(\vec{v}) = B\vec{v}$. The jth column of AB is $(AB)\vec{e}_j$. Now $\vec{e}_j$ is the jth standard basis vector of $\mathbb{R}^n$, so $B\vec{e}_j = \vec{b}_j$ (the jth column of B). Thus $(AB)\vec{e}_j = A(B\vec{e}_j) = A\vec{b}_j$. So the jth column of AB is $A\vec{b}_j$, giving $AB = \begin{pmatrix} A\vec{b}_1 & A\vec{b}_2 & \cdots & A\vec{b}_n \end{pmatrix}$. ■

Entry-wise formula. The entry in row i, column j of AB is:

$$(AB)_{ij} = \sum_{k=1}^{m} a_{ik}b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{im}b_{mj}.$$

(Row i of A dotted with column j of B.)

Definition. The $n \times n$ *identity matrix* is

$$I_n = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{pmatrix}$$

(1's on diagonal, 0's elsewhere). It is the standard matrix of $\mathrm{id}_{\mathbb{R}^n}(\vec{v}) = \vec{v}$.

Proposition. If A is $m \times n$ then $I_m A = A = AI_n$.

Proof. For $I_m A$: Let $\vec{a}_1, \ldots, \vec{a}_n$ be the columns of A. By the column-wise formula, the jth column of $I_m A$ is $I_m\vec{a}_j$. Since I_m is the identity, $I_m\vec{a}_j = \vec{a}_j$ (the jth column of A). So $I_m A = \begin{pmatrix} \vec{a}_1 & \cdots & \vec{a}_n \end{pmatrix} = A$. For AI_n: the jth column of AI_n is $A\vec{e}_j$ where $\vec{e}_j$ is the jth standard basis vector of $\mathbb{R}^n$. By the fact that $A\vec{e}_j$ is the jth column of A, we get $AI_n = \begin{pmatrix} A\vec{e}_1 & \cdots & A\vec{e}_n \end{pmatrix} = \begin{pmatrix} \vec{a}_1 & \cdots & \vec{a}_n \end{pmatrix} = A$. ■

3.5 Properties of Matrix Multiplication

Proposition (Associativity). If A is $m \times n$, B is $n \times l$, C is $l \times k$, then $A(BC) = (AB)C$.

Proof (via function composition). Let $T_A(\vec{v}) = A\vec{v}$, $T_B(\vec{v}) = B\vec{v}$, $T_C(\vec{v}) = C\vec{v}$ be the corresponding linear transformations. The matrix BC represents $T_B \circ T_C$ and AB represents $T_A \circ T_B$. Then $A(BC)$ represents $T_A \circ (T_B \circ T_C)$, and $(AB)C$ represents $(T_A \circ T_B) \circ T_C$. But function composition is associative: $T_A \circ (T_B \circ T_C) = (T_A \circ T_B) \circ T_C$, since both evaluate to $T_A(T_B(T_C(\vec{v})))$ for any $\vec{v}$. Since the two products represent the same linear transformation and the standard matrix is unique, $A(BC) = (AB)C$. ■

Proposition (Distributivity). For matrices of compatible sizes and $r \in \mathbb{R}$:

- $A(B + C) = AB + AC$.
- $(A + B)C = AC + BC$.
- $r(AB) = (rA)B = A(rB)$.

Proof. We prove each. Let A be $m \times n$.

1. $A(B + C) = AB + AC$: Let $\vec{c}_1, \ldots, \vec{c}_k$ be the columns of C and $\vec{b}_1, \ldots, \vec{b}_k$ the columns of B. The jth column of $B + C$ is $\vec{b}_j + \vec{c}_j$, so the jth column of $A(B + C)$ is $A(\vec{b}_j + \vec{c}_j) = A\vec{b}_j + A\vec{c}_j$ (using linearity of matrix-vector product). This equals the jth column of AB plus the jth column of AC, i.e., the jth column of $AB + AC$. So $A(B + C) = AB + AC$.
2. $(A + B)C = AC + BC$: The jth column of $(A + B)C$ is $(A + B)\vec{c}_j = A\vec{c}_j + B\vec{c}_j$ (by definition of matrix addition), which is the jth column of AC plus the jth column of BC, i.e., the jth column of $AC + BC$. So $(A + B)C = AC + BC$.
3. $r(AB) = (rA)B = A(rB)$: The jth column of $r(AB)$ is $r(A\vec{b}_j) = (rA)\vec{b}_j$ (linearity), which is the jth column of $(rA)B$. Also $r(A\vec{b}_j) = A(r\vec{b}_j)$ (linearity), which is the jth column of $A(rB)$ (whose jth column is $r\vec{b}_j$). So $r(AB) = (rA)B = A(rB)$.

■

Practice Problem #4.5 (Binomial expansion for nilpotent B)

Problem. Let B be a square matrix satisfying $B^4 = 0$ but $B^3 \neq 0$ (nilpotent of index 4). Prove that for any positive integer n,

$$(I + B)^n = I + nB + \binom{n}{2}B^2 + \binom{n}{3}B^3.$$

Solution. Since matrix multiplication is distributive and associative, we can expand $(I + B)^n$ by the binomial theorem:

$$(I + B)^n = \sum_{k=0}^{n} \binom{n}{k} I^{n-k} B^k = \sum_{k=0}^{n} \binom{n}{k} B^k.$$

But $B^k = 0$ for all $k \geq 4$ (since $B^4 = 0$ implies $B^5 = B^4 B = 0$, etc.). So only the terms $k = 0, 1, 2, 3$ survive:

$$(I + B)^n = \binom{n}{0}B^0 + \binom{n}{1}B^1 + \binom{n}{2}B^2 + \binom{n}{3}B^3 = I + nB + \binom{n}{2}B^2 + \binom{n}{3}B^3.$$

Clever trick: nilpotent truncates the binomial theorem

When B is nilpotent of index k, the expansion of $(I + B)^n$ terminates at B^{k-1}. This is the matrix analogue of $(1 + x)^n$ for $x^k = 0$. The binomial theorem for matrices holds whenever I and B commute, which they always do.

Practice Problem #4.8 (Diagonalization gives closed form for A^n)

Problem. Let $A = \begin{pmatrix} a & b \\ b & a \end{pmatrix}$. Diagonalize A and derive a closed formula for A^n.

Solution. The eigenvalues of A are $a+b$ and $a-b$ (since $\det(A-\lambda I) = (a-\lambda)^2 - b^2 = 0$ gives $\lambda = a \pm b$). Corresponding eigenvectors are $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ for $\lambda = a+b$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ for $\lambda = a-b$. So with $P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ and $D = \begin{pmatrix} a+b & 0 \\ 0 & a-b \end{pmatrix}$, we have $A = PDP^{-1}$. Compute $P^{-1} = \frac{1}{-2}\begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. Then

$$A^n = PD^nP^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} (a+b)^n & 0 \\ 0 & (a-b)^n \end{pmatrix}\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Multiplying out:

$$A^n = \frac{1}{2}\begin{pmatrix} (a+b)^n + (a-b)^n & (a+b)^n - (a-b)^n \\ (a+b)^n - (a-b)^n & (a+b)^n + (a-b)^n \end{pmatrix}.$$

Clever trick: diagonalization turns powers into diagonal powers

If $A = PDP^{-1}$, then $A^n = PD^nP^{-1}$. The diagonal matrix D^n is trivial: just raise each diagonal entry to the nth power. This is the standard recipe for closed-form matrix powers.

Practice Problem #4.12 (Commutator of upper unipotent 4×4 matrices)

Problem. Let A and B be 4×4 upper unipotent matrices (diagonal entries all 1, entries below the diagonal all 0). Define the commutator $[A, B] = ABA^{-1}B^{-1}$. Prove that $[A, B]$ has diagonal entries 1, super-diagonal entries 0, and is thus upper unipotent with "one more zero band."

Solution. For upper unipotent matrices, the inverse is also upper unipotent (the RREF of an upper triangular matrix with diagonal 1 is I, so the inverse computed via row reduction stays upper triangular with diagonal 1). So A^{-1} and B^{-1} are also upper unipotent.

Now, the product of two upper triangular matrices is upper triangular, and the diagonal entries of the product are the products of the diagonal entries. Since all four matrices have diagonal 1, the product $ABA^{-1}B^{-1}$ has diagonal $1 \cdot 1 \cdot 1 \cdot 1 = 1$, so it is upper unipotent.

For the super-diagonal: let U denote the super-diagonal part. Write $A = I + A_1 + A_2 + A_3$ where A_k has nonzero entries only on the kth super-diagonal (A_1 is super-diagonal, A_2 is the next, etc.). For upper triangular matrices, the $(i, i+1)$ entry of a product depends only on the first super-diagonal entries of the factors. In particular, $(AB)_{i,i+1} = A_{i,i}B_{i,i+1} + A_{i,i+1}B_{i+1,i+1} = B_{i,i+1} + A_{i,i+1}$ (since $A_{i,i} = B_{i+1,i+1} = 1$). So the super-diagonal of AB is the sum of the super-diagonals of A and B.

Now A^{-1} has super-diagonal $-A_1$ (from $(I + A_1)(I - A_1) = I - A_1^2$ and adjusting; more generally, $(I+M)^{-1} = I - M + M^2 - M^3 + \cdots$ which terminates for nilpotent M). Similarly B^{-1} has super-diagonal $-B_1$. So ABA^{-1} has super-diagonal $A_1 + B_1 + (-A_1) = B_1$, and $ABA^{-1}B^{-1}$ has super-diagonal $B_1 + (-B_1) = 0$. Hence the commutator has zero super-diagonal.

Clever trick: keep track of diagonal bands

When dealing with upper triangular matrices, work band by band. The diagonal, first super-diagonal, second super-diagonal, etc., each evolve independently at the

leading order. For upper unipotent matrices, taking the inverse negates the first super-diagonal. The commutator then kills the first super-diagonal.

Block matrices: rules of the game

A *block matrix* is a matrix viewed as an array of smaller matrices (blocks). All the rules below are consequences of the ordinary definition of matrix multiplication; the only requirement is always *compatible block sizes.*

1. *Addition:* blocks of the same position must have the same size; then add blockwise,
$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} + \begin{pmatrix} A' & B' \\ C' & D' \end{pmatrix} = \begin{pmatrix} A+A' & B+B' \\ C+C' & D+D' \end{pmatrix}.$$
2. *Multiplication (block row by block column):* if the column partition of the left factor matches the row partition of the right factor, multiply as if the blocks were scalars,
$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} E & F \\ G & H \end{pmatrix} = \begin{pmatrix} AE+BG & AF+BH \\ CE+DG & CF+DH \end{pmatrix}.$$
The order of the factors inside each product must be preserved (AE, not EA).
3. *Column view:* $A\begin{pmatrix} \vec{b}_1 & \cdots & \vec{b}_n \end{pmatrix} = \begin{pmatrix} A\vec{b}_1 & \cdots & A\vec{b}_n \end{pmatrix}$; each column of AB is a linear combination of the columns of A with coefficients from the corresponding column of B.
4. *Row view:* each row of AB is a linear combination of the rows of B with coefficients from the corresponding row of A; hence $\operatorname{Col}(AB) \subseteq \operatorname{Col} A$ and $\operatorname{Row}(AB) \subseteq \operatorname{Row} B$.
5. *Transpose reverses block positions:*
$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}^\top = \begin{pmatrix} A^\top & C^\top \\ B^\top & D^\top \end{pmatrix}.$$
6. *Block diagonal matrices* $\operatorname{diag}(D_1, \ldots, D_k)$ multiply, invert, and exponentiate block by block:
$$\operatorname{diag}(D_1, \ldots, D_k)\ \operatorname{diag}(E_1, \ldots, E_k) = \operatorname{diag}(D_1E_1, \ldots, D_kE_k),$$
$$\operatorname{diag}(D_1, \ldots, D_k)^{-1} = \operatorname{diag}(D_1^{-1}, \ldots, D_k^{-1}),$$
with the inverse existing iff every D_i is invertible. Also $\det \operatorname{diag}(D_1, \ldots, D_k) = \det D_1 \cdots \det D_k$ and $\operatorname{rank} = \sum_i \operatorname{rank} D_i$.
7. *Block triangular matrices:* if $X = \begin{pmatrix} A & B \\ 0 & D \end{pmatrix}$ with A, D square, then $\det X = \det A \cdot \det D$; X is invertible iff both A and D are invertible, and then
$$X^{-1} = \begin{pmatrix} A^{-1} & -A^{-1}BD^{-1} \\ 0 & D^{-1} \end{pmatrix}.$$
8. *Off-diagonal blocks:* $\begin{pmatrix} 0 & C_1 \\ C_2 & 0 \end{pmatrix}^{-1} = \begin{pmatrix} 0 & C_2^{-1} \\ C_1^{-1} & 0 \end{pmatrix}$ when C_1, C_2 are square invertible — the inverse *swaps* the two blocks (Practice #4.7).
9. *Scalar blocks:* a block cI behaves exactly like the scalar c; e.g. $(aI + bJ)^2$ expands by ordinary algebra (Midterm 2021 #7).
10. *Block row operations = block elementary matrices:* left-multiplying by $\begin{pmatrix} I & X \\ 0 & I \end{pmatrix}$ adds X times the second block row to the first, leaving det unchanged; block row swaps and invertible block scalings multiply det by ± 1 and det of the scaling block respectively. This is the rigorous way to evaluate determinants of 2×2 block matrices by elimination.

11. *Rank relations:* $\operatorname{rank}\begin{pmatrix} A & B \end{pmatrix} \leq \operatorname{rank} A + \operatorname{rank} B$, $\operatorname{rank}\begin{pmatrix} A \\ B \end{pmatrix} \leq \operatorname{rank} A + \operatorname{rank} B$, and $\operatorname{rank}\begin{pmatrix} A & 0 \\ 0 & D \end{pmatrix} = \operatorname{rank} A + \operatorname{rank} D$ (Practice #3.3, #4.3).
12. *Kernel of a stacked matrix:* $\operatorname{Nul}\begin{pmatrix} A \\ B \end{pmatrix} = \operatorname{Nul} A \cap \operatorname{Nul} B$; $\operatorname{Col}\begin{pmatrix} A & B \end{pmatrix} = \operatorname{Col} A + \operatorname{Col} B$.

3.6 Pathologies of Matrix Multiplication

What matrix multiplication does NOT satisfy

1. AB is defined only if #columns of A = #rows of B.
2. Even if AB and BA are both defined, generally $AB \neq BA$.
3. $AB = AC$ does **not** imply $B = C$ (no cancellation).
4. $AB = 0$ can happen even when $A \neq 0$ and $B \neq 0$ (zero divisors).

Definition. Square matrices A and B of the same size *commute* if $AB = BA$.

3.7 Elementary Matrices and Transpose

Definition. An *elementary matrix* is a matrix formed by performing exactly one row operation on the identity matrix I_n. If E is elementary, then EA is obtained from A by doing that row operation.

Row operations = left multiplication by elementary matrices

Every row operation on A corresponds to left-multiplying A by an elementary matrix E.

Definition. The *transpose* of an $m \times n$ matrix A is the $n \times m$ matrix $A^\top$ formed by flipping A across the main diagonal (rows become columns).

Properties of transpose:

- $(A^\top)^\top = A$.
- $(A+B)^\top = A^\top + B^\top$ and $(cA)^\top = c(A^\top)$.
- $(AB)^\top = B^\top A^\top$ (order reverses).

Proof. We prove each property.

1. $(A^\top)^\top = A$: The (i,j) entry of $A^\top$ is a_{ji}. So the (i,j) entry of $(A^\top)^\top$ is $(A^\top)_{ji} = a_{ij}$. Hence $(A^\top)^\top = A$.
2. $(A+B)^\top = A^\top + B^\top$ and $(cA)^\top = c(A^\top)$: The (i,j) entry of $(A+B)^\top$ is $(A+B)_{ji} = a_{ji} + b_{ji}$, which equals $a_{ji} + b_{ji} = (A^\top)_{ij} + (B^\top)_{ij} = (A^\top + B^\top)_{ij}$. So $(A+B)^\top = A^\top + B^\top$. Similarly, the (i,j) entry of $(cA)^\top$ is $(cA)_{ji} = c\,a_{ji} = c(A^\top)_{ij} = (cA^\top)_{ij}$, so $(cA)^\top = c(A^\top)$.
3. $(AB)^\top = B^\top A^\top$: Let A be $m \times n$ and B be $n \times p$. Then AB is $m \times p$, so $(AB)^\top$ is $p \times m$. Also $B^\top$ is $p \times n$ and $A^\top$ is $n \times m$, so $B^\top A^\top$ is $p \times m$ (sizes match). The (i,j) entry of $(AB)^\top$ is $(AB)_{ji} = \sum_{k=1}^{n} a_{jk} b_{ki}$. The (i,j) entry of $B^\top A^\top$ is $\sum_{k=1}^{n} (B^\top)_{ik}(A^\top)_{kj} = \sum_{k=1}^{n} b_{ki} a_{jk} = \sum_{k=1}^{n} a_{jk} b_{ki}$. These are equal, so $(AB)^\top = B^\top A^\top$.

■

Practice Problem #4.2 (Column operations via right multiplication)

Problem. Explain how to perform elementary *column* operations on a matrix A using matrix multiplication. Specifically: given an elementary row operation represented by matrix E, what right-multiplication achieves the corresponding column operation?

Solution. Elementary row operations correspond to left multiplication: EA performs the row operation on A. To perform the same operation on columns, we use *right* multiplication by the transpose of the elementary matrix.

Let E be an elementary matrix. Then:

- EA performs the row operation represented by E on A.
- $AE^\top$ performs the corresponding *column* operation on A.

For example: if E is the matrix that swaps rows i and j of I, then $E^\top = E$ (swapping rows i and j is symmetric), and $AE^\top = AE$ swaps columns i and j of A. If E is the matrix that adds c times row j to row i of I, then $E^\top$ adds c times *column* j to *column* i of I, so $AE^\top$ adds c times column i to column j of A (note the index swap). In general, $AE^\top$ has the effect on columns that E has on rows, with row indices becoming column indices.

Clever trick: row ops = left, column ops = right with $E^\top$

Left-multiplication by E acts on rows; right-multiplication by $E^\top$ acts on columns. This duality is one of the most useful algebraic facts about elementary operations.

Practice Problem #4.15 (Anti-transpose and its properties)

Problem. Let J be the $n \times n$ matrix with 1's on the anti-diagonal (so $J_{i,n+1-i} = 1$ and all other entries 0). Define the *anti-transpose* of A by $A^\dagger = JA^\top J$. Prove that $(AB)^\dagger = B^\dagger A^\dagger$.

Solution. Compute directly using the defining property $(XY)^\top = Y^\top X^\top$:

$$(AB)^\dagger = J(AB)^\top J = J(B^\top A^\top)J.$$

Now insert $J^2 = I$ between $B^\top$ and $A^\top$ (since J is a permutation matrix of order 2, applying the anti-diagonal flip twice gives the identity):

$$JB^\top A^\top J = JB^\top IA^\top J = JB^\top (JJ)A^\top J = (JB^\top J)(JA^\top J) = B^\dagger A^\dagger.$$

Hence $(AB)^\dagger = B^\dagger A^\dagger$, same order reversal as the ordinary transpose.

Clever trick: $J^2 = I$ is the key

Inserting $I = J^2$ at the right moment lets us group the terms into $(JB^\top J)$ and $(JA^\top J)$, which are exactly $B^\dagger$ and $A^\dagger$. The anti-transpose is a "flipped" version of the transpose, and it reverses order in the same way.

3.8 Invertible Matrices

Definition. $f : X \to Y$ is *invertible* (or *bijective*) if f is both onto and one-to-one.

Proposition. f is invertible $\Leftrightarrow$ for each $b \in Y$ there is exactly one $a \in X$ with $f(a) = b$.

Proof. ($\Rightarrow$) Suppose f is invertible (bijective), so f is one-to-one and onto. Since f is onto, for each $b \in Y$ there exists $a \in X$ with $f(a) = b$. Since f is one-to-one, this a is unique. ($\Leftarrow$) Suppose for each $b \in Y$ there is exactly one $a \in X$ with $f(a) = b$.

Then f is onto (every b has a preimage) and one-to-one (if $f(a_1) = f(a_2)$, then both a_1, a_2 are preimages of the same b, so $a_1 = a_2$ by uniqueness). So f is bijective, i.e., invertible. ■

Proposition. f is invertible $\Leftrightarrow$ there exists a unique function $f^{-1} : Y \to X$ (the *inverse*) such that $f \circ f^{-1} = \mathrm{id}_Y$ and $f^{-1} \circ f = \mathrm{id}_X$.

Proof. ($\Rightarrow$) If f is invertible, for each $b \in Y$ define $f^{-1}(b)$ to be the unique $a \in X$ with $f(a) = b$ (exists and unique by the previous proposition). Then $f(f^{-1}(b)) = b$ so $f \circ f^{-1} = \mathrm{id}_Y$, and $f^{-1}(f(a)) = a$ so $f^{-1} \circ f = \mathrm{id}_X$. For uniqueness: if $g : Y \to X$ satisfies $f \circ g = \mathrm{id}_Y$, then for any $b \in Y$, $f(g(b)) = b$, so $g(b)$ is a preimage of b under f, which is unique; hence $g(b) = f^{-1}(b)$. So f^{-1} is unique. ($\Leftarrow$) If such f^{-1} exists, then for each $b \in Y$, $f(f^{-1}(b)) = b$ shows f is onto, and if $f(a_1) = f(a_2)$, then applying f^{-1} gives $a_1 = f^{-1}(f(a_1)) = f^{-1}(f(a_2)) = a_2$, so f is one-to-one. Thus f is invertible. ■

Proposition. If $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear and invertible, then $n = m$ and T^{-1} is linear.

Proof. Since T is invertible, it is one-to-one and onto. The fact T is onto requires $n \geq m$ (a linear $T : \mathbb{R}^n \to \mathbb{R}^m$ can be onto only if $n \geq m$, since the columns of the standard matrix must span $\mathbb{R}^m$, requiring at least m columns). The fact T is one-to-one requires $n \leq m$ (the columns of the standard matrix must be linearly independent, requiring at most m columns in $\mathbb{R}^m$). Combining, $n = m$. Now we show $T^{-1} : \mathbb{R}^m \to \mathbb{R}^m$ is linear. For $\vec{u}, \vec{v} \in \mathbb{R}^m$, let $\vec{a} = T^{-1}(\vec{u})$ and $\vec{b} = T^{-1}(\vec{v})$, so $T(\vec{a}) = \vec{u}$ and $T(\vec{b}) = \vec{v}$. Then $T(\vec{a} + \vec{b}) = T(\vec{a}) + T(\vec{b}) = \vec{u} + \vec{v}$ (linearity of T), so $T^{-1}(\vec{u} + \vec{v}) = \vec{a} + \vec{b} = T^{-1}(\vec{u}) + T^{-1}(\vec{v})$. Similarly, $T(c\vec{a}) = cT(\vec{a}) = c\vec{u}$, so $T^{-1}(c\vec{u}) = c\vec{a} = cT^{-1}(\vec{u})$. Hence T^{-1} satisfies both linearity conditions. ■

Definition. An $n \times n$ matrix A is *invertible* (or *non-singular*) if the linear function $T(\vec{x}) = A\vec{x}$ is invertible. The *inverse* of A is the unique matrix A^{-1} such that $T^{-1}(\vec{x}) = A^{-1}\vec{x}$.

Equivalent conditions for invertibility

Proposition. Let A be $n \times n$. The following are equivalent:
1. A is invertible.
2. There exists A^{-1} with $AA^{-1} = A^{-1}A = I_n$.
3. For each $\vec{b} \in \mathbb{R}^n$, $A\vec{x} = \vec{b}$ has a unique solution.
4. $\mathrm{RREF}(A) = I_n$.

Proof. We show (1) $\Leftrightarrow$ (2) $\Leftrightarrow$ (3) $\Leftrightarrow$ (4).

(1) $\Leftrightarrow$ (2): (1) says $T(\vec{x}) = A\vec{x}$ is invertible as a function. By the proposition on invertible functions, this is equivalent to the existence of T^{-1} with $T \circ T^{-1} = T^{-1} \circ T = \mathrm{id}$. Since T^{-1} is linear (by the previous proposition) with standard matrix A^{-1}, this means $AA^{-1} = A^{-1}A = I_n$, which is (2).

(2) $\Leftrightarrow$ (3): If (2), then for any $\vec{b}$, $\vec{x} = A^{-1}\vec{b}$ satisfies $A\vec{x} = A(A^{-1}\vec{b}) = (AA^{-1})\vec{b} = I_n\vec{b} = \vec{b}$, so a solution exists. For uniqueness, if $A\vec{x} = \vec{b}$, then multiplying by A^{-1} on the left gives $\vec{x} = A^{-1}\vec{b}$. So the solution is unique. Conversely, if (3), then $A\vec{x} = \vec{e}_j$ has a unique solution $\vec{c}_j$ for each j. Let $C = \begin{pmatrix} \vec{c}_1 & \cdots & \vec{c}_n \end{pmatrix}$. Then $AC = \begin{pmatrix} A\vec{c}_1 & \cdots & A\vec{c}_n \end{pmatrix} = \begin{pmatrix} \vec{e}_1 & \cdots & \vec{e}_n \end{pmatrix} = I_n$. So $AC = I_n$, and (for square matrices) this gives $C = A^{-1}$, i.e., $AA^{-1} = I_n$.

(3) $\Leftrightarrow$ (4): $A\vec{x} = \vec{b}$ has a unique solution for each $\vec{b}$ iff the augmented matrix $\begin{pmatrix} A & \vec{b} \end{pmatrix}$ has a unique solution for each $\vec{b}$. By the RREF criterion, this requires no free variables (every column of A is a pivot column) and the last column never being a pivot.

No free variables means every column of A is a pivot column, i.e., $\mathrm{RREF}(A)$ has n pivots. Since A is $n \times n$ with n pivots (one in each column, hence one in each row), $\mathrm{RREF}(A) = I_n$. This is (4).

■

2×2 inverse formula

Theorem. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

1. A is invertible if and only if $ad - bc \neq 0$.
2. If $ad - bc \neq 0$, then $A^{-1} = \dfrac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

Proof. We first verify (2) by direct computation, assuming $ad - bc \neq 0$. Let $B = \dfrac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. Then

$$AB = \frac{1}{ad-bc} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{ad-bc} \begin{pmatrix} ad-bc & -ab+ba \\ cd-dc & -cb+da \end{pmatrix}$$
$$= \frac{1}{ad-bc} \begin{pmatrix} ad-bc & 0 \\ 0 & ad-bc \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2.$$

Similarly, $BA = \dfrac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \dfrac{1}{ad-bc} \begin{pmatrix} da-bc & db-bd \\ -ca+ac & -cb+ad \end{pmatrix} = I_2$. So $B = A^{-1}$, confirming (2). For (1): If $ad - bc \neq 0$, the computation above gives an explicit A^{-1}, so A is invertible. Conversely, suppose A is invertible, so there exists A^{-1} with $AA^{-1} = I_2$. Then $\det(AA^{-1}) = \det(I_2) = 1$, and if we know $\det(AB) = (\det A)(\det B)$, we get $(\det A)(\det A^{-1}) = 1$, so $\det A \neq 0$. Alternatively, a direct proof: if $ad - bc = 0$, then the columns $\begin{pmatrix} a \\ c \end{pmatrix}$ and $\begin{pmatrix} b \\ d \end{pmatrix}$ satisfy $a \cdot d - b \cdot c = 0$, meaning they are linearly dependent (one is a scalar multiple of the other, or both are zero). If the columns are linearly dependent, then $A\vec{x} = \vec{0}$ has a nontrivial solution, so $T(\vec{x}) = A\vec{x}$ is not one-to-one, hence not invertible. So A invertible implies $ad-bc \neq 0$.
■

Recipe for 2×2 inverse

1. Swap diagonal entries.
2. Negate off-diagonal entries.
3. Divide by $ad - bc$.

Theorem. Let A and B be $n \times n$ matrices.

1. If A is invertible, then $(A^{-1})^{-1} = A$.
2. If A and B are invertible, then AB is invertible with $(AB)^{-1} = B^{-1}A^{-1}$.
3. If A is invertible, then $A^\top$ is invertible with $(A^\top)^{-1} = (A^{-1})^\top$.

Proof. We verify each by checking the defining property of the inverse (i.e., finding a matrix C with $AC = CA = I_n$).

1. Since A is invertible, $AA^{-1} = A^{-1}A = I_n$. Reading this as "(the inverse of A^{-1}) times A^{-1} equals I" and "A^{-1} times (the inverse of A^{-1}) equals I", we see A satisfies the property of being the inverse of A^{-1}. By uniqueness of the inverse, $(A^{-1})^{-1} = A$.
2. Compute:
$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = A\,I_n\,A^{-1} = AA^{-1} = I_n,$$
$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}I_nB = B^{-1}B = I_n.$$

So $B^{-1}A^{-1}$ serves as the inverse of AB. By uniqueness, $(AB)^{-1} = B^{-1}A^{-1}$.

3. We use $(XY)^\top = Y^\top X^\top$. Take $X = A$ and $Y = A^{-1}$: $(AA^{-1})^\top = (A^{-1})^\top A^\top$. But $AA^{-1} = I_n$, so $(I_n)^\top = I_n = (A^{-1})^\top A^\top$. Similarly, $(A^{-1}A)^\top = A^\top (A^{-1})^\top = I_n$. So $(A^{-1})^\top$ serves as the inverse of $A^\top$. By uniqueness, $(A^\top)^{-1} = (A^{-1})^\top$.

■

General formula for products

$(ABC\cdots Z)^{-1} = Z^{-1}\cdots C^{-1}B^{-1}A^{-1}$ (inverses in reverse order).

3.9 Computing A^{-1}

Algorithm for A^{-1}

Form the $n \times 2n$ matrix $(A \quad I_n)$ and row reduce to reduced echelon form. If A is invertible, then

$$\text{RREF}\left((A \quad I_n)\right) = \left(I_n \quad A^{-1}\right).$$

Take the last n columns as A^{-1}. If $\text{RREF}(A) \neq I_n$, then A is not invertible.

Only square matrices can be invertible

If A is not square, it is possible to have $AB = I$ for some B, but A is still not invertible (there is no A' with both $A'A = I$ and $AA' = I$). For square matrices, $AB = I_n$ suffices to conclude $B = A^{-1}$.

4 Linear Transformations

4.1 Linear Independence

Definition. A list of p vectors $\vec{v}_1, \ldots, \vec{v}_p \in \mathbb{R}^n$ is *linearly independent* if

$$\vec{v}_i \notin \text{span}\{\vec{v}_1, \ldots, \vec{v}_{i-1}, \vec{v}_{i+1}, \ldots, \vec{v}_p\} \quad \text{for all } i = 1, \ldots, p.$$

Special cases:

- Empty list: always linearly independent (by convention).
- One vector $\vec{v}_1$: independent iff $\vec{v}_1 \neq \vec{0}$.
- Two vectors $\vec{v}_1, \vec{v}_2$: independent iff neither is a scalar multiple of the other.
- Three vectors: independent iff no one is in the span of the other two.

Equivalent characterization

Proposition. $\vec{v}_1, \ldots, \vec{v}_p$ are linearly independent if and only if the vector equation

$$\boxed{x_1\vec{v}_1 + x_2\vec{v}_2 + \cdots + x_p\vec{v}_p = \vec{0}}$$

has **no other solutions** besides the trivial solution $x_1 = x_2 = \cdots = x_p = 0$. A nontrivial solution is called a *linear dependence* among the vectors.

Proof. We prove both directions.

($\Rightarrow$) Suppose $\vec{v}_1, \ldots, \vec{v}_p$ are linearly independent. By definition, $\vec{v}_i \notin \text{span}\{\vec{v}_j\}_{j\neq i}$ for all i. Suppose for contradiction that $x_1\vec{v}_1 + \cdots + x_p\vec{v}_p = \vec{0}$ has a nontrivial solution,

i.e., some $x_i \neq 0$. Then

$$\vec{v}_i = -\sum_{j \neq i} \frac{x_j}{x_i} \vec{v}_j,$$

which expresses $\vec{v}_i$ as a linear combination of the other vectors. This means $\vec{v}_i \in \text{span}\{\vec{v}_1, \ldots, \vec{v}_{i-1}, \vec{v}_{i+1}, \ldots, \vec{v}_p\}$, contradicting the linear independence. So only the trivial solution exists. ■

($\Leftarrow$) Suppose the only solution to $x_1\vec{v}_1 + \cdots + x_p\vec{v}_p = \vec{0}$ is the trivial one. We show each $\vec{v}_i \notin \text{span}\{\vec{v}_1, \ldots, \vec{v}_{i-1}, \vec{v}_{i+1}, \ldots, \vec{v}_p\}$. Suppose for contradiction $\vec{v}_i = \sum_{j \neq i} a_j \vec{v}_j$ for some scalars a_j. Then setting $x_i = -1$ and $x_j = a_j$ for $j \neq i$ gives

$$\begin{aligned} x_1\vec{v}_1 + \cdots + x_p\vec{v}_p &= -\vec{v}_i + \sum_{j \neq i} a_j \vec{v}_j \\ &= -\vec{v}_i + \vec{v}_i = \vec{0}, \end{aligned}$$

a nontrivial solution (since $x_i = -1 \neq 0$). Contradiction. So the list is linearly independent.

■

Detecting linear independence

Corollary. The columns of a matrix A are linearly independent if and only if A has a pivot position in **every column**.

Proof. Let $A = (\vec{v}_1 \ \cdots \ \vec{v}_p)$. By the proposition, the columns are linearly independent iff $x_1\vec{v}_1 + \cdots + x_p\vec{v}_p = \vec{0}$ has only the trivial solution, i.e., $A\vec{x} = \vec{0}$ has only the trivial solution $\vec{x} = \vec{0}$. By the RREF criterion, this homogeneous system has a unique solution iff there are no free variables, i.e., every column of A is a pivot column. So the columns of A are linearly independent iff A has a pivot in every column. ■

Algorithm: Form the matrix $A = (\vec{v}_1 \ \ \vec{v}_2 \ \ \cdots \ \ \vec{v}_p)$, reduce to echelon form, and check if every column has a pivot.

Practice Problem #3.1 (Row-deletion linear transformations)

Problem. Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be the linear transformation that deletes the second entry of a vector:

$$T\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_3 \\ x_4 \end{pmatrix}.$$

Find the standard matrix of T and determine whether T is one-to-one, onto, or neither.

Solution. The standard matrix A has columns $T(\vec{e}_1), T(\vec{e}_2), T(\vec{e}_3), T(\vec{e}_4)$.

$$A = (T(\vec{e}_1) \ \ T(\vec{e}_2) \ \ T(\vec{e}_3) \ \ T(\vec{e}_4)) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

This 3×4 matrix has pivots in columns 1, 3, 4 but **not** column 2. So:

- T is **not** one-to-one (there is a free variable x_2, so $A\vec{x} = \vec{0}$ has nontrivial solutions such as $\vec{x} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$).
- T **is** onto (every row has a pivot, so $\text{Col}\, A = \mathbb{R}^3$).

Practice Problem #3.2 (Reflection and composition)

Problem. Let $\vec{v} \in \mathbb{R}^n$ be a nonzero vector. The *reflection* across the hyperplane perpendicular to $\vec{v}$ is the linear map

$$r_{\vec{v}}(\vec{x}) = \vec{x} - 2\frac{\vec{v}^\top \vec{x}}{\vec{v}^\top \vec{v}}\vec{v}.$$

Prove that $r_{\vec{v}}$ is its own inverse (i.e., $r_{\vec{v}}^2 = \mathrm{id}$), and that the composition of two reflections is a rotation.

Solution. To verify $r_{\vec{v}}^2 = \mathrm{id}$, apply $r_{\vec{v}}$ twice:

$$\begin{aligned} r_{\vec{v}}(r_{\vec{v}}(\vec{x})) &= r_{\vec{v}}\left(\vec{x} - 2\frac{\vec{v}^\top \vec{x}}{\vec{v}^\top \vec{v}}\vec{v}\right) \\ &= \left(\vec{x} - 2\frac{\vec{v}^\top \vec{x}}{\vec{v}^\top \vec{v}}\vec{v}\right) - 2\frac{\vec{v}^\top\left(\vec{x} - 2\frac{\vec{v}^\top \vec{x}}{\vec{v}^\top \vec{v}}\vec{v}\right)}{\vec{v}^\top \vec{v}}\vec{v} \\ &= \vec{x} - 2\frac{\vec{v}^\top \vec{x}}{\vec{v}^\top \vec{v}}\vec{v} - 2\frac{\vec{v}^\top \vec{x} - 2\frac{\vec{v}^\top \vec{x}}{\vec{v}^\top \vec{v}}\vec{v}^\top \vec{v}}{\vec{v}^\top \vec{v}}\vec{v} \\ &= \vec{x} - 2\frac{\vec{v}^\top \vec{x}}{\vec{v}^\top \vec{v}}\vec{v} - 2\frac{\vec{v}^\top \vec{x} - 2\vec{v}^\top \vec{x}}{\vec{v}^\top \vec{v}}\vec{v} \\ &= \vec{x} - 2\frac{\vec{v}^\top \vec{x}}{\vec{v}^\top \vec{v}}\vec{v} - 2\frac{-\vec{v}^\top \vec{x}}{\vec{v}^\top \vec{v}}\vec{v} = \vec{x}. \end{aligned}$$

So $r_{\vec{v}}^2 = \mathrm{id}$, meaning a reflection is invertible and is its own inverse.

For two reflections $r_{\vec{u}}$ and $r_{\vec{v}}$: each has determinant -1 (a reflection flips orientation). The composition $r_{\vec{u}} r_{\vec{v}}$ has determinant $\det(r_{\vec{u}})\det(r_{\vec{v}}) = (-1)(-1) = 1$. Orthogonal transformations with determinant 1 are rotations. More explicitly, in $\mathbb{R}^2$ the composition of two reflections across lines at angle θ produces a rotation by angle 2θ.

Clever trick: reflections are involutions

A reflection satisfies $r^2 = \mathrm{id}$ and $\det r = -1$. Composing two gives determinant $+1$, restoring orientation, which is a rotation in odd dimensions (and an even-dimensional rotation, possibly plus a sign flip, in even dimensions).

Practice Problem #3.10 (Finite vs linear: one-to-one + bounded dimension)

Problem. Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be linear.

1. Prove that if T is one-to-one and $m \leq n$, then T is onto.
2. Prove that if T is onto and $m \leq n$, then T is one-to-one.

Solution. Let A be the standard matrix of T (size $m \times n$).

(1) One-to-one means the columns of A are linearly independent, so every column of A is a pivot column. Since there are n columns and at most m rows, $n \leq m$. But we are given $m \leq n$, so $n = m$. Thus A is square with pivots in all columns. In a square matrix, pivots in all columns $\Leftrightarrow$ pivots in all rows $\Leftrightarrow$ A is invertible $\Leftrightarrow$ T is onto.

(2) Onto means A has a pivot in every row. There are m rows, so there are at least m pivot columns. But there are only n columns total, so $m \leq n$ (which is given). If $n > m$, there could be non-pivot columns (free variables), so T need not be one-to-one. But the statement as written requires $m \leq n$; when $m < n$ it is possible to have pivots in all rows but not all columns (e.g., $A = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}$ for $T : \mathbb{R}^3 \to \mathbb{R}$ is onto but not one-to-one). So part (2) as stated needs $m = n$: for a linear map between spaces of equal dimension, onto $\Leftrightarrow$ one-to-one $\Leftrightarrow$ invertible.

Practice Problem #3.12 (Linear map sending unit square to parallelogram)

Problem. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be linear. Suppose T maps the unit square $\{\begin{pmatrix} x \\ y \end{pmatrix} : 0 \leq x, y \leq 1\}$ bijectively onto a parallelogram with vertices at 0, $\vec{p}$, $\vec{q}$, and $\vec{p} + \vec{q}$. What must be the columns of the standard matrix of T?

Solution. The unit square has corners $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Since T is linear, $T\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ must map to the origin vertex. Now $T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \vec{p}$ and $T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \vec{q}$ (or vice versa), because the edges adjacent to the origin must map to the edges adjacent to the origin in the image parallelogram.
The standard matrix $A = (T(\vec{e}_1) \quad T(\vec{e}_2))$ must therefore have columns either $(\vec{p} \quad \vec{q})$ or $(\vec{q} \quad \vec{p})$. Since T is bijective, the columns must be linearly independent (which ensures the parallelogram is non-degenerate, i.e., $\vec{p}$ and $\vec{q}$ are not collinear).

Clever trick: use the origin

Since linear maps preserve the origin, the corner $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ of the unit square must map to the origin of the target parallelogram. This pins down which standard basis vector maps to which edge vector.

Practice Problem #6.1 (a linear map is determined by its values on a basis)

Problem. Suppose $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n$ and $\vec{w}_1, \vec{w}_2, \ldots, \vec{w}_n$ are vectors in $\mathbb{R}^n$, and put

$$A = (\vec{v}_1 \quad \vec{v}_2 \quad \cdots \quad \vec{v}_n), \qquad B = (\vec{w}_1 \quad \vec{w}_2 \quad \cdots \quad \vec{w}_n).$$

Suppose A is invertible. Explain why there is a unique linear function $f : \mathbb{R}^n \to \mathbb{R}^n$ with $f(\vec{v}_i) = \vec{w}_i$ for all $i = 1, 2, \ldots, n$, and find its standard matrix. As an application, find the standard matrix of the unique linear function $f : \mathbb{R}^3 \to \mathbb{R}^3$ with

$$f\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}, \qquad f\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \qquad f\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}.$$

Solution. Let M be the standard matrix of a linear map $f : \mathbb{R}^n \to \mathbb{R}^n$. The condition $f(\vec{v}_i) = \vec{w}_i$ says exactly that $M\vec{v}_i = \vec{w}_i$ for every i, i.e. that

$$MA = M(\vec{v}_1 \quad \cdots \quad \vec{v}_n) = (M\vec{v}_1 \quad \cdots \quad M\vec{v}_n) = (\vec{w}_1 \quad \cdots \quad \vec{w}_n) = B.$$

Uniqueness. Since A is invertible, multiplying $MA = B$ on the right by A^{-1} forces $M = BA^{-1}$. So there is at most one candidate for the standard matrix.
Existence. Conversely, if $M = BA^{-1}$ then $MA = (BA^{-1})A = B$, so the linear map with standard matrix BA^{-1} does satisfy $M\vec{v}_i = \vec{w}_i$ for all i. Hence a unique such f exists, and

$$\text{standard matrix of } f = BA^{-1}.$$

Application. Here

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 0 & 1 \end{pmatrix}.$$

The identities

$$\vec{e}_1 = \vec{v}_1 - \vec{v}_2, \qquad \vec{e}_2 = \vec{v}_1 - \vec{v}_3, \qquad \vec{e}_3 = -\vec{v}_1 + \vec{v}_2 + \vec{v}_3$$

are checked by adding the columns of A; they show that the $\vec{v}_i$ span $\mathbb{R}^3$. Reading off

the coefficients as columns gives

$$A^{-1} = \begin{pmatrix} 1 & 1 & -1 \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{pmatrix}, \qquad \det A = 1.$$

By linearity, the same relations give the values of f on the standard basis:

$$f(\vec{e}_1) = \vec{w}_1 - \vec{w}_2 = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \qquad f(\vec{e}_2) = \vec{w}_1 - \vec{w}_3 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix},$$

$$f(\vec{e}_3) = -\vec{w}_1 + \vec{w}_2 + \vec{w}_3 = \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix}.$$

These are the columns of the answer:

$$BA^{-1} = \begin{pmatrix} 1 & 0 & 1 \\ -1 & -1 & 3 \\ 2 & 1 & -1 \end{pmatrix}.$$

Practice Problem #6.2 (if A is singular the interpolating map is never unique)

Problem. Define A and B as in the previous problem, and suppose A is not invertible. Explain why there is never a unique linear function $f : \mathbb{R}^n \to \mathbb{R}^n$ with $f(\vec{v}_i) = \vec{w}_i$ for all i: there is either no such function, or more than one.

Solution. As in Problem 6.1, a standard matrix M must satisfy the matrix equation $MA = B$.

Case 1: the equation has no solution. Then no linear map with the required values exists at all.

Case 2: the equation has a solution M_0, i.e. $M_0A = B$. Since A is not invertible, neither is $A^\top$, so there is a nonzero vector $\vec{y} \in \mathbb{R}^n$ with $A^\top \vec{y} = \vec{0}$. Transposing, $\vec{y}^\top A = \vec{0}^\top$.

Choose any nonzero vector $\vec{u} \in \mathbb{R}^n$ and, for each $t \in \mathbb{R}$, set $M_t = M_0 + t\,\vec{u}\vec{y}^\top$. Then

$$M_tA = M_0A + t\,\vec{u}(\vec{y}^\top A) = B + t\,\vec{u}\,\vec{0}^\top = B,$$

so every M_t is a solution. The matrix $\vec{u}\vec{y}^\top$ is nonzero, since the product of a nonzero entry of $\vec{u}$ with a nonzero entry of $\vec{y}$ is a nonzero entry; hence the matrices M_t are pairwise distinct as t varies. So there are infinitely many such linear maps.

Combining the two cases: when A is singular there are either no solutions or infinitely many, never exactly one. A unique f exists precisely when A is invertible, and then f has standard matrix BA^{-1}.

4.2 Linear Dependence

Definition. A list is *linearly dependent* if it is not linearly independent.

Sufficient conditions for dependence

Proposition. If $p > n$ then $\vec{v}_1, \ldots, \vec{v}_p \in \mathbb{R}^n$ are linearly dependent (more vectors than dimensions).

> **Proof.** Form the matrix $A = (\vec{v}_1 \quad \cdots \quad \vec{v}_p)$ of size $n \times p$. Since $p > n$, A has more columns than rows. In any echelon form, each pivot is in a distinct row, so there are at most n pivots. Since there are $p > n$ columns, at least one column is not a pivot column, i.e., there is a free variable. So $A\vec{x} = \vec{0}$ has a nontrivial solution, meaning the columns are linearly dependent. ■

Proposition. If any $\vec{v}_i = \vec{0}$ then the list is linearly dependent.

Proof. Suppose $\vec{v}_i = \vec{0}$ for some i. Set $x_i = 1$ and $x_j = 0$ for $j \neq i$. Then $x_1\vec{v}_1 + \cdots + x_p\vec{v}_p = \vec{v}_i = \vec{0}$, a nontrivial solution (since $x_i = 1 \neq 0$) to $x_1\vec{v}_1 + \cdots + x_p\vec{v}_p = \vec{0}$. By the equivalent characterization, the list is linearly dependent. ■

4.3 Linear Transformations

Definition. A function $f : X \to Y$ takes inputs from the *domain* X and produces outputs in the *codomain* Y. Not every element of Y needs to be an output.

Definition. Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a function. T is a *linear transformation* if:

1. $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$ for all $\vec{u}, \vec{v} \in \mathbb{R}^n$.
2. $T(c\vec{v}) = cT(\vec{v})$ for all $\vec{v} \in \mathbb{R}^n$, $c \in \mathbb{R}$.

Proposition. If T is linear, then:

- $T(\vec{0}) = \vec{0}$.
- $T(\vec{u} - \vec{v}) = T(\vec{u}) - T(\vec{v})$.
- $T(a_1\vec{v}_1 + \cdots + a_p\vec{v}_p) = a_1T(\vec{v}_1) + \cdots + a_pT(\vec{v}_p)$ (compatible with linear combinations).

Proof. We prove each property using the two linearity axioms: $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$ and $T(c\vec{v}) = cT(\vec{v})$.

1. $T(\vec{0}) = T(\vec{0} + \vec{0}) = T(\vec{0}) + T(\vec{0})$ (using additivity). Subtracting $T(\vec{0})$ from both sides gives $\vec{0} = T(\vec{0})$.
2. $T(\vec{u} - \vec{v}) = T(\vec{u} + (-1)\vec{v}) = T(\vec{u}) + T((-1)\vec{v}) = T(\vec{u}) + (-1)T(\vec{v}) = T(\vec{u}) - T(\vec{v})$ (using additivity and homogeneity).
3. By induction on p. Base case $p = 1$: $T(a_1\vec{v}_1) = a_1T(\vec{v}_1)$ (homogeneity). Inductive step: suppose it holds for $p - 1$ vectors. Then $T(a_1\vec{v}_1 + \cdots + a_p\vec{v}_p) = T\big((a_1\vec{v}_1 + \cdots + a_{p-1}\vec{v}_{p-1}) + a_p\vec{v}_p\big) = T(a_1\vec{v}_1 + \cdots + a_{p-1}\vec{v}_{p-1}) + T(a_p\vec{v}_p)$ (additivity). By induction, this is $a_1T(\vec{v}_1) + \cdots + a_{p-1}T(\vec{v}_{p-1}) + a_pT(\vec{v}_p)$. Done.

■

Definition. The *standard basis* vectors of $\mathbb{R}^n$ are $\vec{e}_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \vec{e}_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \ldots, \vec{e}_n = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$.

Fact. If A is $m \times n$, then $A\vec{e}_i$ is the ith column of A.

Proof: Let $A = (\vec{a}_1 \quad \vec{a}_2 \quad \cdots \quad \vec{a}_n)$ where $\vec{a}_j \in \mathbb{R}^m$ is the jth column. The standard basis vector $\vec{e}_i \in \mathbb{R}^n$ has ith entry equal to 1 and all other entries equal to 0. By the definition of the matrix-vector product,

$$A\vec{e}_i = (\vec{a}_1 \quad \vec{a}_2 \quad \cdots \quad \vec{a}_n) \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} = 0 \cdot \vec{a}_1 + \cdots + 1 \cdot \vec{a}_i + \cdots + 0 \cdot \vec{a}_n = \vec{a}_i.$$

So $A\vec{e}_i = \vec{a}_i$, the ith column of A. ■

Fundamental theorem of linear transformations

Theorem. If $T : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation, then there is a **unique** $m \times n$ matrix A such that $T(\vec{v}) = A\vec{v}$ for all $\vec{v} \in \mathbb{R}^n$. This matrix is

$$\boxed{A = \begin{pmatrix} T(\vec{e}_1) & T(\vec{e}_2) & \cdots & T(\vec{e}_n) \end{pmatrix}.}$$

We call A the *standard matrix* of T.

Proof. Existence: Define $A = \begin{pmatrix} T(\vec{e}_1) & T(\vec{e}_2) & \cdots & T(\vec{e}_n) \end{pmatrix}$, an $m \times n$ matrix (each $T(\vec{e}_i) \in \mathbb{R}^m$). For any $\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{R}^n$, we have $\vec{v} = v_1\vec{e}_1 + v_2\vec{e}_2 + \cdots + v_n\vec{e}_n$ (this is how $\vec{v}$ decomposes in the standard basis). By linearity of T (repeated use of additivity and homogeneity, i.e., the previous proposition), $T(\vec{v}) = T(v_1\vec{e}_1 + \cdots + v_n\vec{e}_n) = v_1T(\vec{e}_1) + \cdots + v_nT(\vec{e}_n)$. By the definition of matrix-vector product, $v_1T(\vec{e}_1) + \cdots + v_nT(\vec{e}_n) = \begin{pmatrix} T(\vec{e}_1) & \cdots & T(\vec{e}_n) \end{pmatrix} \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = A\vec{v}$. So $T(\vec{v}) = A\vec{v}$ for all $\vec{v}$, as needed. **Uniqueness:** Suppose B is another $m \times n$ matrix with $T(\vec{v}) = B\vec{v}$ for all $\vec{v} \in \mathbb{R}^n$. Then for each standard basis vector $\vec{e}_i$, $T(\vec{e}_i) = B\vec{e}_i$. But $B\vec{e}_i$ is the ith column of B. So the ith column of B equals $T(\vec{e}_i)$, which is the ith column of A. Hence $A = B$, proving uniqueness. ■

Rotations, dilations, and powers of a rotation

Let R_θ be counterclockwise rotation of $\mathbb{R}^2$ through the angle θ. Its standard matrix is

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix},$$

since $R_\theta\vec{e}_1 = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$ and $R_\theta\vec{e}_2 = \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}$. The facts used repeatedly:

- *Composition is matrix multiplication:* $R_\theta R_\varphi = R_{\theta+\varphi}$, hence $R_\theta^k = R_{k\theta}$ for every integer $k \geq 0$.
- *Inverse:* $R_\theta^{-1} = R_{-\theta} = R_\theta^\top$ (a rotation matrix is orthogonal).
- *Dilation:* scaling every vector by $c > 0$ is the map cI_2; "rotate by θ, then dilate by c" has standard matrix cR_θ.
- *Identity test:* $R_\theta = I_2$ iff $\theta \equiv 0 \pmod{2\pi}$. Hence $R_\theta^m = I_2$ iff $m\theta \equiv 0 \pmod{2\pi}$; if $\theta = \dfrac{2\pi p}{q}$ in lowest terms, then R_θ has *order* q, i.e. $R_\theta^0, \ldots, R_\theta^{q-1}$ are distinct and $R_\theta^q = I_2$.
- *Distinct images:* $R_\theta\vec{v} = R_\varphi\vec{v}$ for some nonzero $\vec{v}$ iff $\theta - \varphi \equiv 0 \pmod{2\pi}$.
- *No nonzero fixed vector:* for $R_\theta \neq I_2$ the equation $R_\theta\vec{w} = \vec{w}$ forces $\vec{w} = \vec{0}$, because $\det(R_\theta - I_2) = (\cos\theta - 1)^2 + \sin^2\theta = 2 - 2\cos\theta \neq 0$.

Midterm 2021 Problem 3 (rotation–dilation, and powers of a rotation)

Problem. (a) Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformation that rotates a vector counterclockwise by $45°$ and then doubles its length. Find the standard matrix A of T.
(b) Let M be a 2×2 rotation matrix not equal to the identity matrix. Suppose $M^{-1} = M^5$ and $\vec{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. How many different vectors could the set $S = \{\vec{v}, M\vec{v}, M^2\vec{v}, M^3\vec{v}, M^4\vec{v}, M^5\vec{v}\}$ contain? For each possibility, describe the pic-

ture of S and compute $\vec{v} + M\vec{v} + M^2\vec{v} + M^3\vec{v} + M^4\vec{v} + M^5\vec{v}$.

Solution (a). The map is $T = D \circ R_{45^\circ}$, where $D(\vec{x}) = 2\vec{x}$ has matrix $2I_2$. By the fundamental theorem, the standard matrix is built from the images of the standard basis vectors:

$$T(\vec{e}_1) = 2R_{45^\circ}\vec{e}_1 = 2\begin{pmatrix}\cos 45^\circ \\ \sin 45^\circ\end{pmatrix} = \begin{pmatrix}\sqrt{2} \\ \sqrt{2}\end{pmatrix},$$

$$T(\vec{e}_2) = 2R_{45^\circ}\vec{e}_2 = 2\begin{pmatrix}-\sin 45^\circ \\ \cos 45^\circ\end{pmatrix} = \begin{pmatrix}-\sqrt{2} \\ \sqrt{2}\end{pmatrix}.$$

Hence

$$A = \begin{pmatrix}\sqrt{2} & -\sqrt{2} \\ \sqrt{2} & \sqrt{2}\end{pmatrix} = \sqrt{2}\, R_{45^\circ} = \begin{pmatrix}1 & -1 \\ 1 & 1\end{pmatrix}.$$

Check: $\det A = 2$, the area-scaling factor of a rotation composed with a dilation of ratio 2.

Solution (b). Write $M = R_\theta$ with $\theta \not\equiv 0 \pmod{2\pi}$ (since $M \neq I_2$). By the properties above,

$$M^{-1} = M^5 \iff R_{-\theta} = R_{5\theta} \iff 6\theta \equiv 0 \pmod{2\pi},$$

so $\theta = k \cdot 60^\circ$ with $k = 1, \ldots, 5$. Thus $M^6 = I_2$, and the number of distinct vectors is governed by the order of M, i.e. the least $q \geq 1$ with $q\theta \equiv 0 \pmod{2\pi}$. The five angles collapse to three cases:

- $\theta = 60^\circ$ or 300° (order 6): S has 6 distinct vectors — the vertices of a regular hexagon centred at the origin, lying on the circle of radius $\|\vec{v}\| = \sqrt{5}$ at the angles $\arg\vec{v} + 0^\circ, 60^\circ, \ldots, 300^\circ$.
- $\theta = 120^\circ$ or 240° (order 3): S has 3 distinct vectors — an equilateral triangle inscribed in the same circle, at the angles $0^\circ, 120^\circ, 240^\circ$ offset by $\arg\vec{v}$.
- $\theta = 180^\circ$ (order 2): S has 2 distinct vectors, namely $\vec{v}$ and $-\vec{v}$ (antipodal points).

So the possible sizes are $|S| \in \{2, 3, 6\}$.

The sum. In every case the sum is $\vec{0}$. Let $\vec{w} = \vec{v} + M\vec{v} + \cdots + M^5\vec{v}$. Multiplying by M and using $M^6 = I_2$,

$$M\vec{w} = M\vec{v} + M^2\vec{v} + \cdots + M^5\vec{v} + M^6\vec{v} = M\vec{v} + \cdots + M^5\vec{v} + \vec{v} = \vec{w},$$

so $(M - I_2)\vec{w} = \vec{0}$. But $M \neq I_2$ is a rotation, so $\det(M - I_2) = 2 - 2\cos\theta \neq 0$ and $M - I_2$ is invertible; hence $\vec{w} = \vec{0}$.

Equivalently, in matrix form $(I_2 + M + \cdots + M^5)(M - I_2) = M^6 - I_2 = 0$, and since $M - I_2$ is invertible, $I_2 + M + \cdots + M^5 = 0$; the sum vanishes for *every* starting vector $\vec{v}$. Geometrically: the vectors of S are equally spaced around a circle, so their centroid is the origin.

Clever trick: a vanishing geometric sum

A relation $M^6 = I$ together with invertibility of $M - I$ gives $I + M + \cdots + M^5 = (M^6 - I)(M - I)^{-1} = 0$ at once, without knowing θ. The only analytic input is that a nonzero rotation moves every nonzero vector, i.e. $\det(M - I) = 2 - 2\cos\theta \neq 0$.

Not every function is linear

$T(\vec{v}) = v_1^2 + \cdots + v_n^2$ is **not** linear since $T(2\vec{v}) = 4T(\vec{v}) \neq 2T(\vec{v})$. Functions of the form $f(x) = ax + b$ with $b \neq 0$ are **not** linear (they are "affine").

4.4 Functions and Range

Definition. For $f : X \to Y$:

- The *image* of x under f is $f(x)$.
- The *range* (or *image*) of f is $\text{range}(f) = \{f(x) : x \in X\} \subseteq Y$.

Two functions are the same only when domain, codomain, and formula are all equal.

4.5 One-to-one (Injective)

Definition. $f : X \to Y$ is *one-to-one* (or *injective*) if $f(a) = f(b)$ implies $a = b$. Equivalently, f does not send two different inputs to the same output.

Detecting one-to-one for linear T

Theorem. Suppose $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear with standard matrix A $(m \times n)$. The following are equivalent:

(a) T is one-to-one.
(b) The only solution to $T(\vec{x}) = \vec{0}$ is $\vec{x} = \vec{0}$.
(c) The columns of A are linearly independent.
(d) Every column of A contains a pivot position.

Proof. We show (a) $\Leftrightarrow$ (b) $\Leftrightarrow$ (c) $\Leftrightarrow$ (d).

(a) $\Leftrightarrow$ (b): ($\Rightarrow$) If T is one-to-one and $T(\vec{x}) = \vec{0}$, since $T(\vec{0}) = \vec{0}$ (by linearity), we have $T(\vec{x}) = T(\vec{0})$, so one-to-one gives $\vec{x} = \vec{0}$. ($\Leftarrow$) Suppose the only solution to $T(\vec{x}) = \vec{0}$ is $\vec{x} = \vec{0}$. If $T(\vec{x}) = T(\vec{y})$, then by linearity $T(\vec{x} - \vec{y}) = T(\vec{x}) - T(\vec{y}) = \vec{0}$. By assumption, $\vec{x} - \vec{y} = \vec{0}$, so $\vec{x} = \vec{y}$. Thus T is one-to-one.

(b) $\Leftrightarrow$ (c): $T(\vec{x}) = \vec{0}$ is $A\vec{x} = \vec{0}$, i.e., $x_1\vec{a}_1 + \cdots + x_n\vec{a}_n = \vec{0}$. By the linear independence characterization, this has only the trivial solution iff the columns $\vec{a}_1, \ldots, \vec{a}_n$ are linearly independent. So (b) $\Leftrightarrow$ (c).

(c) $\Leftrightarrow$ (d): By the corollary in Section 4 (detecting linear independence), the columns of A are linearly independent iff A has a pivot in every column. So (c) $\Leftrightarrow$ (d).

■

4.6 Onto (Surjective)

Definition. $f : X \to Y$ is *onto* (or *surjective*) if $\text{range}(f) = Y$. Equivalently, for each $y \in Y$ there exists $x \in X$ with $f(x) = y$.

Detecting onto for linear T

Theorem. Suppose $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear with standard matrix A $(m \times n)$. The following are equivalent:

(a) T is onto.
(b) $A\vec{x} = \vec{b}$ has a solution for each $\vec{b} \in \mathbb{R}^m$.
(c) The span of the columns of A is $\mathbb{R}^m$.
(d) Every row of A contains a pivot position.

Proof: (a) $\Leftrightarrow$ (b): By definition, T is onto means $\text{range}(T) = \mathbb{R}^m$. But $\text{range}(T) = \{T(\vec{x}) : \vec{x} \in \mathbb{R}^n\} = \{A\vec{x} : \vec{x} \in \mathbb{R}^n\}$. So T is onto iff for every $\vec{b} \in \mathbb{R}^m$ there exists $\vec{x} \in \mathbb{R}^n$ with $A\vec{x} = \vec{b}$, which is exactly (b).

(b) $\Leftrightarrow$ (c): The span of the columns of A is $\{x_1\vec{a}_1 + \cdots + x_n\vec{a}_n : x_i \in \mathbb{R}\} = \{A\vec{x} : \vec{x} \in \mathbb{R}^n\}$. So (c) says $\{A\vec{x} : \vec{x} \in \mathbb{R}^n\} = \mathbb{R}^m$, which is exactly (b).

(b) $\Leftrightarrow$ (d): The system $A\vec{x} = \vec{b}$ is consistent iff the augmented matrix $\begin{pmatrix} A & \vec{b} \end{pmatrix}$ has no row of the form $(0 \ \cdots \ 0 \mid b)$ with $b \neq 0$ after row reduction. This is equivalent

to: every row of A has a pivot position (since if a row of A has no pivot, then for some $\vec{b}$ — namely a vector with a nonzero entry in that position — the augmented column becomes a pivot and the system is inconsistent). Thus $A\vec{x} = \vec{b}$ is solvable for **every** $\vec{b}$ iff every row of A has a pivot, which is (d). ■

4.7 Size Constraints

Dimension constraints

Corollary. For a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$:
- T can be one-to-one only if $n \leq m$ (domain no larger than codomain).
- T can be onto only if $n \geq m$ (domain no smaller than codomain).

Proof: For one-to-one: If T is one-to-one, then every column of the standard matrix A has a pivot (by the one-to-one theorem). Since A is $m \times n$ and each of the n columns has a pivot, we need at least n rows to hold these pivots, so $n \leq m$.

For onto: If T is onto, then every row of A has a pivot (by the onto theorem). Since A is $m \times n$ and each of the m rows has a pivot, we need at least m columns to hold these pivots, so $n \geq m$. ■

One-to-one vs. onto depends on different things

Whether T is one-to-one depends on domain and formula, but **not** on the codomain. Whether T is onto **does** depend on the choice of codomain.

5 Subspaces, Basis, and Dimension

5.1 Invertibility: Summary for Square Matrices

Full characterization (square matrices)

Theorem. When A is $n \times n$, the following are equivalent:
1. A is invertible.
2. The columns of A are linearly independent.
3. The span of the columns of A is $\mathbb{R}^n$.

Corollary. If A and B are both $n \times n$ and $AB = I_n$, then $BA = I_n$. (For square matrices, checking $AB = I$ suffices.)

Proof of the theorem: (1) $\Rightarrow$ (2): Suppose A is invertible with inverse A^{-1}. To show the columns are linearly independent, suppose $A\vec{x} = \vec{0}$. Multiplying both sides by A^{-1} on the left gives $\vec{x} = A^{-1}\vec{0} = \vec{0}$. Thus the only solution to $A\vec{x} = \vec{0}$ is $\vec{x} = \vec{0}$, so the columns are linearly independent.

(2) $\Rightarrow$ (3): Suppose the columns are linearly independent. Since A is $n \times n$, the n columns are linearly independent in $\mathbb{R}^n$. For a list of n linearly independent vectors in $\mathbb{R}^n$ (which has dimension n), they must form a basis, so their span is $\mathbb{R}^n$. (Equivalently, since the columns are independent, every column is a pivot column, so $\operatorname{rank} A = n$, meaning $\dim \operatorname{Col} A = n$, and since $\operatorname{Col} A \subseteq \mathbb{R}^n$ has the same dimension, $\operatorname{Col} A = \mathbb{R}^n$.)

(3) $\Rightarrow$ (1): Suppose the columns span $\mathbb{R}^n$. Then for each standard basis vector $\vec{e}_j$, there exists $\vec{b}_j$ with $A\vec{b}_j = \vec{e}_j$. Let $B = \begin{pmatrix} \vec{b}_1 & \cdots & \vec{b}_n \end{pmatrix}$. Then $AB = \begin{pmatrix} A\vec{b}_1 & \cdots & A\vec{b}_n \end{pmatrix} = \begin{pmatrix} \vec{e}_1 & \cdots & \vec{e}_n \end{pmatrix} = I_n$. By the corollary below (whose proof does not depend on this theorem), $BA = I_n$ as well, so A is invertible. ■

Proof of the corollary: Suppose $AB = I_n$. We show A is invertible, which then gives $BA = I_n$. Since $AB = I_n$, for each $\vec{b} \in \mathbb{R}^n$ we have $A(B\vec{b}) = (AB)\vec{b} = I_n\vec{b} = \vec{b}$, so every $\vec{b} \in \mathbb{R}^n$ is in the span of the columns of A. By (3) $\Rightarrow$ (1) of the full characterization (which requires only the fact that if columns span then $A\vec{x} = \vec{e}_j$ is solvable, giving a right inverse), A is invertible. Then $BA = A^{-1}(AB)A = A^{-1}I_nA = I_n$. More directly: $AB = I_n$ and A invertible give $B = A^{-1}$, so $BA = A^{-1}A = I_n$.

(To avoid circular reasoning: from $AB = I_n$, the columns of A span $\mathbb{R}^n$ as shown above. Since A is $n \times n$ with columns spanning $\mathbb{R}^n$ (dimension n), and there are exactly n columns, they must also be linearly independent — otherwise a dependent list of n vectors would span a space of dimension $< n$. Hence A is invertible by the full characterization (2) $\Leftrightarrow$ (1), and $B = A^{-1}$.) ■

Practice Problem #4.10a (Inverse from polynomial identity)

Problem. Let A be an $n \times n$ matrix satisfying $A^2 = bI + cA$ for some scalars $b \neq 0$ and c. Prove that A is invertible with inverse $A^{-1} = \frac{1}{b}(A - cI)$.

Solution. Start from the given identity $A^2 - cA = bI$, i.e., $A(A - cI) = bI$. Since $b \neq 0$, divide by b:

$$A \cdot \frac{1}{b}(A - cI) = I.$$

Let $C = \frac{1}{b}(A - cI)$. Then $AC = I$, so A has a right inverse. Since A is square, a right inverse is also a left inverse, hence the unique inverse: $A^{-1} = C = \frac{1}{b}(A - cI)$.

Check: $CA = \frac{1}{b}(A - cI)A = \frac{1}{b}(A^2 - cA) = \frac{1}{b} \cdot bI = I$. Done.

Clever trick: rearrange to get A(something) $= I$

Any polynomial relation $p(A) = 0$ can be rearranged as $A \cdot q(A) = I$ whenever the constant term of p is nonzero, giving $A^{-1} = q(A)$ immediately. No row reduction needed.

Practice Problem #4.10b (Rank-1 perturbation: invertibility condition)

Problem. Let J be the $n \times n$ all-ones matrix (every entry is 1). Let $A = (b-c)I + cJ$. Find when A is invertible and give its inverse formula.

Solution. The matrix J is rank 1, so its eigenvalues are n (with eigenvector the all-ones vector $\begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$) and 0 (multiplicity $n - 1$, with eigenvectors orthogonal to the all-ones vector).

Therefore $A = (b - c)I + cJ$ has eigenvalues $(b - c) + cn = b + (n - 1)c$ and $(b - c)$ (multiplicity $n - 1$). So A is invertible iff all eigenvalues are nonzero, i.e., iff both $b - c \neq 0$ and $b + (n - 1)c \neq 0$.

To find the inverse, use the Sherman-Morrison formula for rank-1 updates: $(I + \alpha\vec{u}\vec{v}^\top)^{-1} = I - \frac{\alpha}{1 + \alpha\vec{v}^\top\vec{u}}\vec{u}\vec{v}^\top$. Here $J = \vec{1}\vec{1}^\top$ where $\vec{1} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$, and $A = (b - c)(I +$

$\frac{c}{b-c}J$). So

$$A^{-1} = \frac{1}{b-c}\left(I - \frac{\frac{c}{b-c}}{1+\frac{c}{b-c}\cdot n}J\right) = \frac{1}{b-c}I - \frac{c}{(b-c)(b+(n-1)c)}J.$$

Clever trick: rank-1 matrices are diagonalized by J

The all-ones matrix J is a classic rank-1 example. Its eigenvalues are immediate: $J\vec{1} = n\vec{1}$, and every vector orthogonal to $\vec{1}$ is in $\operatorname{Nul} J$. This makes $(b-c)I+cJ$ trivial to invert using either eigenvalues or the Sherman-Morrison formula.

Practice Problem #4.1 (TREF: two-sided reduced echelon form)

Problem. Define $\mathrm{TREF}(A) = \mathrm{RREF}(\mathrm{RREF}(A)^\top)^\top$. Compute $\mathrm{TREF}(A)$ for $A = \begin{pmatrix} 1 & 2 & 1 \\ -2 & -3 & 1 \\ 3 & 5 & 0 \end{pmatrix}$. Classify all possible TREF forms for general $m \times n$ matrices.

Solution. For the given matrix, $\mathrm{RREF}(A) = \begin{pmatrix} 1 & 0 & -5 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}$. Its transpose $\mathrm{RREF}(A)^\top = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -5 & 3 & 0 \end{pmatrix}$, which RREFs to $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. Transposing gives $\mathrm{TREF}(A) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

For general $m \times n$ matrix with rank r, $\mathrm{TREF}(A)$ is always $\begin{pmatrix} I_r & 0_{r\times(n-r)} \\ 0_{(m-r)\times r} & 0_{(m-r)\times(n-r)} \end{pmatrix}$ where $r = \mathrm{rank}(A)$. Every matrix with this shape (for any $0 \le r \le \min(m,n)$) occurs as the TREF of itself.

Clever trick: TREF isolates the rank

Both row and column operations eliminate zero rows/columns, leaving only the rank. TREF is the canonical form that directly displays the rank as the size of the identity block.

Practice Problem #4.7 (off-diagonal block matrix invertibility)

Problem. Let $A = \begin{pmatrix} 0 & C_1 \\ C_2 & 0 \end{pmatrix}$ where $C_1 = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}$ and $C_2 = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}$. When is A invertible? Find A^{-1}.

Solution. Let $\Delta_i = a_i d_i - b_i c_i$ for $i = 1, 2$. If C_1 has a non-zero kernel $\vec{v}$, then $\begin{pmatrix} 0 \\ \vec{v} \end{pmatrix}$ is a non-zero kernel of A. Similarly for C_2. Conversely, if both C_1 and C_2 are invertible, then $A^{-1} = \begin{pmatrix} 0 & C_2^{-1} \\ C_1^{-1} & 0 \end{pmatrix}$. Check:

$$\begin{pmatrix} 0 & C_1 \\ C_2 & 0 \end{pmatrix}\begin{pmatrix} 0 & C_2^{-1} \\ C_1^{-1} & 0 \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix}.$$

So A is invertible $\iff \Delta_1\Delta_2 \neq 0$, and

$$A^{-1} = \begin{pmatrix} 0 & \frac{1}{\Delta_2}\begin{pmatrix} d_2 & -b_2 \\ -c_2 & a_2 \end{pmatrix} \\ \frac{1}{\Delta_1}\begin{pmatrix} d_1 & -b_1 \\ -c_1 & a_1 \end{pmatrix} & 0 \end{pmatrix}.$$

Clever trick: swap blocks, flip sub-inverse

For off-diagonal 2×2 blocks, the inverse swaps the two blocks and inverts each individually. This pattern holds whenever the diagonal blocks are zero and the off-diagonal blocks are square and invertible.

Practice Problem #4.13 (find ALL right inverses B with $AB = I_2$)

Problem. Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$. Find all 3×2 matrices B with $AB = I_2$.

Solution. Write $B = (\vec{u} \quad \vec{v})$. Then $A\vec{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $A\vec{v} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Row reduce the augmented matrix:

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 0 \\ 4 & 5 & 6 & 0 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & -1 & -\frac{5}{3} & \frac{2}{3} \\ 0 & 1 & 2 & \frac{4}{3} & -\frac{1}{3} \end{pmatrix}.$$

Free variable for the first system: $u_3 = s$, so $u_1 = s - \frac{5}{3}$, $u_2 = \frac{4}{3} - 2s$. Free variable for the second: $v_3 = t$, so $v_1 = t + \frac{2}{3}$, $v_2 = -\frac{1}{3} - 2t$. Hence

$$B = \begin{pmatrix} s - \frac{5}{3} & t + \frac{2}{3} \\ \frac{4}{3} - 2s & -\frac{1}{3} - 2t \\ s & t \end{pmatrix} \quad \text{for any } s, t \in \mathbb{R}.$$

Clever trick: solve both right-hand sides simultaneously

Row-reducing $(A \quad I_m)$ gives you all basis vectors of the right inverse space at once. Each column of I_m becomes the corresponding column of the general right inverse.

Practice Problem #4.14 (algorithm: determine if right inverse exists + construct it)

Problem. Describe an algorithm to determine if there exists an $n \times m$ matrix B with $AB = I_m$, and to construct such B when one exists. Explain why it works.

Solution. Write $B = \begin{pmatrix} \vec{b}_1 & \cdots & \vec{b}_m \end{pmatrix}$. Then $AB = I_m \iff A\vec{b}_j = \vec{e}_j$ for each $j = 1, \ldots, m$. A right inverse exists iff every standard basis vector of $\mathbb{R}^m$ is in $\mathrm{Col}(A)$, i.e., iff A has a pivot in every row.

Algorithm:

1. Row reduce $(A \quad I_m) \to (R \quad E)$, where $EA = R$.
2. If R has a zero row, no right inverse exists (that row of E would give an inconsistent equation for the corresponding $\vec{e}_j$).
3. Otherwise, let $j_1 < \cdots < j_m$ be the pivot columns of R. Form B by putting row i of E into row j_i of B (the pivot row position) and zeros elsewhere.

Then $RB = I_m \iff EAB = I_m \iff AB = I_m$ (since E is invertible).

Clever trick: row-reducing $(A \;\; I_m)$ gives both existence and construction at once

The augmented block I_m tracks all m right-hand sides simultaneously. If every row is consistent, the I_m block after reduction encodes the inverse transformation needed to build B.

Practice Problem #5.6 (4×4 matrix with parameter x — invertibility and A^{-1})

Problem. Let $A = \begin{pmatrix} 2 & 0 & 4 & 0 \\ 1 & 8 & 0 & 5 \\ 2 & -8 & 4 & -4 \\ 0 & 0 & x & -9 \end{pmatrix}$. Determine for which x the matrix A is invertible, and find A^{-1}.

Solution. Reduce $Az = y$ for general $y = (y_1, y_2, y_3, y_4)^\top$. The equations are:

1. $2z_1 + 4z_3 = y_1$,
2. $z_1 + 8z_2 + 5z_4 = y_2$,
3. $2z_1 - 8z_2 + 4z_3 - 4z_4 = y_3$,
4. $xz_3 - 9z_4 = y_4$.

Adding (2) + (3) -3(1) gives $2z_4 - z_3 = \dfrac{y_2 + y_3 - 3y_1}{2}$, so $z_3 = 2z_4 - \dfrac{y_2 + y_3 - 3y_1}{2}$.

Substitute into (4): $(x - 18)z_4 = y_4 + \dfrac{9(y_2 + y_3 - 3y_1)}{2}$.

If $x \neq 18$, then $z_4 = \dfrac{2y_4 + 9y_2 + 9y_3 - 27y_1}{2(x - 18)}$, and all variables are determined uniquely. The inverse is

$$A^{-1} = \frac{1}{8(x-18)} \begin{pmatrix} 4(x+36) & -144 & -144 & -16 \\ 7x - 18 & -4x & 18 - 5x & -8 \\ -108 & 72 & 72 & 8 \\ -12x & 8x & 8x & 16 \end{pmatrix}.$$

If $x = 18$, the kernel is spanned by $\begin{pmatrix} -2 \\ -1 \\ 1 \\ 2 \end{pmatrix}$, so A is not invertible.

Clever trick: solve symbolically, not numerically

Treat the parameter x as symbolic during row reduction to see exactly when coefficients vanish. This reveals the critical value $x = 18$ before any numerical computation, and gives the inverse formula directly as a rational function of x.

5.2 Subspaces of $\mathbb{R}^n$

Definition. A subset $H \subseteq \mathbb{R}^n$ is a *subspace* if:

1. $\vec{0} \in H$.
2. $\vec{u} + \vec{v} \in H$ for all $\vec{u}, \vec{v} \in H$ (closed under addition).
3. $c\vec{v} \in H$ for all $c \in \mathbb{R}$, $\vec{v} \in H$ (closed under scalar multiplication).

Equivalently, a subspace is a nonempty subset closed under linear combinations.

Examples of subspaces

- $\mathbb{R}^n$ itself.
- $\{\vec{0}\}$ (the zero subspace).
- The span of any set of vectors in $\mathbb{R}^n$.

- The range of a linear function $T : \mathbb{R}^k \to \mathbb{R}^n$.
- $\{\vec{v} : T(\vec{v}) = \vec{0}\}$ for a linear $T : \mathbb{R}^n \to \mathbb{R}^k$.

Subspaces of $\mathbb{R}^2$: $\{\vec{0}\}$, lines through origin, or $\mathbb{R}^2$. Subspaces of $\mathbb{R}^3$: $\{\vec{0}\}$, lines through origin, planes through origin, or $\mathbb{R}^3$.

Proof that $\text{span}\{\vec{v}_1, \ldots, \vec{v}_p\}$ **is a subspace:** Let $H = \text{span}\{\vec{v}_1, \ldots, \vec{v}_p\} \subseteq \mathbb{R}^n$. We verify the three subspace conditions.

(i) $\vec{0} \in H$: Write $\vec{0} = 0 \cdot \vec{v}_1 + 0 \cdot \vec{v}_2 + \cdots + 0 \cdot \vec{v}_p$, which is a linear combination of $\vec{v}_1, \ldots, \vec{v}_p$ (with all coefficients 0). So $\vec{0} \in H$.

(ii) Closed under addition: Let $\vec{u}, \vec{w} \in H$. Then $\vec{u} = a_1\vec{v}_1 + \cdots + a_p\vec{v}_p$ and $\vec{w} = b_1\vec{v}_1 + \cdots + b_p\vec{v}_p$ for some scalars $a_i, b_i \in \mathbb{R}$. Then

$$\vec{u} + \vec{w} = (a_1 + b_1)\vec{v}_1 + \cdots + (a_p + b_p)\vec{v}_p,$$

which is a linear combination of $\vec{v}_1, \ldots, \vec{v}_p$ (with coefficients $a_i + b_i$). So $\vec{u} + \vec{w} \in H$.

(iii) Closed under scalar multiplication: Let $\vec{u} \in H$ and $c \in \mathbb{R}$. Write $\vec{u} = a_1\vec{v}_1 + \cdots + a_p\vec{v}_p$. Then

$$c\vec{u} = (ca_1)\vec{v}_1 + \cdots + (ca_p)\vec{v}_p,$$

which is a linear combination of $\vec{v}_1, \ldots, \vec{v}_p$ (with coefficients ca_i). So $c\vec{u} \in H$.

Hence $H = \text{span}\{\vec{v}_1, \ldots, \vec{v}_p\}$ is a subspace of $\mathbb{R}^n$. ■

Proof that the range of a linear function is a subspace: Let $T : \mathbb{R}^k \to \mathbb{R}^n$ be linear, and let $H = \text{range}(T) = \{T(\vec{x}) : \vec{x} \in \mathbb{R}^k\} \subseteq \mathbb{R}^n$. We verify the three subspace conditions.

(i) $\vec{0} \in H$: Since T is linear, $T(\vec{0}) = \vec{0}$ (proven earlier). So $\vec{0} = T(\vec{0}) \in \text{range}(T) = H$.

(ii) Closed under addition: Let $\vec{u}, \vec{w} \in H$. Then $\vec{u} = T(\vec{x})$ and $\vec{w} = T(\vec{y})$ for some $\vec{x}, \vec{y} \in \mathbb{R}^k$. By linearity of T, $T(\vec{x}+\vec{y}) = T(\vec{x})+T(\vec{y}) = \vec{u}+\vec{w}$. So $\vec{u}+\vec{w} = T(\vec{x}+\vec{y}) \in H$ (since $\vec{x} + \vec{y} \in \mathbb{R}^k$).

(iii) Closed under scalar multiplication: Let $\vec{u} \in H$ and $c \in \mathbb{R}$. Write $\vec{u} = T(\vec{x})$ for some $\vec{x} \in \mathbb{R}^k$. By linearity of T, $T(c\vec{x}) = cT(\vec{x}) = c\vec{u}$. So $c\vec{u} = T(c\vec{x}) \in H$ (since $c\vec{x} \in \mathbb{R}^k$).

Hence range(T) is a subspace of $\mathbb{R}^n$. ■

Not a subspace

The set $\{\vec{v} \in \mathbb{R}^3 : v_1 + v_2 + v_3 = 1\}$ is **not** a subspace (does not contain $\vec{0}$). But $\{\vec{v} \in \mathbb{R}^3 : v_1 + v_2 + v_3 = 0\}$ **is** a subspace.

5.3 Column Space and Null Space

Definition. The *column space* of an $m \times n$ matrix A is

$$\text{Col}\, A = \{A\vec{x} : \vec{x} \in \mathbb{R}^n\} \subseteq \mathbb{R}^m.$$

This is the span of the columns of A.

Definition. The *null space* of an $m \times n$ matrix A is

$$\text{Nul}\, A = \{\vec{v} \in \mathbb{R}^n : A\vec{v} = \vec{0}\} \subseteq \mathbb{R}^n.$$

This is the set of solutions to $A\vec{x} = \vec{0}$.

Key facts

- $\operatorname{Col} A$ is a subspace of $\mathbb{R}^m$ ($m = \#$rows of A).
- $\operatorname{Nul} A$ is a subspace of $\mathbb{R}^n$ ($n = \#$columns of A).
- If $T(\vec{x}) = A\vec{x}$, then $\operatorname{Col} A = \operatorname{range}(T)$ and $\operatorname{Nul} A = \{\vec{x} : T(\vec{x}) = \vec{0}\}$.
- $\vec{b} \in \operatorname{Col} A \Leftrightarrow A\vec{x} = \vec{b}$ has a solution.
- $\operatorname{Col} A = \mathbb{R}^m \Leftrightarrow A\vec{x} = \vec{b}$ has a solution for each $\vec{b}$.

Proof that $\operatorname{Nul} A$ is a subspace: We verify the three subspace conditions. (i) $\vec{0} \in \operatorname{Nul} A$: by linearity of matrix multiplication, $A\vec{0} = \vec{0}$, so $\vec{0} \in \operatorname{Nul} A$.

(ii) Closed under addition: Let $\vec{u}, \vec{v} \in \operatorname{Nul} A$, so $A\vec{u} = \vec{0}$ and $A\vec{v} = \vec{0}$. By linearity of matrix multiplication, $A(\vec{u} + \vec{v}) = A\vec{u} + A\vec{v} = \vec{0} + \vec{0} = \vec{0}$, so $\vec{u} + \vec{v} \in \operatorname{Nul} A$.

(iii) Closed under scalar multiplication: Let $\vec{v} \in \operatorname{Nul} A$ and $c \in \mathbb{R}$. Then $A(c\vec{v}) = c(A\vec{v}) = c\vec{0} = \vec{0}$, so $c\vec{v} \in \operatorname{Nul} A$.

Hence $\operatorname{Nul} A$ is a subspace of $\mathbb{R}^n$. ■

Proof that $\operatorname{Col} A$ is a subspace: Let A be an $m \times n$ matrix. Since $\operatorname{Col} A = \{A\vec{x} : \vec{x} \in \mathbb{R}^n\} = \operatorname{range}(T)$ where $T(\vec{x}) = A\vec{x}$ is a linear transformation, and the range of any linear transformation is a subspace (proven above), $\operatorname{Col} A$ is a subspace of $\mathbb{R}^m$.

Alternatively, since $\operatorname{Col} A = \operatorname{span}\{\vec{a}_1, \ldots, \vec{a}_n\}$ (the span of the columns of A), and the span of any set of vectors is a subspace (proven above), $\operatorname{Col} A$ is a subspace of $\mathbb{R}^m$. ■

Practice Problem #5.3 (Col and Nul bases for a one-row matrix)

Problem. Let $A = \begin{pmatrix} a_1 & a_2 & \cdots & a_n \end{pmatrix}$ be a $1 \times n$ row matrix. Find bases for $\operatorname{Col} A$ and $\operatorname{Nul} A$, considering cases.

Solution. There are two cases.

Case 1: $A = \begin{pmatrix} 0 & 0 & \cdots & 0 \end{pmatrix}$ (all zero row). Then $\operatorname{Col} A = \{\vec{0}\}$ (since $A\vec{x} = 0 \cdot x_1 + \cdots + 0 \cdot x_n = 0$ always), so the empty set is a basis. And $\operatorname{Nul} A = \mathbb{R}^n$ (all vectors satisfy $A\vec{x} = 0$), with standard basis $\{\vec{e}_1, \ldots, \vec{e}_n\}$.

Case 2: $A \neq 0$. Suppose $a_k \neq 0$ (some entry nonzero). Then:

- $\operatorname{Col} A$ is a subspace of $\mathbb{R}^1$ containing the nonzero number a_k. Since the only nonzero subspaces of $\mathbb{R}$ are $\mathbb{R}$ itself, $\operatorname{Col} A = \mathbb{R}$, and $\{1\}$ is a basis (or use a_k itself).
- $\operatorname{Nul} A = \{\vec{x} \in \mathbb{R}^n : a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0\}$. This is a hyperplane of dimension $n - 1$. To find a basis, free variables are all x_i except x_k (which solves to $x_k = -\dfrac{1}{a_k}(a_1x_1 + \cdots + \widehat{a_kx_k} + \cdots + a_nx_n)$). Setting each free variable to 1 in turn and the others to 0 gives $n - 1$ basis vectors: for each $i \neq k$,

$$\vec{b}_i = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \\ -\dfrac{a_i}{a_k} \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad (1 \text{ at position } i,\ -\frac{a_i}{a_k} \text{ at position } k).$$

These form a basis for $\operatorname{Nul} A$, confirming $\dim \operatorname{Col} A + \dim \operatorname{Nul} A = 1 + (n-1) = n$

(rank-nullity).

Midterm 2024 Problem 8 (matrix game: forcing a null vector)

Problem. Alice and Bob take turns (Alice first) filling entries of an empty 4×4 matrix A until all 16 positions are occupied. Bob wins if $(1 \quad 2 \quad 3 \quad 4)^\top \in \operatorname{Nul} A$; otherwise Alice wins. Does either player have a winning strategy?
Solution. Bob always wins. Pairing strategy: whenever Alice writes a number a in position (i, j), Bob chooses any still-free position (i, k) in the *same row* and writes $-\frac{j}{k}a$ there. The pair then contributes $a \cdot j + \left(-\frac{j}{k}a\right) \cdot k = aj - aj = 0$ to the ith entry of $A\,(1 \quad 2 \quad 3 \quad 4)^\top$.
The required free cell always exists: before each Alice move, every row contains an even number of filled entries (initially 0; each round fills two cells of one row, preserving even parity). Alice must play in a row with 0 or 2 filled cells, leaving at least one free cell for Bob. At the end, each row consists of two canceling pairs, so all four row sums are zero; hence $A\,(1 \quad 2 \quad 3 \quad 4)^\top = \vec{0}$.

Clever trick: pairing strategy with an even-parity invariant

To force a prescribed linear relation among the entries of a jointly filled matrix, pair positions inside each row so that each pair contributes a prescribed weighted sum of 0. The weights come from the target null vector. An even number of cells per row (here 4) and moving second make the pairing always available.

5.4 Basis

Definition. A *basis* for a subspace $H \subseteq \mathbb{R}^n$ is a set of vectors $\vec{v}_1, \ldots, \vec{v}_k \in H$ that are linearly independent and whose span is H. The empty set $\emptyset$ is a basis for $\{\vec{0}\}$.

Standard basis

The vectors $\vec{e}_1, \vec{e}_2, \ldots, \vec{e}_n$ form the *standard basis* of $\mathbb{R}^n$. They are linearly independent (since $\operatorname{RREF}(I_n) = I_n$ has pivots in all columns) and span $\mathbb{R}^n$.

Theorem. Every subspace H of $\mathbb{R}^n$ has a basis of size at most n.

Proof: If $H = \{\vec{0}\}$, the empty set is a basis. Otherwise pick any nonzero $\vec{v}_1 \in H$; this is a linearly independent set. If $\operatorname{span}\{\vec{v}_1\} = H$ we are done. Otherwise pick $\vec{v}_2 \in H \setminus \operatorname{span}\{\vec{v}_1\}$. Then $\{\vec{v}_1, \vec{v}_2\}$ is linearly independent: if $c_1\vec{v}_1 + c_2\vec{v}_2 = \vec{0}$ with $c_2 \neq 0$, then $\vec{v}_2 = -\frac{c_1}{c_2}\vec{v}_1 \in \operatorname{span}\{\vec{v}_1\}$, a contradiction. Continuing, if $\{\vec{v}_1, \ldots, \vec{v}_k\}$ is independent and does not span H, pick $\vec{v}_{k+1} \in H \setminus \operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_k\}$; the same argument shows $\{\vec{v}_1, \ldots, \vec{v}_{k+1}\}$ is independent. This process must terminate with $k \leq n$ since any list of more than n vectors in $\mathbb{R}^n$ is linearly dependent (more vectors than dimensions). When it terminates we have an independent set that spans H, i.e., a basis. ■

Finding a Basis for $\operatorname{Nul} A$

1. Row reduce A to find $\operatorname{RREF}(A)$ and solve $A\vec{x} = \vec{0}$.
2. Express each basic variable in terms of free variables.
3. Write $\vec{x}$ as a linear combination of vectors $\vec{b}_1, \ldots, \vec{b}_k$ (one per free variable).
4. These vectors form a basis for $\operatorname{Nul} A$.

Why the basis vectors are linearly independent

Look at the rows of the coefficient matrix that correspond to the free variables. Each basis vector has a 1 in the position of one free variable and 0 in the positions of the other free variables, making the set linearly independent.

Finding a Basis for $\operatorname{Col} A$

Pivot columns form a basis

Proposition. The pivot columns of A form a basis for $\operatorname{Col} A$.
Note: use the pivot columns of the **original** matrix A, not the pivot columns of $\operatorname{RREF}(A)$.

Proof: Let $\vec{a}_{i_1}, \ldots, \vec{a}_{i_k}$ be the pivot columns of A (in the original matrix), corresponding to the pivot columns $i_1 < \cdots < i_k$ of $\operatorname{RREF}(A)$. We must show they span $\operatorname{Col} A$ and are linearly independent.

Spanning: Let $\vec{a}_j$ be any non-pivot column of A. In $\operatorname{RREF}(A)$, column j is a linear combination of the pivot columns to its left (this is the defining property of RREF — each non-pivot column's entries are determined by the leading 1's in pivot positions in the same rows). Since row operations are linear combinations of rows and preserve linear relations among **columns**, the same relation holds for A: $\vec{a}_j = \sum_{i_l<j} c_l \vec{a}_{i_l}$. Thus every non-pivot column is in $\operatorname{span}\{\vec{a}_{i_1}, \ldots, \vec{a}_{i_k}\}$, and so $\operatorname{Col} A = \operatorname{span}\{\vec{a}_{i_1}, \ldots, \vec{a}_{i_k}\}$.

Linear independence: Suppose $\sum_l c_l \vec{a}_{i_l} = \vec{0}$. This same relation holds in $\operatorname{RREF}(A)$: $\sum_l c_l \vec{r}_{i_l} = \vec{0}$, where $\vec{r}_{i_l}$ are the corresponding pivot columns of $\operatorname{RREF}(A)$. But the pivot columns of $\operatorname{RREF}(A)$ are standard basis vectors $\vec{e}_1, \ldots, \vec{e}_k$ (in distinct rows), which are linearly independent. Hence all $c_l = 0$.

Thus the pivot columns of A form a basis for $\operatorname{Col} A$. ■

Practice Problem #2.9 (Hyperplane: sum-of-coordinates-zero)

Problem. Let $H = \{ \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n : x_1 + x_2 + \cdots + x_n = 0 \}$. Prove that H is a subspace with basis $\{\vec{e}_1 - \vec{e}_n, \vec{e}_2 - \vec{e}_n, \ldots, \vec{e}_{n-1} - \vec{e}_n\}$.
Solution. H is the null space of the $1 \times n$ matrix $\begin{pmatrix} 1 & 1 & \cdots & 1 \end{pmatrix}$, hence a subspace by the null-space proposition.

Now consider the set $S = \{\vec{e}_1 - \vec{e}_n, \vec{e}_2 - \vec{e}_n, \ldots, \vec{e}_{n-1} - \vec{e}_n\}$. Each vector $\vec{e}_i - \vec{e}_n = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \\ -1 \end{pmatrix}$ has sum of entries $1 + (-1) = 0$, so it lies in H.

Linear independence: Suppose $\sum_{i=1}^{n-1} c_i(\vec{e}_i - \vec{e}_n) = \vec{0}$. Then $\sum_{i=1}^{n-1} c_i \vec{e}_i - \left(\sum_{i=1}^{n-1} c_i \right) \vec{e}_n = \vec{0}$. Since $\vec{e}_1, \ldots, \vec{e}_n$ are linearly independent, the coefficient of each $\vec{e}_i$ must be zero:

$c_i = 0$ for $i = 1, \ldots, n-1$, and $\sum c_i = 0$ (which is automatically satisfied). So S is linearly independent.

Spanning: Let $\vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in H$, so $\sum x_i = 0$, i.e., $x_n = -\sum_{i=1}^{n-1} x_i$. Then

$$\vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_{n-1} \\ -\sum_{i=1}^{n-1} x_i \end{pmatrix} = \sum_{i=1}^{n-1} x_i \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \\ -1 \end{pmatrix} = \sum_{i=1}^{n-1} x_i(\vec{e}_i - \vec{e}_n).$$

So every $\vec{x} \in H$ is a linear combination of vectors in S. Hence S is a basis, and $\dim H = n - 1$.

Clever trick: one vector per free variable

For the hyperplane $\sum x_i = 0$, set x_n as the dependent variable (solve $x_n = -(x_1 + \cdots + x_{n-1})$) and the first $n-1$ variables as free. Each free variable gives one basis vector: set that variable to 1, the other free variables to 0, and read off the dependent variable.

Practice Problem #5.4 (Extend independent vectors to a basis)

Problem. Let $\vec{v}_1, \ldots, \vec{v}_k \in \mathbb{R}^n$ be linearly independent. Describe an algorithm that extends them to a basis of $\mathbb{R}^n$ using the RREF of $(V \quad I_n)$, where $V = (\vec{v}_1 \quad \cdots \quad \vec{v}_k)$.
Solution. Form the augmented matrix $(V \quad I_n)$ (size $n \times (k+n)$) and row reduce to RREF. The algorithm then identifies which columns of I_n to keep:

1. Row reduce $(V \quad I_n)$ to RREF.
2. The first k columns correspond to $\vec{v}_1, \ldots, \vec{v}_k$; by independence they are all pivot columns.
3. Look at the remaining n columns (from I_n). For each row, if the pivot of that row *does not* lie inside V (i.e., it lies in the I_n block), then the corresponding column of I_n (the standard basis vector $\vec{e}_j$) should be added to our set.
4. The combined set $\{\vec{v}_1, \ldots, \vec{v}_k, \vec{e}_{j_1}, \ldots, \vec{e}_{j_{n-k}}\}$ is a basis for $\mathbb{R}^n$.

Why does this work? In RREF $(V \quad I_n)$, the non-pivot columns in the I_n block are linear combinations of earlier pivot columns (which come from V). Those standard basis vectors $\vec{e}_j$ are already in $\operatorname{Col} V = \operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_k\}$, so they would be redundant. The pivot columns from the I_n block are *not* combinations of the $\vec{v}_i$, so adding them preserves independence and increases the span.

Clever trick: use I_n as a reservoir

By appending I_n after V, row reduction reveals exactly which standard basis vectors are already in $\operatorname{Col} V$ (non-pivot columns) and which are not (pivot columns). The latter fill out the basis.

5.5 Operations on Subspaces

Definition. The *sum* of two subspaces U, V is $U + V = \{\vec{u} + \vec{v} : \vec{u} \in U, \vec{v} \in V\}$ (a subspace).

The *intersection* $U \cap V$ is always a subspace.

Union is not a subspace

The union $U \cup V$ of two subspaces is generally **not** a subspace (e.g., two different lines through the origin in $\mathbb{R}^2$).

Properties: If A, B have the same number of rows, $\operatorname{Col}\begin{pmatrix} A & B \end{pmatrix} = \operatorname{Col} A + \operatorname{Col} B$. If A, B have the same number of columns, $\operatorname{Nul}\begin{pmatrix} A \\ B \end{pmatrix} = \operatorname{Nul} A \cap \operatorname{Nul} B$.

Basis for $U + V$ (Span of the Union)

Method: basis for the sum

Since $U + V = \operatorname{span}(U \cup V)$, put the basis vectors of U and of V together as the columns of one matrix M; the pivot columns of the **original** M form a basis for $U + V$.

- *Column spaces*: $\operatorname{Col} A + \operatorname{Col} B = \operatorname{Col}\begin{pmatrix} A & B \end{pmatrix}$; take the pivot columns of the original $\begin{pmatrix} A & B \end{pmatrix}$.
- *Null spaces*: find bases of $\operatorname{Nul} A$ and $\operatorname{Nul} B$ (by solving $A\vec{x} = \vec{0}$ and $B\vec{x} = \vec{0}$), place these basis vectors as the columns of M, and take the pivot columns of the original M.

Proof: $U + V = \{\vec{u} + \vec{v} : \vec{u} \in U, \vec{v} \in V\} = \operatorname{span}(U \cup V)$. If $\{\vec{u}_i\}$ spans U and $\{\vec{v}_j\}$ spans V, then $\operatorname{span}(U \cup V) = \operatorname{span}\{\vec{u}_i, \vec{v}_j\} = \operatorname{Col} M$, and the pivot columns of the original M form a basis for $\operatorname{Col} M$ (proven above). For column spaces, the columns of $\begin{pmatrix} A & B \end{pmatrix}$ are exactly the columns of A together with those of B, whose span is $\operatorname{Col} A + \operatorname{Col} B$. ■

Basis for $\operatorname{Col} A \cap \operatorname{Col} B$

Method: intersection of column spaces

Let A be $m \times n$ and B be $m \times p$.

1. $\vec{w} \in \operatorname{Col} A \cap \operatorname{Col} B \Leftrightarrow \vec{w} = A\vec{x} = B\vec{y}$ for some $\vec{x} \in \mathbb{R}^n$, $\vec{y} \in \mathbb{R}^p$, i.e.,
$$\begin{pmatrix} A & -B \end{pmatrix} \begin{pmatrix} \vec{x} \\ \vec{y} \end{pmatrix} = \vec{0}.$$
2. Find a basis of $\operatorname{Nul}\begin{pmatrix} A & -B \end{pmatrix}$; write each basis vector as $\begin{pmatrix} \vec{x}_i \\ \vec{y}_i \end{pmatrix}$ with $\vec{x}_i \in \mathbb{R}^n$, $\vec{y}_i \in \mathbb{R}^p$.
3. Compute $\vec{w}_i = A\vec{x}_i$ $(= B\vec{y}_i)$. Then $\{\vec{w}_1, \ldots, \vec{w}_k\}$ **spans** $\operatorname{Col} A \cap \operatorname{Col} B$.
4. The $\vec{w}_i$ need not be independent: row reduce $\begin{pmatrix} \vec{w}_1 & \cdots & \vec{w}_k \end{pmatrix}$ and take its pivot columns. They form a basis for $\operatorname{Col} A \cap \operatorname{Col} B$.

Proof: *Step 1:* $\vec{w} = A\vec{x} = B\vec{y} \Leftrightarrow A\vec{x} - B\vec{y} = \vec{0} \Leftrightarrow \begin{pmatrix} A & -B \end{pmatrix} \begin{pmatrix} \vec{x} \\ \vec{y} \end{pmatrix} = \vec{0}$.

Steps 2–3: By step 1, the intersection equals $\{A\vec{x} : \begin{pmatrix} \vec{x} \\ \vec{y} \end{pmatrix} \in \operatorname{Nul}\begin{pmatrix} A & -B \end{pmatrix}\}$. Let $P = \{\vec{x} \in \mathbb{R}^n : A\vec{x} = B\vec{y} \text{ for some } \vec{y}\}$. Every $\vec{x} \in P$ comes from some $\begin{pmatrix} \vec{x} \\ \vec{y} \end{pmatrix} = \sum_i c_i \begin{pmatrix} \vec{x}_i \\ \vec{y}_i \end{pmatrix}$, and comparing the top blocks gives $\vec{x} = \sum_i c_i \vec{x}_i$, so $\{\vec{x}_i\}$ spans P. Hence the intersection is $\{A\vec{x} : \vec{x} \in P\} = \operatorname{span}\{A\vec{x}_i\} = \operatorname{span}\{\vec{w}_i\}$.

Step 4: The $\vec{w}_i$ may be linearly dependent (e.g., if $\operatorname{Nul} A \neq \{\vec{0}\}$, any $\begin{pmatrix} \vec{x} \\ \vec{0} \end{pmatrix}$ with

$\vec{x} \in \operatorname{Nul} A$, $\vec{x} \neq \vec{0}$, gives $\vec{w} = A\vec{x} = \vec{0}$). Discarding non-pivot columns removes only vectors that are linear combinations of the pivot columns, so the span is unchanged and the remaining pivot columns form a basis. ■

Basis for $\operatorname{Nul} A \cap \operatorname{Nul} B$

Method: intersection of null spaces

Stack the matrices: $\operatorname{Nul} A \cap \operatorname{Nul} B = \operatorname{Nul} \begin{pmatrix} A \\ B \end{pmatrix}$. Solve $\begin{pmatrix} A \\ B \end{pmatrix} \vec{x} = \vec{0}$ by the standard null-space method; the resulting vectors form a basis for the intersection.

Proof: $\vec{x} \in \operatorname{Nul} A \cap \operatorname{Nul} B \Leftrightarrow A\vec{x} = \vec{0}$ and $B\vec{x} = \vec{0} \Leftrightarrow \begin{pmatrix} A \\ B \end{pmatrix} \vec{x} = \begin{pmatrix} A\vec{x} \\ B\vec{x} \end{pmatrix} = \vec{0} \Leftrightarrow \vec{x} \in \operatorname{Nul} \begin{pmatrix} A \\ B \end{pmatrix}$. ■

Dimension formula

For any two subspaces $U, V \subseteq \mathbb{R}^n$:

$$\dim(U + V) = \dim U + \dim V - \dim(U \cap V).$$

Proof: Let $\{\vec{w}_1, \ldots, \vec{w}_k\}$ be a basis of $U \cap V$. Extend it to a basis of U by adding vectors $\vec{u}_1, \ldots, \vec{u}_p$ and to a basis of V by adding vectors $\vec{v}_1, \ldots, \vec{v}_q$ (so $\dim U = k + p$ and $\dim V = k + q$). We claim

$$\mathcal{B} = \{\vec{w}_1, \ldots, \vec{w}_k, \vec{u}_1, \ldots, \vec{u}_p, \vec{v}_1, \ldots, \vec{v}_q\}$$

is a basis of $U + V$.

Spanning: any $\vec{u} + \vec{v} \in U + V$ is a combination of the $\vec{w}$'s and $\vec{u}$'s (for $\vec{u}$) plus a combination of the $\vec{w}$'s and $\vec{v}$'s (for $\vec{v}$), hence a combination of the vectors in $\mathcal{B}$.

Independence: suppose

$$\sum_i a_i \vec{w}_i + \sum_j b_j \vec{u}_j + \sum_l c_l \vec{v}_l = \vec{0}.$$

Then $\sum_l c_l \vec{v}_l = -\left(\sum_i a_i \vec{w}_i + \sum_j b_j \vec{u}_j \right) \in U$; being a combination of the $\vec{v}$'s, it is also in V, hence in $U \cap V$. So $\sum_l c_l \vec{v}_l = \sum_i d_i \vec{w}_i$ for some scalars d_i. Since $\{\vec{w}_1, \ldots, \vec{w}_k, \vec{v}_1, \ldots, \vec{v}_q\}$ is a basis of V (linearly independent), all $c_l = 0$. The relation reduces to $\sum_i a_i \vec{w}_i + \sum_j b_j \vec{u}_j = \vec{0}$, and since $\{\vec{w}_1, \ldots, \vec{w}_k, \vec{u}_1, \ldots, \vec{u}_p\}$ is a basis of U, all $a_i = 0$ and all $b_j = 0$. Hence $\mathcal{B}$ is linearly independent.

Therefore $\dim(U + V) = k + p + q = (k + p) + (k + q) - k = \dim U + \dim V - \dim(U \cap V)$. ■

Example. Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$.

Sum: $(A \quad B) = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ is already in RREF with pivots in columns $1, 2, 4$, so

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\} \text{ is a basis of } \operatorname{Col} A + \operatorname{Col} B = \mathbb{R}^3.$$

Intersection: row reduce

$$(A \quad -B) = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

The free variable is $y_1 = t$, giving $x_1 = t$, $x_2 = t$, $y_2 = 0$, so $\operatorname{Nul}(A \quad -B) = \operatorname{span}\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix} \right\}$.

The $\vec{x}$-part is $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$, and $A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$. Hence

$$\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\} \text{ is a basis of } \operatorname{Col} A \cap \operatorname{Col} B.$$

Dimension check: $3 = 2 + 2 - 1$, consistent with the dimension formula.

Practice Problem #2.8 (Span of $\sum c_i \vec{a}_i = 0$ subspace)

Problem. Let $\vec{a}_1, \vec{a}_2, \vec{a}_3 \in \mathbb{R}^n$ be three vectors, and let $W = \left\{ \sum_{i=1}^{3} c_i \vec{a}_i : c_1\vec{a}_1 + c_2\vec{a}_2 + c_3\vec{a}_3 = \vec{0} \right\}$. Describe W in all possible cases (8 cases based on the linear dependence of the $\vec{a}_i$).

Solution. The subspace of coefficient vectors $\vec{c} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$ satisfying $\sum c_i \vec{a}_i = \vec{0}$ is the null space of the $n \times 3$ matrix $A = (\vec{a}_1 \quad \vec{a}_2 \quad \vec{a}_3)$. Then $W = \{A\vec{c} : \vec{c} \in \operatorname{Nul} A\}$.
We consider cases based on $\dim \operatorname{Nul} A = 3 - \operatorname{rank} A$ (rank-nullity):

- $\operatorname{rank} A = 0$ (all $\vec{a}_i = \vec{0}$): Then $\operatorname{Nul} A = \mathbb{R}^3$, but $A\vec{c} = \vec{0}$ always, so $W = \{\vec{0}\}$.
- $\operatorname{rank} A = 1$ (all three nonzero vectors are scalar multiples): Suppose $\vec{a}_1 \neq 0$, $\vec{a}_2 = s\vec{a}_1$, $\vec{a}_3 = t\vec{a}_1$. Then $\operatorname{Nul} A = \{ \begin{pmatrix} -sc_2 - tc_3 \\ c_2 \\ c_3 \end{pmatrix} : c_2, c_3 \in \mathbb{R} \}$ (dimension 2). Then $A\vec{c} = c_1\vec{a}_1 + c_2 s\vec{a}_1 + c_3 t\vec{a}_1 = (c_1 + sc_2 + tc_3)\vec{a}_1 = 0 \cdot \vec{a}_1 = \vec{0}$ always. So $W = \{\vec{0}\}$.
- $\operatorname{rank} A = 2$ (two independent, third dependent): Suppose $\vec{a}_1, \vec{a}_2$ are independent, $\vec{a}_3 = \alpha\vec{a}_1 + \beta\vec{a}_2$. Then $\operatorname{Nul} A$ is 1-dimensional: spanned by $\begin{pmatrix} \alpha \\ \beta \\ -1 \end{pmatrix}$. Then $A \begin{pmatrix} \alpha \\ \beta \\ -1 \end{pmatrix} = \alpha\vec{a}_1 + \beta\vec{a}_2 - \vec{a}_3 = \vec{0}$. So $W = \{\vec{0}\}$ (trivially, since the null space maps to zero).
- $\operatorname{rank} A = 3$ (all independent): $\operatorname{Nul} A = \{\vec{0}\}$, so $W = \{A\vec{0}\} = \{\vec{0}\}$.

In *all* cases, $W = \{\vec{0}\}$! This is because any $\vec{c}$ with $A\vec{c} = \vec{0}$ gives $A\vec{c} = \vec{0}$, which is

exactly the definition of $\operatorname{Nul} A$. So $W = \operatorname{Col} A|_{\operatorname{Nul} A} = \{\vec{0}\}$ by definition.

Clever trick: $W = \operatorname{Col} A$ restricted to $\operatorname{Nul} A$

The problem looks like it has many cases, but it collapses immediately: $W = A(\operatorname{Nul} A) = \{\vec{0}\}$ by the very definition of null space. The rank cases only matter if you want to *understand* why the coefficient condition is always achieved — but the result is always zero.

Practice Problem #3.8 (Algorithm: check span equality)

Problem. Let $V = \{\vec{v}_1, \ldots, \vec{v}_p\}$ and $W = \{\vec{w}_1, \ldots, \vec{w}_q\}$ be two sets of vectors in $\mathbb{R}^n$. Describe a concrete row-reduction algorithm to decide whether $\operatorname{span} V = \operatorname{span} W$.
Solution. The key fact: $\operatorname{span} V = \operatorname{span} W$ if and only if *every vector in V is in* $\operatorname{span} W$ *and every vector in W is in* $\operatorname{span} V$. We check both in one go:

1. Form the matrix $M = \begin{pmatrix} \vec{w}_1 & \cdots & \vec{w}_q & \vec{v}_1 & \cdots & \vec{v}_p \end{pmatrix}$ (put W first, then V).
2. Row reduce to RREF. Check whether the last p columns (those from V) are **not** pivot columns. If any of them *is* a pivot column, then some $\vec{v}_i$ is not in $\operatorname{span} W$, so $\operatorname{span} V \neq \operatorname{span} W$.
3. Now swap the order: form $M' = \begin{pmatrix} \vec{v}_1 & \cdots & \vec{v}_p & \vec{w}_1 & \cdots & \vec{w}_q \end{pmatrix}$ and row reduce. Check that the last q columns are **not** pivot columns. If any $\vec{w}_j$ is a pivot column, then $\vec{w}_j \notin \operatorname{span} V$.
4. If both checks pass, $\operatorname{span} V = \operatorname{span} W$.

Why this works: a vector $\vec{v}_i$ is in $\operatorname{span} W$ iff the system $W\vec{x} = \vec{v}_i$ is consistent. In the augmented matrix $\begin{pmatrix} W & \vec{v}_i \end{pmatrix}$, consistency means the last column is *not* a pivot column. By checking M in step 2 we simultaneously check all $\vec{v}_i$ against the span of all $\vec{w}_j$.

Clever trick: one big augmented matrix checks all memberships

Instead of solving p separate systems, put everything into one big augmented matrix. Each non-pivot column on the right of the divide is automatically a linear combination of the pivot columns on the left.

5.6 Coordinate Systems

Theorem. Let H be a subspace with basis $\mathcal{B} = (\vec{b}_1, \ldots, \vec{b}_k)$. For each $\vec{v} \in H$, there exist **unique** coefficients $c_1, \ldots, c_k \in \mathbb{R}$ such that $\vec{v} = c_1\vec{b}_1 + \cdots + c_k\vec{b}_k$.

Proof: *Existence:* Since $\mathcal{B}$ is a basis, it spans H. Since $\vec{v} \in H$, there exist $c_1, \ldots, c_k \in \mathbb{R}$ such that $\vec{v} = c_1\vec{b}_1 + \cdots + c_k\vec{b}_k$.

Uniqueness: Suppose $\vec{v} = c_1\vec{b}_1 + \cdots + c_k\vec{b}_k = d_1\vec{b}_1 + \cdots + d_k\vec{b}_k$. Subtracting, $(c_1 - d_1)\vec{b}_1 + \cdots + (c_k - d_k)\vec{b}_k = \vec{0}$. Since $\mathcal{B}$ is a basis, the $\vec{b}_i$ are linearly independent, so $c_i - d_i = 0$ for all i, i.e., $c_i = d_i$. Hence the coefficients are unique. ■

Definition. The *coordinate vector* of $\vec{v}$ in the basis $\mathcal{B}$ is

$$[\vec{v}]_{\mathcal{B}} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_k \end{pmatrix} \in \mathbb{R}^k$$

where $\vec{v} = c_1\vec{b}_1 + \cdots + c_k\vec{b}_k$. Equivalently, $[\vec{v}]_{\mathcal{B}}$ is the unique solution to $\begin{pmatrix} \vec{b}_1 & \cdots & \vec{b}_k \end{pmatrix}\vec{x} = \vec{v}$.

The map $\vec{v} \mapsto [\vec{v}]_{\mathcal{B}}$ is linear and invertible

The function $H \to \mathbb{R}^k$ given by $\vec{v} \mapsto [\vec{v}]_{\mathcal{B}}$ is a linear transformation and is invertible. Thus H "looks the same as" $\mathbb{R}^k$.

5.7 Dimension

All bases have the same size

Theorem. Every basis of a subspace $H \subseteq \mathbb{R}^n$ has the same number of elements.

Proof: Suppose for contradiction that $\mathcal{B} = (\vec{b}_1, \ldots, \vec{b}_k)$ and $\mathcal{B}' = (\vec{b}'_1, \ldots, \vec{b}'_l)$ are two bases of H with $k < l$.

Consider the coordinate vectors $[\vec{b}'_1]_{\mathcal{B}}, \ldots, [\vec{b}'_l]_{\mathcal{B}}$, which are vectors in $\mathbb{R}^k$. Since $l > k$, this is a list of more than k vectors in $\mathbb{R}^k$, so by the proposition "$p > n \Rightarrow$ linearly dependent" (applied with $n = k$, $p = l$), these vectors are linearly dependent. Thus there exist scalars $c_1, \ldots, c_l$, not all zero, such that

$$c_1[\vec{b}'_1]_{\mathcal{B}} + \cdots + c_l[\vec{b}'_l]_{\mathcal{B}} = \vec{0} \in \mathbb{R}^k.$$

Since the coordinate map $\vec{v} \mapsto [\vec{v}]_{\mathcal{B}}$ is linear and invertible (hence injective), this implies

$$c_1\vec{b}'_1 + \cdots + c_l\vec{b}'_l = \vec{0}$$

with not all $c_i = 0$. This contradicts the linear independence of $\mathcal{B}'$. Hence $l \leq k$. By symmetry (exchanging $\mathcal{B}$ and $\mathcal{B}'$), $k \leq l$. Therefore $k = l$. ■

Definition. The *dimension* of a subspace H, denoted $\dim H$, is the number of vectors in any basis of H. We have $\dim \mathbb{R}^n = n$ and $\dim\{\vec{0}\} = 0$.

Basis theorem

Theorem (Basis theorem). If H is a subspace of $\mathbb{R}^n$ with $\dim H = p$, then:
1. Any set of p linearly independent vectors in H is a basis for H.
2. Any set of p vectors in H whose span is H is a basis for H.

Proof: (i) Let $\vec{v}_1, \ldots, \vec{v}_p$ be linearly independent vectors in H, where $p = \dim H$. We show they span H. Let $\mathcal{B} = (\vec{b}_1, \ldots, \vec{b}_p)$ be a basis of H (which has p elements). Suppose $\operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_p\} \neq H$. Then there exists $\vec{v}_{p+1} \in H \setminus \operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_p\}$. By the standard argument (used in the proof that every subspace has a basis), $\{\vec{v}_1, \ldots, \vec{v}_p, \vec{v}_{p+1}\}$ is linearly independent, giving $p+1$ linearly independent vectors in H. But then $\{[\vec{v}_1]_{\mathcal{B}}, \ldots, [\vec{v}_{p+1}]_{\mathcal{B}}\}$ is a list of $p+1$ vectors in $\mathbb{R}^p$, hence linearly dependent — leading to a nontrivial relation among $\vec{v}_1, \ldots, \vec{v}_{p+1}$ (since the coordinate map is injective), contradicting their independence. So $\vec{v}_1, \ldots, \vec{v}_p$ span H, and being independent they form a basis. ■

(ii) Let $\vec{v}_1, \ldots, \vec{v}_p$ be vectors in H with $\operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_p\} = H$. Suppose they are linearly dependent. Then some $\vec{v}_j$ is a linear combination of the others, so

$$\operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_{j-1}, \vec{v}_{j+1}, \ldots, \vec{v}_p\} = H.$$

This gives a spanning set of H with $p-1$ vectors, from which we can extract a basis of size $\leq p-1$. But every basis of H has size $p = \dim H$ by the previous theorem, a contradiction. Hence $\vec{v}_1, \ldots, \vec{v}_p$ are linearly independent, and being a spanning set, they form a basis. ■

5.8 Rank

Definition. The *rank* of a matrix A is $\operatorname{rank} A = \dim \operatorname{Col} A$ (the number of pivot columns).

Rank-nullity theorem

Theorem (Rank-nullity theorem). If A is an $m \times n$ matrix, then

$$\boxed{\operatorname{rank} A + \dim \operatorname{Nul} A = n.}$$

Proof: Row reduce A to $\operatorname{RREF}(A)$. The solutions to $A\vec{x} = \vec{0}$ are the same as the solutions to $\operatorname{RREF}(A)\vec{x} = \vec{0}$ (row equivalence preserves solutions). In $\operatorname{RREF}(A)$, each column is either:

- a *pivot column*: corresponds to a **basic variable**, determined by the free variables; or
- a *non-pivot column*: corresponds to a **free variable**, which can take any value in $\mathbb{R}$.

The number of pivot columns is $\operatorname{rank} A$ (since the pivot columns of A form a basis for $\operatorname{Col} A$, the number of pivots is $\dim \operatorname{Col} A$). The number of non-pivot (free) columns is $n - \operatorname{rank} A$.

Each free variable gives one basis vector for $\operatorname{Nul} A$ (obtained by setting that free variable to 1 and the other free variables to 0). Thus $\dim \operatorname{Nul} A$ = number of free variables = number of non-pivot columns $= n - \operatorname{rank} A$. Therefore

$$\operatorname{rank} A + \dim \operatorname{Nul} A = \operatorname{rank} A + (n - \operatorname{rank} A) = n.$$

■

Invertibility characterization via rank

Corollary. For an $n \times n$ matrix A, the following are equivalent:

1. A is invertible.
2. The columns of A form a basis for $\mathbb{R}^n$.
3. $\operatorname{rank} A = n$.
4. $\dim \operatorname{Nul} A = 0$.

Proof: (1) $\Leftrightarrow$ (2): A is invertible iff its columns are linearly independent and span $\mathbb{R}^n$ (by the full characterization theorem for square matrices), which is exactly the definition of the columns forming a basis for $\mathbb{R}^n$.

(2) $\Leftrightarrow$ (3): The columns form a basis for $\mathbb{R}^n$ iff they are linearly independent and span $\mathbb{R}^n$. By the proposition that pivot columns of A form a basis for $\operatorname{Col} A$, $\operatorname{rank} A = \dim \operatorname{Col} A$. So the columns span $\mathbb{R}^n$ iff $\operatorname{Col} A = \mathbb{R}^n$ iff $\dim \operatorname{Col} A = n$ iff $\operatorname{rank} A = n$. And if $\operatorname{rank} A = n$, all n columns are pivot columns, so the columns are linearly independent. Hence (2) $\Leftrightarrow$ (3).

(3) $\Leftrightarrow$ (4): By the rank-nullity theorem, $\operatorname{rank} A + \dim \operatorname{Nul} A = n$. So $\operatorname{rank} A = n$ iff $\dim \operatorname{Nul} A = 0$. ■

Proposition. If $U \subseteq V$ are subspaces of $\mathbb{R}^n$, then $\dim U \leq \dim V \leq n$. Moreover, if $U \subsetneq V$ then $\dim U < \dim V$, and if $\dim U = \dim V$ then $U = V$.

Proof: Let $\dim V = k$ with basis $\mathcal{B}_V = (\vec{b}_1, \ldots, \vec{b}_k)$. Since $U \subseteq V$, any linearly independent set in U is also a linearly independent set in $V \cong \mathbb{R}^k$ (via the coordinate map). By the proposition "$p > n \Rightarrow$ linearly dependent," any list of more than k vectors in V is dependent. Hence any basis of U has at most k elements: $\dim U \leq k = \dim V$. Also $\dim V \leq n$ since $V \subseteq \mathbb{R}^n$ and any basis of V is an independent list in $\mathbb{R}^n$ of size at most n.

If $U \subsetneq V$: pick $\vec{v} \in V \setminus U$. If $\mathcal{B}_U = (\vec{u}_1, \ldots, \vec{u}_l)$ is a basis of U, then $\vec{v} \notin \operatorname{span}\{\vec{u}_1, \ldots, \vec{u}_l\}$, so $\{\vec{u}_1, \ldots, \vec{u}_l, \vec{v}\}$ is linearly independent in V. Thus $l + 1 \leq \dim V$, i.e., $\dim U < \dim V$.

If $\dim U = \dim V = k$: let $\mathcal{B}_U = (\vec{u}_1, \ldots, \vec{u}_k)$ be a basis of U. This is a list of $k = \dim V$ linearly independent vectors in V. By the Basis theorem (part 1), this set is a basis for V. So $U = \operatorname{span}\{\vec{u}_1, \ldots, \vec{u}_k\} = V$. ■

Image of a subspace under a linear map

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be linear with standard matrix A ($m \times n$), let $H \subseteq \mathbb{R}^n$ be a subspace with $\dim H = k$, and let B be an $n \times k$ matrix whose columns form a basis of H, so $H = \operatorname{Col} B$. Set $T(H) = \{T(\vec{v}) : \vec{v} \in H\}$.

1. $T(H) = \operatorname{Col}(AB)$; hence $T(H)$ is a subspace of $\mathbb{R}^m$.
2. $\dim T(H) = \operatorname{rank}(AB) = k - \dim(H \cap \operatorname{Nul} A)$. In particular $\dim T(H) = k$ iff $H \cap \operatorname{Nul} A = \{\vec{0}\}$.
3. $\dim T(H) \le \min\{k,\ m,\ \operatorname{rank} A\}$.
4. If T is onto ($\operatorname{rank} A = m$, so $\dim \operatorname{Nul} A = n - m$), then
$$\max\{k - (n - m),\ 0\} \ \le\ \dim T(H) \ \le\ \min\{k,\ m\},$$
and every integer in this interval is attained by some k-dimensional H.
5. If T is one-to-one ($\operatorname{Nul} A = \{\vec{0}\}$), then $\dim T(H) = k$ exactly; this requires $k \le n \le m$.

Proof: (1) Every $\vec{h} \in H$ has the form $\vec{h} = B\vec{x}$ with $\vec{x} \in \mathbb{R}^k$, so $T(\vec{h}) = A\vec{h} = (AB)\vec{x}$, giving $T(H) = \{(AB)\vec{x} : \vec{x} \in \mathbb{R}^k\} = \operatorname{Col}(AB)$. (Directly: $\vec{0} = T(\vec{0}) \in T(H)$; $T(\vec{h}_1) + T(\vec{h}_2) = T(\vec{h}_1 + \vec{h}_2) \in T(H)$; and $c\,T(\vec{h}) = T(c\vec{h}) \in T(H)$.)

(2) For $\vec{x} \in \mathbb{R}^k$: $\vec{x} \in \operatorname{Nul}(AB) \iff AB\vec{x} = \vec{0} \iff B\vec{x} \in \operatorname{Nul} A$, while $B\vec{x} \in \operatorname{Col} B = H$ always; so $\vec{x} \in \operatorname{Nul}(AB) \iff B\vec{x} \in H \cap \operatorname{Nul} A$. Since B has linearly independent columns, $\vec{x} \mapsto B\vec{x}$ is one-to-one, hence $\dim \operatorname{Nul}(AB) = \dim(H \cap \operatorname{Nul} A)$. The matrix AB has k columns, so rank-nullity gives $\operatorname{rank}(AB) = k - \dim \operatorname{Nul}(AB)$, and $\dim T(H) = \dim \operatorname{Col}(AB) = \operatorname{rank}(AB)$.

(3) By (2), $\dim T(H) \le k$; since $T(H) \subseteq \mathbb{R}^m$, $\dim T(H) \le m$; and since $\operatorname{Col}(AB) \subseteq \operatorname{Col} A$, $\dim T(H) \le \operatorname{rank} A$.

(4) With $j = \dim(H \cap \operatorname{Nul} A)$, (2) gives $\dim T(H) = k - j$. As $H \cap \operatorname{Nul} A$ lies in both H and $\operatorname{Nul} A$, $j \le \min\{k, n - m\}$, so
$$\dim T(H) = k - j \ge k - \min\{k, n - m\} = \max\{k - (n - m), 0\},$$
and also $\dim T(H) \le \min\{k, m\}$. Conversely fix any j with $\max\{0, k - m\} \le j \le \min\{k, n - m\}$. Take j independent vectors of $\operatorname{Nul} A$, and inside a complement of $\operatorname{Nul} A$ (on which A acts as an isomorphism onto $\mathbb{R}^m$, since $\operatorname{rank} A = m$) take $k - j$ further vectors with linearly independent images. Their span H has dimension k, with $\dim(H \cap \operatorname{Nul} A) = j$ and $\dim T(H) = k - j$. Letting j vary gives every integer in the stated interval.

(5) If T is one-to-one then $\operatorname{Nul} A = \{\vec{0}\}$, so $H \cap \operatorname{Nul} A = \{\vec{0}\}$ and (2) gives $\dim T(H) = k$. (One-to-one forces $n \le m$, and $k \le n$, so $k \le m$.) ■

Midterm 2021 Problem 5 (dimension of $T(H)$)

Problem. Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation and let H be a k-dimensional subspace of $\mathbb{R}^n$. Define $T(H) = \{T(\vec{v}) : \vec{v} \in H\}$. (a) Explain why $T(H)$ is a subspace of $\mathbb{R}^m$. (b) If T is onto, what are the possibilities for $\dim T(H)$? Justify the answer in terms of k, m, n. (c) If T is one-to-one, what are the possibilities for $\dim T(H)$? Justify the answer in terms of k, m, n.

Solution (a). Let A be the standard matrix of T and let B be an $n \times k$ matrix whose columns are a basis of H. Then $H = \operatorname{Col} B$ and
$$T(H) = \{(AB)\vec{x} : \vec{x} \in \mathbb{R}^k\} = \operatorname{Col}(AB),$$

which is a subspace of $\mathbb{R}^m$. (Axiom check: $T(\vec{0}) = \vec{0} \in T(H)$; if $\vec{u}, \vec{w} \in H$ then $T(\vec{u}) + T(\vec{w}) = T(\vec{u} + \vec{w}) \in T(H)$; and $c\,T(\vec{u}) = T(c\vec{u}) \in T(H)$.)

Solution (b). By (2) above, $\dim T(H) = k - j$ with $j = \dim(H \cap \operatorname{Nul} A)$. Since T is onto, $\operatorname{rank} A = m \leq n$ and $\dim \operatorname{Nul} A = n - \operatorname{rank} A = n - m$. Because $H \cap \operatorname{Nul} A$ lies in both H and $\operatorname{Nul} A$,

$$0 \leq j \leq \min\{k,\, n - m\},$$

and therefore

$$\max\{k - (n - m),\, 0\} \;\leq\; \dim T(H) = k - j \;\leq\; \min\{k,\, m\}.$$

The bound $\dim T(H) \leq m$ is simply $T(H) \subseteq \mathbb{R}^m$. Every integer in the interval occurs: choosing H to contain exactly j independent directions of $\operatorname{Nul} A$ (and $k - j$ independent directions outside it) gives $\dim T(H) = k - j$ for each admissible j.

Example. Take $n = 3$, $m = 2$, $T(x_1, x_2, x_3) = (x_1, x_2)$ (onto) and $k = 1$: the interval is $\max\{1 - 1, 0\} \leq \dim T(H) \leq \min\{1, 2\}$, i.e. $0 \leq \dim T(H) \leq 1$. Indeed $H = \operatorname{span}\{\vec{e}_3\} \subseteq \operatorname{Nul} A$ gives $T(H) = \{\vec{0}\}$ (dimension 0), while $H = \operatorname{span}\{\vec{e}_1\}$ gives $T(H) = \operatorname{span}\{(1, 0)\}$ (dimension 1).

Solution (c). If T is one-to-one then $\operatorname{Nul} A = \{\vec{0}\}$, so $H \cap \operatorname{Nul} A = \{\vec{0}\}$ and $j = 0$. Hence $\dim T(H) = k$ is the only possibility. Since one-to-one forces $n \leq m$ and $k \leq n$, we have $k \leq m$, so this value is compatible with $T(H) \subseteq \mathbb{R}^m$. Geometrically, a one-to-one linear map carries a k-dimensional subspace to a k-dimensional subspace without collapsing any direction.

Clever trick: count what the kernel eats

$\dim T(H) = \dim H - \dim(H \cap \operatorname{Nul} A)$: the image loses exactly the directions of H that the map sends to zero. Parts (b) and (c) are then only questions about bounding $\dim(H \cap \operatorname{Nul} A)$ — by $\dim \operatorname{Nul} A = n - m$ when T is onto, and by 0 when T is one-to-one.

Practice Problem #5.5 (Rank-1 characterization: $A = \vec{v}\vec{w}^{\top}$)

Problem. Prove that an $m \times n$ matrix A has rank exactly 1 if and only if $A = \vec{v}\vec{w}^{\top}$ for some nonzero vectors $\vec{v} \in \mathbb{R}^m$ and $\vec{w} \in \mathbb{R}^n$.

Solution. ($\Rightarrow$) Suppose $\operatorname{rank} A = 1$. Then $\operatorname{Col} A$ is 1-dimensional, so all columns of A are scalar multiples of some fixed nonzero column $\vec{v}$: column j is $w_j\vec{v}$ for some scalar w_j. Hence

$$A = \begin{pmatrix} w_1\vec{v} & w_2\vec{v} & \cdots & w_n\vec{v} \end{pmatrix} = \vec{v}\begin{pmatrix} w_1 & w_2 & \cdots & w_n \end{pmatrix} = \vec{v}\vec{w}^{\top},$$

where $\vec{w} = \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix}$. Since $\operatorname{rank} A = 1 > 0$, neither $\vec{v}$ nor $\vec{w}$ is zero.

($\Leftarrow$) Suppose $A = \vec{v}\vec{w}^{\top}$ with $\vec{v} \neq \vec{0}$, $\vec{w} \neq \vec{0}$. The jth column of A is $\vec{v}w_j$, which is a scalar multiple of $\vec{v}$. So every column of A lies in $\operatorname{span}\{\vec{v}\}$, meaning $\operatorname{Col} A \subseteq \operatorname{span}\{\vec{v}\}$, hence $\operatorname{rank} A = \dim \operatorname{Col} A \leq 1$. Since $\vec{v} \neq \vec{0}$ and $\vec{w} \neq \vec{0}$, at least one entry $w_j \neq 0$, so column j of A is $w_j\vec{v} \neq \vec{0}$, hence $\operatorname{rank} A \geq 1$. So $\operatorname{rank} A = 1$.

Practice Problem #3.3 (Block matrix $\begin{pmatrix} A & B \end{pmatrix}$ rank relations)

Problem. Let A and B have the same number of rows. Prove that $\operatorname{rank}\begin{pmatrix} A & B \end{pmatrix} \leq \operatorname{rank} A + \operatorname{rank} B$.

Solution. Let $r_A = \operatorname{rank} A$ and $r_B = \operatorname{rank} B$. Let $\mathcal{B}_A = \{\vec{a}_{i_1}, \ldots, \vec{a}_{i_{r_A}}\}$ be a basis for $\operatorname{Col} A$ (pivot columns of A) and $\mathcal{B}_B = \{\vec{b}_{j_1}, \ldots, \vec{b}_{j_{r_B}}\}$ be a basis for $\operatorname{Col} B$ (pivot

columns of B). Now every column of $(A \quad B)$ is either a column of A (hence in $\operatorname{span}\mathcal{B}_A$) or a column of B (hence in $\operatorname{span}\mathcal{B}_B$). Therefore all columns of $(A \quad B)$ lie in $\operatorname{span}(\mathcal{B}_A \cup \mathcal{B}_B)$, which has dimension at most $|\mathcal{B}_A| + |\mathcal{B}_B| = r_A + r_B$. Hence $\operatorname{rank}(A \quad B) = \dim \operatorname{Col}(A \quad B) \leq r_A + r_B$.

Clever trick: union of bases spans the sum

$\operatorname{Col}(A \quad B) = \operatorname{Col} A + \operatorname{Col} B$ by definition (every column of the block matrix is in one of the two column spaces). The sum of two subspaces has dimension at most the sum of their dimensions, which is exactly the inequality we proved.

Practice Problem #3.4 (Stacked matrix $\begin{pmatrix} A \\ C \end{pmatrix}$ rank relations)

Problem. Let A and C have the same number of columns. Prove that $\operatorname{rank}\begin{pmatrix} A \\ C \end{pmatrix} \leq \operatorname{rank} A + \operatorname{rank} C$.

Solution. Let $r_A = \operatorname{rank} A$ and $r_C = \operatorname{rank} C$. The row space of $\begin{pmatrix} A \\ C \end{pmatrix}$ is the span of the rows of A together with the rows of C. The row space of A has dimension r_A and the row space of C has dimension r_C. The combined row space is the sum of these two subspaces, hence has dimension at most $r_A + r_C$. But for any matrix, the rank equals the dimension of the row space (row rank = column rank). Therefore $\operatorname{rank}\begin{pmatrix} A \\ C \end{pmatrix} \leq r_A + r_C = \operatorname{rank} A + \operatorname{rank} C$.

Practice Problem #4.11 (If $\vec{v}^\top A\vec{w} = \vec{v}^\top B\vec{w}$ for all $\vec{v}, \vec{w}$, then $A = B$)

Problem. Let A and B be $m \times n$ matrices. Suppose $\vec{v}^\top A\vec{w} = \vec{v}^\top B\vec{w}$ for all $\vec{v} \in \mathbb{R}^m$ and $\vec{w} \in \mathbb{R}^n$. Prove that $A = B$.

Solution. Set $\vec{v} = \vec{e}_i$ (the ith standard basis vector of $\mathbb{R}^m$) and $\vec{w} = \vec{e}_j$ (the jth standard basis vector of $\mathbb{R}^n$). Then

$$\vec{e}_i^\top A\vec{e}_j = (\vec{e}_i^\top A)\vec{e}_j = (\text{the } i\text{th row of } A) \text{ dotted with } \vec{e}_j = A_{ij}.$$

Similarly $\vec{e}_i^\top B\vec{e}_j = B_{ij}$. By assumption, $A_{ij} = B_{ij}$ for all i, j, so every entry of A equals the corresponding entry of B. Hence $A = B$.

Clever trick: test on standard basis vectors

Any identity that holds for all vectors must in particular hold for the standard basis vectors, which gives you every matrix entry one at a time. This is the "foundation test": if a matrix identity works on all basis vectors, it works everywhere by linearity.

Practice Problem #4.4 (Transpose of linear function = $A^\top$, uniqueness via $\vec{e}_i$)

Problem. Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be linear with standard matrix A. Define the *transpose* of T to be the map $T^\top : \mathbb{R}^m \to \mathbb{R}^n$ given by $(T^\top(\vec{v}))^\top \vec{w} = \vec{v}^\top T(\vec{w})$ for all $\vec{v} \in \mathbb{R}^m$, $\vec{w} \in \mathbb{R}^n$. Prove that $T^\top$ is linear with standard matrix $A^\top$.

Solution. Let B be the standard matrix of $T^\top$, so $T^\top(\vec{v}) = B\vec{v}$ for all $\vec{v}$. The defining equation says $(B\vec{v})^\top \vec{w} = \vec{v}^\top(A\vec{w})$, i.e., $\vec{v}^\top B^\top \vec{w} = \vec{v}^\top A\vec{w}$ for all $\vec{v}, \vec{w}$. By the uniqueness problem (P4.11), this forces $B^\top = A$, so $B = A^\top$.

To verify $T^\top$ is linear: $(T^\top(\vec{u}+\vec{v}))^\top \vec{w} = (\vec{u}+\vec{v})^\top A\vec{w} = \vec{u}^\top A\vec{w} + \vec{v}^\top A\vec{w} = (T^\top(\vec{u}))^\top \vec{w} + (T^\top(\vec{v}))^\top \vec{w} = (T^\top(\vec{u}) + T^\top(\vec{v}))^\top \vec{w}$ for all $\vec{w}$, hence $T^\top(\vec{u} + \vec{v}) = T^\top(\vec{u}) + T^\top(\vec{v})$. Homogeneity is similar.

Practice Problem #5.15 ($\mathrm{Nul}(A^\top A) = \mathrm{Nul}\, A$, so $\mathrm{rank}(A^\top A) = \mathrm{rank}\, A$)

Problem. Let A be an $m \times n$ matrix. Prove that $\mathrm{Nul}(A^\top A) = \mathrm{Nul}\, A$, and deduce that $\mathrm{rank}(A^\top A) = \mathrm{rank}\, A$.
Solution. We show both inclusions.
($\mathrm{Nul}\, A \subseteq \mathrm{Nul}(A^\top A)$): If $A\vec{x} = \vec{0}$, then $(A^\top A)\vec{x} = A^\top(A\vec{x}) = A^\top\vec{0} = \vec{0}$, so $\vec{x} \in \mathrm{Nul}(A^\top A)$.
($\mathrm{Nul}(A^\top A) \subseteq \mathrm{Nul}\, A$): If $(A^\top A)\vec{x} = \vec{0}$, then premultiply by $\vec{x}^\top$:

$$\vec{x}^\top(A^\top A)\vec{x} = 0 \quad \Rightarrow \quad (A\vec{x})^\top(A\vec{x}) = 0 \quad \Rightarrow \quad \|A\vec{x}\|^2 = 0.$$

The squared norm of a real vector is zero iff the vector itself is zero, so $A\vec{x} = \vec{0}$. Hence $\vec{x} \in \mathrm{Nul}\, A$.
So $\mathrm{Nul}(A^\top A) = \mathrm{Nul}\, A$. Now apply rank-nullity: $\mathrm{rank}(A^\top A) + \dim \mathrm{Nul}(A^\top A) = n$ and $\mathrm{rank}\, A + \dim \mathrm{Nul}\, A = n$. Since the null-space dimensions are equal, the ranks must be equal: $\mathrm{rank}(A^\top A) = \mathrm{rank}\, A$.

Clever trick: use the squared norm

$(A^\top A)\vec{x} = \vec{0}$ is a matrix equation, but the trick is to multiply by $\vec{x}^\top$ to get a scalar equation: $\|A\vec{x}\|^2 = 0$. In real vector spaces, this forces $A\vec{x} = \vec{0}$ immediately. This argument uses the positivity of the inner product and does not generalize to complex matrices without a hermitian conjugate.

Practice Problem #5.8 (If $\mathrm{rank}\, A = n$ then $\mathrm{rank}(AB) = \mathrm{rank}\, B$)

Problem. Let A be an $m \times n$ matrix and B be $n \times p$. Suppose $\mathrm{rank}\, A = n$. Prove that $\mathrm{rank}(AB) = \mathrm{rank}\, B$.
Solution. We show that the columns of B and the columns of AB have the same linear independence. Let $\vec{b}_1, \ldots, \vec{b}_p$ be the columns of B, so the columns of AB are $A\vec{b}_1, \ldots, A\vec{b}_p$.

Linear relation preservation: Suppose $\sum_{j=1}^{p} c_j\vec{b}_j = \vec{0}$. Apply A: $A\left(\sum c_j\vec{b}_j\right) = \sum c_j A\vec{b}_j = \vec{0}$. So any dependence among the $\vec{b}_j$ gives a dependence among the $A\vec{b}_j$.

Conversely: Suppose $\sum c_j A\vec{b}_j = \vec{0}$, i.e., $A\left(\sum c_j\vec{b}_j\right) = \vec{0}$. Let $\vec{x} = \sum c_j\vec{b}_j$. Then $A\vec{x} = \vec{0}$, so $\vec{x} \in \mathrm{Nul}\, A$. But $\mathrm{rank}\, A = n$ means $\dim \mathrm{Nul}\, A = n - n = 0$, so $\mathrm{Nul}\, A = \{\vec{0}\}$. Hence $\vec{x} = \sum c_j\vec{b}_j = \vec{0}$.

Therefore the same coefficients produce a dependence among $\{\vec{b}_j\}$ and among $\{A\vec{b}_j\}$. So the two sets have the same dimension of their spans, i.e., $\mathrm{rank}\, B = \mathrm{rank}(AB)$.

Practice Problem #5.7 (Right inverse iff pivot in every row; left inverse iff pivot in every column)

Problem. Let A be $m \times n$. Prove:
1. A has a *right inverse* (i.e., there exists B with $AB = I_m$) iff $\mathrm{rank}\, A = m$ (pivot in every row).
2. A has a *left inverse* (i.e., there exists C with $CA = I_n$) iff $\mathrm{rank}\, A = n$ (pivot in every column).

Solution.
(1) ($\Rightarrow$) Suppose $AB = I_m$. Then for any $\vec{v} \in \mathbb{R}^m$, $\vec{v} = I_m\vec{v} = A(B\vec{v})$, so $\vec{v} \in \mathrm{Col}\, A$.

Hence $\operatorname{Col} A = \mathbb{R}^m$, which means $\operatorname{rank} A = m$ (every row has a pivot).
($\Leftarrow$) Suppose $\operatorname{rank} A = m$. Then $\operatorname{Col} A = \mathbb{R}^m$, so for each standard basis vector $\vec{e}_j \in \mathbb{R}^m$, the system $A\vec{x} = \vec{e}_j$ is consistent. Let $\vec{b}_j$ be a solution, and set $B = \begin{pmatrix}\vec{b}_1 & \cdots & \vec{b}_m\end{pmatrix}$. Then $AB = \begin{pmatrix}A\vec{b}_1 & \cdots & A\vec{b}_m\end{pmatrix} = (\vec{e}_1 \ \cdots \ \vec{e}_m) = I_m$.
(2) ($\Rightarrow$) Suppose $CA = I_n$. Then $(CA)^\top = A^\top C^\top = I_n^\top = I_n$, so $A^\top$ has a right inverse $C^\top$. By part (1), applied to $A^\top$, we have $\operatorname{rank} A^\top = n$. But $\operatorname{rank} A^\top = \operatorname{rank} A$, so $\operatorname{rank} A = n$.
($\Leftarrow$) Suppose $\operatorname{rank} A = n$. Then $\operatorname{rank} A^\top = n$ also. By part (1), applied to $A^\top$, there exists D such that $A^\top D = I_n$. Then $(A^\top D)^\top = D^\top A = I_n^\top = I_n$, so $C = D^\top$ is a left inverse of A.

Clever trick: transpose turns left into right

Notice that $CA = I$ iff $A^\top C^\top = I$, so left-inverse questions reduce to right-inverse questions about the transpose. This avoids having to repeat both proofs from scratch.

Practice Problem #5.9 ($\operatorname{Nul} A \cap \operatorname{Col} A \supseteq \{\vec{v}, \vec{w}\} \Rightarrow \vec{w} = t\vec{v}$)

Problem. Let A be $n \times n$. Suppose $\vec{v}, \vec{w} \in \operatorname{Nul} A \cap \operatorname{Col} A$. Prove that if $\vec{v} \neq \vec{0}$ and $\vec{v} \neq \vec{w}$, then $\vec{w} = t\vec{v}$ for some scalar t.
Solution. Since $\vec{v} \in \operatorname{Col} A$, there exists $\vec{x}$ with $\vec{v} = A\vec{x}$. Since $\vec{v} \in \operatorname{Nul} A$, we have $A\vec{v} = A(A\vec{x}) = \vec{0}$, so $\vec{x} \in \operatorname{Nul}(A^2)$. Similarly, $\vec{w} = A\vec{y}$ for some $\vec{y}$ with $A\vec{y} = \vec{0}$.
Now consider the subspace $\operatorname{Nul}(A^2) \cap \operatorname{Col} A$. Since $\vec{v} = A\vec{x}$ with $\vec{x} \in \operatorname{Nul}(A^2)$, we have $\vec{v} \in A(\operatorname{Nul}(A^2)) = \operatorname{Col} A|_{\operatorname{Nul}(A^2)}$. Now $\operatorname{Nul}(A^2)$ is larger than or equal to $\operatorname{Nul} A$, so its image under A is contained in $\operatorname{Nul} A \cap \operatorname{Col} A$.
More directly: suppose there are two linearly independent vectors $\vec{v}, \vec{w} \in \operatorname{Nul} A \cap \operatorname{Col} A$. Then $\dim(\operatorname{Nul} A \cap \operatorname{Col} A) \geq 2$. By the dimension formula, this forces $\dim \operatorname{Nul} A + \dim \operatorname{Col} A > n$, contradicting rank-nullity (which says $\dim \operatorname{Nul} A + \dim \operatorname{Col} A = n$). So $\operatorname{Nul} A \cap \operatorname{Col} A$ has dimension at most 1. Therefore any two nonzero vectors in it are scalar multiples of each other.

Practice Problem #5.10 (Generalized $\operatorname{Col}^+ A$ and $\operatorname{Nul}^+ A$ are subspaces)

Problem. Define $\operatorname{Col}^+ A = \bigcup_{k \geq 1} \operatorname{Col}(A^k)$ and $\operatorname{Nul}^+ A = \bigcup_{k \geq 1} \operatorname{Nul}(A^k)$ (generalized column space and generalized null space of a square matrix A). Prove that both are subspaces.
Solution. These are *nested* unions: $\operatorname{Col} A \subseteq \operatorname{Col}(A^2) \subseteq \operatorname{Col}(A^3) \subseteq \cdots$ (because $\operatorname{Col}(A^k) = \{A^k\vec{x}\} = \{A^{k-1}(A\vec{x})\} \subseteq \{A^{k-1}\vec{y}\} = \operatorname{Col}(A^{k-1})$ ... actually, the inclusion goes the other way: $\operatorname{Col}(A^k) \subseteq \operatorname{Col}(A^{k-1})$ since every $A^k\vec{x}$ is $A^{k-1}(A\vec{x}) \in \operatorname{Col}(A^{k-1})$. So $\operatorname{Col} A \supseteq \operatorname{Col}(A^2) \supseteq \operatorname{Col}(A^3) \supseteq \cdots$ is a descending chain, and $\operatorname{Col}^+ A = \bigcap_{k \geq 1} \operatorname{Col}(A^k)$ is the *intersection* of these subspaces (in older notation, sometimes called the eventual column space).
For the null spaces: $\operatorname{Nul} A \subseteq \operatorname{Nul}(A^2) \subseteq \operatorname{Nul}(A^3) \subseteq \cdots$ (if $A^k\vec{x} = \vec{0}$ then $A^{k+1}\vec{x} = A(A^k\vec{x}) = \vec{0}$). So $\operatorname{Nul}^+ A = \bigcup_{k \geq 1} \operatorname{Nul}(A^k)$ is an *ascending* union of subspaces.
For the ascending union: let $\vec{u}, \vec{v} \in \operatorname{Nul}^+ A$. Then $\vec{u} \in \operatorname{Nul}(A^p)$ and $\vec{v} \in \operatorname{Nul}(A^q)$ for some p, q. Let $k = \max(p, q)$. Since the chain is ascending, both $\vec{u}, \vec{v} \in \operatorname{Nul}(A^k)$, which is a subspace, so $\vec{u} + \vec{v} \in \operatorname{Nul}(A^k) \subseteq \operatorname{Nul}^+ A$ and $c\vec{u} \in \operatorname{Nul}(A^k) \subseteq \operatorname{Nul}^+ A$. Also $\vec{0} \in \operatorname{Nul} A \subseteq \operatorname{Nul}^+ A$. Hence $\operatorname{Nul}^+ A$ is a subspace.

For the intersection $\mathrm{Col}^+ A = \bigcap_{k\geq 1} \mathrm{Col}(A^k)$: the intersection of any collection of subspaces is always a subspace (contains $\vec{0}$, closed under addition and scalar multiplication because each $\mathrm{Col}(A^k)$ is closed and $\vec{0}$ is in all of them).

Practice Problem #5.11 (A acts invertibly on $\mathrm{Col}^+ A$, and $\mathrm{Col}^+ A \cap \mathrm{Nul}^+ A = \{\vec{0}\}$)

Problem. Let A be $n \times n$. Define $\mathrm{Col}^+ A$ as the intersection of $\mathrm{Col}(A^k)$ and $\mathrm{Nul}^+ A$ as the union of $\mathrm{Nul}(A^k)$ (as in P5.10). Prove:
1. The restriction of A to $\mathrm{Col}^+ A$ is invertible.
2. $\mathrm{Col}^+ A \cap \mathrm{Nul}^+ A = \{\vec{0}\}$.

Solution.
(1) Let $\vec{v} \in \mathrm{Col}^+ A$. Then $\vec{v} \in \mathrm{Col}(A^k)$ for all k, so in particular $\vec{v} \in \mathrm{Col}(A^{k+1})$ for all k, meaning $\vec{v} = A^{k+1}\vec{x}_k$ for some $\vec{x}_k$. But $\vec{v} = A(A^k\vec{x}_k)$, so $\vec{v} \in A(\mathbb{R}^n) = \mathrm{Col}\, A$. Actually the cleanest argument: for any $\vec{v} \in \mathrm{Col}^+ A \subseteq \mathrm{Col}\, A$, there exists $\vec{u}$ with $\vec{v} = A\vec{u}$. Now $\vec{v} \in \mathrm{Col}(A^k)$ for all k, so $\vec{u} \in \mathrm{Col}(A^{k-1})$ for all k (since $A\vec{u} \in \mathrm{Col}(A^k) \Rightarrow \vec{u} \in \mathrm{Col}(A^{k-1})$). Hence $\vec{u} \in \mathrm{Col}^+ A$ also. So $A|_{\mathrm{Col}^+ A} : \mathrm{Col}^+ A \to \mathrm{Col}^+ A$ is surjective. Since it is a surjective linear map on a finite-dimensional space, it is invertible.
(2) Let $\vec{v} \in \mathrm{Col}^+ A \cap \mathrm{Nul}^+ A$. Since $\vec{v} \in \mathrm{Nul}^+ A$, there exists some k with $A^k\vec{v} = \vec{0}$. Since $\vec{v} \in \mathrm{Col}^+ A \subseteq \mathrm{Col}(A^k)$, there exists $\vec{x}$ with $\vec{v} = A^k\vec{x}$. Now apply A^k: $A^k\vec{v} = A^{2k}\vec{x} = \vec{0}$, so $\vec{x} \in \mathrm{Nul}(A^{2k})$. But we need $\vec{v} = \vec{0}$. Actually: $\vec{v} \in \mathrm{Col}^+ A$ means $\vec{v} = A^k\vec{x}_k$ for all k, while $\vec{v} \in \mathrm{Nul}(A^m)$ for some m. Then $\vec{v} = A^m\vec{x}_m$ with $A^m\vec{v} = \vec{0}$, so $A^{2m}\vec{x}_m = \vec{0}$. But $\vec{v} = A^m\vec{x}_m \in \mathrm{Nul}(A^m)$, which means $A^m(A^m\vec{x}_m) = A^{2m}\vec{x}_m = \vec{0}$. Since $A|_{\mathrm{Col}^+ A}$ is invertible and $\vec{v} = A^m\vec{x}_m \in \mathrm{Col}^+ A$, applying A^m to $\vec{v}$ gives $\vec{0}$, which forces $\vec{v} = \vec{0}$ (because invertible maps are one-to-one: if $A^m\vec{v} = \vec{0}$ and $\vec{v} \in \mathrm{Col}^+ A$, then $\vec{v}$ must be $\vec{0}$). Hence the intersection is trivial.

Practice Problem #5.12 (Unique decomposition $\vec{u} = \vec{v} + \vec{w}$, $\vec{v} \in \mathrm{Col}^+ A$, $\vec{w} \in \mathrm{Nul}^+ A$)

Problem. Let A be $n \times n$. Prove that every vector $\vec{u} \in \mathbb{R}^n$ can be written uniquely as $\vec{u} = \vec{v} + \vec{w}$ where $\vec{v} \in \mathrm{Col}^+ A$ and $\vec{w} \in \mathrm{Nul}^+ A$.
Solution. By rank-nullity, $\dim \mathrm{Col}(A^k) + \dim \mathrm{Nul}(A^k) = n$ for each k. The dimensions $\dim \mathrm{Col}(A^k)$ are a non-increasing sequence of nonnegative integers, so they stabilize at some index k_0: for $k \geq k_0$, $\dim \mathrm{Col}(A^k) = \dim \mathrm{Col}(A^{k_0}) =: r_+$, and correspondingly $\dim \mathrm{Nul}(A^k) = n - r_+$.
Now $\mathrm{Col}^+ A = \mathrm{Col}(A^{k_0})$ (once the rank stabilizes, $\mathrm{Col}(A^{k+1}) = \mathrm{Col}(A^k)$), and $\mathrm{Nul}^+ A = \mathrm{Nul}(A^{k_0})$ (since the null-space dimensions are also stable from k_0 onward). From P5.11, $\mathrm{Col}^+ A \cap \mathrm{Nul}^+ A = \{\vec{0}\}$. Also $\dim \mathrm{Col}^+ A + \dim \mathrm{Nul}^+ A = r_+ + (n - r_+) = n$. Two subspaces whose dimensions add up to n and whose intersection is $\{\vec{0}\}$ must have $\mathrm{Col}^+ A + \mathrm{Nul}^+ A = \mathbb{R}^n$. Hence every $\vec{u}$ can be written as $\vec{u} = \vec{v} + \vec{w}$ with $\vec{v} \in \mathrm{Col}^+ A$ and $\vec{w} \in \mathrm{Nul}^+ A$.
Uniqueness: Suppose $\vec{u} = \vec{v}_1 + \vec{w}_1 = \vec{v}_2 + \vec{w}_2$. Then $\vec{v}_1 - \vec{v}_2 = \vec{w}_2 - \vec{w}_1$. The left side is in $\mathrm{Col}^+ A$, the right side is in $\mathrm{Nul}^+ A$. Since $\mathrm{Col}^+ A \cap \mathrm{Nul}^+ A = \{\vec{0}\}$, we must have $\vec{v}_1 - \vec{v}_2 = \vec{0}$ and $\vec{w}_2 - \vec{w}_1 = \vec{0}$, i.e., $\vec{v}_1 = \vec{v}_2$ and $\vec{w}_1 = \vec{w}_2$.

Clever trick: the Fitting decomposition

This is the *Fitting decomposition* for a linear map on a finite-dimensional vector space. The eventual column space (the stable image) and eventual null space (the stable

kernel) form a direct sum decomposition of the whole space. It is the cleanest way to understand the Jordan form and the action of powers of a matrix.

Practice Problem #4.3 (rank relationships for row/column block matrices)

Problem. Let A be $m \times n$, B be $m \times p$, C be $q \times n$. Prove:

(a) $\operatorname{rank}(A) \leq \operatorname{rank}\begin{pmatrix} A & B \end{pmatrix}$.

(b) $\operatorname{rank}(B) \leq \operatorname{rank}\begin{pmatrix} A & B \end{pmatrix}$.

(c) $\operatorname{rank}(A) \leq \operatorname{rank}\begin{pmatrix} A \\ C \end{pmatrix}$.

(d) $\operatorname{rank}\begin{pmatrix} A & B \end{pmatrix} = \operatorname{rank}\begin{pmatrix} B & A \end{pmatrix}$.

Solution.

(a) Every pivot of A remains a pivot in the larger row-reduced matrix, so the rank can only increase. The first n columns of $\operatorname{RREF}\begin{pmatrix} A & B \end{pmatrix}$ contain exactly $\operatorname{RREF}(A)$ (possibly with additional zero rows appended), so they have at least as many pivots as $\operatorname{RREF}(A)$. Hence $\operatorname{rank}(A) \leq \operatorname{rank}\begin{pmatrix} A & B \end{pmatrix}$.

(b) Follows from (a) by swapping the block labels A and B. Also, $\operatorname{rank}\begin{pmatrix} A & B \end{pmatrix} = \operatorname{rank}\begin{pmatrix} B & A \end{pmatrix}$ by part (d).

(c) The rows of A are a subset of the rows of $\begin{pmatrix} A \\ C \end{pmatrix}$. Row operations can only add pivots, never remove them from an already-pivoted row position. So the stacked matrix has at least as many row pivots as A. *Warning:* the RREF of the stacked matrix is **not** a simple vertical extension of $\operatorname{RREF}(A)$. Counterexample: $A = \begin{pmatrix} 1 & 1 \end{pmatrix}$, $C = \begin{pmatrix} 1 & 2 \end{pmatrix}$, both already RREF, but $\operatorname{RREF}\begin{pmatrix} A \\ C \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

(d) Swapping block columns is a permutation of the columns of the matrix. Rank is invariant under column permutations (permutation is a bijection on the column set, so it preserves linear independence and spanning properties). Hence $\operatorname{rank}\begin{pmatrix} A & B \end{pmatrix} = \operatorname{rank}\begin{pmatrix} B & A \end{pmatrix}$.

Clever trick: rank is permutation-invariant

Swapping rows or columns of a matrix does not change the rank. But swapping them *while row-reducing* does affect the RREF. This distinction is crucial: the rank is a robust invariant, while the specific reduced form depends on column ordering.

Practice Problem #5.14 (quadratic forms correspond to unique symmetric matrices)

Problem.

(a) Explain why $f(v) = v^\top A v$ is homogeneous of degree 2 and is a polynomial.

(b) Show every homogeneous degree-2 polynomial has a **unique** symmetric matrix A with $f(v) = v^\top A v$.

Solution.

(a) Homogeneity: $f(cv) = (cv)^\top A(cv) = c^2 v^\top A v = c^2 f(v)$, so f is homogeneous of degree 2. Writing $v = (x_1, \ldots, x_n)^\top$ and $A = (a_{ij})$:

$$f(v) = \sum_{i,j=1}^{n} a_{ij} x_i x_j = \sum_i a_{ii} x_i^2 + \sum_{i<j} (a_{ij} + a_{ji}) x_i x_j,$$

which is a polynomial in the variables $x_1, \ldots, x_n$.

(b) Write a general homogeneous degree-2 polynomial as

$$f = \sum_i d_i x_i^2 + \sum_{i<j} b_{ij} x_i x_j.$$

Define A by $a_{ii} = d_i$ and $a_{ij} = a_{ji} = \dfrac{b_{ij}}{2}$ for $i < j$. Then A is symmetric and $v^\top A v = f(v)$.

For uniqueness, suppose S is another symmetric matrix with $v^\top S v = f(v)$. At $v = e_i$ (the ith standard basis vector): $s_{ii} = e_i^\top S e_i = f(e_i) = d_i$. At $v = e_i + e_j$ with $i < j$:

$$s_{ii} + 2s_{ij} + s_{jj} = (e_i + e_j)^\top S(e_i + e_j) = f(e_i + e_j) = d_i + b_{ij} + d_j.$$

Substituting the known diagonal entries gives $d_i + 2s_{ij} + d_j = d_i + b_{ij} + d_j$, so $s_{ij} = \dfrac{b_{ij}}{2}$, matching A.

Example. $f(x_1, x_2) = 2x_1^2 + 6x_1x_2 - 5x_2^2$ has standard matrix $\begin{pmatrix} 2 & 3 \\ 3 & -5 \end{pmatrix}$ (the cross coefficient 6 is divided by 2).

Clever trick: symmetrize!

The entry a_{ij} gets **half** the cross-term coefficient because both (i, j) and (j, i) entries of A contribute to the $x_i x_j$ term in $v^\top A v$. This symmetrization is the canonical way to pass from a quadratic form to its matrix representation.

Practice Problem #6.7 (a nonsingular $k \times k$ submatrix forces $\operatorname{rank} A \geq k$)

Problem. Let A be a matrix. A $k \times l$ *submatrix* of A is formed by choosing k (not necessarily adjacent) rows and l (not necessarily adjacent) columns of A and using the entries of A in these rows and columns as the entries of a $k \times l$ matrix. For example, the 3×3 matrix

$$A = \begin{pmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ 2 & -3 & 0 \end{pmatrix}$$

has the 2×2 submatrices $\begin{pmatrix} -1 & 0 \\ 2 & -3 \end{pmatrix}$ (rows $2, 3$ and columns $1, 2$) and $\begin{pmatrix} 0 & -2 \\ 3 & 0 \end{pmatrix}$ (rows $1, 2$ and columns $2, 3$), among others. Explain why $\operatorname{rank}(A) \geq k$ if A has a $k \times k$ submatrix with nonzero determinant.

Solution. Suppose the submatrix C uses rows $i_1 < i_2 < \cdots < i_k$ and columns $j_1 < j_2 < \cdots < j_k$, and that $\det C \neq 0$. Let $\vec{v}_1, \ldots, \vec{v}_k$ be the corresponding columns $j_1, \ldots, j_k$ of A; thus C is obtained from these columns by keeping only the selected k rows.

We show that $\vec{v}_1, \ldots, \vec{v}_k$ are linearly independent. Suppose $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k = \vec{0}$. This says the same linear combination of the entries of these columns vanishes in <u>every</u> row of A. Restricting the equation to the k selected rows leaves

$$C \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_k \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Since $\det C \neq 0$, the matrix C is invertible, so $c_1 = \cdots = c_k = 0$. Hence the k columns $\vec{v}_1, \ldots, \vec{v}_k$ of A are linearly independent.

The rank of A is the dimension of its column space, i.e. the largest number of linearly

independent columns. Since we have found k of them,

$$\operatorname{rank}(A) \geq k.$$

(Equivalently, the nonzero determinant of C means the rows $i_1, \ldots, i_k$ of A are linearly independent, so also $\operatorname{rank}(A) \geq k$ by the row version of the same argument.)

Practice Problem #6.8 (if every $k \times k$ submatrix is singular then $\operatorname{rank} A < k$)

Problem. Prove that $\operatorname{rank}(A) < k$ if every $k \times k$ submatrix of A has zero determinant.

Solution. We prove the contrapositive: if $\operatorname{rank}(A) \geq k$, then some $k \times k$ submatrix of A is invertible.

Suppose $\operatorname{rank}(A) = r \geq k$. By definition of rank, A has r linearly independent columns; choose k of them, say $\vec{c}_1, \ldots, \vec{c}_k$, and let B be the $n \times k$ matrix with these columns. The columns of B are linearly independent, so $\operatorname{rank}(B) = k$.

Now use row rank = column rank: the row space of B has dimension k, so the n rows of B, regarded as vectors in $\mathbb{R}^k$, span a k-dimensional space. Hence some k of them, say the rows $i_1 < i_2 < \cdots < i_k$, are linearly independent. Let M be the $k \times k$ matrix formed by these k rows of B. Its k rows are linearly independent vectors in $\mathbb{R}^k$, so M is invertible and $\det M \neq 0$.

But the entries of M are exactly the entries of A lying in the rows $i_1, \ldots, i_k$ and in the k chosen columns, so M is a $k \times k$ submatrix of A with nonzero determinant.

Therefore, if every $k \times k$ submatrix of A has determinant 0, the inequality $\operatorname{rank}(A) \geq k$ is impossible, so $\operatorname{rank}(A) < k$. Combined with Problem 6.7 this gives the useful criterion

$$\operatorname{rank}(A) \geq k \iff A \text{ has a } k \times k \text{ submatrix with nonzero determinant,}$$

i.e. $\operatorname{rank}(A)$ is the size of the largest invertible submatrix of A.

6 Determinants

6.1 Defining Properties of the Determinant

Definition. The *determinant* is the unique function $\det : \{n \times n \text{ matrices}\} \to \mathbb{R}$ satisfying:

1. $\det I_n = 1$.
2. Interchanging two columns reverses the sign:

$$\det(\cdots \quad \vec{a}_i \quad \cdots \quad \vec{a}_j \quad \cdots) = -\det(\cdots \quad \vec{a}_j \quad \cdots \quad \vec{a}_i \quad \cdots).$$

3. Fixing all columns except the first, det is a linear function of the first column:

$$\det(\vec{u} + \vec{v} \quad \vec{a}_2 \quad \cdots \quad \vec{a}_n) = \det(\vec{u} \quad \vec{a}_2 \quad \cdots \quad \vec{a}_n) + \det(\vec{v} \quad \vec{a}_2 \quad \cdots \quad \vec{a}_n),$$
$$\det(c\vec{v} \quad \vec{a}_2 \quad \cdots \quad \vec{a}_n) = c \cdot \det(\vec{v} \quad \vec{a}_2 \quad \cdots \quad \vec{a}_n).$$

Linearity in any column

Property (3) extends to any column via property (2): det is linear as a function of any single column.

Proof: linearity in every column

Lemma. For every j, if all columns except column j are held fixed, then det is a linear function of column j.

Proof: Fix j and let M be the matrix obtained from A by moving column j to the first position, keeping the relative order of the other columns. This takes

$j-1$ adjacent column swaps, so property (2) gives $\det M = (-1)^{j-1}\det A$, i.e. $\det A = (-1)^{j-1}\det M$. In M, the variable column (formerly column j of A) is the first column, while all other columns are the fixed columns of A in permuted order. By property (3), $\det M$ is linear in the first column, i.e. in column j of A; multiplying by the constant $(-1)^{j-1}$ preserves linearity. Hence $\det A$ is linear in column j. ■

Midterm Review 2026 Problem 19 (the determinant is linear in each column)

Problem. Is

$$T\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \det\begin{pmatrix} 1 & v_1 & 3 \\ 4 & v_2 & 6 \\ 7 & v_3 & 9 \end{pmatrix}$$

a linear transformation $\mathbb{R}^3 \to \mathbb{R}$? If it is, compute its standard matrix.

Solution. Yes. The input (v_1, v_2, v_3) occupies exactly one column (column 2), and det is linear as a function of a single column, so T is linear. Expanding along the first row,

$$T\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = 1\cdot(9v_2 - 6v_3) - v_1(36-42) + 3(4v_3 - 7v_2) = 6v_1 - 12v_2 + 6v_3.$$

So the standard matrix of T is $A = \begin{pmatrix} 6 & -12 & 6 \end{pmatrix}$.

Clever trick: a matrix with one variable column gives a linear map

If one full column of a matrix is a variable vector and all other entries are constants, then det is a linear function of that column vector — this is a defining property of the determinant. Such expressions can always be written as a matrix product $A\vec{v}$, so A is the standard matrix of the map.

6.2 Key Lemmas

Lemma. If A has two equal columns, then $\det A = 0$.

Proof: Suppose columns i and j of A are equal. Swapping these two columns produces the same matrix A (since they are equal). But by defining property (2), swapping two columns negates the determinant. Let A' be the matrix after the swap. Since $A' = A$, we have

$$\det A = \det A' = -\det A.$$

Adding $\det A$ to both sides gives $2\det A = 0$, so $\det A = 0$. ■

Corollary. If the columns of A are linearly dependent, then $\det A = 0$.

Proof: If the columns of A are linearly dependent, then one column, say column j, is a linear combination of the others: $\vec{a}_j = c_1\vec{a}_1 + \cdots + c_{j-1}\vec{a}_{j-1} + c_{j+1}\vec{a}_{j+1} + \cdots + c_n\vec{a}_n$. By linearity of det in column j (property (3)), $\det A$ equals det of the matrix with $\vec{a}_j$ replaced by each term. For the term $c_i\vec{a}_i$ ($i \neq j$), the resulting matrix has two equal columns (columns i and j are both $c_i\vec{a}_i$). By the lemma, this contributes 0. Summing all such contributions, $\det A = 0$. ■

Corollary. If A is not invertible, then $\det A = 0$.

Proof: If A is not invertible, then (by the full characterization for square matrices) its columns are linearly dependent. By the previous corollary, $\det A = 0$. ■

6.3 Permutation Matrices

Definition. A *permutation matrix* is an $n \times n$ matrix formed by rearranging the columns of I_n, with entries in $\{0, 1\}$ having exactly one nonzero entry in each row and each column. Let S_n be the set of $n \times n$ permutation matrices. $|S_n| = n!$. If A is $n \times n$ and $X \in S_n$, then AX permutes the columns of A.

Definition. For $X \in S_n$, define $\text{inv}(X)$ as the number of 2×2 submatrices of X equal to $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Lemma. If $X \in S_n$, then $\det X = (-1)^{\text{inv}(X)}$.

Proof: We prove this by induction on $\text{inv}(X)$. The base case is $\text{inv}(X) = 0$, which means X has no $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ submatrix; this forces $X = I_n$ (the only permutation matrix with no inversions is the identity). By property (1), $\det I_n = 1 = (-1)^0$.

For the inductive step: suppose $\text{inv}(X) = k > 0$. Then $X \neq I_n$, so X has a column j that is not $\vec{e}_j$. There must exist a 2×2 submatrix of the form $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$; concretely, there are columns $i < j$ of X that are "out of order." Let X' be obtained from X by swapping two adjacent columns that are out of order (i.e., swap columns $i, i+1$ where column i has its 1 in a row below the 1 of column $i+1$). This swap changes inv by exactly ± 1 (it changes the status of exactly one 2×2 submatrix), so $\text{inv}(X') = k - 1$. By the induction hypothesis, $\det X' = (-1)^{k-1}$. By property (2), swapping two columns negates the determinant, so $\det X = -\det X' = -(-1)^{k-1} = (-1)^k = (-1)^{\text{inv}(X)}$. ■

Lemma. If X has 0/1 entries with one nonzero per column but is **not** a permutation matrix (i.e., $X \notin S_n$), then $\det X = 0$.

Proof: If X has 0/1 entries with one nonzero entry per column but $X \notin S_n$, then by pigeonhole some row must contain two or more nonzero entries (since n columns with one nonzero each, but not one per row, forces a collision). So there exist columns i, j and a row r where $X_{ri} = X_{rj} = 1$. Since each column has exactly one nonzero entry, this means two columns of X have their 1 in the same row. But two columns being $\vec{e}_r$ means they are equal — contradicting the condition that X is a permutation matrix. Hence the columns of X are linearly dependent (two equal columns), so by the lemma (two equal columns $\Rightarrow \det = 0$), $\det X = 0$.

More precisely: the columns of X are a list of n standard basis vectors (with one nonzero entry each). If two columns have their 1 in the same row, those two columns are equal, so X has two equal columns, giving $\det X = 0$ by the lemma. ■

6.4 General Formula for $\det A$

Definition. For a matrix X with 0/1 entries and a matrix A, define $\text{prod}(X, A)$ as the product of the entries of A in the nonzero positions of X.

Determinant formula

Theorem. For any $n \times n$ matrix A,

$$\boxed{\det A = \sum_{X \in S_n} \text{prod}(X, A) \cdot (-1)^{\text{inv}(X)}.}$$

Proof: Let $A = (\vec{a}_1 \quad \vec{a}_2 \quad \cdots \quad \vec{a}_n)$ with each $\vec{a}_j = \sum_{i=1}^{n} a_{ij} \vec{e}_i$. By multilinearity (prop-

erty (3) applied to each column), $\det A$ expands as

$$\det A = \sum_{i_1=1}^{n} \sum_{i_2=1}^{n} \cdots \sum_{i_n=1}^{n} a_{i_1 1} a_{i_2 2} \cdots a_{i_n n} \cdot \det \begin{pmatrix} \vec{e}_{i_1} & \vec{e}_{i_2} & \cdots & \vec{e}_{i_n} \end{pmatrix}.$$

The matrix $\begin{pmatrix} \vec{e}_{i_1} & \cdots & \vec{e}_{i_n} \end{pmatrix}$ is a 0/1 matrix with one nonzero per column. If it is a permutation matrix $X \in S_n$ (i.e., the $i_1, \ldots, i_n$ are all distinct), then by the lemma $\det X = (-1)^{\mathrm{inv}(X)}$; otherwise (two indices equal) it has two equal rows and $\det = 0$ by the lemma on non-permutation 0/1 matrices. So only the $X \in S_n$ terms survive, and for those, $a_{i_1 1} \cdots a_{i_n n} = \mathrm{prod}(X, A)$. Thus

$$\det A = \sum_{X \in S_n} \mathrm{prod}(X, A) \cdot (-1)^{\mathrm{inv}(X)}.$$

■

Theorem. The determinant is the unique function with properties (1) $\det I_n = 1$, (2) column swaps negate the determinant, (3) linearity in each column.

Proof: *Existence:* The formula $\det A = \sum_{X \in S_n} \mathrm{prod}(X, A) \cdot (-1)^{\mathrm{inv}(X)}$ (previous theorem) defines a function. One verifies it satisfies properties (1)–(3): (1) For $A = I_n$, $\mathrm{prod}(X, I_n) = 0$ for $X \neq I_n$ (some entry of I_n outside X's nonzero positions is needed), and $\mathrm{prod}(I_n, I_n) = 1$ with $\mathrm{inv}(I_n) = 0$, so $\det I_n = 1$. (2) Swapping columns i, j of A swaps the corresponding rows of each X, which changes $\mathrm{inv}(X)$ by an odd number (sign flip), giving the $-$ sign. (3) $\mathrm{prod}(X, A)$ is linear in each column of A, hence so is the sum.

Uniqueness: Suppose f and g both satisfy (1)–(3). By the same multilinear expansion as in the previous proof, any such f must satisfy

$$f(A) = \sum_{X \in S_n} \mathrm{prod}(X, A) \cdot f(X).$$

By the lemma, $f(X) = (-1)^{\mathrm{inv}(X)}$ for $X \in S_n$ (the proof of that lemma uses only properties (1) and (2), which f satisfies). So $f(A) = \sum_{X \in S_n} \mathrm{prod}(X, A) \cdot (-1)^{\mathrm{inv}(X)}$, which is the same for any f. Hence $f = g = \det$. ■

Small cases.

- $n = 1$: $\det(a) = a$.
- $n = 2$: $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$.
- $n = 3$: $\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg)$.

Geometric meaning of determinant

For a 2×2 matrix, $|\det A|$ is the area of the parallelogram spanned by the columns of A. For $n \times n$, $|\det A|$ is the volume of the n-dimensional parallelepiped spanned by the columns.

Examples: computing $\det A$ with the general formula

The general formula is

$$\det A = \sum_{X \in S_n} \mathrm{prod}(X, A) \cdot (-1)^{\mathrm{inv}(X)},$$

a sum of exactly $n!$ terms, one for each permutation matrix $X \in S_n$. Read each factor as follows:

- X **chooses which entries to use.** Since X has exactly one 1 in every row and every column, it selects n entries of A: writing i_j for the row of the 1 in column j, the chosen entries are $a_{i_1 1}, a_{i_2 2}, \ldots, a_{i_n n}$ (one from each row and each column), where $(i_1, \ldots, i_n)$ is a permutation of $1, \ldots, n$.
- $\mathrm{prod}(X, A)$ **multiplies them:** $\mathrm{prod}(X, A) = a_{i_1 1} a_{i_2 2} \cdots a_{i_n n}$. If any chosen entry is 0, that whole term is 0.
- $\mathrm{inv}(X)$ **fixes the sign.** It counts the out-of-order pairs of the permutation $(i_1, \ldots, i_n)$ (equivalently, the number of 2×2 blocks of X equal to $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$), and the term carries the factor $(-1)^{\mathrm{inv}(X)}$.

(a) $n = 2$**.** Here $S_2 = \{I_2, P\}$ with $P = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$:

X	(i_1, i_2)	$\mathrm{prod}(X, A)$	$\mathrm{inv}(X)$	signed term
I_2	$(1,2)$	$a_{11}a_{22} = ad$	0	$+ad$
$\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$	$(2,1)$	$a_{21}a_{12} = bc$	1	$-bc$

Hence $\det \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = ad - bc$. The off-diagonal X has the single inversion $(2, 1)$, which produces the minus sign.

(b) $n = 3$**.** Abbreviate $a_{11}, a_{12}, a_{13}, a_{21}, \ldots, a_{33}$ by $a, b, c, d, e, f, g, h, i$. The six permutation matrices are encoded by triples (i_1, i_2, i_3), and each contributes one product whose sign comes from its inversion count:

(i_1, i_2, i_3)	$\mathrm{prod}(X, A)$	$\mathrm{inv}(X)$	sign	term
$(1,2,3)$	$a_{11}a_{22}a_{33} = aei$	0	$+$	$+aei$
$(1,3,2)$	$a_{11}a_{32}a_{23} = afh$	1	$-$	$-afh$
$(2,1,3)$	$a_{21}a_{12}a_{33} = bdi$	1	$-$	$-bdi$
$(2,3,1)$	$a_{21}a_{32}a_{13} = cdh$	2	$+$	$+cdh$
$(3,1,2)$	$a_{31}a_{12}a_{23} = bfg$	2	$+$	$+bfg$
$(3,2,1)$	$a_{31}a_{22}a_{13} = ceg$	3	$-$	$-ceg$

Summing the last column,

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = aei - afh - bdi + cdh + bfg - ceg = a(ei - fh) - b(di - fg) + c(dh - eg),$$

the familiar cofactor expansion. For instance, the row $(3, 1, 2)$ says: the permutation matrix with its 1's in column 1 at row 3, column 2 at row 1 and column 3 at row 2 selects a_{31}, a_{12}, a_{23}, i.e. g, b, f, so prod $= bfg$; the permutation $(3, 1, 2)$ has two inversions, so its sign is $+$.

Why $\mathrm{inv}(X)$ **controls the sign.** Swapping two columns of a matrix flips the sign of its determinant (defining property (2)). Sorting the permutation $(i_1, \ldots, i_n)$ into increasing order takes exactly $\mathrm{inv}(X)$ adjacent swaps, and each swap flips the sign once; so the accumulated sign is $(-1)^{\mathrm{inv}(X)}$. In particular the identity permutation (no inversions) gives $+$, while a single transposition gives $-$.

(c) A sparse 4×4 **example.** When most entries are zero, only the permutation matrices whose 1's land on nonzero entries contribute. This is worked out in the next box.

Midterm Review 2026 Problem 22 (a sparse antidiagonal matrix)

Problem. Compute $\det A$ for $A = \begin{pmatrix} 0 & x & 0 & 0 \\ y & 0 & 0 & 0 \\ 0 & 0 & z & 0 \\ 0 & 0 & 0 & w \end{pmatrix}$.

Solution. The formula sums over all $4! = 24$ permutation matrices $X \in S_4$. A term contributes only if every 1 of X sits on a nonzero entry of A, i.e. $\operatorname{prod}(X, A) \neq 0$. The nonzero entries of A are exactly $a_{12} = x$, $a_{21} = y$, $a_{33} = z$, $a_{44} = w$ (there are just four of them, one in each row and each column). So X must have its 1 in column 1 at row 2, in column 2 at row 1, in column 3 at row 3, and in column 4 at row 4; writing the row indices as the permutation (i_1, i_2, i_3, i_4), this forces $(i_1, i_2, i_3, i_4) = (2, 1, 3, 4)$. Hence the only surviving permutation matrix is

$$X = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Reading the two factors:

- $\operatorname{prod}(X, A) = a_{2,1}a_{1,2}a_{3,3}a_{4,4} = y \cdot x \cdot z \cdot w = xyzw$ (the four chosen entries are precisely the nonzero ones).
- $\operatorname{inv}(X)$: the permutation $(2, 1, 3, 4)$ has exactly one out-of-order pair, namely $(2, 1)$; equivalently X contains the single 2×2 block $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ in rows and columns $1, 2$ and nothing else, so $\operatorname{inv}(X) = 1$ and the sign is $(-1)^1 = -1$.

Therefore

$$\det A = \operatorname{prod}(X, A) \cdot (-1)^{\operatorname{inv}(X)} = xyzw \cdot (-1) = -xyzw.$$

The other 23 permutation matrices each use at least one zero entry of A, so their products vanish and they drop out. Note that the result is the product along the "antidiagonal block" with the minus sign coming from the single row swap in the top-left 2×2 block.

Clever trick: keep only the matching permutation matrices

For a sparse matrix, list the nonzero entries and keep the permutation matrices whose 1's can be placed exactly on those entries. For each surviving permutation matrix, read off the sign from its inversion count $(-1)^{\operatorname{inv}(X)}$.

Practice Problem #6.5 (there are no nonzero alternating multilinear maps when $k > n$)

Problem. The set of alternating multilinear maps $\mathbb{R}^{n \times k} \to \mathbb{R}$ is a vector space for any $k \geq 1$. Explain why this space is 0-dimensional if $k > n$.

Solution. Let $f : \mathbb{R}^{n \times k} \to \mathbb{R}$ be alternating and multilinear, and assume $k > n$. Write the jth column of an $n \times k$ input matrix as $\vec{u}_j = \sum_{i=1}^{n} a_{ij}\vec{e}_i$. Multilinearity allows us to expand in each of the k columns separately:

$$f(\vec{u}_1, \ldots, \vec{u}_k) = \sum_{i_1=1}^{n} \sum_{i_2=1}^{n} \cdots \sum_{i_k=1}^{n} a_{i_1 1} a_{i_2 2} \cdots a_{i_k k}\, f(\vec{e}_{i_1}, \vec{e}_{i_2}, \ldots, \vec{e}_{i_k}).$$

Each term is indexed by a k-tuple $(i_1, \ldots, i_k)$ of indices chosen from the n possibilities $1, \ldots, n$. Since $k > n$, the pigeonhole principle produces $p \neq q$ with $i_p = i_q$. In that term the pth and qth input columns are the same vector $\vec{e}_{i_p}$, so interchanging them

does not change the input matrix, while alternation multiplies the value by -1:

$$f(\ldots,\vec{e}_{i_p},\ldots,\vec{e}_{i_q},\ldots) = -f(\ldots,\vec{e}_{i_p},\ldots,\vec{e}_{i_q},\ldots),$$

hence that value is 0. Every term of the sum vanishes, so f is the zero map. Therefore the space is $\{0\}$, whose dimension is 0 and whose basis is the empty set (not the set $\{0\}$, which is not linearly independent).

Practice Problem #6.6 (a basis of the alternating multilinear maps $\mathbb{R}^{n\times 2}\to\mathbb{R}$)

Problem. Assume $n \geq 2$. Check that the function $\mathbb{R}^{n\times 2}\to\mathbb{R}$ given by

$$f\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ \vdots & \vdots \\ a_{n1} & a_{n2} \end{pmatrix} = \det\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

is alternating and multilinear. Then find a basis for the vector space of alternating multilinear maps $\mathbb{R}^{n\times 2}\to\mathbb{R}$.

Solution. For $1 \leq i < j \leq n$ and vectors $\vec{u},\vec{v}\in\mathbb{R}^n$, define

$$F_{ij}\left(\vec{u} \quad \vec{v}\right) = \det\begin{pmatrix} u_i & v_i \\ u_j & v_j \end{pmatrix} = u_iv_j - u_jv_i.$$

Multilinearity. For scalars α,β and a further vector $\vec{w}$,

$$F_{ij}\left(\alpha\vec{u}+\beta\vec{w} \quad \vec{v}\right) = (\alpha u_i+\beta w_i)v_j - (\alpha u_j+\beta w_j)v_i = \alpha F_{ij}\left(\vec{u} \quad \vec{v}\right) + \beta F_{ij}\left(\vec{w} \quad \vec{v}\right),$$

and the same computation in the second column shows linearity there too.
Alternation. $F_{ij}\left(\vec{v} \quad \vec{u}\right) = v_iu_j - v_ju_i = -(u_iv_j - u_jv_i) = -F_{ij}\left(\vec{u} \quad \vec{v}\right)$.
The function f in the problem is exactly F_{12}, since only the first two rows enter, so it is alternating and multilinear.
A basis. We claim that $\{F_{ij} : 1 \leq i < j \leq n\}$ is a basis of the space A_2 of alternating bilinear maps $\mathbb{R}^{n\times 2}\to\mathbb{R}$.

Spanning. Let $g\in A_2$. Bilinearity gives $g(\vec{u},\vec{v}) = \displaystyle\sum_{i=1}^{n}\sum_{j=1}^{n} u_iv_j\, g(\vec{e}_i,\vec{e}_j)$. The diagonal terms vanish because $g(\vec{e}_i,\vec{e}_i) = -g(\vec{e}_i,\vec{e}_i)$, and the two terms with indices (i,j) and (j,i), $i<j$, combine as

$$u_iv_j\, g(\vec{e}_i,\vec{e}_j) + u_jv_i\, g(\vec{e}_j,\vec{e}_i) = (u_iv_j - u_jv_i)\, g(\vec{e}_i,\vec{e}_j).$$

Hence

$$g(\vec{u},\vec{v}) = \sum_{1\leq i<j\leq n} g(\vec{e}_i,\vec{e}_j)\, F_{ij}\left(\vec{u} \quad \vec{v}\right),$$

i.e. $g = \displaystyle\sum_{i<j} g(\vec{e}_i,\vec{e}_j)F_{ij}$ is a linear combination of the F_{ij}.

Independence. Suppose $\displaystyle\sum_{i<j} c_{ij}F_{ij} = 0$ as a function. Fix $p<q$ and evaluate on the matrix with columns $\vec{e}_p,\vec{e}_q$. Then $F_{pq}(\vec{e}_p,\vec{e}_q) = 1$, while $F_{ij}(\vec{e}_p,\vec{e}_q) = 0$ for every other pair (some coordinate is forced to be 0). So $c_{pq} = 0$, and this holds for every $p<q$. Therefore $\{F_{ij}\}$ is a basis and

$$\dim A_2 = \binom{n}{2} = \frac{n(n-1)}{2}.$$

For $n=2$ this says the alternating bilinear maps on $\mathbb{R}^{2\times 2}$ form a 1-dimensional space with basis $\{\det\}$, consistent with the fact that det is the unique alternating multilinear function normalized by $\det I_2 = 1$.

Practice Problem #6.4 (the alternating multilinear maps $\mathbb{R}^{n\times n} \to \mathbb{R}$ form a 1-dimensional space)

Problem. Let $\mathbb{R}^{n\times k}$ be the set of $n \times k$ matrices. A function $f : \mathbb{R}^{n\times n} \to \mathbb{R}$ is *alternating* if switching two columns of the input multiplies its value by -1, and *multilinear* if it is linear as a function of the ith column for every fixed index i; such functions form a vector space. Show that this vector space is 1-dimensional with basis $\{\det\}$, in two steps:

(a) if f is alternating and multilinear with $f(I) \neq 0$, then f is a scalar multiple of det;

(b) if f is alternating and multilinear with $f(I) = 0$, then $f(A) = 0$ for every A.

Solution. Both parts follow from the single identity

$$(\star) \qquad f(A) = f(I)\det(A),$$

which we now prove for an arbitrary alternating multilinear f. Write $\vec{a}_j = \sum_{i=1}^{n} a_{ij}\vec{e}_i$ for the jth column of A. Expanding f in each of the n columns (multilinearity),

$$f(A) = \sum_{i_1=1}^{n} \cdots \sum_{i_n=1}^{n} a_{i_1 1} \cdots a_{i_n n}\, f(\vec{e}_{i_1}, \ldots, \vec{e}_{i_n}).$$

If $i_p = i_q$ for some $p \neq q$, then the input has two equal columns; switching them does not change the input but multiplies the value by -1, so that value is 0 and the term vanishes. The surviving tuples $(i_1, \ldots, i_n)$ are therefore precisely the permutations $\sigma \in S_n$. For such a tuple, moving $\vec{e}_{\sigma(1)}, \ldots, \vec{e}_{\sigma(n)}$ into the order $\vec{e}_1, \ldots, \vec{e}_n$ needs $\operatorname{sgn}(\sigma)$ switches of adjacent columns, so alternation gives

$$f(\vec{e}_{\sigma(1)}, \ldots, \vec{e}_{\sigma(n)}) = \operatorname{sgn}(\sigma)\, f(\vec{e}_1, \ldots, \vec{e}_n) = \operatorname{sgn}(\sigma)\, f(I).$$

Substituting,

$$f(A) = f(I) \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma)\, a_{\sigma(1)1} \cdots a_{\sigma(n)n} = f(I)\det(A),$$

which is $(\star)$.

(a) If $f(I) = c \neq 0$, then $f(A) = c\det(A)$ for every A, i.e. $f = c\det$ is a scalar multiple of det.

(b) If $f(I) = 0$, then $(\star)$ gives $f(A) = 0 \cdot \det(A) = 0$ for every A, so f is the zero map.

Conclusion. Every alternating multilinear f equals $f(I)\det$, so $\{\det\}$ spans the space. Also $\det I = 1 \neq 0$, so det is not the zero map and $c\det = 0$ forces $c = 0$; hence $\{\det\}$ is linearly independent. Therefore the space is 1-dimensional with basis $\{\det\}$. (Note that $(\star)$ holds for every such f, so the space is spanned by a single element: there is no room for a second independent alternating multilinear function.)

6.5 Determinant of Transpose

Lemma. If $X \in S_n$ then $X^\top \in S_n$ and $\operatorname{inv}(X) = \operatorname{inv}(X^\top)$.

Proof: If $X \in S_n$, then X has exactly one nonzero entry in each row and each column. Transposing swaps rows and columns, so $X^\top$ also has exactly one nonzero entry in each row and each column. Hence $X^\top \in S_n$.

For the inversion count: a 2×2 submatrix of X at rows $i < j$ and columns $k < l$ equals $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ iff in X, rows i, j and columns k, l have entries $X_{il} = X_{jk} = 1$ and $X_{ik} = X_{jl} = 0$. Under transpose, this submatrix becomes the submatrix of $X^\top$ at

rows $k < l$ and columns $i < j$, with entries $(X^\top)_{ki} = X_{ik} = 0$, $(X^\top)_{lj} = X_{jl} = 0$, $(X^\top)_{li} = X_{il} = 1$, $(X^\top)_{kj} = X_{jk} = 1$. So the transposed submatrix is also $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Thus transposition gives a bijection between the inversion submatrices of X and of $X^\top$, so $\operatorname{inv}(X) = \operatorname{inv}(X^\top)$. ■

Determinant of transpose

Corollary. $\det A = \det(A^\top)$ for any square matrix A.

Proof: Using the general formula,

$$\det(A^\top) = \sum_{X \in S_n} \operatorname{prod}(X, A^\top) \cdot (-1)^{\operatorname{inv}(X)}.$$

Since $X \in S_n \Leftrightarrow X^\top \in S_n$ and $\operatorname{inv}(X) = \operatorname{inv}(X^\top)$ (previous lemma), we can re-index the sum by $Y = X^\top$:

$$\det(A^\top) = \sum_{Y \in S_n} \operatorname{prod}(Y^\top, A^\top) \cdot (-1)^{\operatorname{inv}(Y)}.$$

Now $\operatorname{prod}(Y^\top, A^\top)$ is the product of entries of $A^\top$ in the nonzero positions of $Y^\top$. But the nonzero positions of $Y^\top$ are the transposes of the nonzero positions of Y, and $(A^\top)_{ji} = A_{ij}$. So $\operatorname{prod}(Y^\top, A^\top) = \operatorname{prod}(Y, A)$. Hence

$$\det(A^\top) = \sum_{Y \in S_n} \operatorname{prod}(Y, A) \cdot (-1)^{\operatorname{inv}(Y)} = \det A.$$

■

Corollary. If A has two equal rows, then $\det A = 0$.

Proof: If A has two equal rows, then $A^\top$ has two equal columns. By the lemma (two equal columns $\Rightarrow$ det $= 0$), $\det(A^\top) = 0$. By the corollary $\det A = \det(A^\top)$, so $\det A = 0$. ■

Practice Problem #6.3 (determinant of a skew-symmetric matrix)

Problem. A matrix A is *skew-symmetric* if $A^\top = -A$; equivalently A is square with $a_{ii} = 0$ and $a_{ji} = -a_{ij}$ for all $i \neq j$. Suppose A is an $n \times n$ skew-symmetric matrix with all integer entries.

(a) Prove that if n is odd then $\det(A) = 0$.

(b) It can be shown that if n is even then $\det(A)$ is a perfect square. For example, if $A = \begin{pmatrix} 0 & -a \\ a & 0 \end{pmatrix}$ then $\det(A) = a^2$. Check that this also holds when $n = 4$.

Solution. (a) Transposition preserves the determinant, and multiplying all n rows of a matrix by -1 multiplies its determinant by $(-1)^n$. Using $A^\top = -A$,

$$\det(A) = \det(A^\top) = \det(-A) = (-1)^n \det(A).$$

If n is odd this reads $\det(A) = -\det(A)$, hence $2\det(A) = 0$ and $\det(A) = 0$. (Only skew-symmetry is used here; integrality of the entries is not needed.)

(b) Every 4×4 skew-symmetric matrix with integer entries has the form

$$A = \begin{pmatrix} 0 & a & b & c \\ -a & 0 & d & e \\ -b & -d & 0 & f \\ -c & -e & -f & 0 \end{pmatrix}, \qquad a, b, c, d, e, f \in \mathbb{Z}.$$

Put $q = af - be + cd$. Expanding along the first row, $\det A = a_{11}C_{11} + a_{12}C_{12} +$

$a_{13}C_{13} + a_{14}C_{14}$ with the cofactors $C_{1j} = (-1)^{1+j}M_{1j}$:

$$\det A = -a\det\begin{pmatrix}-a & d & e\\ -b & 0 & f\\ -c & -f & 0\end{pmatrix} + b\det\begin{pmatrix}-a & 0 & e\\ -b & -d & f\\ -c & -e & 0\end{pmatrix} - c\det\begin{pmatrix}-a & 0 & d\\ -b & -d & 0\\ -c & -e & -f\end{pmatrix}.$$

The three 3×3 determinants are computed in the same way:

$$\det\begin{pmatrix}-a & d & e\\ -b & 0 & f\\ -c & -f & 0\end{pmatrix} = -a(f^2) - d(cf) + e(bf) = -f(af - be + cd) = -fq,$$

$$\det\begin{pmatrix}-a & 0 & e\\ -b & -d & f\\ -c & -e & 0\end{pmatrix} = -a(fe) + e(be - dc) = -e(af - be + cd) = -eq,$$

$$\det\begin{pmatrix}-a & 0 & d\\ -b & -d & 0\\ -c & -e & -f\end{pmatrix} = -a(df) + d(be - dc) = -d(af - be + cd) = -dq.$$

Substituting,

$$\det A = -a(-fq) + b(-eq) - c(-dq) = afq - beq + cdq = (af - be + cd)\,q = q^2.$$

So

$$\det A = (af - be + cd)^2,$$

a perfect square of an integer, as claimed. (For instance, all entries 1 give $q = 1 - 1 + 1 = 1$ and $\det A = 1$; $a, b, c, d, e, f = 1, 2, 3, 4, 5, 6$ give $q = 6 - 10 + 12 = 8$ and $\det A = 64$.)

6.6 Determinant of Products

$\det(AB) = (\det A)(\det B)$

Lemma (case $\det A \neq 0$). Define $f(M) = \dfrac{\det(AM)}{\det A}$. Then f satisfies the three defining properties of the determinant, so $f = \det$. Thus $\dfrac{\det(AB)}{\det A} = \det B$.

Theorem. For any $n \times n$ matrices A and B: $\det(AB) = (\det A)(\det B)$.

Lemma. If A or B is not invertible, then AB is not invertible.

Proof of the lemma (case $\det A \neq 0$): We verify $f(M) = \dfrac{\det(AM)}{\det A}$ satisfies the three defining properties of the determinant (as a function of M):

1. $f(I_n) = \dfrac{\det(AI_n)}{\det A} = \dfrac{\det A}{\det A} = 1$.
2. Swapping two columns i, j of M swaps the corresponding columns of AM (since $AM = (A\vec{m}_1 \;\cdots\; A\vec{m}_n)$ depends column-wise on M). By property (2) of det, $\det(AM') = -\det(AM)$, so $f(M') = -f(M)$.
3. Fixing all columns of M except the first: $AM = (A\vec{m}_1 \quad A\vec{m}_2 \quad \cdots \quad A\vec{m}_n)$. If $\vec{m}_1 = \vec{u} + \vec{v}$, then $A\vec{m}_1 = A\vec{u} + A\vec{v}$, and by linearity of det in column 1, $\det(AM_{\vec{u}+\vec{v}}) = \det(AM_{\vec{u}}) + \det(AM_{\vec{v}})$. Dividing by $\det A$, f is linear in column 1. Similarly for scalar multiplication.

Since f satisfies (1)–(3), by the uniqueness theorem $f = \det$. Hence $\dfrac{\det(AB)}{\det A} = f(B) = \det B$, giving $\det(AB) = (\det A)(\det B)$ when $\det A \neq 0$. ■

Proof of the lemma (non-invertible $\Rightarrow$ AB non-invertible): If A is not invert-

ible, then the columns of A do not span $\mathbb{R}^n$ (by the full characterization for square matrices), so $\operatorname{Col} A \subsetneq \mathbb{R}^n$. Then $\operatorname{Col}(AB) = \{AB\vec{x} : \vec{x} \in \mathbb{R}^n\} = \{A\vec{y} : \vec{y} \in \operatorname{Col} B\} \subseteq \operatorname{Col} A \subsetneq \mathbb{R}^n$. So the columns of AB do not span $\mathbb{R}^n$, hence AB is not invertible.

If B is not invertible, then the columns of B are linearly dependent. So there exists $\vec{x} \neq \vec{0}$ with $B\vec{x} = \vec{0}$. Then $(AB)\vec{x} = A(B\vec{x}) = A\vec{0} = \vec{0}$, so the columns of AB are also linearly dependent, hence AB is not invertible. ■

Proof of the theorem: *Case 1:* $\det A \neq 0$. By the lemma (case $\det A \neq 0$), $\det(AB) = (\det A)(\det B)$.

Case 2: $\det A = 0$. Then A is not invertible. By the lemma, AB is not invertible, so by the corollary (non-invertible $\Rightarrow \det = 0$), $\det(AB) = 0$. And $(\det A)(\det B) = 0 \cdot (\det B) = 0$. So $\det(AB) = 0 = (\det A)(\det B)$.

In both cases, $\det(AB) = (\det A)(\det B)$. ■

Midterm Review 2026 Problem 13 (determinant of a block-diagonal matrix)

Problem. Let

$$A = \begin{pmatrix} a & b & 0 & 0 & 0 \\ c & d & 0 & 0 & 0 \\ 0 & 0 & e & 0 & 0 \\ 0 & 0 & 0 & f & g \\ 0 & 0 & 0 & h & i \end{pmatrix}.$$

Compute $\det A$, determine when A is invertible, and (when it is) give A^{-1}.

Solution. Factor $A = BCD$ where each of B, C, D differs from I_5 in a single block:

$$B = \begin{pmatrix} a & b & 0 & 0 & 0 \\ c & d & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & e & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & f & g \\ 0 & 0 & 0 & h & i \end{pmatrix}.$$

By cofactor expansion, $\det B = ad - bc$ (the 2×2 block), $\det C = e$, and $\det D = fi - gh$. Multiplicativity of det gives

$$\det A = (\det B)(\det C)(\det D) = (ad - bc)\, e\, (fi - gh).$$

Hence A is invertible $\iff$ $ad - bc \neq 0$, $e \neq 0$, and $fi - gh \neq 0$; then

$$A^{-1} = \begin{pmatrix} \dfrac{d}{ad-bc} & \dfrac{-b}{ad-bc} & 0 & 0 & 0 \\ \dfrac{-c}{ad-bc} & \dfrac{a}{ad-bc} & 0 & 0 & 0 \\ 0 & 0 & \dfrac{1}{e} & 0 & 0 \\ 0 & 0 & 0 & \dfrac{i}{fi-gh} & \dfrac{-g}{fi-gh} \\ 0 & 0 & 0 & \dfrac{-h}{fi-gh} & \dfrac{f}{fi-gh} \end{pmatrix}.$$

Clever trick: split a block matrix into single-block factors

A block-diagonal matrix can be written as a product $A = BCD$ of matrices each equal to the identity outside one block. Then $\det A$ is the product of the block determinants, and A^{-1} is assembled from the inverses of those blocks — using $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \dfrac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ for the 2×2 blocks.

6.7 Triangular Matrices

Definition. A matrix is *upper-triangular* if all entries below the main diagonal are zero. It is *lower-triangular* if all entries above the main diagonal are zero. A matrix is *triangular* if it is either upper- or lower-triangular. A matrix is *diagonal* if it is both upper- and lower-triangular (nonzero entries only on the diagonal).

Determinant of triangular matrices

Proposition. If A is triangular, then $\det A$ is the product of the diagonal entries: $\det A = a_{11}a_{22}\cdots a_{nn}$.

Proof: *Case 1: A is upper-triangular* (so $a_{ij} = 0$ for $i > j$). Using the general formula $\det A = \sum_{X \in S_n} \text{prod}(X, A) \cdot (-1)^{\text{inv}(X)}$, we analyze each term. For $X = I_n$, $\text{prod}(I_n, A) = a_{11}a_{22}\cdots a_{nn}$ (the diagonal entries) and $(-1)^{\text{inv}(I_n)} = 1$.

For $X \neq I_n$ (any other permutation matrix), X has a nonzero entry in some position (i, j) with $i > j$ (i.e., below the diagonal) — because X being different from I_n means some column's 1 is not on the diagonal, and by counting (since each row and column has exactly one 1), at least one such 1 must be below the diagonal. Then $\text{prod}(X, A)$ includes the factor a_{ij} with $i > j$, which is 0 (since A is upper-triangular). So $\text{prod}(X, A) = 0$ for $X \neq I_n$.

Therefore $\det A = \text{prod}(I_n, A) \cdot 1 = a_{11}a_{22}\cdots a_{nn}$.

Case 2: A is lower-triangular. Then $A^\top$ is upper-triangular, and by Case 1, $\det(A^\top) = a_{11}a_{22}\cdots a_{nn}$ (the diagonal of $A^\top$ equals the diagonal of A). By $\det A = \det(A^\top)$, we get $\det A = a_{11}a_{22}\cdots a_{nn}$. ■

Midterm 2024 Problem 4 (inverse of a repeated-column lower-triangular matrix)

Problem. Let $a_1, a_2, \ldots, a_n \in \mathbb{R}$ and let A be the lower-triangular matrix with $A_{ij} = a_j$ whenever $i \geq j$:

$$A = \begin{pmatrix} a_1 & 0 & 0 & \cdots & 0 \\ a_1 & a_2 & 0 & & 0 \\ a_1 & a_2 & a_3 & & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_1 & a_2 & a_3 & \cdots & a_n \end{pmatrix}.$$

Determine when A is invertible and describe A^{-1} explicitly.
Solution. Since A is triangular, $\det A = a_1 a_2 \cdots a_n$, so A is invertible **if and only if every** $a_i \neq 0$.
Row reduce $(A \quad I_n)$. Subtracting row 1 from every lower row creates zeros in column 1 below the pivot and writes -1's below position $(1, n+1)$ of the augmented part; subtracting row 2 from every lower row does the same for the next block. Continuing for $i = 1, \ldots, n-1$ yields $\text{diag}(a_1, \ldots, a_n)$ on the left and I_n with -1 just below each diagonal entry on the right. Dividing row i by a_i gives the inverse explicitly:

$$A^{-1} = \begin{pmatrix} a_1^{-1} & 0 & 0 & \cdots & 0 \\ -a_2^{-1} & a_2^{-1} & 0 & & 0 \\ 0 & -a_3^{-1} & a_3^{-1} & & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & -a_n^{-1} & a_n^{-1} \end{pmatrix},$$

i.e., $(A^{-1})_{ii} = a_i^{-1}$ and $(A^{-1})_{i,i-1} = -a_i^{-1}$ for $i = 2, \ldots, n$, with all other entries zero.

Clever trick: difference rows first, scale last

For matrices whose columns are constant below the diagonal, one round of "subtract the pivot row from all lower rows" eliminates an entire column *and* simultaneously builds the off-diagonal part of the inverse in the augmented block. Only a final diagonal scaling remains.

6.8 Invertibility and Determinant

A invertible $\Leftrightarrow \det A \neq 0$

Lemma. $\det A$ is a nonzero multiple of $\det(\mathrm{RREF}(A))$. Each row operation multiplies det by a nonzero scalar:
- Row swap: multiply by -1.
- Row scaling by $\lambda \neq 0$: multiply by λ.
- Row replacement (add multiple of one row to another): no change.

Theorem. A is invertible if and only if $\det A \neq 0$.

Proof of the lemma: By $\det A = \det(A^\top)$, the effect of a **row** operation on det equals the effect of the corresponding **column** operation. We know the column operations behave as:

- *Column swap*: by defining property (2), this multiplies det by -1. Hence row swap multiplies det by -1.
- *Column scaling by* $\lambda \neq 0$: by linearity (property (3)),
$$\det(\cdots \quad \lambda\vec{a}_j \quad \cdots) = \lambda \det(\cdots \quad \vec{a}_j \quad \cdots).$$
Hence row scaling multiplies det by $\lambda \neq 0$.
- *Column replacement* (add λ times column j to column i): by linearity in column i,
$$\det(\cdots \quad \vec{a}_i + \lambda\vec{a}_j \quad \cdots) = \det(\cdots \quad \vec{a}_i \quad \cdots) + \lambda \det(\cdots \quad \vec{a}_j \quad \cdots).$$
The second term has two equal columns (columns i and j both equal $\vec{a}_j$), so it is 0. Hence det is unchanged. The same holds for row replacement.

Since each row operation multiplies det by a nonzero scalar (-1 or $\lambda \neq 0$ or 1), the overall factor between $\det A$ and $\det(\mathrm{RREF}(A))$ is a product of nonzero scalars, hence nonzero. ■

Proof of the theorem: ($\Leftarrow$) Suppose $\det A \neq 0$. Write $R = \mathrm{RREF}(A)$. By the lemma, $\det A = c \cdot \det R$ for some nonzero scalar c. So $\det R = \dfrac{\det A}{c} \neq 0$. Now R is in reduced row echelon form. If R had a row of zeros (no pivot in that row), then by the proposition on triangular matrices (since R is upper-triangular with some zero on the diagonal), $\det R = 0$, a contradiction. So every row has a pivot. Since R is $n \times n$ square and has n pivots (one per row, and at most one per column, so exactly n), we have $R = I_n$. Hence A is row equivalent to I_n, so A is invertible.

($\Rightarrow$) Suppose A is invertible. Then $\mathrm{RREF}(A) = I_n$, so $\det(\mathrm{RREF}(A)) = \det I_n = 1 \neq 0$. By the lemma, $\det A = c \cdot \det(\mathrm{RREF}(A)) = c \cdot 1 = c$ for some nonzero c, so $\det A \neq 0$.

(Equivalently, for ($\Rightarrow$): if A is invertible, its columns are linearly independent. By the corollary (dependent columns $\Rightarrow$ det $= 0$), the contrapositive gives $\det A \neq 0$.) ■

Corollary. If A has a zero row or a zero column, then $\det A = 0$.

Proof: If A has a zero column, then the columns of A are linearly dependent (any set containing the zero vector is linearly dependent). By the corollary (linearly dependent

columns $\Rightarrow \det = 0$), $\det A = 0$.

If A has a zero row, then $A^\top$ has a zero column. By the above argument, $\det(A^\top) = 0$. Since $\det A = \det(A^\top)$, we get $\det A = 0$. ■

Corollary. For any $n \times n$ matrix A and scalar $c \in \mathbb{R}$: $\det(cA) = c^n \det A$.

Proof: The matrix cA is obtained from A by scaling each of the n rows by c. By the lemma on row operations, scaling a single row by c multiplies the determinant by c. Applying this n times (once for each row), we get

$$\det(cA) = \underbrace{c \cdot c \cdots c}_{n \text{ times}} \det A = c^n \det A.$$

(Equivalently: cA scales each of the n columns by c, and by multilinearity (property (3)) in each column, each scaling contributes a factor of c, giving $c^n \det A$.) ■

Corollary. If A is invertible, then $\det(A^{-1}) = \dfrac{1}{\det A}$.

Proof: Since A is invertible, A^{-1} exists and $AA^{-1} = I_n$. By the theorem $\det(AB) = (\det A)(\det B)$, we have

$$\det(AA^{-1}) = (\det A)(\det A^{-1}).$$

But $AA^{-1} = I_n$, so $\det(AA^{-1}) = \det I_n = 1$. Therefore

$$(\det A)(\det A^{-1}) = 1.$$

Since A is invertible, $\det A \neq 0$ (by the previous theorem). Dividing both sides by $\det A$ gives

$$\det(A^{-1}) = \frac{1}{\det A}.$$

■

Midterm 2024 Problem 9 (GCD of determinants of 4×4 matrices with entries ± 1)

Problem. What is the greatest common divisor of all numbers occurring as $\det A$, as A ranges over the invertible 4×4 matrices whose entries are all $+1$ or -1?
Solution. The GCD is **8**.
Upper bound (8 is attained). Row replacements leave the determinant unchanged, so

$$\det \begin{pmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{pmatrix} = \det \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix} = 8,$$

by adding the first row to rows $2, 3, 4$. Hence the GCD divides 8, so it is at most 8.
Lower bound (8 divides every such determinant). Let A be any such matrix and form B by adding the first row of A to rows $2, 3, 4$. Then $\det A = \det B$ (row replacements). The entries in row 1 of B are still ± 1, while every entry below row 1 is a sum of two numbers from $\{\pm 1\}$, hence belongs to $\{0, 2, -2\}$. In the permutation formula $\det B = \sum_{X \in S_4} \operatorname{prod}(X, B)(-1)^{\operatorname{inv}(X)}$, each product contains one factor ± 1 from row 1 and three factors, each 0 or ± 2, from the other rows. Every summand is therefore 0 or ± 8, and a sum of multiples of 8 is divisible by 8. Thus $8 \mid \det B = \det A$ for every allowed A.
Combining the two bounds, the GCD is exactly 8.

Clever trick: normalize against the first row, then read off a universal divisor

For ± 1 matrices, adding the first row to every other row turns all entries below it into multiples of 2 (values $0, \pm 2$). Each permutation term then automatically carries a factor 2^{n-1}. Exhibiting one matrix whose determinant equals exactly 2^{n-1} pins down the GCD. This is the $n = 4$ instance of a general fact (Hadamard-type bounds on determinant divisibility).

Midterm 2022 Problem 3 (which properties make A invertible?)

Problem. Suppose A is a (not necessarily square) $m \times n$ matrix. Consider the following possible properties of A:

(1) A is square ($m = n$).
(2) every row of A has a pivot position.
(3) every column of A has a pivot position.
(4) $A\vec{x} = \vec{b}$ has a solution for each $\vec{b} \in \mathbb{R}^m$.
(5) $A\vec{x} = \vec{b}$ has a unique solution for each $\vec{b} \in \mathbb{R}^m$.
(6) $A\vec{x} = \vec{b}$ has a unique solution for some $\vec{b} \in \mathbb{R}^m$.
(7) $A\vec{x} = \vec{0}$ has infinitely many solutions.
(8) $A\vec{x} = \vec{0}$ has exactly one solution.
(9) $A\vec{x} = \vec{0}$ has no solution.
(10) $\operatorname{Col} A = \mathbb{R}^m$.
(11) $\operatorname{Nul} A = \mathbb{R}^n$.
(12) $\operatorname{Col} A = \{\vec{0}\}$.
(13) $\operatorname{Nul} A = \{\vec{0}\}$.
(14) the columns of A are linearly independent.
(15) the span of the columns of A is $\mathbb{R}^m$.
(16) the reduced echelon form of A is an identity matrix.
(17) AB is an identity matrix for some $n \times m$ matrix B.
(18) $\operatorname{rank}(A) = \min\{m, n\}$.
(19) $\det(A^\top A) \neq 0$.
(20) $(A^\top A)^k$ is an identity matrix for some integer $k \geq 1$.

(a) List all properties that A *must* have if A is invertible.
(b) List all properties that imply by themselves that A is invertible.
(c) List all properties that imply A is invertible if we *also* assume A is square ($m = n$).
(d) Compute the inverse of $A = \begin{pmatrix} 0 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 \\ 4 & 4 & 4 & 4 \end{pmatrix}$.

Solution. **(a)** All of $1, 2, 3, 4, 5, 6, 8, 10, 13, 14, 15, 16, 17, 18, 19$ (but not $7, 9, 11, 12, 20$). An invertible A is square (1), so it has a pivot in every row and column (2,3); the system $A\vec{x} = \vec{b}$ is solvable, and uniquely so, for every $\vec{b}$ (4,5,6); $A\vec{x} = \vec{0}$ has only the trivial solution (8, not 7); $\operatorname{Col} A = \mathbb{R}^n$ (10, not 12) and $\operatorname{Nul} A = \{\vec{0}\}$ (13, not 11); the columns are linearly independent (14) and span $\mathbb{R}^n$ (15); $\operatorname{RREF}(A) = I_n$ (16); A has a two-sided inverse $B = A^{-1}$ (17); and $\operatorname{rank}(A) = n = \min\{n, n\}$ (18). For (19), $\det(A^\top A) = \det(A^\top)\det(A) = (\det A)^2 \neq 0$. Property (20) need not hold: $A = (2)$ is invertible but $(A^\top A)^k = 4^k \neq 1$ for all $k \geq 1$. Property (9) never holds,

since $\vec{x} = \vec{0}$ always solves $A\vec{x} = \vec{0}$.

(b) Only 5 and 16. Property (5) forces A square: consistency for every $\vec{b}$ gives $m \leq n$, while uniqueness gives $n \leq m$ (linearly independent columns), so $m = n$ and A is invertible. Property (16) forces A square with full rank, hence invertible. Every other property is satisfied by some non-square matrix; e.g. $A = \begin{pmatrix} 1 & 0 \end{pmatrix}$ has a pivot in every row (2), spans $\mathbb{R}^1$ (10, 15) and has $\det(A^\top A) = 1 \neq 0$ (19), yet is not invertible.

(c) All except $1, 7, 9, 11, 12$; i.e. $2, 3, 4, 5, 6, 8, 10, 13, 14, 15, 16, 17, 18, 19, 20$. Once A is square, each of these forces full rank: for (19), $\det(A^\top A) = (\det A)^2$, so $\det(A^\top A) \neq 0 \Rightarrow \det A \neq 0$; for (20), $(A^\top A)^k = I$ makes $A^\top A$ invertible, hence A invertible, with $A^{-1} = (A^\top A)^{k-1} A^\top$.

(d) Reduce $(A \mid I_4)$: subtract row 1 from row 4, subtract twice row 3 from row 4, subtract twice row 2 from row 3, rearrange the rows into order $1, 2, 3, 4$, then divide row 2 by 2, row 3 by 4 and row 4 by 4. This gives $\mathrm{RREF}(A) = I_4$ (so A is invertible) and

$$A^{-1} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & -1 & \frac{1}{2} & 0 \\ -\frac{1}{4} & 0 & -\frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & 0 & 0 & 0 \end{pmatrix}.$$

Clever trick: $A^\top A$ and the non-square traps

For square A, $\det(A^\top A) = (\det A)^2$, so $\det(A^\top A) \neq 0 \iff \det A \neq 0$. For a non-square A, most "invertibility-like" properties only pin down one inequality ($m \leq n$ for spanning/onto, $n \leq m$ for linear independence), so they cannot guarantee invertibility; only a property forcing $m = n$ *and* full rank (such as 5, or $\mathrm{RREF}(A) = I$) does.

6.9 Computing Determinants

Algorithm 1: Row Reduction

Row reduction algorithm for $\det A$

1. Set `denom` $= 1$.
2. Row reduce A to echelon form E (upper-triangular suffices). Update `denom`:
   - Row swap: `denom` $\leftarrow -$`denom`.
   - Scale row by $\lambda \neq 0$: `denom` $\leftarrow \lambda \cdot$ `denom`.
   - Row replacement: no change to `denom`.
3. $\det E =$ product of diagonal entries of E.
4. $\det A = \dfrac{\det E}{\texttt{denom}}$.

Powers, polynomial identities, and the all-ones matrix

- *Powers:* $A^0 = I$ and $A^p A^q = A^{p+q}$; all powers of one matrix commute, so a polynomial identity in A may be manipulated like ordinary algebra.
- *Inverse from a polynomial identity:* if $A^2 = bI + cA$ with $b \neq 0$, then $A(A - cI) = bI$ and $A^{-1} = \dfrac{1}{b}(A - cI)$. More generally, $A^s + c_{s-1}A^{s-1} + \cdots + c_1 A + c_0 I = 0$ with

$c_0 \neq 0$ gives $A \cdot \left[-\frac{1}{c_0}\left(A^{s-1} + \cdots + c_1 I\right)\right] = I$.

- *All-ones matrix:* if J is the $n \times n$ matrix all of whose entries equal 1, then $J^2 = nJ$ (each entry of J^2 is a sum of n ones), $\operatorname{rank} J = 1$, $J\vec{x} = \left(\sum_i x_i\right)\vec{1}$, and $\operatorname{Nul} J = \{\vec{x} : \sum_i x_i = 0\}$ has dimension $n-1$.
- *Effect of row operations on* det*:* row replacement leaves det unchanged, a row swap negates it, and scaling a row by λ multiplies it by λ; the determinant of a triangular matrix is the product of its diagonal entries.

Midterm 2021 Problem 7 (0 on the diagonal, 1 elsewhere)

Problem. Let $n \geq 2$ be an integer and let A be the $n \times n$ matrix with 0's on the main diagonal and 1's everywhere else (for $n = 4$, every off-diagonal entry is 1). Let $I = I_n$. (a) Find numbers b and c (depending on n) such that $A^2 = bI + cA$. (b) Compute a formula for the inverse of A. (c) Compute a formula for $\det A$.

Solution (a). Write J for the $n \times n$ all-ones matrix, so that

$$A = J - I.$$

Since $J^2 = nJ$,

$$\begin{aligned} A^2 &= (J-I)^2 = J^2 - 2J + I = nJ - 2J + I \\ &= (n-2)J + I = (n-2)(A+I) + I = (n-2)A + (n-1)I. \end{aligned}$$

Hence $b = n-1$ and $c = n-2$.

Entry check. The diagonal entries of A^2 are $\sum_{k \neq i} A_{ik}A_{ki} = n-1$ (there are $n-1$ nonzero terms, each equal to 1). For $i \neq j$, $(A^2)_{ij} = \sum_k A_{ik}A_{kj}$, where the terms with $k = i$ or $k = j$ vanish, leaving $n-2$ terms equal to 1. Thus $A^2 = (n-1)I + (n-2)A$, as claimed.

Solution (b). By (a), $A^2 - (n-2)A = (n-1)I$, i.e.

$$A\big(A - (n-2)I\big) = (n-1)I.$$

Since $n \geq 2$, we have $n - 1 \neq 0$, so

$$A \cdot \frac{1}{n-1}\big(A - (n-2)I\big) = I.$$

As A is square, a one-sided inverse is the two-sided inverse, hence

$$A^{-1} = \frac{1}{n-1}\big(A - (n-2)I\big) = \frac{1}{n-1}A + \frac{2-n}{n-1}I.$$

Check: $A \cdot \frac{1}{n-1}\big(A-(n-2)I\big) = \frac{1}{n-1}\big(A^2-(n-2)A\big) = \frac{(n-1)I}{n-1} = I$. So A^{-1} has all diagonal entries $\frac{2-n}{n-1}$ and all off-diagonal entries $\frac{1}{n-1}$. For $n = 2$, $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and the formula gives $A^{-1} = A$, as expected.

Solution (c). Use only determinant-preserving row replacements, so det never changes.

Step 1: replace row 1 by the sum of all n rows. Every entry of the new first row equals $0 + \underbrace{1 + \cdots + 1}_{n-1} = n-1$; the other rows are unchanged.

Step 2: for $i = 2, \ldots, n$, subtract $\dfrac{1}{n-1}$ times the new row 1 from row i. In row i the entry in column i becomes $0 - \dfrac{n-1}{n-1} = -1$, and every other entry becomes $1 - 1 = 0$. The result is

$$\begin{pmatrix} n-1 & n-1 & \cdots & n-1 \\ 0 & -1 & & 0 \\ \vdots & & \ddots & \\ 0 & & & -1 \end{pmatrix},$$

which is triangular. Therefore

$$\det A = (n-1)(-1)^{n-1}.$$

Checks: $n = 2$ gives -1 (indeed $\det \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = -1$), $n = 3$ gives 2, $n = 4$ gives -3. Since $n - 1 \neq 0$, $\det A \neq 0$, consistent with the invertibility established in (b). (Equivalently, A has eigenvalues $n-1$ and -1 with multiplicity $n-1$, whose product is $(n-1)(-1)^{n-1}$.)

Clever trick: recognize $A = J - I$

Once A is written as "all-ones matrix minus I", both questions collapse: $J^2 = nJ$ gives the polynomial identity $A^2 = (n-2)A + (n-1)I$, which produces A^{-1} by pure algebra with no row reduction; and for the determinant, adding all rows to the first row and then clearing below it gives a triangular matrix in two determinant-preserving steps. The same computation shows A is invertible for every $n \geq 2$.

Algorithm 2: Cofactor Expansion

Definition. Let $A_{(i,j)}$ be the submatrix formed by deleting row i and column j from A.

Cofactor expansion

Theorem. For an $n \times n$ matrix A with entry a_{ij} in row i, column j:

$$\det A = a_{11} \det A_{(1,1)} - a_{12} \det A_{(1,2)} + a_{13} \det A_{(1,3)} - \cdots + (-1)^{1+n} a_{1n} \det A_{(1,n)}$$

$$\det A = a_{11} \det A_{(1,1)} - a_{21} \det A_{(2,1)} + a_{31} \det A_{(3,1)} - \cdots + (-1)^{n+1} a_{n1} \det A_{(n,1)}$$

(Expansion along the first row and first column, respectively.)

Proof: We prove the row expansion (expansion along the first row); the column expansion follows from $\det A = \det(A^\top)$.

Write A with columns $\vec{a}_1, \ldots, \vec{a}_n$. By linearity of det in the first column, write $\vec{a}_1 = a_{11}\vec{e}_1 + a_{21}\vec{e}_2 + \cdots + a_{n1}\vec{e}_n$. Then

$$\det A = \sum_{i=1}^{n} a_{i1} \det \begin{pmatrix} \vec{e}_i & \vec{a}_2 & \cdots & \vec{a}_n \end{pmatrix}.$$

Let $M_i = \begin{pmatrix} \vec{e}_i & \vec{a}_2 & \cdots & \vec{a}_n \end{pmatrix}$. We compute $\det M_i$.

In M_i, the first column is $\vec{e}_i$. If $i \neq 1$, swap row 1 with row i (using $i-1$ adjacent row swaps) to move the 1 to the top-left corner. Each swap multiplies det by -1, so $\det M_i = (-1)^{i-1} \det M_i'$ where M_i' has first column $\vec{e}_1$ and the other columns are $\vec{a}_2, \ldots, \vec{a}_n$ (with rows 1 and i swapped). Now in M_i', the first row is $\begin{pmatrix} 1 & 0 & \cdots & 0 \end{pmatrix}$. By the general formula, only the term $X = I_n$ (or X with first column $= \vec{e}_1$) contributes, and the remaining determinant is exactly that of the submatrix $A_{(i,1)}$ (delete row i

and column 1 from A, with appropriate relabeling). So $\det M_i' = \det A_{(i,1)}$, giving $\det M_i = (-1)^{i-1} \det A_{(i,1)}$.

Hence $\det A = \sum_{i=1}^{n} a_{i1}(-1)^{i-1} \det A_{(i,1)}$, which is the expansion along column 1:

$$\det A = a_{11} \det A_{(1,1)} - a_{21} \det A_{(2,1)} + a_{31} \det A_{(3,1)} - \cdots + (-1)^{n+1} a_{n1} \det A_{(n,1)}.$$

For expansion along the first row: apply the column expansion to $A^\top$ (whose first column is the first row of A), then use $\det A = \det(A^\top)$:

$$\det A = \det(A^\top) = \sum_{j=1}^{n} a_{1j}(-1)^{j-1} \det(A^\top)_{(j,1)} = \sum_{j=1}^{n} (-1)^{j+1} a_{1j} \det A_{(1,j)},$$

since $(A^\top)_{(j,1)} = (A_{(1,j)})^\top$ and $\det(A_{(1,j)})^\top = \det A_{(1,j)}$. ■

When to use which algorithm

- Row reduction: efficient for general matrices, especially $n > 3$.
- Cofactor expansion: useful when A has many zero entries (few terms) or $n \leq 3$.

Example. For $n = 3$:

$$\begin{aligned} \det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} &= a \det \begin{pmatrix} e & f \\ h & i \end{pmatrix} - b \det \begin{pmatrix} d & f \\ g & i \end{pmatrix} + c \det \begin{pmatrix} d & e \\ g & h \end{pmatrix} \\ &= a(ei - fh) - b(di - fg) + c(dh - eg). \end{aligned}$$

Practice Problem #5.6 (cofactor expansion on a parametrized matrix)

Problem. Let

$$A = \begin{pmatrix} 2 & 0 & 4 & 0 \\ 1 & 8 & 0 & 5 \\ 2 & -8 & 4 & -4 \\ 0 & 0 & x & -9 \end{pmatrix}.$$

Determine the values of x such that A is invertible, and find a formula for A^{-1} in that case.

Solution — Method 1 (row reduction, sketched). The exercise solution rows-reduces the augmented system $A\vec{z} = \vec{y}$ for arbitrary $\vec{y} \in \mathbb{R}^4$, arriving at the condition $x \neq 18$, then collects the coefficients of each y_i into the columns of A^{-1}.

Solution — Method 2 (cofactor expansion, featured here). Row 4 and column 2 each contain two zeros — both are excellent choices for expansion. We expand along row 4 (only $a_{43} = x$ and $a_{44} = -9$ are nonzero), so

$$\det A = a_{43}(-1)^{4+3} \det A_{(4,3)} + a_{44}(-1)^{4+4} \det A_{(4,4)}.$$

The submatrix $A_{(4,3)}$ deletes row 4 and column 3:

$$A_{(4,3)} = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 8 & 5 \\ 2 & -8 & -4 \end{pmatrix}.$$

Expanding this 3×3 along its first row (two zeros),

$$\det A_{(4,3)} = 2 \det \begin{pmatrix} 8 & 5 \\ -8 & -4 \end{pmatrix} = 2\big(8(-4) - 5(-8)\big) = 2(-32 + 40) = 16.$$

The submatrix $A_{(4,4)}$ deletes row 4 and column 4:

$$A_{(4,4)} = \begin{pmatrix} 2 & 0 & 4 \\ 1 & 8 & 0 \\ 2 & -8 & 4 \end{pmatrix}.$$

Expanding this along its first row,

$$\det A_{(4,4)} = 2\det\begin{pmatrix} 8 & 0 \\ -8 & 4 \end{pmatrix} + 4\det\begin{pmatrix} 1 & 8 \\ 2 & -8 \end{pmatrix} = 2(32) + 4(-24) = 64 - 96 = -32.$$

Combining:

$$\det A = x \cdot (-1) \cdot 16 + (-9) \cdot 1 \cdot (-32) = -16x + 288 = -16(x - 18).$$

Thus $\det A \neq 0 \iff x \neq 18$, and A is invertible exactly when $x \neq 18$. At $x = 18$ we have $\det A = 0$; the nonzero null vector $\vec{z} = (-2 \quad -1 \quad 1 \quad 2)^\top$ satisfies $A\vec{z} = \vec{0}$, as one checks directly.

Formula for A^{-1} via the adjugate. When $x \neq 18$, computing all sixteen cofactors $C_{ij} = (-1)^{i+j}\det A_{(i,j)}$ and transposing gives the adjugate matrix; the identity $A \cdot \operatorname{adj} A = (\det A)I$ (proved below) then yields

$$A^{-1} = \frac{1}{\det A}\operatorname{adj} A = \frac{1}{8(x-18)}\begin{pmatrix} 4(x+36) & -144 & -144 & -16 \\ 7x - 18 & -4x & 18 - 5x & -8 \\ -108 & 72 & 72 & 8 \\ -12x & 8x & 8x & 16 \end{pmatrix}.$$

The exercise solution obtains the same expression by row-reducing $(A \mid I_4)$ and collecting coefficients of the right-hand side $\vec{y}$; this "collect coefficients" approach is typically faster than computing all cofactors for an invertible matrix (row reduction is $O(n^3)$, cofactor expansion is $O(n!)$), but cofactor expansion excels when A has enough zeros to give only n or fewer surviving terms, as here.

Proof of the adjugate formula. Let $C = (C_{ij})$ be the cofactor matrix. By the cofactor expansion along row i, $\sum_j a_{ij}C_{ij} = \det A$. If we replace row i of A by row k, the resulting matrix $A_{i\leftarrow k}$ has two identical rows, so $\det A_{i\leftarrow k} = 0$; cofactor expansion along row i of this matrix gives $\sum_j a_{kj}C_{ij} = 0$ when $k \neq i$. Hence

$$AC^\top = (\det A)\, I_n,$$

and multiplying both sides by $\dfrac{1}{\det A}$ (valid when $\det A \neq 0$) gives $A^{-1} = \dfrac{1}{\det A}C^\top = \dfrac{1}{\det A}\operatorname{adj} A$.

Summary of determinant properties

- $\det I_n = 1$.
- $\det A = \det(A^\top)$.
- $\det(AB) = (\det A)(\det B)$.
- A invertible $\Leftrightarrow \det A \neq 0$.
- If A is triangular, $\det A =$ product of diagonal entries.
- $(A^{-1})^{-1} = A$; $(AB)^{-1} = B^{-1}A^{-1}$; $(A^\top)^{-1} = (A^{-1})^\top$.

Practice Problem #6.12 (a game on 3×3 matrices: Bob can force $\det A = 0$)

Problem. Alice and Bob start with an empty 3×3 matrix and take turns writing numbers in the nine positions, Alice going first. Once the matrix is filled, Bob wins

if $\det(A) = 0$ and Alice wins otherwise. Which player has a winning strategy, and what is it?

Solution. Bob wins. He writes 0 on every one of his turns; only the positions he chooses matter.

Step 1: which four positions force $\det A = 0$? Suppose the four cells owned by Bob contain 0. Since

$$\det A = \sum_{\sigma \in S_3} \operatorname{sgn}(\sigma)\, a_{1\sigma(1)} a_{2\sigma(2)} a_{3\sigma(3)},$$

the determinant is a sum of six monomials, each a product of one entry from each row and each column. If every permutation σ has at least one of its three cells among Bob's cells, then every monomial contains a 0 and $\det A = 0$ whatever Alice wrote. Conversely, if some σ uses only Alice's cells, then the corresponding monomial is a product of three of Alice's numbers, and no other monomial of the sum is the same product of the same three positions; so when Alice's five numbers are distinct and nonzero, $\det A \neq 0$. Hence Bob wins exactly when his four cells meet all six triples

$$P_\sigma = \{(1, \sigma(1)), (2, \sigma(2)), (3, \sigma(3))\} \qquad (\sigma \in S_3),$$

namely

$$P_1 = \{(1,1),(2,2),(3,3)\}, \quad P_2 = \{(1,1),(2,3),(3,2)\}, \quad P_3 = \{(1,2),(2,1),(3,3)\},$$
$$P_4 = \{(1,2),(2,3),(3,1)\}, \quad P_5 = \{(1,3),(2,1),(3,2)\}, \quad P_6 = \{(1,3),(2,2),(3,1)\}.$$

Step 2: which four cells meet all six triples? Checking the $\binom{9}{4} = 126$ four-element subsets of the board gives a short list: a four-element set meets all six triples exactly when it contains

(i) a whole line (a row, a column, or a diagonal), or

(ii) a 2×2 block, i.e. all four cells lying in two chosen rows and two chosen columns.

For (ii): a permutation uses two cells in the two chosen rows and, since only one column lies outside the two chosen columns, at least one of these two cells lies in a chosen column, so it belongs to the block. For (i) the claim is immediate. Configurations of the two types therefore kill the determinant.

Step 3: Bob's strategy. Bob places his zeros so that at the end his four cells contain one of the configurations of Step 2. His first zero goes into a cell lying neither in Alice's row nor in Alice's column (four possibilities); each of his zeros then threatens both its entire row and its entire column, i.e. it creates two simultaneous threats that Alice would have to answer separately. With only five moves Alice cannot cover all three rows, all three columns and the nine 2×2 blocks at once, so Bob always completes one of the configurations: a short case check on the three possible shapes of Alice's first move (corner, edge-centre, centre) shows that Bob can maintain the invariant that his remaining zeros can still be completed to a line or a 2×2 block. Once Bob has a full line of zeros, that row (or column) is a zero row, so $\det A = 0$; once he has a 2×2 block of zeros, say in rows $1, 2$ and columns $1, 2$, then every monomial of the determinant uses at least two entries from rows $1, 2$ in columns $1, 2$, one of which is 0, so again $\det A = 0$. In both cases Bob wins.

Practice Problem #6.13 (largest determinant of a 3×3 binary matrix)

Problem. A *binary matrix* is a matrix whose entries are each 0 or 1. What is the maximum value of $\det A$ as A ranges over all 3×3 binary matrices? Find all binary matrices A with $\det A$ equal to this maximum, and justify that they are the only ones.

Solution. For a binary matrix $A = (a_{ij})$, the general formula reads

$$\det A = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32}.$$

Every one of these six products is 0 or 1. Let p be the number of positive products equal to 1 and q the number of negative products equal to 1; then $\det A = p - q$ with

$0 \le p, q \le 3$.
Step 1: $p = 3$ forces $\det A = 0$. The three positive products involve the nine entries

$$a_{11}, a_{22}, a_{33} \mid a_{12}, a_{23}, a_{31} \mid a_{13}, a_{21}, a_{32},$$

which are exactly the nine entries of A, each appearing once. So all three products are 1 only if every entry of A is 1. But then every negative product is also 1, i.e. $q = 3$, giving $\det A = 3 - 3 = 0$.
Step 2: otherwise $\det A \le 2$. If $p \le 2$ then $\det A = p - q \le p \le 2$.
Step 3: the value 2 is attained. The three matrices

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

are binary and each has determinant 2 (e.g. for the first, $p = 2$ and $q = 0$).
Step 4: completeness. Suppose $\det A = 2$. Step 1 excludes $p = 3$, so $p \le 2$; but $2 = \det A = p - q \le p$ forces $p = 2$ and $q = 0$. There are exactly three ways to choose which two of the three positive products equal 1. If, say, $a_{11}a_{22}a_{33} = a_{12}a_{23}a_{31} = 1$, then the six entries appearing there are all 1, so

$$A = \begin{pmatrix} 1 & 1 & u \\ v & 1 & 1 \\ 1 & w & 1 \end{pmatrix}, \qquad u, v, w \in \{0, 1\},$$

and the three negative products are exactly u, v, w. The condition $q = 0$ forces $u = v = w = 0$, which is the third matrix displayed above. The other two choices of the surviving positive pair give, by the identical argument, the second and first matrices. Hence

$$\max \det A = 2,$$

attained by exactly the three binary matrices listed above.

Practice Problem #6.14 (the ratio $f(n+1)/g(n)$ is 2^n)

Problem. Let $f(n)$ be the maximum value of $\det A$ as A ranges over all $n \times n$ matrices whose entries are each -1 or 1, and let $g(n)$ be the maximum value of $\det A$ as A ranges over all $n \times n$ binary matrices. What is $f(n+1)/g(n)$? Justify your answer.
Solution. Multiplying a single row of A by -1 changes the sign of $\det A$ but not its absolute value, so $f(n)$ and $g(n)$ may be computed as maximum absolute values; we use this freely below. The answer is

$$\frac{f(n+1)}{g(n)} = 2^n.$$

Step 1 (normalization). Let B be an $(n+1) \times (n+1)$ matrix with all entries ± 1. Multiplying the jth column by $b_{1j} \in \{\pm 1\}$ makes the first row equal to $(1, 1, \ldots, 1)$; then multiplying the ith row ($i \ge 2$) by $b_{i1} \in \{\pm 1\}$ makes the first column equal to $(1, 1, \ldots, 1)^\top$ without disturbing the first row. Each of these operations is multiplication by an invertible diagonal matrix of determinant ± 1, so $\det B$ is only multiplied by ± 1 and $|\det B|$ is unchanged. The result has the block form

$$B = \begin{pmatrix} 1 & \mathbf{1}^\top \\ \mathbf{1} & A \end{pmatrix},$$

where A is an $n \times n$ matrix with entries ± 1 and $\mathbf{1}$ is the all-ones column vector.
Step 2 (block elimination). Subtracting the first row from each of the other n rows does not change the determinant, so

$$\det B = \det \begin{pmatrix} 1 & \mathbf{1}^\top \\ \mathbf{0} & A - \mathbf{1}\mathbf{1}^\top \end{pmatrix} = \det\bigl(A - \mathbf{1}\mathbf{1}^\top\bigr).$$

Every entry of $\mathbf{1}\mathbf{1}^{\top}$ equals 1, so the entries of $A - \mathbf{1}\mathbf{1}^{\top}$ are $1-1=0$ (where A has a 1) and $-1-1=-2$ (where A has a -1). Hence

$$A - \mathbf{1}\mathbf{1}^{\top} = -2C, \qquad C = \tfrac{1}{2}(\mathbf{1}\mathbf{1}^{\top} - A) \text{ binary.}$$

Since $\det(-2C) = (-2)^n \det C$,

$$|\det B| = \left|\det\left(A - \mathbf{1}\mathbf{1}^{\top}\right)\right| = 2^n |\det C| \le 2^n g(n).$$

As B was an arbitrary $(n+1)\times(n+1)$ matrix with entries ± 1, this gives $f(n+1) \le 2^n g(n)$.

Step 3 (the other inequality). Conversely, let C be a binary $n \times n$ matrix. Put $A = \mathbf{1}\mathbf{1}^{\top} - 2C$, whose entries are ± 1, and

$$B = \begin{pmatrix} 1 & \mathbf{1}^{\top} \\ \mathbf{1} & A \end{pmatrix},$$

which has entries ± 1. By Step 2, $\det B = \det(A - \mathbf{1}\mathbf{1}^{\top}) = \det(-2C) = (-2)^n \det C$, so $|\det B| = 2^n |\det C|$. Choosing C with $|\det C| = g(n)$ gives $f(n+1) \ge 2^n g(n)$. Combining the two inequalities, $f(n+1) = 2^n g(n)$ and therefore

$$\frac{f(n+1)}{g(n)} = 2^n.$$

(The identity also explains why the extremal ± 1 matrices of order $n+1$ are exactly the normalized matrices $\begin{pmatrix} 1 & \mathbf{1}^{\top} \\ \mathbf{1} & \mathbf{1}\mathbf{1}^{\top} - 2C \end{pmatrix}$ obtained from the extremal binary matrices C of order n, up to multiplying rows and columns by -1.)

6.10 Geometric Interpretation of the Determinant

Proposition. For an $n \times n$ matrix A, the number $|\det A|$ is the volume of the n-dimensional parallelepiped

$$\boxed{P(A) = \{A\vec{v} : \vec{v} \in \mathbb{R}^n \text{ with } 0 \le v_i \le 1 \text{ for all } i = 1, 2, \ldots, n\},}$$

i.e. the image of the unit cube under the map $\vec{v} \mapsto A\vec{v}$.

Proof for $n = 2$: Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $a, b, c, d > 0$. Then $P(A)$ is the parallelogram spanned by the columns $\begin{pmatrix} a \\ c \end{pmatrix}$ and $\begin{pmatrix} b \\ d \end{pmatrix}$. Enclose it in the axis-aligned rectangle R_0 of width $a+b$ and height $c+d$. The complement of $P(A)$ inside R_0 consists of two $b \times c$ rectangles R_1, R_2 and four right triangles forming two pairs with total areas ac and bd. Hence

$$\operatorname{area} P(A) = \underbrace{(a+b)(c+d)}_{R_0} - \underbrace{2bc}_{R_1+R_2} - \underbrace{ac}_{T_1+T_2} - \underbrace{bd}_{T_3+T_4} = ad - bc = \det A.$$

For general signs of the entries the same argument gives $\operatorname{area} P(A) = |\det A|$. ∎

Corollary. If $T : \mathbb{R}^n \to \mathbb{R}^n$ is linear with standard matrix A, and $S \subseteq \mathbb{R}^n$ is any region with finite volume, then

$$\boxed{\operatorname{volume}\big(T(S)\big) = |\det A| \cdot \operatorname{volume}(S),}$$

since $T(S)$ is the union of the images of small cubes tiling S, each scaled by $|\det A|$.

Sign and orientation. The sign of $\det A$ records whether T preserves or reverses orientation. For a 2×2 matrix with columns $\vec{u}, \vec{v}$: $\det A > 0$ when $\vec{v}$ lies counterclockwise

from $\vec{u}$; $\det A < 0$ when clockwise — in the second case the map flips the plane over.

7 Abstract Vector Spaces

7.1 Definition of a Vector Space

So far the course has focused on $\mathbb{R}^n$ and its subspaces. These are examples of (real) *vector spaces*. (There is also a notion of *complex* vector space, where scalars come from $\mathbb{C}$ instead of $\mathbb{R}$; essentially all of the theory is the same.)

Definition. A *vector space* is a nonempty set V with two operations, *vector addition* and *scalar multiplication*, satisfying the conditions below. The elements of V are called *vectors*.

Vector addition assigns to each pair $\vec{u}, \vec{v} \in V$ an output $\vec{u} + \vec{v} \in V$ such that

1. $\vec{u} + \vec{v} = \vec{v} + \vec{u}$ (commutative);
2. $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$ (associative);
3. there exists a unique *zero vector* $\vec{0} \in V$ with $\vec{0} + \vec{v} = \vec{v}$ for all $\vec{v} \in V$.

Scalar multiplication assigns to each scalar $c \in \mathbb{R}$ and $\vec{v} \in V$ an output $c\vec{v} \in V$ such that

1. $\vec{v} + (-1)\vec{v} = \vec{0}$;
2. $c(\vec{u} + \vec{v}) = c\vec{u} + c\vec{v}$ (distributivity over vector addition);
3. $(c + d)\vec{v} = c\vec{v} + d\vec{v}$ for $c, d \in \mathbb{R}$;
4. $c(d\vec{v}) = (cd)\vec{v}$ for $c, d \in \mathbb{R}$;
5. $1\vec{v} = \vec{v}$.

Notation. For $\vec{v} \in V$ define $-\vec{v} = (-1)\vec{v}$ and $\vec{u} - \vec{v} = \vec{u} + (-\vec{v})$.

Example. $\mathbb{R}^n$ and any subspace of $\mathbb{R}^n$ are vector spaces, with the usual operations of vector addition and scalar multiplication.

Example. The set $\mathbb{R}^{m \times n}$ of $m \times n$ matrices is a vector space. Its zero vector is the zero $m \times n$ matrix.

Example. The set of linear functions $f : \mathbb{R}^n \to \mathbb{R}^m$ is a vector space. Its zero vector is the function with formula $f(\vec{x}) = \vec{0}$ for all $\vec{x} \in \mathbb{R}^n$. This example and the previous one are related: taking standard matrices gives an invertible linear map between them.

Example (a nonstandard one). The set of positive real numbers $\mathbb{R}_+ = \{x \in \mathbb{R} : x > 0\}$ is a vector space, but *not* for the usual addition and multiplication. Instead define new operations

$$x \oplus y = xy \quad \text{for } x, y \in \mathbb{R}_+, \qquad c \otimes x = x^c \quad \text{for } c \in \mathbb{R},\ x \in \mathbb{R}_+.$$

The zero vector of $(\mathbb{R}_+, \oplus, \otimes)$ is the number 1. We also have $-x = \dfrac{1}{x}$ and $x - y = \dfrac{x}{y}$. This example comes from the bijection $x \mapsto e^x$, which turns ordinary addition into multiplication. (Here the "vectors" are numbers, so we keep them unadorned rather than writing $\vec{x}$.)

Example (the most important source). Let X be a set and let $\text{Functions}(X, \mathbb{R})$ be the set of functions $f : X \to \mathbb{R}$. Given $f, g \in \text{Functions}(X, \mathbb{R})$ and $c \in \mathbb{R}$, define $f + g$ and cf by the pointwise formulas

$$(f + g)(x) = f(x) + g(x), \qquad (cf)(x) = cf(x) \qquad \text{for } x \in X.$$

Then $\text{Functions}(X, \mathbb{R})$ is a vector space; its zero vector is the function with formula $f(x) = 0$ for all $x \in X$.

More generally, if V is any vector space, the set $\text{Functions}(X, V) = \{f : X \to V\}$ is a vector space for the analogous pointwise operations. Note that $\mathbb{R}^n$ "is the same as"

Functions($\{1, 2, \ldots, n\}, \mathbb{R}$): an n-entry list is a function from $\{1, \ldots, n\}$ to $\mathbb{R}$.

7.2 First Properties from the Axioms

It is rarely necessary to check the vector space axioms in detail: typically every vector space we meet is a special case of the general constructions above. But the axioms alone imply the familiar identities:

Proposition. If V is a vector space then $0\vec{v} = \vec{0}$ and $c\vec{0} = \vec{0}$ for all $c \in \mathbb{R}$ and $\vec{v} \in V$.

Proof: We have $0\vec{v} = (0 + 0)\vec{v} = 0\vec{v} + 0\vec{v}$, so

$$\vec{0} = 0\vec{v} - 0\vec{v} = (0\vec{v} + 0\vec{v}) - 0\vec{v} = 0\vec{v} + (0\vec{v} - 0\vec{v}) = 0\vec{v} + \vec{0} = 0\vec{v}.$$

Similarly, $c\vec{0} = c(\vec{0} + \vec{0}) = c\vec{0} + c\vec{0}$, so

$$\vec{0} = c\vec{0} - c\vec{0} = (c\vec{0} + c\vec{0}) - c\vec{0} = c\vec{0} + (c\vec{0} - c\vec{0}) = c\vec{0} + \vec{0} = c\vec{0}.$$

■

7.3 Subspaces, Bases, and Dimension (Abstract)

The notions of subspaces, bases, and dimension for general vector spaces are the same as for $\mathbb{R}^n$.

Definition. A *subspace* of a vector space V is a subset $H \subseteq V$ containing the zero vector of V, such that if $\vec{u}, \vec{v} \in H$ and $c \in \mathbb{R}$ then $\vec{u} + \vec{v} \in H$ and $c\vec{v} \in H$. A subspace H is itself a vector space with the same operations.

Example. V is a subspace of itself, and $\{\vec{0}\} \subseteq V$ is a subspace.

Example. $\mathbb{R}^2$ is technically not a subspace of $\mathbb{R}^3$, since $\mathbb{R}^2$ is not a subset of $\mathbb{R}^3$. Subspaces of $\mathbb{R}^3$ that "look like" $\mathbb{R}^2$ include the xy-plane, the xz-plane, and the yz-plane; nothing intrinsic makes one of these more natural than the others.

Example. Let X be any set and $Y \subseteq X$ a subset. The set H of functions $f : X \to \mathbb{R}$ with $f(y) = 0$ for all $y \in Y$ is a subspace of Functions$(X, \mathbb{R})$.

Example. The set Functions$(\mathbb{R}^n, \mathbb{R}^m)$ is a vector space, and the subset of linear functions $f : \mathbb{R}^n \to \mathbb{R}^m$ is a subspace of it.

Definition. A *linear combination* of $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k \in V$ is a vector of the form $c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k$ for scalars $c_i \in \mathbb{R}$. A linear combination of an *infinite* set means a linear combination of some *finite* subset — by definition only finitely many vectors (equivalently, finitely many nonzero coefficients) are involved.

Definition. The *span* of a set of vectors is the set of all linear combinations formed from the vectors. The span of any set of vectors in V is a subspace of V.

Example. Let $V =$ Functions$(\mathbb{R}, \mathbb{R})$. The span of the infinite set $\{1, x, x^2, x^3, \ldots\}$ is the subspace of *polynomial functions*: each polynomial $c_n x^n + \cdots + c_1 x + c_0$ uses only finitely many monomials. An infinite sum like $1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \cdots$ (which converges to e^x) is *not* a polynomial, so e^x is not in this span.

Definition. A finite set of vectors $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k \in V$ is *linearly independent* if $c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k = \vec{0}$ is possible only when $c_1 = c_2 = \cdots = c_k = 0$. An infinite set is linearly independent if every finite subset is linearly independent.

Definition. A *basis* of a vector space V is a subset of linearly independent vectors whose span is V. Saying $\vec{b}_1, \vec{b}_2, \vec{b}_3, \ldots$ is a basis for V is the same as saying that each $\vec{v} \in V$ can be expressed as a *unique* linear combination of basis elements.

Theorem. Let V be a vector space.

1. V has at least one basis.

2. Every basis of V has the same size (possibly infinite).
3. If A is a linearly independent subset of V, then V has a basis B with $A \subseteq B$.
4. If C is a subset of V whose span is V, then V has a basis B with $B \subseteq C$.

Definition. The *dimension* of a vector space V, denoted $\dim V$, is the common number of elements in any of its bases.

Corollary. If $H \subseteq V$ is a subspace then $\dim H \leq \dim V$. Moreover, if $H \subseteq V$ is a subspace with $\dim H = \dim V < \infty$ then $H = V$.

Example. If X is a finite set then $\dim \text{Functions}(X, \mathbb{R}) = |X|$. A basis is given by the *delta functions* $\delta_y : X \to \mathbb{R}$ for $y \in X$, defined by

$$\delta_y(x) = \begin{cases} 1 & \text{if } x = y, \\ 0 & \text{if } x \neq y. \end{cases}$$

Proof: *Spanning:* any $f : X \to \mathbb{R}$ equals $f = \sum_{y \in X} f(y)\delta_y$, since both sides take the same value at every $x \in X$. *Independence:* if $\sum_{y \in X} c_y \delta_y = 0$ (the zero function), then evaluating at $x \in X$ gives $c_x = 0$ for every x. ■

Example. $\dim \mathbb{R}^{m \times n} = mn$: a basis is given by the matrices E_{ij} having a single nonzero entry 1 in position (i, j). For instance when $m = n = 2$,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = aE_{11} + bE_{12} + cE_{21} + dE_{22},$$

which shows spanning; the only way to make the zero matrix is $a = b = c = d = 0$, which shows independence.

Practice Problem #6.10 (a basis of the solution space of $f'' + f = 0$)

Problem. The set V of twice differentiable functions $f : \mathbb{R} \to \mathbb{R}$ satisfying $f'' + f = 0$ is a subspace of the vector space of all functions $\mathbb{R} \to \mathbb{R}$. Find a basis for V and compute $\dim V$.

Solution. V is a subspace. The zero function lies in V, and if $f, g \in V$ then $(af + bg)'' + (af + bg) = a(f'' + f) + b(g'' + g) = 0$ by linearity of differentiation, so $af + bg \in V$.

Two elements. The functions $\cos x$ and $\sin x$ satisfy $f'' = -f$, hence lie in V.

They span V. Let $f \in V$ and set $a = f(0)$, $b = f'(0)$, and

$$h(x) = f(x) - a\cos x - b\sin x.$$

Then $h \in V$ and $h(0) = a - a = 0$, $h'(0) = b - b = 0$. Consider the nonnegative function $E(x) = h(x)^2 + h'(x)^2$. Its derivative is

$$E'(x) = 2h(x)h'(x) + 2h'(x)h''(x) = 2h'(x)\big(h(x) + h''(x)\big) = 0,$$

since $h'' + h = 0$. Hence E is constant on $\mathbb{R}$ (mean value theorem), and $E(x) = E(0) = 0$ for all x. So $h(x)^2 + h'(x)^2 = 0$, which forces $h(x) = 0$ for every x. Therefore

$$f(x) = f(0)\cos x + f'(0)\sin x,$$

so every element of V is a linear combination of $\cos x$ and $\sin x$.

They are independent. If $a\cos x + b\sin x$ is the zero function, evaluating at $x = 0$ gives $a = 0$, and then evaluating the derivative $-a\sin x + b\cos x$ at $x = 0$ gives $b = 0$. Consequently $B_V = \{\cos x, \sin x\}$ is a basis and $\dim V = 2$. Note that the data $f(0), f'(0)$ can be prescribed arbitrarily, which is another way to see that V is 2-dimensional.

7.4 Linear Functions Between Vector Spaces

Suppose U and V are vector spaces.

Definition. A function $f : U \to V$ is *linear* if $f(\vec{u}+\vec{v}) = f(\vec{u})+f(\vec{v})$ and $f(c\vec{v}) = cf(\vec{v})$ for all $c \in \mathbb{R}$ and $\vec{u}, \vec{v} \in U$.

Proposition. If U, V, W are vector spaces and $g : U \to V$ and $f : V \to W$ are linear, then the composition $f \circ g : U \to W$ (defined by $(f \circ g)(\vec{x}) = f(g(\vec{x}))$) is also linear.

Proof: The proof is the same as when $U = \mathbb{R}^n$, $V = \mathbb{R}^m$, $W = \mathbb{R}^p$: $(f \circ g)(\vec{u} + \vec{v}) = f(g(\vec{u}) + g(\vec{v})) = f(g(\vec{u})) + f(g(\vec{v}))$, and similarly $(f \circ g)(c\vec{v}) = c(f \circ g)(\vec{v})$. ■

Definition. Let $\mathrm{Lin}(U, V)$ be the subspace of linear functions $f : U \to V$ inside $\mathrm{Functions}(U, V)$.

Proposition. If V is a finite-dimensional vector space then $\dim \mathrm{Lin}(V, \mathbb{R}) = \dim V$.

Proof: Suppose $\vec{b}_1, \vec{b}_2, \ldots, \vec{b}_n$ is a basis for V. A basis for $\mathrm{Lin}(V, \mathbb{R})$ is given by the linear functions $\beta_1, \beta_2, \ldots, \beta_n : V \to \mathbb{R}$ with

$$\beta_i(x_1\vec{b}_1 + x_2\vec{b}_2 + \cdots + x_n\vec{b}_n) = x_i \qquad \text{for } x_1, \ldots, x_n \in \mathbb{R}.$$

The unique way to express any linear $f : V \to \mathbb{R}$ as a combination of these is $f = f(\vec{b}_1)\beta_1 + f(\vec{b}_2)\beta_2 + \cdots + f(\vec{b}_n)\beta_n$ (both sides agree on every basis vector, hence on all of V); independence follows by evaluating $\sum_i c_i\beta_i = 0$ at each $\vec{b}_j$, giving $c_j = 0$.

■

When $V = \mathbb{R}^n$, we can think of $\mathrm{Lin}(\mathbb{R}^n, \mathbb{R})$ as the vector space of $1 \times n$ matrices: a row vector $\vec{w} = \begin{pmatrix} w_1 & \cdots & w_n \end{pmatrix}$ defines the linear function $\vec{x} \mapsto \vec{w}\vec{x}$. If $\vec{b}_i = \vec{e}_i$ is the standard basis, then β_i corresponds to the row $\vec{e}_i^{\top}$.

Definition. For a linear function $f : U \to V$ define

$$\mathrm{range}(f) = \{f(\vec{x}) : \vec{x} \in U\} \subseteq V, \qquad \mathrm{kernel}(f) = \{\vec{x} \in U : f(\vec{x}) = \vec{0}\} \subseteq U.$$

These subspaces generalize the column space and null space of a matrix: if $f(\vec{x}) = A\vec{x}$ then $\mathrm{range}(f) = \mathrm{Col}\, A$ and $\mathrm{kernel}(f) = \mathrm{Nul}\, A$.

Theorem (Rank–Nullity, abstract version). If $\dim U < \infty$ then

$$\boxed{\dim \mathrm{range}(f) + \dim \mathrm{kernel}(f) = \dim U.}$$

This specializes to the earlier rank theorem for matrices when $U = \mathbb{R}^n$ and $V = \mathbb{R}^m$.

Example: symmetric and skew-symmetric matrices via rank–nullity

Define $f : \mathbb{R}^{3\times 3} \to \mathbb{R}^{3\times 3}$ by $f(X) = X + X^{\top}$; this is linear. Its kernel consists of the *skew-symmetric* matrices ($X^{\top} = -X$), with basis

$$\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}, \qquad \dim \mathrm{kernel}(f) = 3.$$

Its range is contained in the *symmetric* matrices ($S^{\top} = S$), and conversely every symmetric S satisfies $f\left(\frac{1}{2}S\right) = S$, so the range is exactly the symmetric subspace, with basis $E_{11}, E_{22}, E_{33}, E_{12} + E_{21}, E_{13} + E_{31}, E_{23} + E_{32}$, giving $\dim \mathrm{range}(f) = 6$. As predicted by rank–nullity:

$$\dim \mathrm{range}(f) + \dim \mathrm{kernel}(f) = 6 + 3 = 9 = \dim \mathbb{R}^{3\times 3}.$$

Practice Problem #6.9 (the map $L(A) = A - A^\top$ on 3×3 matrices)

Problem. Let V be the vector space of 3×3 matrices and define $L : V \to V$ by $L(A) = A - A^\top$.
(a) Describe a basis for V and compute $\dim V$.
(b) Find a basis for the subspace $N = \{A \in V : L(A) = 0\}$ and compute $\dim N$.
(c) Find a basis for the subspace $R = \{L(A) : A \in V\}$ and compute $\dim R$.
(d) Find $\lambda, \mu \in \mathbb{R}$ and nonzero $A, B \in V$ with $L(A) = \lambda A$ and $L(B) = \mu B$.

Solution. Let E_{ij} denote the 3×3 matrix with 1 in position (i, j) and 0 elsewhere.

(a) Every $A = (a_{ij})$ has the unique expression $A = \sum_{i=1}^{3} \sum_{j=1}^{3} a_{ij} E_{ij}$, so

$$B_V = \{E_{11}, E_{12}, E_{13}, E_{21}, E_{22}, E_{23}, E_{31}, E_{32}, E_{33}\}$$

is a basis and $\dim V = 9$.
(b) $L(A) = 0 \iff A = A^\top$, so N is the space of symmetric matrices. Each element has the form

$$\begin{pmatrix} a & b & c \\ b & d & e \\ c & e & f \end{pmatrix} = aE_{11} + dE_{22} + fE_{33} + b(E_{12} + E_{21}) + c(E_{13} + E_{31}) + e(E_{23} + E_{32}),$$

and the six coefficients are read off uniquely from the entries. Hence

$$B_N = \{E_{11}, E_{22}, E_{33}, E_{12} + E_{21}, E_{13} + E_{31}, E_{23} + E_{32}\}$$

is a basis and $\dim N = 6$.
(c) For every A, $L(A)^\top = (A - A^\top)^\top = A^\top - A = -L(A)$, so $L(A)$ is always skew-symmetric. Conversely, if $S^\top = -S$ then $L(S/2) = \frac{S}{2} - \frac{S^\top}{2} = \frac{S}{2} + \frac{S}{2} = S$. Hence R is exactly the space of skew-symmetric matrices. Its elements are

$$\begin{pmatrix} 0 & p & q \\ -p & 0 & r \\ -q & -r & 0 \end{pmatrix} = p(E_{12} - E_{21}) + q(E_{13} - E_{31}) + r(E_{23} - E_{32}),$$

with p, q, r uniquely determined, so

$$B_R = \{E_{12} - E_{21}, E_{13} - E_{31}, E_{23} - E_{32}\}$$

is a basis and $\dim R = 3$. As a check, rank–nullity holds: $\dim N + \dim R = 6 + 3 = 9 = \dim V$.
(d) Take a nonzero symmetric matrix and a nonzero skew-symmetric matrix, e.g.

$$A = E_{11} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad B = E_{12} - E_{21} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Then $L(A) = A - A^\top = 0 = 0 \cdot A$ and $L(B) = B - B^\top = B - (-B) = 2B$, so $\lambda = 0$ and $\mu = 2$. In other words, N is the eigenspace of L for the eigenvalue 0 and R is the eigenspace for the eigenvalue 2.

Practice Problem #6.11 (the double dual $V \to \mathrm{Lin}(\mathrm{Lin}(V, \mathbb{R}), \mathbb{R})$ is an isomorphism)

Problem. Suppose V is a vector space and recall that $\mathrm{Lin}(V, \mathbb{R})$ is the vector space of linear functions $V \to \mathbb{R}$. For $v \in V$ and $f \in \mathrm{Lin}(V, \mathbb{R})$ define $v^*(f) = f(v)$. Show that if $\dim V < \infty$ then $v \mapsto v^*$ is an invertible linear function $V \to \mathrm{Lin}(\mathrm{Lin}(V, \mathbb{R}), \mathbb{R})$.

Solution. Write $V^* = \mathrm{Lin}(V, \mathbb{R})$ and $V^{**} = \mathrm{Lin}(V^*, \mathbb{R})$, and let $\Phi : V \to V^{**}$ be the map $\Phi(v)(f) = f(v)$.

$\Phi(v)$ is linear. For $f, g \in V^*$ and $a, b \in \mathbb{R}$,

$$\Phi(v)(af + bg) = (af + bg)(v) = af(v) + bg(v) = a\Phi(v)(f) + b\Phi(v)(g),$$

so $\Phi(v) \in V^{**}$.
Φ is linear. For $v, w \in V$, $a, b \in \mathbb{R}$ and every $f \in V^*$,

$$\Phi(av + bw)(f) = f(av + bw) = af(v) + bf(w) = a\Phi(v)(f) + b\Phi(w)(f),$$

so $\Phi(av + bw) = a\Phi(v) + b\Phi(w)$.
Now assume $\dim V = d < \infty$ and choose a basis $\vec{b}_1, \ldots, \vec{b}_d$ of V, with coordinate functionals $\varepsilon_1, \ldots, \varepsilon_d \in V^*$ defined by

$$\varepsilon_i\Big(\sum_{j=1}^{d} a_j \vec{b}_j\Big) = a_i.$$

Two elementary facts: $\{\varepsilon_i\}$ is a basis of V^* (so $\dim V^* = d$), and every $f \in V^*$ satisfies $f = \sum_{i=1}^{d} f(\vec{b}_i)\varepsilon_i$, because both sides agree on every basis vector.
Φ is one-to-one. If $\Phi(v) = 0$ then $\varepsilon_i(v) = \Phi(v)(\varepsilon_i) = 0$ for every i, i.e. every coordinate of v is zero, so $v = \vec{0}$.
Φ is onto. Let $F \in V^{**}$ and set $v = \sum_{i=1}^{d} F(\varepsilon_i)\vec{b}_i \in V$. For every $f \in V^*$, using $f = \sum_i f(\vec{b}_i)\varepsilon_i$ and the linearity of F,

$$\Phi(v)(f) = f(v) = \sum_{i=1}^{d} F(\varepsilon_i) f(\vec{b}_i) = F\Big(\sum_{i=1}^{d} f(\vec{b}_i)\varepsilon_i\Big) = F(f).$$

Hence $\Phi(v) = F$.
Therefore Φ is a bijective linear map, with inverse $\Phi^{-1}(F) = \sum_{i=1}^{d} F(\varepsilon_i)\vec{b}_i$. In particular $V \cong V^{**}$, and both are isomorphic to V^*; the isomorphism Φ is defined without choosing a basis, although the proof of surjectivity uses one (together with $\dim V < \infty$).

Practice Problem #6.15 (X and D on polynomials: $D \circ X - X \circ D = \mathrm{id}$)

Problem. Interpret a vector $\vec{f} = (a, b, c, \ldots) \in \mathbb{R}^\infty$ as the polynomial $f(x) = a + bx + cx^2 + \cdots$, and let $P \subset \mathbb{R}^\infty$ be the subspace of vectors with only finitely many nonzero entries. The elementary vectors $\vec{e}_1, \vec{e}_2, \vec{e}_3, \ldots \in \mathbb{R}^\infty$ form a basis for P.
(a) Write down a sensible definition of the standard matrix of a linear function $P \to P$.
(b) Find the standard matrices of $X : P \to P$ and $D : P \to P$ given by $X(\vec{f}) =$ the vector representing $xf(x)$, and $D(\vec{f}) =$ the vector representing $\dfrac{d}{dx}f(x)$.
(c) Compute the standard matrix of $D \circ X - X \circ D$.
Solution. (a) Here $\vec{e}_j$ represents the monomial x^{j-1}. For a linear map $T : P \to P$ define its standard matrix $[T] = (t_{ij})_{i,j \geq 1}$ to be the infinite matrix whose jth column

is the coefficient vector of $T(\vec{e}_j)$:

$$T(\vec{e}_j) = \sum_{i \geq 1} t_{ij} \vec{e}_i.$$

Each column must have only finitely many nonzero entries, because $T(\vec{e}_j) \in P$; such matrices are called column-finite. Conversely, every column-finite infinite matrix defines a linear map $P \to P$ by $T(\vec{v}) = \sum_{j=1}^{N} v_j T(\vec{e}_j)$ when $\vec{v} = \sum_{j=1}^{N} v_j \vec{e}_j$: only finitely many columns occur and each has finite support, so the output again lies in P. This is ordinary matrix multiplication, with no infinite sums of nonzero terms to evaluate. There is no requirement that a row have finite support, so the infinite product is not symmetric in the two factors.
(b) Multiplication by x and differentiation act on the basis by

$$X(\vec{e}_j) = \vec{e}_{j+1} \ (j \geq 1), \qquad D(\vec{e}_1) = \vec{0}, \qquad D(\vec{e}_j) = (j-1)\vec{e}_{j-1} \ (j \geq 2).$$

Hence

$$[X] = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & \cdots \\ 1 & 0 & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 1 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \qquad [D] = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 2 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 3 & 0 & \cdots \\ 0 & 0 & 0 & 0 & 4 & \cdots \\ 0 & 0 & 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

Both are column-finite, as required.
(c) On each basis vector,

$$(D \circ X)(\vec{e}_j) = D(\vec{e}_{j+1}) = j\,\vec{e}_j, \qquad (X \circ D)(\vec{e}_j) = X\big((j-1)\vec{e}_{j-1}\big) = (j-1)\vec{e}_j,$$

where the second formula also holds for $j = 1$ since $D(\vec{e}_1) = \vec{0}$. Their difference is $\vec{e}_j$ for every j, so $D \circ X - X \circ D = \mathrm{id}_P$, whose standard matrix is the infinite identity matrix. At the level of matrices this reads

$$[D][X] = \mathrm{diag}(1, 2, 3, \ldots), \qquad [X][D] = \mathrm{diag}(0, 1, 2, \ldots),$$

and indeed the difference is I. The same identity is the product rule: $(xf(x))' - xf'(x) = f(x)$.

8 Eigenvalues and Eigenvectors

8.1 Eigenvectors and Eigenvalues

Let A be a square $n \times n$ matrix.

Definition. An *eigenvector* of A is a *nonzero* vector $\vec{v} \in \mathbb{R}^n$ such that

$$A\vec{v} = \lambda\vec{v}$$

for a number $\lambda \in \mathbb{R}$ (λ is the Greek letter "lambda"). The number λ is called the *eigenvalue* of A for the eigenvector $\vec{v}$.

Why require $\vec{v} \neq \vec{0}$?

If $\vec{v} = \vec{0}$ then $A\vec{v} = \lambda\vec{v} = \vec{0}$ holds for *every* number $\lambda \in \mathbb{R}$, so the zero vector carries no information about A. The number 0, however, *is* allowed to be an eigenvalue: the eigenvectors of A with eigenvalue 0 are precisely the nonzero vectors $\vec{v}$ with $A\vec{v} = \vec{0}$, i.e. the nonzero elements of $\mathrm{Nul}\, A$. Geometrically, $\vec{v}$ is an eigenvector exactly when $\vec{v}$

and $A\vec{v}$ lie on the same line through the origin.

Example (rotations). Let $A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ be rotation by θ radians. For $\vec{v}$ to be an eigenvector, $A\vec{v}$ must be a scalar multiple of $\vec{v}$, i.e. the rotated vector must stay on the same line. If θ is not a multiple of π this never happens: A has no (real) eigenvectors. If $\theta = 0$ then $A = I$ and *every* nonzero vector is an eigenvector with eigenvalue 1; if $\theta = \pi$ then $A = -I$ with eigenvalue -1.

Example. Given A and $\vec{v}$, one can check whether $\vec{v}$ is an eigenvector just by computing $A\vec{v}$. If $A = \begin{pmatrix} 1 & 6 \\ 5 & 2 \end{pmatrix}$ and $\vec{v} = \begin{pmatrix} 6 \\ -5 \end{pmatrix}$ then

$$A\vec{v} = \begin{pmatrix} 1 & 6 \\ 5 & 2 \end{pmatrix} \begin{pmatrix} 6 \\ -5 \end{pmatrix} = \begin{pmatrix} 6 - 30 \\ 30 - 10 \end{pmatrix} = \begin{pmatrix} -24 \\ 20 \end{pmatrix} = -4\vec{v},$$

so $\vec{v}$ is an eigenvector of A with eigenvalue $\lambda = -4$.

Example. What are the eigenvectors of the shift matrix $A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$? If $\vec{v} \in \mathbb{R}^4$ were an eigenvector with eigenvalue λ then

$$A\vec{v} = \begin{pmatrix} v_2 \\ v_3 \\ v_4 \\ 0 \end{pmatrix} = \lambda\vec{v} = \begin{pmatrix} \lambda v_1 \\ \lambda v_2 \\ \lambda v_3 \\ \lambda v_4 \end{pmatrix}.$$

Comparing entries from the bottom up: $0 = \lambda v_4$, $v_4 = \lambda v_3$, $v_3 = \lambda v_2$, $v_2 = \lambda v_1$, so $0 = \lambda v_4 = \lambda^2 v_3 = \lambda^3 v_2 = \lambda^4 v_1$. If $\lambda \neq 0$ this forces $v_1 = v_2 = v_3 = v_4 = 0$, contradicting $\vec{v} \neq \vec{0}$. Hence the only possible eigenvalue is $\lambda = 0$, and its eigenvectors are

$$\vec{v} = \begin{pmatrix} x \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \text{with } x \neq 0.$$

8.2 Finding Eigenvalues and Eigenspaces

Recall that $I = I_n$ denotes the $n \times n$ identity matrix.

Proposition. A number $\lambda \in \mathbb{R}$ is an eigenvalue of A if and only if $A - \lambda I$ is not invertible.

Proof: $A\vec{x} = \lambda\vec{x}$ has a nonzero solution $\vec{x} \in \mathbb{R}^n$ if and only if $(A - \lambda I)\vec{x} = \vec{0}$ has a nonzero solution, which occurs if and only if $A - \lambda I$ is not invertible. ■

Corollary. A number $\lambda \in \mathbb{R}$ is an eigenvalue of A if and only if $\det(A - \lambda I) = 0$.

Proof: $A - \lambda I$ is not invertible if and only if $\det(A - \lambda I) = 0$. ■

Definition. The eigenvectors of A with eigenvalue λ are precisely the nonzero elements of the null space $\mathrm{Nul}(A - \lambda I)$. One calls

$$\mathrm{Nul}(A - \lambda I) = \{\vec{v} \in \mathbb{R}^n : A\vec{v} = \lambda\vec{v}\}$$

the *eigenspace* of A for λ, or the *λ-eigenspace*. A number λ is an eigenvalue of A if and only if the corresponding eigenspace is nonzero (contains a nonzero vector). Since we know how to construct a basis for the null space of any matrix, we know how to find *all* eigenvectors for any given eigenvalue.

Example. For $A = \begin{pmatrix} 1 & 6 \\ 5 & 2 \end{pmatrix}$ and $\lambda = 7$:

$$A - 7I = \begin{pmatrix} 1-7 & 6 \\ 5 & 2-7 \end{pmatrix} = \begin{pmatrix} -6 & 6 \\ 5 & -5 \end{pmatrix} \sim \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} = \mathrm{RREF}(A - 7I).$$

Since $\text{RREF}(A-7I) \neq I$, the matrix $A-7I$ is not invertible, so 7 is an eigenvalue of A. Solving $(A-7I)\vec{x} = \vec{0}$: the free variable is x_2 and $x_1 = x_2$, so $\vec{x} = x_2\begin{pmatrix}1\\1\end{pmatrix}$. Thus $\begin{pmatrix}1\\1\end{pmatrix}$ is a basis for the 7-eigenspace, and every eigenvector with eigenvalue 7 has the form $a\begin{pmatrix}1\\1\end{pmatrix}$ for $a \neq 0$. Check: $\begin{pmatrix}1 & 6\\5 & 2\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}7\\7\end{pmatrix} = 7\begin{pmatrix}1\\1\end{pmatrix}$.

Example. Suppose we are told that $A = \begin{pmatrix}4 & 1 & 6\\2 & 3 & 6\\2 & 1 & 8\end{pmatrix}$ has 2 as an eigenvalue. To find a basis for the 2-eigenspace, row reduce

$$A - 2I = \begin{pmatrix}2 & 1 & 6\\2 & 1 & 6\\2 & 1 & 6\end{pmatrix} \sim \begin{pmatrix}1 & \frac{1}{2} & 3\\0 & 0 & 0\\0 & 0 & 0\end{pmatrix} = \text{RREF}(A-2I).$$

Thus $A\vec{x} = 2\vec{x}$ if and only if $x_1 + \frac{1}{2}x_2 + 3x_3 = 0$, i.e. $x_1 = -\frac{1}{2}x_2 - 3x_3$ with x_2, x_3 free:

$$\vec{x} = x_2\begin{pmatrix}-\frac{1}{2}\\1\\0\end{pmatrix} + x_3\begin{pmatrix}-3\\0\\1\end{pmatrix}.$$

So a basis for the 2-eigenspace is

$$\begin{pmatrix}-\frac{1}{2}\\1\\0\end{pmatrix}, \quad \begin{pmatrix}-3\\0\\1\end{pmatrix} \quad \text{(or, clearing denominators, } \begin{pmatrix}-1\\2\\0\end{pmatrix}, \begin{pmatrix}-3\\0\\1\end{pmatrix}\text{)},$$

and all eigenvectors with eigenvalue 2 are the nontrivial linear combinations of these.

Theorem. The eigenvalues of a triangular square matrix A are its diagonal entries.

Proof: If A has diagonal entries $d_1, d_2, \ldots, d_n$ then $A-\lambda I$ is triangular with diagonal entries $d_1-\lambda, d_2-\lambda, \ldots, d_n-\lambda$, so

$$\det(A-\lambda I) = (d_1-\lambda)(d_2-\lambda)\cdots(d_n-\lambda),$$

which is zero if and only if $\lambda \in \{d_1, d_2, \ldots, d_n\}$. ■

Example. The eigenvalues of $\begin{pmatrix}3 & 6 & 8\\0 & 0 & 6\\0 & 0 & 2\end{pmatrix}$ are 3, 0, and 2.

Eigenvalues of permutation matrices

Permutation matrices are another family with "easy" eigenvalue restrictions. Every permutation matrix X has *finite order*: its powers $X, X^2, X^3, \ldots$ all lie in the finite set of $n!$ permutation matrices, so two powers coincide, $X^i = X^j$ with $i < j$, giving $X^k = I$ for $k = j - i$. If $X\vec{v} = \lambda\vec{v}$ for a real eigenvalue λ and $\vec{v} \neq \vec{0}$, then

$$\vec{v} = I\vec{v} = X^k\vec{v} = \lambda^k\vec{v},$$

so $\lambda^k = 1$, which for real λ forces $\lambda = 1$ (if k is odd) or $\lambda = \pm 1$ (if k is even).

Summary: eigenvalues and eigenvectors

- $\vec{v} \neq \vec{0}$ is an eigenvector of A with eigenvalue $\lambda \iff A\vec{v} = \lambda\vec{v} \iff \vec{v} \in \text{Nul}(A-\lambda I)$.
- λ is an eigenvalue of $A \iff A - \lambda I$ is not invertible $\iff \det(A-\lambda I) = 0 \iff$ the λ-eigenspace $\text{Nul}(A-\lambda I)$ is nonzero.

- Eigenvalue 0 occurs $\iff$ A is not invertible.
- To find all eigenvectors for a known λ: compute a basis of $\mathrm{Nul}(A - \lambda I)$ by row reduction.
- Triangular matrices: eigenvalues are the diagonal entries.
- Rotation by θ: no real eigenvectors unless θ is a multiple of π.

8.3 Characteristic Polynomial and Algebraic Multiplicity

Let x be a variable. The *characteristic polynomial* of an $n \times n$ matrix A is

$$\det(A - xI),$$

which is a polynomial in x of degree exactly n (because the diagonal entries produce the term $(a_{11} - x)\cdots(a_{nn} - x)$ and every other term in the general formula has degree $\leq n - 2$). The *characteristic equation* of A is

$$\det(A - xI) = 0,$$

and the eigenvalues of A are precisely its solutions.

Example. The matrix

$$A = \begin{pmatrix} 5 & -2 & 6 & -1 \\ 0 & 3 & -8 & 0 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

is upper triangular, so

$$\det(A - xI) = (5 - x)(3 - x)(5 - x)(1 - x) = (5 - x)^2(3 - x)(1 - x).$$

Since $(5 - x)^2$ divides $\det(A - xI)$ but $(5 - x)^3$ does not, we say that 5 is an eigenvalue of *algebraic multiplicity* 2; the eigenvalues 1 and 3 each have algebraic multiplicity 1.

Definition. The *algebraic multiplicity* of an eigenvalue λ of a square matrix A is the unique integer $m \geq 1$ such that $(\lambda - x)^m$ divides $\det(A - xI)$ but $(\lambda - x)^{m+1}$ does not.

Example (a full computation). Consider

$$A = \begin{pmatrix} 1 & 5 & 4 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Since A is triangular, $\det(A - xI) = (1 - x)(2 - x)(3 - x)$ and the eigenvalues are $1, 2, 3$, each of algebraic multiplicity 1. We compute the three eigenspaces.

The 1*-eigenspace.*

$$A - I = \begin{pmatrix} 0 & 5 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \sim \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \mathrm{RREF}(A - I).$$

This shows that $\vec{x} \in \mathrm{Nul}(A - I)$ if and only if $x_2 = x_3 = 0$, so $\vec{x} = x_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$. Hence $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ is a basis for $\mathrm{Nul}(A - I)$, and the eigenvectors with eigenvalue 1 are its nonzero scalar multiples.

The 2*-eigenspace.*

$$A - 2I = \begin{pmatrix} -1 & 5 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & -5 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \mathrm{RREF}(A - 2I).$$

So $x_1 = 5x_2$ and $x_3 = 0$, giving $\vec{x} = x_2 \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix}$; a basis is $\begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix}$.

The 3-eigenspace.

$$A-3I=\begin{pmatrix}-2 & 5 & 4\\ 0 & -1 & 0\\ 0 & 0 & 0\end{pmatrix}\sim\begin{pmatrix}1 & 0 & -2\\ 0 & 1 & 0\\ 0 & 0 & 0\end{pmatrix}=\mathrm{RREF}(A-3I).$$

So $x_1=2x_3$ and $x_2=0$, giving $\vec{x}=x_3\begin{pmatrix}2\\0\\1\end{pmatrix}$; a basis is $\begin{pmatrix}2\\0\\1\end{pmatrix}$.

Since $1,2,3$ are distinct, the three eigenvectors

$$\begin{pmatrix}1\\0\\0\end{pmatrix},\quad\begin{pmatrix}5\\1\\0\end{pmatrix},\quad\begin{pmatrix}2\\0\\1\end{pmatrix}$$

are linearly independent (proved below). Setting

$$P=\begin{pmatrix}1 & 5 & 2\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix},\qquad D=\begin{pmatrix}1 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & 3\end{pmatrix},$$

one checks column by column, using $P\vec{e}_i=$ the ith column of P and $D\vec{e}_i=\lambda_i\vec{e}_i$, that

$$P^{-1}AP\vec{e}_1=\vec{e}_1,\qquad P^{-1}AP\vec{e}_2=2\vec{e}_2,\qquad P^{-1}AP\vec{e}_3=3\vec{e}_3,$$

so $P^{-1}AP=D$ and therefore $A=PDP^{-1}$.

Reading the diagonalization off the columns

If $P=(\vec{v}_1\ \ \vec{v}_2\ \ \cdots\ \ \vec{v}_n)$ has columns that are eigenvectors with $A\vec{v}_i=\lambda_i\vec{v}_i$, then for the standard basis $\vec{e}_i$,

$$AP\vec{e}_i=A\vec{v}_i=\lambda_i\vec{v}_i=P(\lambda_i\vec{e}_i)=PD\vec{e}_i.$$

Since AP and PD agree on every $\vec{e}_i$, they agree everywhere: $AP=PD$, hence $P^{-1}AP=D$. This is the column-by-column version of the computation above.

8.4 Powers of a Diagonalizable Matrix

Why diagonalizing makes powers easy

Suppose $A=PDP^{-1}$ with D diagonal. Then for every integer $n\geq 0$,

$$A^n=(PDP^{-1})^n=PD^nP^{-1},$$

because the inner factors telescope: $(PDP^{-1})(PDP^{-1})=PD(P^{-1}P)DP^{-1}=PD^2P^{-1}$, and the pattern continues. Moreover

$$D^n=\begin{pmatrix}x_1^n & & \\ & \ddots & \\ & & x_k^n\end{pmatrix}\quad\text{whenever}\quad D=\begin{pmatrix}x_1 & & \\ & \ddots & \\ & & x_k\end{pmatrix},$$

since multiplying diagonal matrices only multiplies the corresponding diagonal entries.

Example. Continuing the example above, $P=\begin{pmatrix}1 & 5 & 2\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}$ has inverse

$$P^{-1}=\begin{pmatrix}1 & -5 & -2\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}$$

(one checks $PP^{-1} = I$ directly). Then

$$A^n = PD^nP^{-1} = \begin{pmatrix} 1 & 5 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2^n & 0 \\ 0 & 0 & 3^n \end{pmatrix} \begin{pmatrix} 1 & -5 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

which multiplies out to

$$A^n = \begin{pmatrix} 1 & 5(2^n-1) & 2(3^n-1) \\ 0 & 2^n & 0 \\ 0 & 0 & 3^n \end{pmatrix}.$$

For instance the $(1,2)$ entry is $-5+5\cdot 2^n = 5(2^n-1)$, and the $(1,3)$ entry is $-2+2\cdot 3^n = 2(3^n-1)$.

Application: a discrete dynamical system

Suppose sequences a_n, b_n, c_n satisfy

$$a_n = a_{n-1} + 5b_{n-1} + 4c_{n-1}, \qquad b_n = 2b_{n-1}, \qquad c_n = 3c_{n-1} \quad (n \geq 1).$$

In matrix form $\begin{pmatrix} a_n \\ b_n \\ c_n \end{pmatrix} = A \begin{pmatrix} a_{n-1} \\ b_{n-1} \\ c_{n-1} \end{pmatrix}$, so $\begin{pmatrix} a_n \\ b_n \\ c_n \end{pmatrix} = A^n \begin{pmatrix} a_0 \\ b_0 \\ c_0 \end{pmatrix}$. With the formula for A^n above,

$$a_n = a_0 + 5(2^n-1)b_0 + 2(3^n-1)c_0, \qquad b_n = 2^n b_0, \qquad c_n = 3^n c_0.$$

Being able to compute $A = PDP^{-1}$ therefore also determines the asymptotic growth rate: for instance

$$\lim_{n\to\infty} \frac{a_n}{3^n} = 2c_0.$$

Practice Problem #7.1 (diagonalizing a 2×2 matrix; $\lim A^n$)

Problem. Let $A = \begin{pmatrix} .4 & -.3 \\ .4 & 1.2 \end{pmatrix}$. Find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$. Then compute $\lim_{n\to\infty} A^n$.

Solution.

Step 1: eigenvalues. $\det(tI - A) = (t-.4)(t-1.2) + .12 = t^2 - 1.6t + .6 = (t-1)(t-\frac{3}{5})$, so the eigenvalues are 1 and $\frac{3}{5}$.

Step 2: eigenvectors. For $\lambda = 1$: $A - I = \begin{pmatrix} -.6 & -.3 \\ .4 & .2 \end{pmatrix}$, with null space spanned by $(1 \quad -2)^\top$. For $\lambda = \frac{3}{5}$: $A - \frac{3}{5}I = \begin{pmatrix} -.2 & -.3 \\ .4 & .6 \end{pmatrix}$, with null space spanned by $(3 \quad -2)^\top$.

Step 3: diagonalization. Take

$$P = \begin{pmatrix} 1 & 3 \\ -2 & -2 \end{pmatrix}, \qquad D = \begin{pmatrix} 1 & 0 \\ 0 & 3/5 \end{pmatrix}.$$

Then $\det P = 4 \neq 0$ and $AP = PD$ column by column, so $A = PDP^{-1}$ with $P^{-1} = \frac{1}{4}\begin{pmatrix} -2 & -3 \\ 2 & 1 \end{pmatrix}$.

Step 4: the limit. Write $q_n = \left(\frac{3}{5}\right)^n$. Then

$$A^n = PD^nP^{-1} = \frac{1}{4}\begin{pmatrix} 1 & 3 \\ -2 & -2 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & q_n \end{pmatrix}\begin{pmatrix} -2 & -3 \\ 2 & 1 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} -2+6q_n & -3+3q_n \\ 4-4q_n & 6-2q_n \end{pmatrix},$$

which correctly gives $A^0 = I$ at $n = 0$. Since $q_n \to 0$,

$$\lim_{n\to\infty} A^n = \frac{1}{4}\begin{pmatrix} -2 & -3 \\ 4 & 6 \end{pmatrix}.$$

Practice Problem #7.12 (mixing milk and coffee)

Problem. Cup #1 starts with 1 liter of milk and cup #2 with 1 liter of coffee; a smaller cup #3 has volume r liters for some $0 < r < 1$. At each step you pour from cup #1 to fill cup #3, pour all of cup #3 into cup #2, mix thoroughly, pour from cup #2 to fill cup #3, and pour cup #3 back into cup #1 and mix. Let a_n be the amount of milk in cup #1 and b_n the amount of coffee in cup #2 after step n. (a) Using symmetry, explain why $a_n = b_n$ for all n. (b) Find an exact formula for $a_n = b_n$ as a function of r and n. (c) If $r = 0.001$, what is the smallest n with $a_n < 0.75$?

Solution.

Step 1: the one-step update. Suppose before a step cup #1 contains a liters of milk; then cup #2 contains $1 - a$ liters of milk (the total is 1). Pouring r liters from cup #1 leaves $(1 - r)a$ in cup #1; cup #2 then holds $1 + r$ liters containing $1 - a + ra$ milk, and the return pour brings back the fraction $\dfrac{r}{1+r}$ of that. Hence

$$a_{n+1} = (1-r)a_n + \frac{r}{1+r}\big(1 - a_n + ra_n\big) = \frac{a_n + r(1-a_n)}{1+r}, \qquad a_0 = 1.$$

Sanity check: $a_1 = \dfrac{1}{1+r} = (1-r) + r \cdot \dfrac{r}{1+r}$, as stated in the problem.

Step 2: part (a), symmetry. Interchanging the two cups and simultaneously interchanging the words "milk" and "coffee" leaves both the process and the initial configuration unchanged, but swaps the quantities a_n and b_n. Therefore $a_n = b_n$ for all $n = 0, 1, 2, \ldots$.

Step 3: part (b), diagonalization. Put $c_n = 1 - a_n$ (the milk in cup #2). Step 1 and its mirror image give

$$\begin{pmatrix} a_{n+1} \\ c_{n+1} \end{pmatrix} = M \begin{pmatrix} a_n \\ c_n \end{pmatrix}, \qquad M = \frac{1}{1+r}\begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}.$$

M has eigenvectors $(1 \quad 1)^\top$ and $(1 \quad -1)^\top$ with eigenvalues 1 and $q = \dfrac{1-r}{1+r}$. Since $(1 \quad 0)^\top = \dfrac{1}{2}(1 \quad 1)^\top + \dfrac{1}{2}(1 \quad -1)^\top$,

$$\begin{pmatrix} a_n \\ c_n \end{pmatrix} = M^n \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{q^n}{2}\begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad\Longrightarrow\quad a_n = b_n = \frac{1}{2}\left(1 + \left(\frac{1-r}{1+r}\right)^n\right).$$

Step 4: part (c). For $r = 0.001$: $a_n < 0.75 \iff q^n < \dfrac{1}{2} \iff n > \dfrac{\ln(1/2)}{\ln(0.999/1.001)} \approx 346.57$, so the smallest such n is $n = 347$.

8.5 Similar Matrices

Definition. Two $n \times n$ matrices X and Y are *similar* if there exists an invertible $n \times n$ matrix P with

$$X = PYP^{-1}.$$

In this case $Y = P^{-1}XP$ as well, so the relation is symmetric in X and Y.

Definition. A square matrix X is *diagonalizable* if X is similar to a diagonal matrix, i.e. if $X = PDP^{-1}$ for some invertible P and diagonal D.

Proposition. Every $n \times n$ matrix A is similar to itself.

Proof: Take $P = I$, so that $A = IAI^{-1}$. ■

Proposition. If A is similar to B and B is similar to C, then A is similar to C.

Proof: Write $A = PBP^{-1}$ and $B = QCQ^{-1}$. Then $R = PQ$ is invertible and

$$A = P(QCQ^{-1})P^{-1} = (PQ)C(PQ)^{-1} = RCR^{-1}.$$

■

Theorem. Similar matrices have the same characteristic polynomial, hence the same eigenvalues (with the same algebraic multiplicities).

Proof: Suppose $A = PBP^{-1}$. Then $A - xI = P(B - xI)P^{-1}$, so by multiplicativity of the determinant,

$$\det(A - xI) = \det(P)\det(B - xI)\det(P^{-1}) = \det(B - xI),$$

since $\det(P)\det(P^{-1}) = \det(PP^{-1}) = \det I = 1$. ■

Remark. Similar matrices usually have *different* eigenvectors. Also, having the same eigenvalues does *not* imply being similar:

Example. The matrices

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

both have the single eigenvalue 2. They are not similar: since $A = 2I$, for every invertible P,

$$PAP^{-1} = 2PIP^{-1} = 2I = A \neq B.$$

Practice Problem #7.13 (rearranged diagonal matrices are similar)

Problem. Show that if D is an $n \times n$ diagonal matrix, then D is similar to any $n \times n$ diagonal matrix E formed by rearranging its diagonal entries.

Solution. Write $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ and $E = \operatorname{diag}(\lambda_{\sigma(1)}, \ldots, \lambda_{\sigma(n)})$ for a permutation σ of $\{1, \ldots, n\}$, and let

$$P = \begin{pmatrix} \vec{e}_{\sigma(1)} & \vec{e}_{\sigma(2)} & \cdots & \vec{e}_{\sigma(n)} \end{pmatrix}.$$

The columns of P are the standard coordinate vectors in some order, so P is invertible (in fact $P^{-1} = P^{\top}$). For each j, the jth columns of DP and PE both equal $\lambda_{\sigma(j)}\vec{e}_{\sigma(j)}$. Hence $DP = PE$, i.e. $E = P^{-1}DP$, which is exactly the required similarity. Repeated diagonal entries cause no difficulty; for the reversal in the problem statement, P has 1 in every anti-diagonal position and 0 elsewhere.

8.6 Diagonalization via Eigenvectors

Theorem. An $n \times n$ matrix A is diagonalizable if and only if $\mathbb{R}^n$ has a basis $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n$ consisting entirely of eigenvectors of A. In this case, if $A\vec{v}_i = \lambda_i \vec{v}_i$, then

$$\boxed{A = PDP^{-1} \quad \text{for} \quad P = \begin{pmatrix} \vec{v}_1 & \vec{v}_2 & \cdots & \vec{v}_n \end{pmatrix}, \quad D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}.}$$

Proof ($\Leftarrow$): If the columns of P are such a basis, then P is invertible and the computation $AP\vec{e}_i = A\vec{v}_i = \lambda_i\vec{v}_i = PD\vec{e}_i$ gives $AP = PD$, hence $P^{-1}AP = D$.

Proof ($\Rightarrow$): If $A = PDP^{-1}$ with D diagonal, then P is invertible so its columns are a basis, and

$$A\vec{v}_i = PDP^{-1}\vec{v}_i = PD\vec{e}_i = P(\lambda_i\vec{e}_i) = \lambda_i\vec{v}_i,$$

so each column of P is an eigenvector with eigenvalue λ_i. ■

Example (not every matrix is diagonalizable). The matrix $B = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ has only the eigenvalue 2, and it is not similar to $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ (previous example), so B is *not* diagonalizable. Concretely, its only eigenvectors solve $(B - 2I)\vec{x} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\vec{x} = \vec{0}$, i.e. $x_2 = 0$, so the 2-eigenspace is one-dimensional and cannot supply a basis of $\mathbb{R}^2$.

Theorem. If an $n \times n$ matrix A has n *distinct* eigenvalues then A is diagonalizable.

Proof: Choosing one eigenvector for each eigenvalue gives n eigenvectors with distinct eigenvalues, which are linearly independent (theorem below). Any n linearly independent vectors in $\mathbb{R}^n$ form a basis, so the previous theorem applies. ■

Example. $A = \begin{pmatrix} 5 & -8 & 1 \\ 0 & 0 & 7 \\ 0 & 0 & -2 \end{pmatrix}$ is triangular with eigenvalues $5, 0, -2$; these are distinct, so A is diagonalizable.

Example. Every diagonal matrix D is diagonalizable, taking $P = I$.

Example. Not all diagonalizable matrices have n distinct eigenvalues: I is diagonal, hence diagonalizable, yet has only the single eigenvalue 1.

Why an upper-triangular matrix $\lambda I + N$ with $N \neq 0$ is not diagonalizable

Let A be upper triangular with every diagonal entry equal to λ. If $A = PDP^{-1}$ with D diagonal, then every diagonal entry of D is an eigenvalue of A; the only eigenvalue is λ, so $D = \lambda I$ and hence

$$A = P(\lambda I)P^{-1} = \lambda PIP^{-1} = \lambda I.$$

Thus the only diagonalizable matrix of this shape is λI itself. This is the general obstruction: a single eigenvalue whose eigenspace is too small.

Practice Problem #7.4 (recovering a diagonalization from three images)

Problem. Suppose A is a 3×3 matrix such that

$$A\begin{pmatrix} 13 \\ 4 \\ 9 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \\ 6 \end{pmatrix}, \qquad A\begin{pmatrix} 11 \\ 12 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \\ -4 \end{pmatrix}, \qquad A\begin{pmatrix} 2 \\ -2 \\ -2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}.$$

Find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$. Then determine if $\lim_{n\to\infty} A^n$ exists and compute its value if it does.

Solution.

Step 1: finding the eigenvectors. Let $\vec{x}, \vec{y}, \vec{z}$ be the three input vectors in the order listed. Subtracting and adding the first two equations gives

$$A(\vec{x} - \vec{y}) = \begin{pmatrix} 2 \\ -8 \\ 10 \end{pmatrix} = \vec{x} - \vec{y}, \qquad A(\vec{x} + \vec{y}) = \begin{pmatrix} 6 \\ 4 \\ 2 \end{pmatrix} = \frac{1}{4}(\vec{x} + \vec{y}),$$

since $\vec{x} - \vec{y} = (2 \quad -8 \quad 10)^\top$ and $\vec{x} + \vec{y} = (24 \quad 16 \quad 8)^\top$; also $A\vec{z} = \frac{1}{2}\vec{z}$. Thus

$$\vec{u}_1 = \tfrac{1}{2}(\vec{x} - \vec{y}) = (1 \quad -4 \quad 5)^\top, \qquad \vec{u}_2 = \tfrac{1}{8}(\vec{x} + \vec{y}) = (3 \quad 2 \quad 1)^\top,$$

$$\vec{u}_3 = \tfrac{1}{2}\vec{z} = (1 \quad -1 \quad -1)^\top$$

are eigenvectors with eigenvalues 1, $\frac{1}{4}$, $\frac{1}{2}$ respectively.

Step 2: diagonalization. Take

$$P = \begin{pmatrix} 1 & 3 & 1 \\ -4 & 2 & -1 \\ 5 & 1 & -1 \end{pmatrix}, \qquad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 1/2 \end{pmatrix}.$$

Expanding along the first row, $\det P = (2 \cdot (-1) - (-1) \cdot 1) - 3((-4)(-1) - (-1) \cdot 5) + ((-4) \cdot 1 - 2 \cdot 5) = -1 - 27 - 14 = -42 \neq 0$, so P is invertible and $A = PDP^{-1}$ with

$$P^{-1} = \begin{pmatrix} 1/42 & -4/42 & 5/42 \\ 3/14 & 2/14 & 1/14 \\ 1/3 & -1/3 & -1/3 \end{pmatrix}.$$

Step 3: the limit. Since $1^n \to 1$ while $(1/4)^n \to 0$ and $(1/2)^n \to 0$, the limit exists and equals $P\,\mathrm{diag}(1,0,0)P^{-1}$, i.e. the first column of P times the first row of P^{-1}:

$$\lim_{n\to\infty} A^n = \begin{pmatrix} 1 \\ -4 \\ 5 \end{pmatrix} (1/42 \quad -4/42 \quad 5/42) = \frac{1}{42} \begin{pmatrix} 1 & -4 & 5 \\ -4 & 16 & -20 \\ 5 & -20 & 25 \end{pmatrix}.$$

8.7 Linear Independence of Eigenvectors

Theorem. Suppose $\lambda_1, \lambda_2, \ldots, \lambda_r$ are *distinct* eigenvalues of A, with corresponding eigenvectors $\vec{v}_1, \ldots, \vec{v}_r$ (so $A\vec{v}_i = \lambda_i \vec{v}_i$). Then $\vec{v}_1, \ldots, \vec{v}_r$ are linearly independent.

Proof: Suppose they were dependent. Then there is an index $p > 0$ such that $\vec{v}_1, \ldots, \vec{v}_p$ are independent while $\vec{v}_{p+1}$ is a combination of them; write

$$\vec{v}_{p+1} = c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_p\vec{v}_p.$$

Applying A to both sides and using $A\vec{v}_i = \lambda_i\vec{v}_i$ gives

$$\lambda_{p+1}\vec{v}_{p+1} = c_1\lambda_1\vec{v}_1 + \cdots + c_p\lambda_p\vec{v}_p.$$

On the other hand, multiplying the original relation by λ_{p+1} gives

$$\lambda_{p+1}\vec{v}_{p+1} = c_1\lambda_{p+1}\vec{v}_1 + \cdots + c_p\lambda_{p+1}\vec{v}_p.$$

Subtracting,

$$\vec{0} = c_1(\lambda_1 - \lambda_{p+1})\vec{v}_1 + \cdots + c_p(\lambda_p - \lambda_{p+1})\vec{v}_p.$$

Since $\vec{v}_1, \ldots, \vec{v}_p$ are independent, every coefficient vanishes: $c_i(\lambda_i - \lambda_{p+1}) = 0$. But $\lambda_i \neq \lambda_{p+1}$ for $i \leq p$, so $c_1 = \cdots = c_p = 0$, which forces $\vec{v}_{p+1} = \vec{0}$ — impossible for an eigenvector. This contradiction proves the theorem. ∎

8.8 Fibonacci Numbers via Diagonalization

The Fibonacci numbers are defined by

$$f_0 = 0, \quad f_1 = 1, \quad f_n = f_{n-1} + f_{n-2} \ \ (n \geq 2),$$

so the sequence begins $0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots$; for instance

$$f_{100} = 354224848179261915075.$$

Define $a_n = f_{2n}$ and $b_n = f_{2n+1}$. For $n > 0$,

$$a_n = f_{2n} = f_{2n-2} + f_{2n-1} = a_{n-1} + b_{n-1},$$
$$b_n = f_{2n+1} = f_{2n-1} + f_{2n} = b_{n-1} + a_n = a_{n-1} + 2b_{n-1}.$$

Hence, with $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$,

$$\begin{pmatrix} a_n \\ b_n \end{pmatrix} = A \begin{pmatrix} a_{n-1} \\ b_{n-1} \end{pmatrix} = A^n \begin{pmatrix} a_0 \\ b_0 \end{pmatrix} = A^n \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

So an exact formula for A^n determines every f_n.

Step 1: eigenvalues.

$$\det(A - xI) = \det \begin{pmatrix} 1-x & 1 \\ 1 & 2-x \end{pmatrix} = (1-x)(2-x) - 1 = x^2 - 3x + 1.$$

By the quadratic formula,

$$\alpha = \frac{3+\sqrt{5}}{2}, \qquad \beta = \frac{3-\sqrt{5}}{2},$$

and $\alpha - \beta = \sqrt{5} \neq 0$, so A is diagonalizable. Note also $\alpha\beta = \dfrac{9-5}{4} = 1$.

Step 2: eigenvectors. Row reducing $A - \alpha I$ and using $(2-\alpha)(1-\alpha) = 1$ gives

$$A - \alpha I = \begin{pmatrix} 1-\alpha & 1 \\ 1 & 2-\alpha \end{pmatrix} \sim \begin{pmatrix} 2-\alpha & 1 \\ 0 & 0 \end{pmatrix},$$

so $\vec{v} = \begin{pmatrix} \alpha - 2 \\ 1 \end{pmatrix}$ is an eigenvector with eigenvalue α; the same computation gives $\vec{w} = \begin{pmatrix} \beta - 2 \\ 1 \end{pmatrix}$ for β.

Step 3: diagonalize. With

$$P = \begin{pmatrix} \alpha - 2 & \beta - 2 \\ 1 & 1 \end{pmatrix}, \qquad D = \begin{pmatrix} \alpha & 0 \\ 0 & \beta \end{pmatrix}, \qquad \det P = \alpha - \beta = \sqrt{5},$$

we have

$$P^{-1} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & 2-\beta \\ -1 & \alpha - 2 \end{pmatrix}, \qquad D^n = \begin{pmatrix} \alpha^n & 0 \\ 0 & \beta^n \end{pmatrix}.$$

Step 4: multiply out. Using $\alpha - 2 = 1 - \beta$ and $\beta - 2 = 1 - \alpha$,

$$\begin{pmatrix} a_n \\ b_n \end{pmatrix} = A^n \begin{pmatrix} 0 \\ 1 \end{pmatrix} = PD^nP^{-1} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1-\beta & 1-\alpha \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \alpha^n & 0 \\ 0 & \beta^n \end{pmatrix} \begin{pmatrix} \alpha - 1 \\ 1 - \beta \end{pmatrix},$$

which simplifies to

$$f_{2n} = a_n = \frac{\alpha^n - \beta^n}{\sqrt{5}}, \qquad f_{2n+1} = b_n = \frac{(\alpha-1)\alpha^n - (\beta-1)\beta^n}{\sqrt{5}}.$$

Step 5: one more identity. Since

$$(\alpha - 1)^2 = \frac{3+\sqrt{5}}{2} = \alpha, \qquad (\beta - 1)^2 = \frac{3-\sqrt{5}}{2} = \beta,$$

the two formulas combine into a single one:

$$\boxed{f_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right) \approx 0.447 \left(1.618^n - (-0.618)^n \right).}$$

Since $\dfrac{1-\sqrt{5}}{2} = -0.618\ldots$, for large n we have $f_n \approx \dfrac{1}{\sqrt{5}} \left(\dfrac{1+\sqrt{5}}{2} \right)^n$.

A curious by-product: 1/89

Summing $\sum_{n\geq 0} f_n 10^{-(n+1)}$ term by term gives

$$\frac{1}{89} = 0.011235955056179\ldots$$

The identity follows from the geometric series $\sum_{n\geq 0} x^n = \dfrac{1}{1-x}$ (valid for $|x| < 1$) applied to $x = \dfrac{1 \pm \sqrt{5}}{20}$; the two geometric sums differ by

$$\frac{1}{1 - \frac{1+\sqrt{5}}{20}} - \frac{1}{1 - \frac{1-\sqrt{5}}{20}} = \frac{10\sqrt{5}}{89}.$$

Practice Problem #7.3 (the Pell recurrence via diagonalization)

Problem. Consider the integer sequence P_n defined by $P_n = 2P_{n-1} + P_{n-2}$ for $n \geq 2$, where $P_0 = 0$ and $P_1 = 1$. Check that

$$\begin{pmatrix} P_{n+1} & P_n \\ P_n & P_{n-1} \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} P_n & P_{n-1} \\ P_{n-1} & P_{n-2} \end{pmatrix}$$

for all integers $n \geq 2$, explain why this implies that $\begin{pmatrix} P_{n+1} & P_n \\ P_n & P_{n-1} \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}^n$ for all integers $n \geq 0$, and use this to derive an exact formula for P_n similar to the formula for the Fibonacci numbers.

Solution.

Step 1: the matrix identity. Multiplying out the right side gives columns $\begin{pmatrix} 2P_n + P_{n-1} & 2P_{n-1} + P_{n-2} \\ P_n & P_{n-1} \end{pmatrix}$, which equals the left side by the recurrence. Iterating,

$$\begin{pmatrix} P_{n+1} & P_n \\ P_n & P_{n-1} \end{pmatrix} = M^2 \begin{pmatrix} P_{n-1} & P_{n-2} \\ P_{n-2} & P_{n-3} \end{pmatrix} = \cdots = M^n \begin{pmatrix} P_1 & P_0 \\ P_0 & P_{-1} \end{pmatrix} = M^n,$$

where $M = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$ and we set $P_{-1} = 1$ (consistent with the recurrence $P_1 = 2P_0 + P_{-1}$) so that the last matrix is I; the identity then holds for all $n \geq 0$.

Step 2: diagonalizing M. $\det(tI - M) = t^2 - 2t - 1$ has roots $\alpha = 1 + \sqrt{2}$ and $\beta = 1 - \sqrt{2}$. The matrix $M - \alpha I$ kills $\begin{pmatrix} 1 & \alpha - 2 \end{pmatrix}^\top = \begin{pmatrix} 1 & \sqrt{2} - 1 \end{pmatrix}^\top$, and similarly for β. So with

$$Q = \begin{pmatrix} 1 & 1 \\ \sqrt{2} - 1 & -\sqrt{2} - 1 \end{pmatrix}, \qquad \det Q = -2\sqrt{2} \neq 0,$$

we have $M = Q\,\mathrm{diag}(\alpha, \beta)Q^{-1}$.

Step 3: the formula. Comparing the $(1,2)$-entries of $\begin{pmatrix} P_{n+1} & P_n \\ P_n & P_{n-1} \end{pmatrix} = M^n = Q\,\mathrm{diag}(\alpha^n, \beta^n)Q^{-1}$ gives

$$P_n = \frac{\alpha^n - \beta^n}{\alpha - \beta} = \frac{(1+\sqrt{2})^n - (1-\sqrt{2})^n}{2\sqrt{2}} \qquad (n \geq 0).$$

Check: $P_1 = 1$, $P_2 = \dfrac{(1+\sqrt{2})^2 - (1-\sqrt{2})^2}{2\sqrt{2}} = \dfrac{4\sqrt{2}}{2\sqrt{2}} = 2$, $P_3 = 2\cdot 2 + 1 = 5$, matching the recurrence.

8.9 Diagonalizing When the Eigenvalues Are Not Distinct

Theorem. Let A be an $n \times n$ matrix with distinct eigenvalues $\lambda_1, \ldots, \lambda_p$, where $p \le n$, and put $d_i = \dim \mathrm{Nul}(A - \lambda_i I)$.

1. $d_i \le$ the algebraic multiplicity m_i of λ_i.
2. A is diagonalizable if and only if $d_1 + d_2 + \cdots + d_p = n$.
3. If this holds and $\mathcal{B}_i$ is a basis of $\mathrm{Nul}(A - \lambda_i I)$, then $\mathcal{B}_1 \cup \cdots \cup \mathcal{B}_p$ is a basis of $\mathbb{R}^n$ consisting of eigenvectors; writing its elements as $\vec{v}_1, \ldots, \vec{v}_n$ gives $A = PDP^{-1}$ with $P = (\vec{v}_1 \ \cdots \ \vec{v}_n)$.

Proof of part (a) (sketch): Fix $\lambda = \lambda_i$ with multiplicity m and $\dim \mathrm{Nul}(A - \lambda I) = d$, and let $\vec{v}_1, \ldots, \vec{v}_d$ be a basis of this eigenspace. Extend it to a basis $\vec{v}_1, \ldots, \vec{v}_n$ of $\mathbb{R}^n$ and set $Q = (\vec{v}_1 \ \cdots \ \vec{v}_n)$, $B = Q^{-1}AQ$. Since $A\vec{v}_j = \lambda \vec{v}_j$ for $j \le d$, the first d columns of B are $\lambda \vec{e}_j$, so B has the block form

$$B = \begin{pmatrix} \lambda I_d & Y \\ 0 & Z \end{pmatrix}.$$

Hence $\det(B - xI) = (\lambda - x)^d \det(Z - xI)$, and since $\det(A - xI) = \det(B - xI)$ this shows $(\lambda - x)^d$ divides $\det(A - xI)$, i.e. $d \le m$. ■

Proof of parts (b) and (c) (sketch): Let $\mathcal{B}_i$ be a basis of the λ_i-eigenspace and write w_i for any linear combination of elements of $\mathcal{B}_i$. If $w_1 + \cdots + w_p = \vec{0}$ with each w_i either zero or an eigenvector (eigenvectors with distinct eigenvalues are independent), then $w_1 = \cdots = w_p = \vec{0}$, and independence within each $\mathcal{B}_i$ forces all coefficients to vanish. Thus $\mathcal{B}_1 \cup \cdots \cup \mathcal{B}_p$ is independent. It has $d_1 + \cdots + d_p$ elements, so it is a basis of $\mathbb{R}^n$ exactly when that sum equals n — which is precisely the condition for A to have n independent eigenvectors, i.e. to be diagonalizable. ■

Example. Consider the lower-triangular matrix

$$A = \begin{pmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 1 & 4 & -3 & 0 \\ -1 & -2 & 0 & -3 \end{pmatrix}, \qquad \det(A - xI) = (5 - x)^2(-3 - x)^2.$$

The eigenvalues are 5 and -3, each of multiplicity 2. Row reducing,

$$A - 5I = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 4 & -8 & 0 \\ -1 & -2 & 0 & -8 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 8 & 16 \\ 0 & 1 & -4 & -4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \mathrm{RREF}(A - 5I),$$

so a basis for $\mathrm{Nul}(A - 5I)$ is $\begin{pmatrix} -8 \\ 4 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -16 \\ 4 \\ 0 \\ 1 \end{pmatrix}$. Similarly

$$A + 3I = \begin{pmatrix} 8 & 0 & 0 & 0 \\ 0 & 8 & 0 & 0 \\ 1 & 4 & 0 & 0 \\ -1 & -2 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \mathrm{RREF}(A + 3I),$$

so a basis for $\mathrm{Nul}(A + 3I)$ is $\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$. The dimensions sum to $2 + 2 = 4 = n$, so A is diagonalizable, with

$$P = \begin{pmatrix} -8 & -16 & 0 & 0 \\ 4 & 4 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \qquad D = \begin{pmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix}, \qquad A = PDP^{-1}.$$

Summary: deciding diagonalizability

- Compute $\det(A - xI)$; factor it over $\mathbb{R}$ to list the distinct eigenvalues $\lambda_1, \ldots, \lambda_p$ and their algebraic multiplicities m_i.
- For each λ_i, row reduce $A - \lambda_i I$ and set $d_i = \dim \operatorname{Nul}(A - \lambda_i I)$ (the geometric multiplicity).
- Always $d_i \leq m_i$ and $d_1 + \cdots + d_p \leq n$; A is diagonalizable $\iff$ $d_1 + \cdots + d_p = n$.
- Shortcut: if there are n distinct eigenvalues, then A is automatically diagonalizable.
- P collects one basis of each eigenspace as columns; D records the eigenvalues in the same order.

Practice Problem #7.2 (diagonalizability with a repeated eigenvalue)

Problem. Determine if $A = \begin{pmatrix} 2 & 0 & 0 \\ -3 & -1 & -2 \\ 3 & 3 & 4 \end{pmatrix}$ is diagonalizable. If A is diagonalizable, find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$.

Solution.

Step 1: eigenvalues. Expanding $\det(tI - A)$ along its first row gives

$$\det(tI - A) = (t-2)\det\begin{pmatrix} t+1 & 2 \\ -3 & t-4 \end{pmatrix} = (t-2)(t^2 - 3t + 2) = (t-1)(t-2)^2.$$

The eigenvalue 2 is repeated, so we must check the dimension of its eigenspace rather than conclude diagonalizability just from the roots.

Step 2: eigenspaces. For $\lambda = 1$: $A - I = \begin{pmatrix} 1 & 0 & 0 \\ -3 & -2 & -2 \\ 3 & 3 & 3 \end{pmatrix}$ has null space given by $x_1 = 0$ and $x_2 = -x_3$, hence $E_1 = \mathbb{R}\text{-span}\left\{\begin{pmatrix} 0 & 1 & -1 \end{pmatrix}^\top\right\}$. For $\lambda = 2$: $A - 2I = \begin{pmatrix} 0 & 0 & 0 \\ -3 & -3 & -2 \\ 3 & 3 & 2 \end{pmatrix}$ has the single independent equation $3x_1 + 3x_2 + 2x_3 = 0$, hence $E_2 = \mathbb{R}\text{-span}\left\{\begin{pmatrix} 1 & -1 & 0 \end{pmatrix}^\top, \begin{pmatrix} 0 & 2 & -3 \end{pmatrix}^\top\right\}$.

Step 3: conclusion. These give three linearly independent eigenvectors. In particular,

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -1 & 2 \\ -1 & 0 & -3 \end{pmatrix}, \qquad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

satisfy $\det P = 1$ and $AP = PD$ column by column. Thus $A = PDP^{-1}$, and A is diagonalizable over $\mathbb{R}$.

Practice Problem #7.7 (real eigenvectors force real eigenvalues)

Problem. Suppose A is a 3×3 real matrix that has exactly two distinct (complex) eigenvalues $\lambda \neq \mu$, and that has $\begin{pmatrix} 0 & 0 & 1 \end{pmatrix}^\top$, $\begin{pmatrix} 1 & 1 & 0 \end{pmatrix}^\top$, $\begin{pmatrix} 2 & -2 & 1 \end{pmatrix}^\top$ as eigenvectors (each with eigenvalue λ or μ). (a) Explain why you must have $\lambda \in \mathbb{R}$ and $\mu \in \mathbb{R}$. (b) What conditions must λ and μ satisfy for A to be diagonalizable? What conditions for A to be non-invertible? (c) Fix a choice of λ and μ satisfying the conditions in (b). Then find two examples of matrices A with the given properties (having only λ and μ as eigenvalues, having the given vectors as eigenvectors, diagonalizable but not invertible) which are not similar.

Solution. Write $\vec{v}_1, \vec{v}_2, \vec{v}_3$ for the three given vectors and form $P = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 1 & -2 \\ 1 & 0 & 1 \end{pmatrix}$. Since $\det P = -4 \neq 0$, these vectors form a basis of $\mathbb{R}^3$. If $A\vec{v}_j = \nu_j \vec{v}_j$ with $\nu_j \in$

$\{\lambda, \mu\}$, then $AP = P\,\mathrm{diag}(\nu_1, \nu_2, \nu_3)$. Both λ and μ occur among the ν_j: otherwise this formula would make A a scalar matrix, with only one eigenvalue.
Part (a). A real matrix cannot have a nonzero real eigenvector with a nonreal eigenvalue: if $A\vec{v} = \nu\vec{v}$ with real $\vec{v} \neq \vec{0}$, then some coordinate $v_i \neq 0$ and $\nu = (A\vec{v})_i / v_i \in \mathbb{R}$. Applying this to the three given vectors shows each ν_j is real, and therefore $\lambda, \mu \in \mathbb{R}$.
Part (b). No additional condition is needed for diagonalizability: the given vectors already form an eigenvector basis, so A is always diagonalizable over $\mathbb{R}$. It is noninvertible precisely when one eigenvalue is zero, that is, $\lambda\mu = 0$ (because $\lambda \neq \mu$, exactly one eigenvalue is zero in this case).
Part (c). Choose $\lambda = 0$ and $\mu = 1$. Using $P^{-1} = \begin{pmatrix} -1/4 & 1/4 & 1 \\ 1/2 & 1/2 & 0 \\ 1/4 & -1/4 & 0 \end{pmatrix}$, define

$$A_1 = P\,\mathrm{diag}(1,0,0)P^{-1} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1/4 & 1/4 & 1 \end{pmatrix},$$

$$A_2 = P\,\mathrm{diag}(0,1,1)P^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1/4 & -1/4 & 0 \end{pmatrix}.$$

For A_1 the vectors $\vec{v}_1, \vec{v}_2, \vec{v}_3$ have eigenvalues $1, 0, 0$; for A_2 they have eigenvalues $0, 1, 1$. Thus both matrices have all the requested properties. They are not similar, because $\operatorname{rank} A_1 = 1 \neq 2 = \operatorname{rank} A_2$, and multiplying on the left or right by an invertible matrix does not change rank; equivalently, their characteristic polynomials are $t^2(t-1)$ and $t(t-1)^2$.

8.10 Complex Numbers

Every eigenvalue we have met so far was real, but a real matrix can easily have a characteristic polynomial with no real roots: for $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ we have $\det(A - xI) = x^2 + 1$. To discuss such eigenvalues we enlarge $\mathbb{R}$ to the field $\mathbb{C}$ of complex numbers. The most economical definition is to treat a complex number as a 2×2 matrix.

Definition. Put

$$i = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad 1 = I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

For $a, b \in \mathbb{R}$, define

$$a + bi = aI_2 + bi = \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} + \begin{pmatrix} 0 & -b \\ b & 0 \end{pmatrix} = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

The set of complex numbers is

$$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\} = \left\{ \begin{pmatrix} a & -b \\ b & a \end{pmatrix} : a, b \in \mathbb{R} \right\},$$

and we view $\mathbb{R}$ as the subset $\{a + 0i : a \in \mathbb{R}\}$ of $\mathbb{C}$.

Since each complex number is literally a 2×2 matrix, its arithmetic is just matrix arithmetic. For $a, b, c, d, x \in \mathbb{R}$:

$$(a + bi) \pm (c + di) = \begin{pmatrix} a \pm c & -(b \pm d) \\ b \pm d & a \pm c \end{pmatrix} = (a \pm c) + (b \pm d)i,$$

$$(a + bi)(c + di) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} c & -d \\ d & c \end{pmatrix} = \begin{pmatrix} ac - bd & -(ad + bc) \\ ad + bc & ac - bd \end{pmatrix} = (ac - bd) + (ad + bc)i,$$

$$x(a + bi) = (a + bi)x = (ax) + (bx)i, \qquad \frac{a + bi}{x} = \frac{a}{x} + \frac{b}{x}i \quad (x \neq 0).$$

Multiplication is commutative, because matrices of the form $aI_2 + bi$ all commute with

each other (each is a polynomial in the single matrix i).

The product rule above is exactly what one obtains by multiplying the formal expressions as polynomials and then substituting $i^2 = -1$:

$$(a+bi)(c+di) = ac + (ad+bc)i + bd\,i^2 = (ac-bd) + (ad+bc)i,$$

$$i^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -1.$$

From now on we use the formal description: $\mathbb{C} = \{a+bi : a, b \in \mathbb{R}\}$ where i is a symbol with $i^2 = -1$, and expressions are manipulated like polynomials, replacing each i^2 by -1. The matrix description proves that this manipulation is legitimate, while the formal description is faster in computations.

Inverses and division. A complex number $a+bi$ is nonzero iff $a \neq 0$ or $b \neq 0$, and

$$\det(a+bi) = \det \begin{pmatrix} a & -b \\ b & a \end{pmatrix} = a^2 + b^2,$$

which vanishes only when $a = b = 0$. Hence every nonzero complex number is invertible, with

$$(a+bi)^{-1} = \frac{1}{a^2+b^2} \begin{pmatrix} a & b \\ -b & a \end{pmatrix} = \frac{a-bi}{a^2+b^2}.$$

Therefore we may divide by any nonzero complex number:

$$\frac{a+bi}{c+di} = (a+bi)(c+di)^{-1} = \frac{(a+bi)(c-di)}{c^2+d^2} \qquad (c+di \neq 0).$$

Example.

$$\frac{3-4i}{2+i} = \frac{(3-4i)(2-i)}{(2+i)(2-i)} = \frac{6-3i-8i+4i^2}{4-i^2} = \frac{6-11i-4}{5} = \frac{2}{5} - \frac{11}{5}i.$$

The trick is always to multiply numerator and denominator by the conjugate of the denominator, so that the new denominator c^2+d^2 is a real number.

Definition. The complex conjugate of $a+bi$ is

$$\overline{a+bi} = (a+bi)^\top = a-bi,$$

i.e. the transpose of the corresponding 2×2 matrix. For $y, z \in \mathbb{C}$ one has

$$\overline{y+z} = \overline{y} + \overline{z}, \qquad \overline{yz} = \overline{y}\,\overline{z}, \qquad \overline{z^{-1}} = (\overline{z})^{-1},$$

because transposing reverses the order of a matrix product and $(y+z)^\top = y^\top + z^\top$. Also

$$z\overline{z} = (a+bi)(a-bi) = a^2+b^2 \in \mathbb{R}, \qquad z = \overline{z} \iff z \in \mathbb{R},$$

and for $c+di \neq 0$,

$$(c+di)^{-1} = \frac{1}{c^2+d^2} \begin{pmatrix} c & d \\ -d & c \end{pmatrix} = \frac{\overline{c+di}}{c^2+d^2}.$$

Real and imaginary parts. For $z = a+bi \in \mathbb{C}$ we call $\Re(z) = a$ the real part and $\Im(z) = b$ the imaginary part of z. It is often useful to draw z as the vector $\begin{pmatrix} a \\ b \end{pmatrix} \in \mathbb{R}^2$. Under this picture, multiplication by i acts as

$$i(a+bi) = -b + ai \quad \longleftrightarrow \quad \begin{pmatrix} -b \\ a \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix},$$

that is, multiplication by i rotates the plane ninety degrees counterclockwise. This is the same geometric observation we made about the matrix i in the discussion of orthogonal vectors in $\mathbb{R}^2$.

Complex arithmetic at a glance

- $i^2 = -1$; adding and multiplying complex numbers is polynomial arithmetic with i^2 replaced by -1.
- $z = a + bi$ corresponds to the matrix $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$; conjugation is transposition and $\det z = |z|^2 = a^2 + b^2$.
- $z^{-1} = \overline{z}/|z|^2$, so division is multiplication by the conjugate of the denominator.
- Every nonzero complex number is invertible; $\mathbb{C}$ is a field containing $\mathbb{R}$.

Theorem (Euler's identity). Define the exponential function on $\mathbb{C}$ by the power series

$$\boxed{e^x = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \cdots}$$

(which converges for every $x \in \mathbb{C}$). Then $e = 2.71828\ldots$ and $e^{i\pi} + 1 = 0$.

Proof: The powers of i cycle with period 4:

$$i^0 = 1, \qquad i^1 = i, \qquad i^2 = -1, \qquad i^3 = -i, \qquad i^{n+4} = i^n.$$

Substituting $x = i\pi$ into the series and collecting the terms with i^0 and those with i^1 gives

$$e^{i\pi} = \left(1 - \frac{\pi^2}{2!} + \frac{\pi^4}{4!} - \frac{\pi^6}{6!} + \cdots\right) + i\left(\pi - \frac{\pi^3}{3!} + \frac{\pi^5}{5!} - \frac{\pi^7}{7!} + \cdots\right).$$

The two real series are the Taylor series of $\cos\pi$ and $\sin\pi$, so

$$e^{i\pi} = \cos\pi + i\sin\pi = -1 + 0i = -1,$$

and hence $e^{i\pi} + 1 = 0$. ■

8.11 The Fundamental Theorem of Algebra

The reason complex numbers are indispensable in linear algebra is the following fact about polynomials, which cannot be proved with elementary tools and which we therefore quote.

Theorem (Fundamental Theorem of Algebra). Let

$$\boxed{p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0}$$

with coefficients $a_0, a_1, \ldots, a_n \in \mathbb{C}$ and $a_n \neq 0$ (so that p has degree n). Then there exist n (not necessarily distinct) complex numbers $r_1, r_2, \ldots, r_n \in \mathbb{C}$ such that

$$\boxed{p(x) = a_n(x - r_1)(x - r_2)\cdots(x - r_n).}$$

The numbers $r_1, \ldots, r_n$ are the roots of p; they give all solutions of $p(x) = 0$. A root r has multiplicity m if exactly m of the numbers $r_1, \ldots, r_n$ are equal to r.

Remarks.

- Complex numbers are essential: the statement is false over $\mathbb{R}$. For instance $p(x) = x^2+1$ cannot be written as $(x - r_1)(x - r_2)$ with $r_1, r_2 \in \mathbb{R}$, but over $\mathbb{C}$ we have $x^2 + 1 = (x - i)(x + i)$.
- The theorem asserts existence of the roots, not a method for finding them. In practice one factors small-degree characteristic polynomials by hand or by the quadratic formula.

Example. $9x^2 + 36 = 9(x^2 + 4) = 9(x - 2i)(x + 2i)$, so the roots are $2i$ and $-2i$, each of multiplicity 1.

Why this matters for eigenvalues

The characteristic polynomial $\det(A - xI)$ of an $n \times n$ matrix has degree n with complex coefficients. The fundamental theorem of algebra therefore guarantees that it splits completely over $\mathbb{C}$: every $n \times n$ matrix has exactly n complex eigenvalues, counted with multiplicity. Over $\mathbb{R}$ this can fail, as $x^2 + 1$ shows; the fix is to allow complex eigenvalues, which is exactly what we do next.

8.12 Complex Eigenvalues and Eigenvectors

Definition. Let $\mathbb{C}^n$ be the set of vectors with n rows whose entries lie in $\mathbb{C}$. Since $\mathbb{R} \subset \mathbb{C}$, we have $\mathbb{R}^n \subset \mathbb{C}^n$. Sums $\vec{u} + \vec{v}$ and scalar multiples $c\vec{v}$ for $\vec{u}, \vec{v} \in \mathbb{C}^n$ and $c \in \mathbb{C}$ are defined exactly as in $\mathbb{R}^n$, using the arithmetic of $\mathbb{C}$. If A is an $n \times n$ matrix and $\vec{v} \in \mathbb{C}^n$, the product $A\vec{v}$ is defined by the same formula as before.

Example. With $A = \begin{pmatrix} i & 1 \\ 3 & 2i \end{pmatrix}$ and $\vec{v} = \begin{pmatrix} 1 \\ 1 - i \end{pmatrix}$,

$$A\vec{v} = \begin{pmatrix} i & 1 \\ 3 & 2i \end{pmatrix} \begin{pmatrix} 1 \\ 1 - i \end{pmatrix} = \begin{pmatrix} i + (1 - i) \\ 3 + 2i(1 - i) \end{pmatrix} = \begin{pmatrix} 1 \\ 3 + 2i - 2i^2 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 + 2i \end{pmatrix}.$$

Definition. Let A be an $n \times n$ matrix with entries in $\mathbb{R}$ or $\mathbb{C}$, and let $\lambda \in \mathbb{C}$. The following are equivalent:

- λ is an eigenvalue of A;
- $A\vec{v} = \lambda\vec{v}$ for some nonzero $\vec{v} \in \mathbb{C}^n$ (a complex eigenvector);
- $\det(A - \lambda I) = 0$.

This is the same definition as before; the only change is that λ (and $\vec{v}$) may now be complex.

Example. Let $A = \begin{pmatrix} i & 1 \\ 3 & 2i \end{pmatrix}$. Then

$$\det(A - xI) = \det \begin{pmatrix} i - x & 1 \\ 3 & 2i - x \end{pmatrix} = (i - x)(2i - x) - 3 = x^2 - 3ix - 5,$$

so by the quadratic formula

$$\lambda = \frac{3i \pm \sqrt{(3i)^2 - 4(1)(-5)}}{2} = \frac{3i \pm \sqrt{-9 + 20}}{2} = \pm\frac{\sqrt{11}}{2} + \frac{3}{2}i.$$

Both eigenvalues are genuinely complex, and they are not conjugates of each other because A does not have real entries.

Fact. Let A be an $n \times n$ matrix. Then A has n eigenvalues in $\mathbb{C}$, counted with multiplicity (namely the roots of $\det(A - xI)$, which exist by the fundamental theorem of algebra).

If A is a matrix and $\vec{v} \in \mathbb{C}^n$, let $\overline{A}$ and $\overline{\vec{v}}$ be obtained by replacing every entry with its complex conjugate.

Proposition. Suppose A is an $n \times n$ matrix with real entries, so $\overline{A} = A$. If $\lambda \in \mathbb{C}$ is an eigenvalue of A with eigenvector $\vec{v} \in \mathbb{C}^n$, then $\overline{\vec{v}}$ is an eigenvector of A with eigenvalue $\overline{\lambda}$.

Proof: Since A has real entries and conjugation is additive and multiplicative,

$$A\overline{\vec{v}} = \overline{A\vec{v}} = \overline{\lambda\vec{v}} = \overline{\lambda}\,\overline{\vec{v}}.$$

Also $\overline{\vec{v}} \neq \vec{0}$ because $\vec{v} \neq \vec{0}$. Hence $\overline{\vec{v}}$ is an eigenvector with eigenvalue $\overline{\lambda}$. ■

Corollary. If A has real entries, then its non-real complex eigenvalues occur in conjugate pairs $\lambda, \overline{\lambda}$, with respective eigenvectors $\vec{v}, \overline{\vec{v}}$. In particular the number of non-real eigenvalues is even, and $\det(A - xI)$ has real coefficients.

Working with complex eigenvalues

- Everything is unchanged: eigenvalues solve $\det(A-\lambda I)=0$, eigenvectors solve $(A-\lambda I)\vec{v}=\vec{0}$, and row reduction is carried out over $\mathbb{C}$.
- Over $\mathbb{C}$ a matrix of size n always has n eigenvalues counted with multiplicity; nothing can "fail to exist."
- For a real matrix A, the eigenvalues that are not real come in conjugate pairs, and so do the eigenvectors.

Practice Problem #7.8 (roots of unity and the cyclic permutation matrix)

Problem. For $\theta\in\mathbb{R}$ write $e^{i\theta}=\cos\theta+i\sin\theta$; then $e^{i\theta_1}=e^{i\theta_2}$ iff $\theta_1-\theta_2$ is an integer multiple of 2π. (a) Explain why $e^{i\theta_1}e^{i\theta_2}=e^{i(\theta_1+\theta_2)}$, and use this notation to find all complex solutions of $x^n=1$ and of $x^n=-1$. (b) Find the characteristic polynomial of the $n\times n$ cyclic permutation matrix $A=\begin{pmatrix}\vec{e}_2 & \vec{e}_3 & \cdots & \vec{e}_n & \vec{e}_1\end{pmatrix}$, and use (a) to deduce that A is diagonalizable over $\mathbb{C}$. (c) Explain why a direct sum $\begin{pmatrix}B & 0\\ 0 & C\end{pmatrix}$ of diagonalizable matrices B, C is diagonalizable, and hence why every permutation matrix is diagonalizable over $\mathbb{C}$.

Solution.

Part (a). The sum-angle identities give

$$\begin{aligned}(\cos\theta_1+i\sin\theta_1)(\cos\theta_2+i\sin\theta_2) &= (\cos\theta_1\cos\theta_2-\sin\theta_1\sin\theta_2)\\ &\quad+i(\sin\theta_1\cos\theta_2+\cos\theta_1\sin\theta_2)\\ &=\cos(\theta_1+\theta_2)+i\sin(\theta_1+\theta_2),\end{aligned}$$

so $e^{i\theta_1}e^{i\theta_2}=e^{i(\theta_1+\theta_2)}$ and, by induction, $(e^{i\theta})^n=e^{in\theta}$. If $x^n=\pm1$ then $|x|^n=1$, so $|x|=1$ and $x=e^{i\theta}$. For $x^n=1$ we need $n\theta\in2\pi\mathbb{Z}$; for $x^n=-1=e^{i\pi}$ we need $n\theta-\pi\in2\pi\mathbb{Z}$. The complete lists, without repetitions, are

$$x^n=1:\ x=e^{2\pi ik/n},\ k=0,1,\ldots,n-1;$$

$$x^n=-1:\ x=e^{(2k+1)\pi i/n},\ k=0,1,\ldots,n-1.$$

Part (b). For $n\geq2$, $tI-A$ has t on the diagonal, -1 immediately below the diagonal, and -1 in the top-right corner. In the permutation expansion of $\det(tI-A)$ only two terms can be nonzero: the diagonal term t^n, and the term picking the entries around the cycle, which contributes $(-1)^{n-1}(-1)^n=-1$ (choosing any off-diagonal entry forces the off-diagonal choice in every row, to avoid repeating a column). Hence

$$\chi_A(t)=\det(tI-A)=t^n-1.$$

(For $n=1$, $A=(1)$ and the same formula holds directly.) By (a) the eigenvalues are the n distinct numbers $1,\zeta,\ldots,\zeta^{n-1}$ with $\zeta=e^{2\pi i/n}$, so A is diagonalizable over $\mathbb{C}$. Explicitly, an eigenvector for an nth root of unity λ is

$$\vec{v}(\lambda)=\begin{pmatrix}1 & \lambda^{-1} & \cdots & \lambda^{-(n-1)}\end{pmatrix}^{\top},$$

since A shifts the coordinates down cyclically and $\lambda^{-(n-1)}=\lambda$; thus $P=\begin{pmatrix}\vec{v}(1) & \vec{v}(\zeta) & \cdots & \vec{v}(\zeta^{n-1})\end{pmatrix}$ and $D=\operatorname{diag}(1,\zeta,\ldots,\zeta^{n-1})$ satisfy $A=PDP^{-1}$.

Part (c). If $B=SD_BS^{-1}$ and $C=TD_CT^{-1}$ with D_B, D_C diagonal, then block multiplication gives

$$\begin{pmatrix}B & 0\\ 0 & C\end{pmatrix}=\begin{pmatrix}S & 0\\ 0 & T\end{pmatrix}\begin{pmatrix}D_B & 0\\ 0 & D_C\end{pmatrix}\begin{pmatrix}S^{-1} & 0\\ 0 & T^{-1}\end{pmatrix},$$

where the first and third factors are mutual inverses and the middle factor is diagonal. Every permutation matrix is similar (via a permutation matrix) to a direct sum of

cyclic permutation matrices — one block per cycle of the permutation — so by (b) and (c) every permutation matrix is diagonalizable over $\mathbb{C}$.

Practice Problem #7.11 (the faulty clock)

Problem. The hour hand of a faulty 12-hour clock points to the correct hour at time $n = 0$. Every sixty minutes the hour hand moves forward one hour with probability $\frac{1}{4}$, does not move at all with probability $\frac{1}{2}$, or moves forward two hours with probability $\frac{1}{4}$. Find an exact formula for the probability that the hour hand points to the correct time after n hours.

Solution. (No solution was provided in the official set; here is a full derivation.)

Step 1: a Markov chain on 12 states. Let $\vec{p}^{(n)} \in \mathbb{R}^{12}$ be the distribution of the hand's position after n hours, indexed by hours $0, 1, \ldots, 11$; initially $\vec{p}^{(0)} = \vec{e}_0$. One hour later the position shifts by $+1$ with probability $\frac{1}{4}$, stays with probability $\frac{1}{2}$, and shifts by $+2$ with probability $\frac{1}{4}$, so $\vec{p}^{(n+1)} = M\vec{p}^{(n)}$ where, writing X for the 12-hour cyclic shift matrix of Practice Problem #7.8(b),

$$M = \tfrac{1}{4}I + \tfrac{1}{2}X + \tfrac{1}{4}X^2.$$

The hand points to the correct hour iff its position is 0 mod 12, so the desired probability is $p_n = (M^n)_{00}$.

Step 2: diagonalizing M. From #7.8(b), $X\vec{v}_k = \omega^k \vec{v}_k$ for $\vec{v}_k = \begin{pmatrix} 1 & \omega^{-k} & \cdots & \omega^{-11k} \end{pmatrix}^\top$, where $\omega = e^{2\pi i/12}$ and $k = 0, 1, \ldots, 11$. Hence $\vec{v}_k$ is an eigenvector of M with eigenvalue

$$\lambda_k = \tfrac{1}{4} + \tfrac{1}{2}\omega^k + \tfrac{1}{4}\omega^{2k} = \omega^k \left(\tfrac{\omega^{-k/2} + \omega^{k/2}}{2}\right)^2 = \omega^k \cos^2 \frac{\pi k}{12}.$$

Since $\vec{e}_0 = \dfrac{1}{12}\displaystyle\sum_{k=0}^{11} \vec{v}_k$ (the roots-of-unity filter: $\displaystyle\sum_{k=0}^{11} \omega^{jk} = 0$ for $j \not\equiv 0 \pmod{12}$),

$$p_n = (M^n)_{00} = \frac{1}{12}\sum_{k=0}^{11} \lambda_k^n = \frac{1}{12}\sum_{k=0}^{11} \omega^{kn} \cos^{2n} \frac{\pi k}{12},$$

which is real because the terms k and $12 - k$ are complex conjugates.

Step 3: a binomial form. The total displacement after n hours is $T_n = Y_1 + \cdots + Y_n$ with independent $Y_i \in \{0, 1, 2\}$, and

$$\sum_s \Pr(T_n = s)\, x^s = \left(\tfrac{1}{4} + \tfrac{1}{2}x + \tfrac{1}{4}x^2\right)^n = \frac{(1+x)^{2n}}{4^n} \quad \Longrightarrow \quad \Pr(T_n = s) = \frac{1}{4^n}\binom{2n}{s}.$$

The hand is correct iff $T_n \equiv 0 \pmod{12}$, so

$$p_n = \frac{1}{4^n} \sum_{j \equiv 0 \,(\mathrm{mod}\ 12)} \binom{2n}{j}.$$

Check: $p_1 = \dfrac{1}{4}$ (only staying still keeps the hand correct after one hour) and $p_2 = \dfrac{1}{16}$ (the maximal displacement in two hours is $4 < 12$, so both hours must be stays), agreeing with the sum in Step 2.

8.13 The Trace and the Eigenvalue Identities

We begin with a small algebraic lemma that will be used repeatedly.

Lemma. Suppose a polynomial in x is written in two ways as

$$\boxed{(\lambda_1 - x)(\lambda_2 - x)\cdots(\lambda_n - x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0}$$

with $\lambda_1, \ldots, \lambda_n, a_0, \ldots, a_n \in \mathbb{C}$. Then

$$\boxed{a_n = (-1)^n, \qquad a_{n-1} = (-1)^{n-1}(\lambda_1 + \lambda_2 + \cdots + \lambda_n), \qquad a_0 = \lambda_1\lambda_2\cdots\lambda_n.}$$

Proof: Expanding the product means choosing either λ_i or $-x$ from each of the n factors and multiplying the n chosen terms. There are 2^n such monomials.

- The only monomial of degree n is $(-x)^n = (-1)^n x^n$, so $a_n = (-1)^n$.
- The only monomial of degree 0 is $\lambda_1\lambda_2\cdots\lambda_n$, so $a_0 = \lambda_1\lambda_2\cdots\lambda_n$.
- The monomials of degree $n-1$ arise by choosing $-x$ from all factors except one:

$$\lambda_1(-x)^{n-1} + \lambda_2(-x)^{n-2}(-x) + \cdots + \lambda_n(-x)^{n-1} = (-1)^{n-1}(\lambda_1 + \cdots + \lambda_n)x^{n-1},$$

so $a_{n-1} = (-1)^{n-1}(\lambda_1 + \cdots + \lambda_n)$.

■

Definition. The trace of a square matrix A, written $\operatorname{tr} A$, is the sum of its diagonal entries.

Example. For $A = \begin{pmatrix} 1 & 0 & 7 \\ -1 & 2 & 8 \\ 2 & 4 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 2 & 4 \\ 7 & 8 & 3 \end{pmatrix}$ we get

$$\operatorname{tr} A = 1 + 2 + 3 = 6 = \operatorname{tr} B = \operatorname{tr}(A^\top),$$

since transposition does not move diagonal entries.

Proposition. If A and B are $n \times n$ matrices then

$$\boxed{\operatorname{tr}(A + B) = \operatorname{tr} A + \operatorname{tr} B, \qquad \operatorname{tr}(AB) = \operatorname{tr}(BA).}$$

Proof: The first identity is immediate, since the diagonal entries of $A + B$ are the sums of the corresponding diagonal entries of A and B. For the second, let E_{ij} be the matrix with 1 in position (i, j) and 0 elsewhere; here the symbol i is an integer index, not the complex number. One checks that

$$E_{ij}E_{kl} = \begin{cases} E_{il}, & j = k, \\ 0, & j \neq k, \end{cases} \qquad \operatorname{tr}(E_{ij}) = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Hence $\operatorname{tr}(E_{ij}E_{kl})$ equals 1 exactly when $i = l$ and $j = k$, and 0 otherwise. This condition is symmetric in the two pairs, so $\operatorname{tr}(E_{ij}E_{kl}) = \operatorname{tr}(E_{kl}E_{ij})$. Writing $A = \sum_{i,j} A_{ij}E_{ij}$ and $B = \sum_{k,l} B_{kl}E_{kl}$ and using linearity of the trace gives $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

■

Remark. Unlike the determinant, the trace is not multiplicative: usually $\operatorname{tr}(AB) \neq \operatorname{tr}(A)\operatorname{tr}(B)$. For example with $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = I_2$ we have $\operatorname{tr}(AB) = \operatorname{tr} A = 5$ while $\operatorname{tr} A \operatorname{tr} B = 5 \cdot 2 = 10$.

Theorem. Let A be an $n \times n$ matrix, and suppose its characteristic polynomial factors as

$$\boxed{\det(A - xI) = (\lambda_1 - x)(\lambda_2 - x)\cdots(\lambda_n - x),}$$

so that $\lambda_1, \ldots, \lambda_n$ are the complex eigenvalues of A repeated with multiplicity. Then

$$\boxed{\det A = \lambda_1\lambda_2\cdots\lambda_n, \qquad \operatorname{tr} A = \lambda_1 + \lambda_2 + \cdots + \lambda_n.}$$

In words: the determinant of A is the product of its eigenvalues, and the trace of A is the

sum of its eigenvalues, in both cases repeated with multiplicity.

Remark. The theorem is easy for diagonalizable matrices: if $A = PDP^{-1}$ with D diagonal, then

$$\det A = \det P \det D \det P^{-1} = \det D = \lambda_1 \cdots \lambda_n,$$
$$\operatorname{tr} A = \operatorname{tr}(PDP^{-1}) = \operatorname{tr}(DP^{-1}P) = \operatorname{tr} D = \lambda_1 + \cdots + \lambda_n.$$

The general proof below works for every matrix.

Proof: Write $\det(A - xI) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$. By the lemma it suffices to prove

$$a_0 = \det A, \qquad a_{n-1} = (-1)^{n-1} \operatorname{tr} A.$$

The first claim is immediate: setting $x = 0$ in $\det(A - xI)$ gives $a_0 = \det A$.

For the second, use the permutation formula for the determinant,

$$\det(A - xI) = \sum_{Z \in S_n} (-1)^{\operatorname{inv}(Z)} \operatorname{prod}(Z, A - xI),$$

where $\operatorname{prod}(Z, A - xI)$ is the product of the entries of $A - xI$ in the nonzero positions of the permutation matrix Z. If Z is not the identity, then Z has at most $n-2$ nonzero entries on the diagonal, hence $\operatorname{prod}(Z, A - xI)$ involves at most $n-2$ diagonal entries $d_i - x$ and so has degree at most $n-2$. Therefore only $Z = I$ can contribute to the coefficient of x^{n-1}:

$$\det(A - xI) = (d_1 - x)(d_2 - x)\cdots(d_n - x) + (\text{terms of degree} \le n-2),$$

where $d_1, \ldots, d_n$ are the diagonal entries of A. By the lemma applied to this product, the coefficient of x^{n-1} is

$$(-1)^{n-1}(d_1 + \cdots + d_n) = (-1)^{n-1} \operatorname{tr} A,$$

which proves $a_{n-1} = (-1)^{n-1} \operatorname{tr} A$. Combining with the lemma gives both identities. ■

Example. Let $A = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & i \end{pmatrix}$. One checks that

$$A \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} = \begin{pmatrix} i \\ -1 \\ 0 \end{pmatrix} = i \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix}, \qquad A \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ i \end{pmatrix} = i \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix},$$
$$A \begin{pmatrix} i \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix} = -i \begin{pmatrix} i \\ 1 \\ 0 \end{pmatrix},$$

so the eigenvalues are $i, i, -i$. Consistently,

$$\det(A - xI) = -x^3 + ix^2 - x + i = (i - x)^2(-i - x),$$
$$\det A = i \cdot i \cdot (-i) = i, \qquad \operatorname{tr} A = i + i - i = i.$$

Example. For $A = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ the matrix is block diagonal with blocks i and 2, so

$$\det(A - xI) = (x^2 + 1)(2 - x),$$

with eigenvalues i, $-i$, 2. Then $\det A = i(-i)(2) = 2$ and $\operatorname{tr} A = i - i + 2 = 2$, as the theorem predicts.

Corollary. Let A be a 2×2 matrix, and put $p = \det A$ and $q = \operatorname{tr} A$. Then A has two distinct eigenvalues if and only if $q^2 \neq 4p$.

Proof: Let $a, b \in \mathbb{C}$ be the eigenvalues of A repeated with multiplicity. By the theorem, $ab = p$ and $a + b = q$, so $a(q - a) = qa - a^2 = p$, i.e. $a^2 - qa + p = 0$. The

quadratic formula gives

$$a = \frac{q \pm \sqrt{q^2 - 4p}}{2}, \qquad b = \frac{q \mp \sqrt{q^2 - 4p}}{2},$$

so $a \neq b$ if and only if $q^2 - 4p \neq 0$. ■

Fast sanity checks with tr and det

Whenever you compute eigenvalues, check the two identities immediately: the sum must equal the trace and the product must equal the determinant. For 2×2 matrices this is a complete test: the eigenvalues are the roots of $x^2 - (\operatorname{tr} A)x + \det A$, which are distinct exactly when $(\operatorname{tr} A)^2 \neq 4 \det A$. Also remember that tr is additive but not multiplicative.

Practice Problem #7.9 (classifying 2×2 real matrices by the discriminant)

Problem. Suppose $a, b, c, d \in \mathbb{R}$ and $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. (a) When does A have two distinct real eigenvalues (so A is diagonalizable over $\mathbb{R}$)? (b) When does A have two distinct complex non-real eigenvalues (so A is diagonalizable over $\mathbb{C}$)? (c) When does A have exactly one complex eigenvalue λ? In this case what is λ? (d) When is A not diagonalizable over $\mathbb{C}$? Give precise conditions in terms of a, b, c, d.

Solution. The characteristic polynomial and its discriminant are

$$\chi_A(t) = t^2 - (a+d)t + (ad - bc), \qquad \Delta = (a+d)^2 - 4(ad-bc) = (a-d)^2 + 4bc,$$

and the quadratic formula gives the eigenvalues $\lambda_\pm = \dfrac{(a+d) \pm \sqrt{\Delta}}{2}$.

Part (a). Two distinct real eigenvalues exactly when $(a-d)^2 + 4bc > 0$; eigenvectors for the two distinct eigenvalues give a real eigenvector basis, so A is diagonalizable over $\mathbb{R}$.

Part (b). Two distinct nonreal eigenvalues exactly when $(a-d)^2 + 4bc < 0$; they are the conjugate pair $\dfrac{a+d}{2} \pm \dfrac{i}{2}\sqrt{-\Delta}$. The distinct eigenvalues imply diagonalizability over $\mathbb{C}$, but not over $\mathbb{R}$.

Part (c). Exactly one distinct eigenvalue exactly when $(a-d)^2 + 4bc = 0$, and then $\lambda = \dfrac{a+d}{2}$ (real, with algebraic multiplicity 2).

Part (d). Only the repeated-eigenvalue case can fail: if the only eigenvalue is λ and A is diagonalizable, then its diagonal form must be λI, forcing $A = P(\lambda I)P^{-1} = \lambda I$; conversely a scalar matrix is already diagonal. Hence A is not diagonalizable over $\mathbb{C}$ exactly when

$$(a-d)^2 + 4bc = 0 \quad \text{and} \quad (a-d, b, c) \neq (0, 0, 0).$$

In particular, A is diagonalizable over $\mathbb{R}$ exactly when $\Delta > 0$, or when $a = d$ and $b = c = 0$.

8.14 Further Properties of Eigenvalues

A triangular matrix exposes its eigenvalues directly, and the trace/determinant identities become transparent for it:

$$A = \begin{pmatrix} \lambda_1 & a & b \\ 0 & \lambda_2 & c \\ 0 & 0 & \lambda_3 \end{pmatrix} \implies \det(A - xI) = (\lambda_1 - x)(\lambda_2 - x)(\lambda_3 - x),$$

$$\operatorname{tr} A = \lambda_1 + \lambda_2 + \lambda_3, \qquad \det A = \lambda_1 \lambda_2 \lambda_3.$$

Proposition. If A is a square matrix then A and $A^\top$ have the same eigenvalues (and indeed the same characteristic polynomial).

Proof: Using $\det B = \det(B^\top)$ and $(A - xI)^\top = A^\top - xI$,

$$\det(A - xI) = \det\left((A - xI)^\top\right) = \det(A^\top - xI).$$

■

Proposition. A square matrix A is invertible if and only if 0 is not an eigenvalue of A.

Proof: 0 is an eigenvalue of $A \iff \det(A - 0I) = \det A = 0 \iff A$ is not invertible. ■

Proposition. Suppose A is invertible. Then A and A^{-1} have the same eigenvectors, and $\vec{v}$ is an eigenvector of A with eigenvalue λ if and only if $\vec{v}$ is an eigenvector of A^{-1} with eigenvalue λ^{-1}.

Proof: If $A\vec{v} = \lambda\vec{v}$ with $\vec{v} \neq \vec{0}$, then $\lambda \neq 0$ (otherwise $A\vec{v} = \vec{0}$ with $\vec{v} \neq \vec{0}$, contradicting invertibility). Applying A^{-1},

$$\vec{v} = A^{-1}A\vec{v} = A^{-1}(\lambda\vec{v}) = \lambda A^{-1}\vec{v}, \qquad \text{so} \qquad A^{-1}\vec{v} = \lambda^{-1}\vec{v}.$$

The converse follows by applying the same argument to A^{-1}. ■

Corollary. If A is invertible and diagonalizable, then A^{-1} is diagonalizable.

Proof: $\mathbb{R}^n$ has a basis of eigenvectors of A; by the proposition these vectors are also eigenvectors of A^{-1}, so they form a basis of eigenvectors of A^{-1}. ■

Corollary. If A is diagonalizable then so is $A^\top$.

Proof: Suppose $A = PDP^{-1}$ with D diagonal. Let $Q = (P^{-1})^\top = (P^\top)^{-1}$. Since $D^\top = D$,

$$A^\top = (PDP^{-1})^\top = (P^{-1})^\top D^\top P^\top = QDQ^{-1},$$

which is a diagonalization of $A^\top$. ■

Eigenvalue transfer rules

- A and $A^\top$ share the characteristic polynomial, hence the eigenvalues, trace and determinant; eigenvectors need not be shared.
- 0 is an eigenvalue of $A \iff A$ is singular; so invertibility can be read off from the eigenvalues.
- For invertible A: eigenvalues of A^{-1} are the reciprocals λ^{-1}, with the same eigenvectors.
- Diagonalizability is preserved under transposition and under inversion (the latter only when A is invertible).

Practice Problem #7.5 (diagonalizable vs. triangular vs. diagonal)

Problem. Give an example of each of the following, or explain why no such example exists (all matrices square with real entries; "diagonalizable" means over $\mathbb{C}$): (a) a diagonalizable matrix that is not diagonal; (b) a diagonal matrix that is not diagonalizable; (c) a diagonalizable matrix that is not triangular; (d) a triangular matrix that is not diagonalizable.

Solution. (a) $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$: triangular with distinct diagonal entries $1 \neq 2$, hence two distinct eigenvalues, hence diagonalizable; it is not diagonal.
(b) No such example exists: a diagonal matrix D is already diagonalized as $D = IDI^{-1}$.

(c) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$: the eigenvalues are ± 1 (roots of $t^2 - 1$), which are distinct, so it is diagonalizable; it is neither upper nor lower triangular.
(d) $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$: upper triangular with the single eigenvalue 1, but $A - I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has rank 1, so the eigenspace is only 1-dimensional and no eigenvector basis exists.

Practice Problem #7.6 (invertible vs. triangular vs. diagonalizable)

Problem. Give an example of each of the following, or explain why no such example exists (all matrices square with real entries; "diagonalizable" means over $\mathbb{C}$): (a) an invertible matrix that is not triangular; (b) a triangular matrix that is not invertible; (c) an invertible matrix that is not diagonalizable; (d) a diagonalizable matrix that is not invertible.
Solution. (a) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$: not triangular, and invertible with determinant -1.
(b) $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$: diagonal (hence triangular), not invertible as its determinant is 0.
(c) $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$: determinant 1, hence invertible; as in #7.5(d) it is not diagonalizable.
(d) $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$: already diagonal, hence diagonalizable, but not invertible.

Practice Problem #7.10 (nilpotent matrices)

Problem. An $n \times n$ matrix A is *nilpotent* if $A^k = 0$ for some integer $k > 0$. First, give an example of a nilpotent matrix that is not strictly upper or lower triangular. Next, explain why if A is a nonzero $n \times n$ nilpotent matrix, then (a) the only eigenvalue of A is zero, but (b) the null space of A is not n-dimensional, and therefore (c) A is not diagonalizable.
Solution. An example is

$$N = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}, \qquad N^2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix};$$

both off-diagonal entries are nonzero, so N is neither upper nor lower triangular.
Now let $A \neq 0$ satisfy $A^k = 0$.
Part (a). If $A\vec{v} = \lambda\vec{v}$ for a nonzero (complex) vector $\vec{v}$, then $0 = A^k\vec{v} = \lambda^k\vec{v}$, forcing $\lambda = 0$. Moreover A is not invertible (otherwise A^k would be invertible, whereas it is zero), so $\operatorname{Nul} A \neq \{\vec{0}\}$ and 0 really is an eigenvalue — the only one.
Part (b). Since $A \neq 0$ its rank is at least 1, so rank-nullity gives $\dim \operatorname{Nul} A = n - \operatorname{rank} A \leq n - 1 < n$.
Part (c). All eigenvectors lie in $\operatorname{Nul} A$, which by (b) is too small to contain a basis of $\mathbb{C}^n$. Equivalently, a diagonal form of A could only have the eigenvalue 0 on its diagonal, forcing $A = P0P^{-1} = 0$ and contradicting $A \neq 0$. Hence no nonzero nilpotent matrix is diagonalizable, even over $\mathbb{C}$.

Practice Problem #7.14 (a on the diagonal, b everywhere else)

Problem. Suppose a and b are real numbers and A is the $n \times n$ matrix that has a in each diagonal position and b in all other positions. What are the eigenvalues of A? What are the dimensions of the corresponding eigenspaces? Is A diagonalizable?
Solution. Let $\vec{1} = (1 \ \cdots \ 1)^\top \in \mathbb{R}^n$ and $J = \vec{1}\vec{1}^\top$ the all-ones matrix. Then

$$A = (a - b)I + bJ, \qquad A\vec{x} = (a - b)\vec{x} + b\left(\sum_{i=1}^{n} x_i\right)\vec{1}.$$

In particular $A\vec{1} = (a + (n-1)b)\vec{1}$, and $A\vec{x} = (a-b)\vec{x}$ whenever $\sum_i x_i = 0$. Every vector decomposes uniquely as $\vec{x} = \frac{\sum_i x_i}{n}\vec{1} + \left(\vec{x} - \frac{\sum_i x_i}{n}\vec{1}\right)$, where the second summand has coordinate sum zero, so these two subspaces together give all of $\mathbb{R}^n$.
Case $n \geq 2$ and $b \neq 0$. The two eigenvalues are distinct (their difference is $nb \neq 0$), with

$$\lambda_1 = a + (n-1)b, \qquad E_{\lambda_1} = \mathbb{R}\text{-span}\{\vec{1}\}, \qquad \dim E_{\lambda_1} = 1;$$
$$\lambda_2 = a - b, \qquad E_{\lambda_2} = \left\{\vec{x} : \textstyle\sum_i x_i = 0\right\}, \qquad \dim E_{\lambda_2} = n - 1.$$

(The second space has basis $\vec{e}_1 - \vec{e}_n, \ldots, \vec{e}_{n-1} - \vec{e}_n$.)
Case $b = 0$. Then $A = aI$: one eigenvalue a with eigenspace $\mathbb{R}^n$ of dimension n.
Case $n = 1$. Then $A = (a)$ regardless of b: the only eigenvalue is a, with eigenspace of dimension 1.
In all cases A is diagonalizable over $\mathbb{R}$; for $n \geq 2$ an explicit choice is

$$P = \begin{pmatrix} \vec{1} & \vec{e}_1 - \vec{e}_n & \cdots & \vec{e}_{n-1} - \vec{e}_n \end{pmatrix}, \qquad D = \operatorname{diag}\big(a + (n-1)b,\ a - b,\ \ldots,\ a - b\big),$$

which remains valid when $b = 0$ (then $D = aI$); for $n = 1$ take $P = (1)$ and $D = (a)$.

Practice Problem #7.15 (when is $I + \vec{u}\vec{v}^\top$ diagonalizable?)

Problem. Suppose $\vec{u}, \vec{v} \in \mathbb{R}^n$ are both nonzero. Determine when $I_n + \vec{u}\vec{v}^\top$ is diagonalizable. Justify your answer.
Solution. The answer is:

$$I_n + \vec{u}\vec{v}^\top \text{ is diagonalizable } \iff \vec{v}^\top\vec{u} \neq 0,$$

and whenever it is diagonalizable, it is already diagonalizable over $\mathbb{R}$.
Step 1: two natural eigenspaces. Write $A = I + \vec{u}\vec{v}^\top$. If $\vec{x} \perp \vec{v}$ then $A\vec{x} = \vec{x}$, so the $(n-1)$-dimensional space $\vec{v}^\perp$ is an eigenspace with eigenvalue 1. Also $A\vec{u} = \vec{u} + \vec{u}(\vec{v}^\top\vec{u}) = (1 + \vec{v}^\top\vec{u})\vec{u}$.
Step 2: case $\vec{v}^\top\vec{u} \neq 0$. Then $\vec{u} \notin \vec{v}^\perp$, so $\vec{u}$ together with any basis of $\vec{v}^\perp$ gives n linearly independent real eigenvectors of A (with eigenvalues $1 + \vec{v}^\top\vec{u}$ and 1). Hence A is diagonalizable over $\mathbb{R}$.
Step 3: case $\vec{v}^\top\vec{u} = 0$. Then $N = \vec{u}\vec{v}^\top \neq 0$ is nilpotent: $N^2 = \vec{u}(\vec{v}^\top\vec{u})\vec{v}^\top = 0$. If $A\vec{x} = \lambda\vec{x}$ with $\vec{x} \neq \vec{0}$, then $N\vec{x} = (\lambda - 1)\vec{x}$, so $\lambda - 1$ is an eigenvalue of the nilpotent matrix N, whence $\lambda = 1$ by #7.10(a). So the only eigenvalue of A is 1; if A were diagonalizable, its diagonal form would have to be I itself, forcing $A = PIP^{-1} = I$, i.e. $\vec{u}\vec{v}^\top = 0$ — contradicting $\vec{u}, \vec{v} \neq \vec{0}$. Hence A is not diagonalizable in this case, not even over $\mathbb{C}$.

9 Orthogonality

9.1 Inner Products and Length

For this chapter all vectors lie in $\mathbb{R}^n$ (or $\mathbb{C}^n$ where stated) and all matrices have real (or complex) entries as indicated.

Definition. The inner product (or dot product) of $\vec{u}, \vec{v} \in \mathbb{R}^n$ is the scalar

$$\vec{u} \cdot \vec{v} = u_1v_1 + u_2v_2 + \cdots + u_nv_n = \vec{u}^\top\vec{v} = \vec{v}^\top\vec{u} = \vec{v} \cdot \vec{u}.$$

Example. $\begin{pmatrix}1\\2\\3\end{pmatrix} \cdot \begin{pmatrix}-1\\-10\\-100\end{pmatrix} = -1 - 20 - 300 = -321.$

Example. In $\mathbb{R}^2$, $\begin{pmatrix}a\\b\end{pmatrix} \cdot \begin{pmatrix}-b\\a\end{pmatrix} = -ab + ab = 0$ for all $a, b \in \mathbb{R}$.

Definition. The length of $\vec{v} \in \mathbb{R}^n$ is

$$\|\vec{v}\| = \sqrt{\vec{v} \cdot \vec{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2} \geq 0.$$

The distance between $\vec{u}$ and $\vec{v}$ is $\|\vec{u} - \vec{v}\|$.

Essential properties. For $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^n$ and $c \in \mathbb{R}$:

(a) $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$, $(\vec{u} + \vec{v}) \cdot \vec{w} = \vec{u} \cdot \vec{w} + \vec{v} \cdot \vec{w}$, $(c\vec{v}) \cdot \vec{w} = c(\vec{v} \cdot \vec{w})$, and $\|c\vec{v}\| = |c|\,\|\vec{v}\|$;

(b) $\vec{v} \cdot \vec{v} = v_1^2 + \cdots + v_n^2 \geq 0$ and $\|\vec{v}\| \geq 0$;

(c) $\vec{v} \cdot \vec{v} = 0 \iff \|\vec{v}\| = 0 \iff \vec{v} = \vec{0}$;

(d) $\vec{u} \cdot \vec{v} = \|\vec{u}\|\,\|\vec{v}\| \cos\theta$, where θ is the angle between $\vec{u}$ and $\vec{v}$ (this holds also when one vector is $\vec{0}$, both sides being 0, although then θ is undefined).

The last identity shows that $\vec{u} \cdot \vec{v} < 0$ exactly when $\vec{u}$ and $\vec{v}$ form an obtuse angle, and $\vec{u} \cdot \vec{v} > 0$ exactly when the angle is acute.

Definition. A unit vector is a vector $\vec{u}$ with $\|\vec{u}\| = 1$. If $\vec{v} \neq \vec{0}$ then the unit vector in the direction of $\vec{v}$ is

$$\vec{u} = \frac{1}{\|\vec{v}\|}\vec{v}, \qquad \text{since} \qquad \left\| \frac{1}{\|\vec{v}\|}\vec{v} \right\| = \frac{1}{\|\vec{v}\|}\|\vec{v}\| = 1.$$

Example. The unit vector in the direction of $\vec{v} = \begin{pmatrix}1\\1\\1\\1\end{pmatrix}$ is $\vec{u} = \frac{1}{\sqrt{4}}\vec{v} = \begin{pmatrix}1/2\\1/2\\1/2\\1/2\end{pmatrix}$.

Definition. Two vectors $\vec{u}, \vec{v} \in \mathbb{R}^n$ are orthogonal if $\vec{u} \cdot \vec{v} = 0$; we then write $\vec{u} \perp \vec{v}$. A list of vectors is orthogonal if any two distinct vectors in it are orthogonal.

Proposition. Suppose $\vec{u}, \vec{v} \in \mathbb{R}^2$ are nonzero and orthogonal. Then, drawn as arrows in the plane, they lie on perpendicular lines through the origin. Concretely, if $\vec{u} = \begin{pmatrix}a\\b\end{pmatrix} \neq \vec{0}$ then $\vec{v}$ is a scalar multiple of $\begin{pmatrix}-b\\a\end{pmatrix}$, which is $\vec{u}$ rotated by ninety degrees counterclockwise.

Proof: Write $\vec{u} = \begin{pmatrix}a\\b\end{pmatrix}$ and $\vec{v} = \begin{pmatrix}x\\y\end{pmatrix}$. Then $ax + by = 0$.

- If $a = 0$ then $b \neq 0$, so $y = 0$ and $\vec{v} = \begin{pmatrix}x\\0\end{pmatrix} = \frac{x}{b}\begin{pmatrix}0\\b\end{pmatrix}$.
- If $a \neq 0$ then $x = -\frac{by}{a}$, so $\vec{v} = \begin{pmatrix}-by/a\\y\end{pmatrix} = \frac{y}{a}\begin{pmatrix}-b\\a\end{pmatrix}$.

In both cases $\vec{v}$ is a multiple of $\begin{pmatrix}-b\\a\end{pmatrix}$. Finally

$$\begin{pmatrix}0 & -1\\1 & 0\end{pmatrix}\begin{pmatrix}a\\b\end{pmatrix} = \begin{pmatrix}-b\\a\end{pmatrix},$$

and the matrix on the left rotates the plane ninety degrees counterclockwise; this is precisely the matrix called i in the previous section. ■

Inner product toolkit

- $\vec{u} \cdot \vec{v} = \vec{u}^\top \vec{v}$ is a 1×1 matrix, i.e. a scalar; it is the sum of entrywise products.

- Length is $\|\vec{v}\| = \sqrt{\vec{v}\cdot\vec{v}}$, and $\|\vec{v}\|^2 = \vec{v}\cdot\vec{v}$ is often easier to compute.
- Orthogonality $\vec{u}\cdot\vec{v} = 0$ means "perpendicular" in $\mathbb{R}^2$ and $\mathbb{R}^3$, and it makes sense in every dimension.
- Normalizing a nonzero vector means dividing it by its length; the result is a unit vector with the same direction.

9.2 Orthogonal Vectors and Orthogonal Complements

Definition. Let $V \subseteq \mathbb{R}^n$ be a subspace. The orthogonal complement of V is

$$V^\perp = \{\vec{w} \in \mathbb{R}^n : \vec{v}\cdot\vec{w} = 0 \text{ for all } \vec{v} \in V\}.$$

Proposition. If $V \subseteq \mathbb{R}^n$ is a subspace then $V^\perp$ is also a subspace.

Proof: Since $\vec{v}\cdot\vec{0} = 0$ for all $\vec{v}$, we have $\vec{0} \in V^\perp$, so $V^\perp$ is nonempty. If $\vec{x}, \vec{y} \in V^\perp$ and $c \in \mathbb{R}$, then for every $\vec{v} \in V$,

$$\vec{v}\cdot(c\vec{x}) = c(\vec{v}\cdot\vec{x}) = 0, \qquad \vec{v}\cdot(\vec{x}+\vec{y}) = \vec{v}\cdot\vec{x} + \vec{v}\cdot\vec{y} = 0,$$

so $c\vec{x}$ and $\vec{x}+\vec{y}$ lie in $V^\perp$. Hence $V^\perp$ is closed under addition and scalar multiplication. ■

Theorem. If A is an $m \times n$ matrix then

$$\boxed{(\operatorname{Col} A)^\perp = \operatorname{Nul}(A^\top) \subseteq \mathbb{R}^m.}$$

Proof: Write $A = \begin{pmatrix}\vec{a}_1 & \vec{a}_2 & \cdots & \vec{a}_n\end{pmatrix}$ with $\vec{a}_i \in \mathbb{R}^m$, and let $\vec{v} \in \mathbb{R}^m$. If $\vec{v} \in (\operatorname{Col} A)^\perp$ then $\vec{v}\cdot\vec{a}_i = \vec{a}_i^\top\vec{v} = 0$ for every i. Conversely, if $\vec{v}\cdot\vec{a}_i = 0$ for all i, then for any scalars $c_1, \ldots, c_n$,

$$(c_1\vec{a}_1 + \cdots + c_n\vec{a}_n)\cdot\vec{v} = c_1(\vec{a}_1\cdot\vec{v}) + \cdots + c_n(\vec{a}_n\cdot\vec{v}) = 0,$$

so $\vec{v}$ is orthogonal to every element of $\operatorname{Col} A$. Thus

$$\vec{v} \in (\operatorname{Col} A)^\perp \iff \vec{a}_i^\top\vec{v} = 0 \text{ for all } i \iff A^\top\vec{v} = \begin{pmatrix}\vec{a}_1^\top \\ \vdots \\ \vec{a}_n^\top\end{pmatrix}\vec{v} = \begin{pmatrix}\vec{a}_1\cdot\vec{v} \\ \vdots \\ \vec{a}_n\cdot\vec{v}\end{pmatrix} = \vec{0},$$

which says exactly that $\vec{v} \in \operatorname{Nul}(A^\top)$. ■

Example. Let $V = \operatorname{span}\{\vec{e}_1, \ldots, \vec{e}_i\} \subseteq \mathbb{R}^n$. Then $V^\perp = \operatorname{span}\{\vec{e}_{i+1}, \ldots, \vec{e}_n\}$. In particular $\{\vec{0}\}^\perp = \mathbb{R}^n$ and $(\mathbb{R}^n)^\perp = \{\vec{0}\}$.

Lemma. If $V \subseteq \mathbb{R}^n$ is a subspace and $\vec{w} \in V \cap V^\perp$ then $\vec{w} = \vec{0}$.

Proof: If $\vec{w} \in V$ and $\vec{w} \in V^\perp$ then $\vec{w}\cdot\vec{w} = 0$, hence $\vec{w} = \vec{0}$. ■

Proposition. Let $V \subseteq \mathbb{R}^n$ be a subspace. If $S \subseteq V$ and $T \subseteq V^\perp$ are sets of linearly independent vectors, then $S \cup T$ is linearly independent.

Proof: Suppose a nontrivial linear combination of the elements of $S \cup T$ is $\vec{0}$. Move the terms from S to the left and the terms from T to the right; this gives

$$\underbrace{a_1\vec{v}_1 + \cdots + a_k\vec{v}_k}_{\in V} = \underbrace{b_1\vec{w}_1 + \cdots + b_l\vec{w}_l}_{\in V^\perp}$$

with $\vec{v}_j \in S$, $\vec{w}_j \in T$ and not all coefficients a_j, b_j zero. The common value belongs to $V \cap V^\perp = \{\vec{0}\}$, so each side is $\vec{0}$. But S and T are each independent, so all a_j and all b_j vanish — a contradiction. ■

Corollary. If $V \subseteq \mathbb{R}^n$ is a subspace then $\dim V^\perp \leq n - \dim V$.

Proof: Let S be a basis of V and T a basis of $V^\perp$. Then $S \cup T$ is independent, so

$$\dim V + \dim V^\perp = |S| + |T| = |S \cup T| \leq n.$$

■

Proposition (Generalized Pythagorean theorem). Two vectors $\vec{u}, \vec{v} \in \mathbb{R}^n$ are orthogonal if and only if $\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2$.

Proof: Expanding,

$$\|\vec{u} + \vec{v}\|^2 = (\vec{u} + \vec{v}) \cdot (\vec{u} + \vec{v}) = \|\vec{u}\|^2 + 2(\vec{u} \cdot \vec{v}) + \|\vec{v}\|^2,$$

so $\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2$ if and only if $\vec{u} \cdot \vec{v} = 0$. ■

9.3 Orthogonal Bases

Definition. A basis $\{\vec{u}_1, \ldots, \vec{u}_p\}$ of a subspace V is an orthogonal basis if $\vec{u}_i \cdot \vec{u}_j = 0$ whenever $i \neq j$. It is an orthonormal basis if in addition each $\vec{u}_i$ is a unit vector.

Example. The standard basis $\vec{e}_1, \ldots, \vec{e}_n$ is an orthonormal basis of $\mathbb{R}^n$.

Theorem. Suppose $\vec{u}_1, \ldots, \vec{u}_p \in \mathbb{R}^n$ are orthogonal and all nonzero. Then they are linearly independent (hence an orthogonal basis of the subspace they span).

Proof: Suppose $c_1\vec{u}_1 + \cdots + c_p\vec{u}_p = \vec{0}$. Taking the inner product with $\vec{u}_i$ and using $\vec{u}_j \cdot \vec{u}_i = 0$ for $j \neq i$,

$$0 = (c_1\vec{u}_1 + \cdots + c_p\vec{u}_p) \cdot \vec{u}_i = c_i(\vec{u}_i \cdot \vec{u}_i) = c_i\|\vec{u}_i\|^2.$$

Since $\vec{u}_i \neq \vec{0}$, we have $\|\vec{u}_i\|^2 \neq 0$, so $c_i = 0$. This holds for every i, hence the vectors are independent. ■

Second proof: Let $A = (\vec{u}_1 \ \cdots \ \vec{u}_p)$ and let $d_i = \vec{u}_i \cdot \vec{u}_i > 0$. Then $A^\top A = D = \operatorname{diag}(d_1, \ldots, d_p)$ is invertible. If $A\vec{x} = \vec{0}$ then $A^\top A\vec{x} = \vec{0}$, hence $D\vec{x} = \vec{0}$, hence $\vec{x} = \vec{0}$. So the columns of A are independent. ■

Proposition. Suppose $\vec{u}_1, \ldots, \vec{u}_p$ is an orthogonal basis for a subspace $V \subseteq \mathbb{R}^n$ and $\vec{y} \in V$. Then

$$\boxed{\vec{y} = c_1\vec{u}_1 + c_2\vec{u}_2 + \cdots + c_p\vec{u}_p, \qquad \text{where} \qquad c_i = \frac{\vec{y} \cdot \vec{u}_i}{\vec{u}_i \cdot \vec{u}_i} = \frac{\vec{y} \cdot \vec{u}_i}{\|\vec{u}_i\|^2}.}$$

Proof: Since the $\vec{u}_i$ form a basis, $\vec{y} = c_1\vec{u}_1 + \cdots + c_p\vec{u}_p$ for some coefficients. Taking the inner product with $\vec{u}_i$ and using orthogonality gives $\vec{y} \cdot \vec{u}_i = c_i(\vec{u}_i \cdot \vec{u}_i)$, and $\vec{u}_i \cdot \vec{u}_i \neq 0$ because $\vec{u}_i$ is a basis vector. ■

This is the main computational advantage of an orthogonal basis: to expand a vector we compute inner products, instead of solving a linear system.

Example. The vectors

$$\vec{u}_1 = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}, \qquad \vec{u}_2 = \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}, \qquad \vec{u}_3 = \begin{pmatrix} -1/2 \\ -2 \\ 7/2 \end{pmatrix}$$

are pairwise orthogonal: $\vec{u}_1 \cdot \vec{u}_2 = -3 + 2 + 1 = 0$, $\vec{u}_1 \cdot \vec{u}_3 = -3/2 - 2 + 7/2 = 0$, and $\vec{u}_2 \cdot \vec{u}_3 = 1/2 - 4 + 7/2 = 0$. They are nonzero, hence an orthogonal basis of $\mathbb{R}^3$. For $\vec{y} = \begin{pmatrix} 6 \\ 1 \\ -8 \end{pmatrix}$ we compute

$$\vec{y} \cdot \vec{u}_1 = 18 + 1 - 8 = 11, \quad \vec{y} \cdot \vec{u}_2 = -6 + 2 - 8 = -12, \quad \vec{y} \cdot \vec{u}_3 = -3 - 2 - 28 = -33,$$

$$\vec{u}_1 \cdot \vec{u}_1 = 11, \qquad \vec{u}_2 \cdot \vec{u}_2 = 6, \qquad \vec{u}_3 \cdot \vec{u}_3 = \frac{1}{4} + 4 + \frac{49}{4} = \frac{33}{2},$$

so

$$c_1 = \frac{11}{11} = 1, \qquad c_2 = \frac{-12}{6} = -2, \qquad c_3 = \frac{-33}{33/2} = -2, \qquad \vec{y} = \vec{u}_1 - 2\vec{u}_2 - 2\vec{u}_3.$$

Indeed $\vec{u}_1 - 2\vec{u}_2 - 2\vec{u}_3 = \begin{pmatrix} 3+2+1 \\ 1-4+4 \\ 1-2-7 \end{pmatrix} = \begin{pmatrix} 6 \\ 1 \\ -8 \end{pmatrix}$.

Orthogonal basis = easy coordinates

- Nonzero orthogonal vectors are automatically independent; no row reduction is needed.
- Coefficients in an orthogonal basis are computed one at a time by $c_i = \dfrac{\vec{y} \cdot \vec{u}_i}{\vec{u}_i \cdot \vec{u}_i}$, with no coupling between the equations.
- With an orthonormal basis the denominators disappear: $c_i = \vec{y} \cdot \vec{u}_i$.

9.4 Orthonormal Vectors and Orthogonal Matrices

Definition. A list of vectors $\vec{u}_1, \ldots, \vec{u}_p$ is orthonormal if

$$\vec{u}_i \cdot \vec{u}_j = 0 \ (i \neq j), \qquad \vec{u}_i \cdot \vec{u}_i = 1 \text{ for all } i.$$

An orthonormal basis of a subspace is a basis that is orthonormal. A square matrix with orthonormal columns is called an orthogonal matrix (a confusing convention: "orthonormal matrix" would be the accurate name).

Theorem. Let U be an $m \times n$ matrix. Its columns are orthonormal if and only if $U^\top U = I_n$. If U is square, this holds if and only if $U^\top = U^{-1}$; in that case also $UU^\top = I_n$, so the rows of U are orthonormal as well.

Proof: Write $U = (\vec{u}_1 \ \cdots \ \vec{u}_n)$ with $\vec{u}_i \in \mathbb{R}^m$. The entry in position (i, j) of $U^\top U$ is $\vec{u}_i^\top \vec{u}_j = \vec{u}_i \cdot \vec{u}_j$. Hence $U^\top U = I_n$ exactly when all diagonal entries $\vec{u}_i \cdot \vec{u}_i$ equal 1 and all off-diagonal entries vanish, i.e. exactly when the columns are orthonormal. For square U, $U^\top U = I_n$ means $U^\top$ is a left inverse of U, hence $U^{-1} = U^\top$ and also $UU^\top = I_n$. ■

Corollary. If U is an orthogonal matrix then $\det U \in \{-1, 1\}$.

Proof: $\det(U)^2 = \det(U^\top)\det(U) = \det(U^\top U) = \det I_n = 1$. ■

Theorem. Let U be an $m \times n$ matrix with orthonormal columns and let $\vec{x}, \vec{y} \in \mathbb{R}^n$. Then

$$\boxed{\|U\vec{x}\| = \|\vec{x}\|, \qquad (U\vec{x}) \cdot (U\vec{y}) = \vec{x} \cdot \vec{y}, \qquad (U\vec{x}) \cdot (U\vec{y}) = 0 \iff \vec{x} \cdot \vec{y} = 0.}$$

Proof: The second identity is

$$(U\vec{x}) \cdot (U\vec{y}) = (U\vec{x})^\top (U\vec{y}) = \vec{x}^\top U^\top U \vec{y} = \vec{x}^\top \vec{y} = \vec{x} \cdot \vec{y},$$

and the first and third are special cases of it (take $\vec{y} = \vec{x}$, respectively note that 0 is preserved). ■

So multiplication by a matrix with orthonormal columns preserves lengths, distances, and angles. In particular, an orthogonal matrix is a rigid motion of $\mathbb{R}^n$.

Example. The vectors

$$\frac{1}{\sqrt{11}} \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}, \qquad \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}, \qquad \frac{1}{\sqrt{66}} \begin{pmatrix} -1 \\ -4 \\ 7 \end{pmatrix}$$

form an orthonormal basis of $\mathbb{R}^3$: they are the mutually orthogonal vectors found earlier, each divided by its length ($\sqrt{11}$, $\sqrt{6}$, $\sqrt{66}$).

Example. For every angle θ the rotation matrix $R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ is orthogonal, since its columns are unit vectors and perpendicular.

Orthonormal columns in practice

- $U^\top U = I_n$ is the single test for orthonormal columns; when U is square this is the same as $U^{-1} = U^\top$.
- Orthogonal matrices preserve all lengths and inner products, hence all angles and distances.
- $\det U = \pm 1$: the volume of the unit cube is unchanged, and $|\det U| = 1$ detects that no stretching occurred.
- For square orthogonal U, both the columns and the rows are orthonormal.

9.5 Orthogonal Projection onto a Line

Let $L \subseteq \mathbb{R}^n$ be a one-dimensional subspace, so $L = \text{span}\{\vec{u}\}$ for any nonzero $\vec{u} \in L$, and let $\vec{y} \in \mathbb{R}^n$.

Definition. The orthogonal projection of $\vec{y}$ onto L is

$$\text{proj}_L(\vec{y}) = \frac{\vec{y}\cdot\vec{u}}{\vec{u}\cdot\vec{u}}\vec{u} \qquad \text{for any } \vec{u} \neq \vec{0} \text{ in } L.$$

The value does not depend on the choice of $\vec{u}$: if $\vec{u}' = c\vec{u}$ with $c \neq 0$ then

$$\frac{\vec{y}\cdot\vec{u}'}{\vec{u}'\cdot\vec{u}'}\vec{u}' = \frac{c(\vec{y}\cdot\vec{u})}{c^2(\vec{u}\cdot\vec{u})}c\vec{u} = \frac{\vec{y}\cdot\vec{u}}{\vec{u}\cdot\vec{u}}\vec{u}.$$

The vector $\vec{z} = \vec{y} - \text{proj}_L(\vec{y})$ is the component of $\vec{y}$ orthogonal to L.

Proposition. $\text{proj}_L(\vec{y})$ is the unique vector $\hat{\vec{y}} \in L$ such that $\vec{y} - \hat{\vec{y}} \in L^\perp$.

Proof: Write $\vec{u} \neq \vec{0}$ in L and $\hat{\vec{y}} = \text{proj}_L(\vec{y})$. Then

$$\left(\vec{y} - \frac{\vec{y}\cdot\vec{u}}{\vec{u}\cdot\vec{u}}\vec{u}\right)\cdot\vec{u} = \vec{y}\cdot\vec{u} - \frac{\vec{y}\cdot\vec{u}}{\vec{u}\cdot\vec{u}}(\vec{u}\cdot\vec{u}) = 0,$$

so $\vec{y} - \hat{\vec{y}} \perp \vec{u}$ and hence $\vec{y} - \hat{\vec{y}} \in L^\perp$; clearly $\hat{\vec{y}} \in L$. For uniqueness, suppose $\tilde{\vec{y}} \in L$ also satisfies $\vec{y} - \tilde{\vec{y}} \in L^\perp$. Write $\tilde{\vec{y}} = c\vec{u}$. Then

$$0 = (\vec{y} - c\vec{u})\cdot\vec{u} = \vec{y}\cdot\vec{u} - c(\vec{u}\cdot\vec{u}), \qquad \text{so} \qquad c = \frac{\vec{y}\cdot\vec{u}}{\vec{u}\cdot\vec{u}},$$

i.e. $\tilde{\vec{y}} = \hat{\vec{y}}$. ■

Example. Let $\vec{y} = \begin{pmatrix} 7 \\ 6 \end{pmatrix}$ and $L = \text{span}\left\{\begin{pmatrix} 4 \\ 2 \end{pmatrix}\right\}$. Then

$$\text{proj}_L(\vec{y}) = \frac{28+12}{16+4}\begin{pmatrix} 4 \\ 2 \end{pmatrix} = \frac{40}{20}\begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 8 \\ 4 \end{pmatrix}.$$

Example (distance from a point to a line). The distance from $(7, 6)$ to the line L with equation $y = \frac{1}{2}x$ is

$$\left\|\begin{pmatrix} 7 \\ 6 \end{pmatrix} - \text{proj}_L\begin{pmatrix} 7 \\ 6 \end{pmatrix}\right\| = \left\|\begin{pmatrix} 7 \\ 6 \end{pmatrix} - \begin{pmatrix} 8 \\ 4 \end{pmatrix}\right\| = \left\|\begin{pmatrix} -1 \\ 2 \end{pmatrix}\right\| = \sqrt{5}.$$

In general the distance from a point $\vec{x}$ to L is $\|\vec{x} - \text{proj}_L(\vec{x})\|$.

9.6 Orthogonal Projections onto Subspaces

The line case generalizes to arbitrary subspaces.

Theorem. Let $W \subseteq \mathbb{R}^n$ be any subspace and $\vec{y} \in \mathbb{R}^n$. Then there exist unique vectors $\hat{\vec{y}} \in W$ and $\vec{z} \in W^\perp$ with

$$\boxed{\vec{y} = \hat{\vec{y}} + \vec{z}.}$$

If $\vec{u}_1, \ldots, \vec{u}_p$ is an orthogonal basis of W, then

$$\boxed{\hat{\vec{y}} = \frac{\vec{y}\cdot\vec{u}_1}{\vec{u}_1\cdot\vec{u}_1}\vec{u}_1 + \frac{\vec{y}\cdot\vec{u}_2}{\vec{u}_2\cdot\vec{u}_2}\vec{u}_2 + \cdots + \frac{\vec{y}\cdot\vec{u}_p}{\vec{u}_p\cdot\vec{u}_p}\vec{u}_p, \qquad \vec{z} = \vec{y} - \hat{\vec{y}}.}$$

The result does not depend on which orthogonal basis of W is chosen.

Proof: We may assume W has an orthogonal basis; this is proved below via the Gram–Schmidt process. Choose one, say $\vec{u}_1, \ldots, \vec{u}_p$, and define $\hat{\vec{y}}$ by the displayed formula. Then $\hat{\vec{y}} \in W$, and for each i,

$$(\vec{y} - \hat{\vec{y}}) \cdot \vec{u}_i = \vec{y}\cdot\vec{u}_i - \frac{\vec{y}\cdot\vec{u}_i}{\vec{u}_i\cdot\vec{u}_i}(\vec{u}_i\cdot\vec{u}_i) = 0,$$

so $\vec{z} = \vec{y} - \hat{\vec{y}} \perp \vec{u}_i$ for all i; since the $\vec{u}_i$ span W, this gives $\vec{z} \in W^\perp$. For uniqueness, suppose also $\vec{y} = \vec{u} + \vec{v}$ with $\vec{u} \in W$ and $\vec{v} \in W^\perp$. Then

$$\vec{u} - \hat{\vec{y}} = \vec{z} - \vec{v} \in W \cap W^\perp = \{\vec{0}\},$$

so $\vec{u} = \hat{\vec{y}}$ and $\vec{v} = \vec{z}$. ■

Definition. The vector $\hat{\vec{y}}$ above is the orthogonal projection of $\vec{y}$ onto W, written $\mathrm{proj}_W(\vec{y})$.

Corollary. If $W \subseteq \mathbb{R}^n$ is any subspace then $\dim W^\perp = n - \dim W$, and consequently $W \oplus W^\perp = \mathbb{R}^n$.

Proof: The theorem shows that W and $W^\perp$ together span $\mathbb{R}^n$: every $\vec{y}$ is $\hat{\vec{y}} + \vec{z}$ with one summand in W and the other in $W^\perp$. Hence the union of a basis of W and a basis of $W^\perp$ spans $\mathbb{R}^n$, so $n \le \dim W + \dim W^\perp$. Combined with the earlier inequality $\dim W^\perp \le n - \dim W$, this gives equality. Uniqueness of the decomposition plus $W \cap W^\perp = \{\vec{0}\}$ gives $W \oplus W^\perp = \mathbb{R}^n$. ■

Facts. Let $W \subseteq \mathbb{R}^n$ be a subspace and $\vec{y} \in \mathbb{R}^n$.

- $\vec{y} \in W \iff \mathrm{proj}_W(\vec{y}) = \vec{y}$; $\quad \vec{y} \in W^\perp \iff \mathrm{proj}_W(\vec{y}) = \vec{0}$.

Proposition. If $\vec{v} \in W$ and $\vec{v} \neq \mathrm{proj}_W(\vec{y})$, then

$$\boxed{\|\vec{y} - \mathrm{proj}_W(\vec{y})\| < \|\vec{y} - \vec{v}\|.}$$

In words: $\mathrm{proj}_W(\vec{y})$ is the vector in W closest to $\vec{y}$.

Proof: Let $\hat{\vec{y}} = \mathrm{proj}_W(\vec{y})$. Then

$$\vec{y} - \vec{v} = (\vec{y} - \hat{\vec{y}}) + (\hat{\vec{y}} - \vec{v}),$$

where the first term lies in $W^\perp$ and the second in W. These are orthogonal, so by the Pythagorean theorem

$$\|\vec{y} - \vec{v}\|^2 = \|\vec{y} - \hat{\vec{y}}\|^2 + \|\hat{\vec{y}} - \vec{v}\|^2 > \|\vec{y} - \hat{\vec{y}}\|^2$$

because $\|\hat{\vec{y}} - \vec{v}\| > 0$. ■

Computation via a matrix. Suppose $\vec{u}_1, \ldots, \vec{u}_p$ is an orthonormal basis of W and

$U = (\vec{u}_1 \ \cdots \ \vec{u}_p)$ (an $n \times p$ matrix with orthonormal columns). Then

$$\operatorname{proj}_W(\vec{y}) = (\vec{y}\cdot\vec{u}_1)\vec{u}_1 + \cdots + (\vec{y}\cdot\vec{u}_p)\vec{u}_p = UU^\top\vec{y}.$$

The projection matrix $P = UU^\top$

For an $n \times p$ matrix U with orthonormal columns, $P = UU^\top$ satisfies

- $P^2 = UU^\top UU^\top = UI_pU^\top = P$ (idempotent: projecting twice changes nothing);
- $P^\top = P$ (symmetric);
- $\operatorname{Col} P = W$ and $\operatorname{rank} P = p = \dim W$;
- $I_n - P$ is the projection onto $W^\perp$, so $\vec{y} = P\vec{y} + (I_n - P)\vec{y}$ is the decomposition $\hat{\vec{y}} + \vec{z}$.

9.7 The Gram–Schmidt Process

Theorem. Every nonzero subspace $W \subseteq \mathbb{R}^n$ has an orthogonal basis, and hence also an orthonormal basis.

Gram–Schmidt process. Suppose $\vec{x}_1, \ldots, \vec{x}_p$ is any basis of W. Define vectors $\vec{v}_1, \ldots, \vec{v}_p$ inductively by

$$\begin{aligned}
\vec{v}_1 &= \vec{x}_1,\\
\vec{v}_2 &= \vec{x}_2 - \frac{\vec{x}_2\cdot\vec{v}_1}{\vec{v}_1\cdot\vec{v}_1}\vec{v}_1,\\
\vec{v}_3 &= \vec{x}_3 - \frac{\vec{x}_3\cdot\vec{v}_1}{\vec{v}_1\cdot\vec{v}_1}\vec{v}_1 - \frac{\vec{x}_3\cdot\vec{v}_2}{\vec{v}_2\cdot\vec{v}_2}\vec{v}_2,\\
&\vdots\\
\vec{v}_i &= \vec{x}_i - \sum_{j=1}^{i-1}\frac{\vec{x}_i\cdot\vec{v}_j}{\vec{v}_j\cdot\vec{v}_j}\vec{v}_j, \qquad i = 2, \ldots, p.
\end{aligned}$$

Then $\vec{v}_1, \ldots, \vec{v}_p$ form an orthogonal basis of W.

Proof: Let $W_i = \operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_i\}$. We show by induction on i that $\vec{v}_1, \ldots, \vec{v}_i$ is an orthogonal basis of W_i and that $\vec{v}_{i+1} = \vec{x}_{i+1} - \operatorname{proj}_{W_i}(\vec{x}_{i+1})$.

For $i = 1$, the single nonzero vector $\vec{v}_1 = \vec{x}_1$ is an orthogonal basis of $W_1 = \operatorname{span}\{\vec{x}_1\}$. Now assume the claim for i. The formula for $\vec{v}_{i+1}$ is exactly $\vec{x}_{i+1} - \operatorname{proj}_{W_i}(\vec{x}_{i+1})$ by the projection formula applied to the orthogonal basis $\vec{v}_1, \ldots, \vec{v}_i$. Hence $\vec{v}_{i+1} \in W_i^\perp$ by the theorem on projections, so $\vec{v}_{i+1}$ is orthogonal to each of $\vec{v}_1, \ldots, \vec{v}_i$. Moreover $\vec{v}_{i+1} \neq \vec{0}$: otherwise $\vec{x}_{i+1} = \operatorname{proj}_{W_i}(\vec{x}_{i+1}) \in W_i$, which would make $\vec{x}_1, \ldots, \vec{x}_{i+1}$ dependent. Therefore $\vec{v}_1, \ldots, \vec{v}_{i+1}$ is an orthogonal basis of W_{i+1}. By induction the list $\vec{v}_1, \ldots, \vec{v}_p$ is an orthogonal basis of $W_p = W$. ∎

Example. Let $\vec{x}_1 = \begin{pmatrix}1\\1\\1\\1\end{pmatrix}$, $\vec{x}_2 = \begin{pmatrix}0\\1\\1\\1\end{pmatrix}$, $\vec{x}_3 = \begin{pmatrix}0\\0\\1\\1\end{pmatrix}$; these are independent, hence a basis of some $W \subseteq \mathbb{R}^4$.

$$\vec{v}_1 = \vec{x}_1 = \begin{pmatrix}1\\1\\1\\1\end{pmatrix}, \qquad \vec{x}_2\cdot\vec{v}_1 = 3, \quad \vec{v}_1\cdot\vec{v}_1 = 4,$$

$$\vec{v}_2 = \vec{x}_2 - \frac{3}{4}\vec{v}_1 = \begin{pmatrix}0\\1\\1\\1\end{pmatrix} - \begin{pmatrix}3/4\\3/4\\3/4\\3/4\end{pmatrix} = \begin{pmatrix}-3/4\\1/4\\1/4\\1/4\end{pmatrix},$$

$$\vec{x}_3 \cdot \vec{v}_1 = 2, \qquad \vec{x}_3 \cdot \vec{v}_2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}, \qquad \vec{v}_2 \cdot \vec{v}_2 = \frac{9}{16} + \frac{3}{16} = \frac{3}{4},$$

$$\vec{v}_3 = \vec{x}_3 - \frac{2}{4}\vec{v}_1 - \frac{1/2}{3/4}\vec{v}_2 = \begin{pmatrix}0\\0\\1\\1\end{pmatrix} - \begin{pmatrix}1/2\\1/2\\1/2\\1/2\end{pmatrix} - \frac{2}{3}\begin{pmatrix}-3/4\\1/4\\1/4\\1/4\end{pmatrix} = \begin{pmatrix}0\\-2/3\\1/3\\1/3\end{pmatrix}.$$

One checks directly that $\vec{v}_1, \vec{v}_2, \vec{v}_3$ are pairwise orthogonal.

Example. Let $W = \operatorname{Nul}(1 \quad 1 \quad 1 \quad 1) = \{\vec{w} \in \mathbb{R}^4 : w_1 + w_2 + w_3 + w_4 = 0\}$, with basis

$$\vec{x}_1 = \begin{pmatrix}1\\-1\\0\\0\end{pmatrix}, \qquad \vec{x}_2 = \begin{pmatrix}1\\0\\-1\\0\end{pmatrix}, \qquad \vec{x}_3 = \begin{pmatrix}1\\0\\0\\-1\end{pmatrix}.$$

$$\vec{v}_1 = \vec{x}_1, \qquad \vec{x}_2 \cdot \vec{v}_1 = 1, \quad \vec{v}_1 \cdot \vec{v}_1 = 2, \qquad \vec{v}_2 = \vec{x}_2 - \frac{1}{2}\vec{v}_1 = \begin{pmatrix}1/2\\1/2\\-1\\0\end{pmatrix},$$

$$\vec{x}_3 \cdot \vec{v}_1 = 1, \qquad \vec{x}_3 \cdot \vec{v}_2 = \frac{1}{2}, \qquad \vec{v}_2 \cdot \vec{v}_2 = \frac{1}{4} + \frac{1}{4} + 1 = \frac{3}{2},$$

$$\vec{v}_3 = \vec{x}_3 - \frac{1}{2}\vec{v}_1 - \frac{1/2}{3/2}\vec{v}_2 = \begin{pmatrix}1\\0\\0\\-1\end{pmatrix} - \begin{pmatrix}1/2\\-1/2\\0\\0\end{pmatrix} - \begin{pmatrix}1/6\\1/6\\-1/3\\0\end{pmatrix} = \begin{pmatrix}1/3\\1/3\\1/3\\-1\end{pmatrix}.$$

Multiplying by 3 and 2 respectively, the rescaled vectors $\begin{pmatrix}1\\-1\\0\\0\end{pmatrix}, \begin{pmatrix}1\\1\\-2\\0\end{pmatrix}, \begin{pmatrix}1\\1\\1\\-3\end{pmatrix}$ are also an orthogonal basis of W.

Remark. To get an orthonormal basis, first run Gram–Schmidt to obtain $\vec{v}_1, \ldots, \vec{v}_p$, then replace each $\vec{v}_i$ by $\vec{u}_i = \dfrac{1}{\|\vec{v}_i\|}\vec{v}_i$.

Gram–Schmidt in four steps

- Step 1: $\vec{v}_1 = \vec{x}_1$ (keep the first vector).
- Step 2: subtract from each $\vec{x}_i$ its projection onto the span of the earlier $\vec{v}_j$'s, using $\dfrac{\vec{x}_i \cdot \vec{v}_j}{\vec{v}_j \cdot \vec{v}_j}\vec{v}_j$.
- Step 3: the resulting list is orthogonal, spans the same subspace, and stays a basis (nonzero vectors at every stage).
- Step 4: divide each $\vec{v}_i$ by $\|\vec{v}_i\|$ to obtain an orthonormal basis.

9.8 Least-Squares Problems

Many linear systems that arise in applications are overdetermined (more equations than unknowns, so A has more rows than columns) and often inconsistent: $\vec{b}$ may consist of noisy measurements, and the linear model encoded in the rows of A may be only an approximation. When no exact solution exists, the next best thing is a vector $\vec{x}$ making $A\vec{x}$ as close to $\vec{b} \in \mathbb{R}^m$ as possible in the Euclidean distance $\|\vec{b} - A\vec{x}\|$.

Definition. If A is an $m \times n$ matrix and $\vec{b} \in \mathbb{R}^m$, a least-squares solution to $A\vec{x} = \vec{b}$ is a vector $\vec{s} \in \mathbb{R}^n$ such that

$$\|\vec{b} - A\vec{s}\| \le \|\vec{b} - A\vec{x}\| \qquad \text{for all } \vec{x} \in \mathbb{R}^n,$$

i.e. $\vec{s}$ minimizes $\|\vec{b}-A\vec{s}\|$. (Minimizing $\|\vec{b}-A\vec{s}\|$ is the same as minimizing its square, the sum of the squares of the entries of $\vec{b}-A\vec{s}$, which explains the name.)

Key points:

- If $A\vec{x}=\vec{b}$ is consistent, every least-squares solution is an exact solution.
- There may be more than one least-squares solution.
- Unlike exact solutions, a least-squares solution always exists.

Lemma. The least-squares solutions to $A\vec{x}=\vec{b}$ are precisely the vectors $\vec{s}$ with

$$\boxed{A\vec{s}=\text{proj}_{\text{Col}\,A}(\vec{b}).}$$

Proof: Since $A\vec{s}\in\text{Col}\,A$ always, minimizing $\|A\vec{s}-\vec{b}\|$ over $\vec{s}$ is the same as minimizing $\|\vec{v}-\vec{b}\|$ over $\vec{v}\in\text{Col}\,A$. By the nearest-point property of orthogonal projections, the minimizing $\vec{v}$ is uniquely $\vec{v}=\text{proj}_{\text{Col}\,A}(\vec{b})$. ■

Theorem. The set of least-squares solutions to $A\vec{x}=\vec{b}$ is exactly the set of ordinary solutions of the linear system

$$\boxed{A^\top A\vec{x}=A^\top\vec{b}.}$$

This system (the normal equations) is always consistent.

Proof: Let $\hat{\vec{b}}=\text{proj}_{\text{Col}\,A}(\vec{b})$. Since $\vec{b}-\hat{\vec{b}}\in(\text{Col}\,A)^\perp=\text{Nul}(A^\top)$, we have $A^\top(\vec{b}-\hat{\vec{b}})=\vec{0}$, i.e. $A^\top\hat{\vec{b}}=A^\top\vec{b}$. If $\vec{s}$ satisfies $A\vec{s}=\hat{\vec{b}}$ then $A^\top A\vec{s}=A^\top\hat{\vec{b}}=A^\top\vec{b}$. Conversely, if $A^\top A\vec{s}=A^\top\vec{b}$ then $A^\top(A\vec{s}-\vec{b})=\vec{0}$, so $A\vec{s}-\vec{b}\in\text{Nul}(A^\top)=(\text{Col}\,A)^\perp$; by uniqueness of the orthogonal projection, $A\vec{s}=\hat{\vec{b}}$. Finally $\hat{\vec{b}}\in\text{Col}\,A$, so there is some $\vec{s}$ with $A\vec{s}=\hat{\vec{b}}$, and such $\vec{s}$ solves $A^\top A\vec{s}=A^\top\vec{b}$; hence the normal equations are consistent. ■

Example. Let

$$A=\begin{pmatrix}4&0\\0&2\\1&1\end{pmatrix},\qquad \vec{b}=\begin{pmatrix}2\\0\\11\end{pmatrix}.$$

Then

$$A^\top A=\begin{pmatrix}4&0&1\\0&2&1\end{pmatrix}\begin{pmatrix}4&0\\0&2\\1&1\end{pmatrix}=\begin{pmatrix}17&1\\1&5\end{pmatrix},\qquad A^\top\vec{b}=\begin{pmatrix}4&0&1\\0&2&1\end{pmatrix}\begin{pmatrix}2\\0\\11\end{pmatrix}=\begin{pmatrix}19\\11\end{pmatrix}.$$

Row reducing the augmented matrix,

$$\begin{pmatrix}17&1&19\\1&5&11\end{pmatrix}\sim\begin{pmatrix}1&5&11\\17&1&19\end{pmatrix}\sim\begin{pmatrix}1&5&11\\0&-84&-168\end{pmatrix}\sim\begin{pmatrix}1&0&1\\0&1&2\end{pmatrix},$$

so the unique least-squares solution is $\vec{s}=\begin{pmatrix}1\\2\end{pmatrix}$. Note that $A\vec{s}=\begin{pmatrix}4\\4\\3\end{pmatrix}\neq\vec{b}$, and

$$\|\vec{b}-A\vec{s}\|=\left\|\begin{pmatrix}-2\\-4\\8\end{pmatrix}\right\|=\sqrt{4+16+64}=\sqrt{84}.$$

Geometrically, $A\vec{s}$ is the point of the plane $\text{Col}\,A\subseteq\mathbb{R}^3$ closest to $\vec{b}$.

Theorem. Let A be an $m\times n$ matrix. The following are equivalent:

(a) $A\vec{x}=\vec{b}$ has a unique least-squares solution for each $\vec{b}\in\mathbb{R}^m$;
(b) the columns of A are linearly independent;
(c) $A^\top A$ is invertible.

When these hold, the unique least-squares solution is

$$\boxed{\vec{s} = (A^\top A)^{-1} A^\top \vec{b},}$$

which is the unique solution of the normal equations.

Proof: (a) $\Rightarrow$ (b): if $\vec{s}$ is a least-squares solution and $\vec{v} \in \operatorname{Nul} A$, then $\vec{s} + \vec{v}$ is also a least-squares solution, because $A(\vec{s} + \vec{v}) = A\vec{s}$. So uniqueness forces $\operatorname{Nul} A = \{\vec{0}\}$, i.e. (b).

(b) $\Rightarrow$ (c): if $\vec{v} \in \mathbb{R}^n$ then $A^\top A\vec{v} = \vec{0} \iff A\vec{v} \in \operatorname{Col} A \cap \operatorname{Nul}(A^\top) = \operatorname{Col} A \cap (\operatorname{Col} A)^\perp = \{\vec{0}\} \iff A\vec{v} = \vec{0}$. Hence $\operatorname{Nul}(A^\top A) = \operatorname{Nul} A$, which is $\{\vec{0}\}$ by (b); since $A^\top A$ is square, it is invertible.

(c) $\Rightarrow$ (a): if $A^\top A$ is invertible then $A^\top A\vec{x} = A^\top \vec{b}$ has the unique solution $\vec{s} = (A^\top A)^{-1} A^\top \vec{b}$, and by the previous theorem this is the unique least-squares solution. ■

Remark. In practice one never computes $(A^\top A)^{-1} A^\top \vec{b}$ for a large system; it is cheaper and numerically safer to row reduce the augmented matrix $\begin{pmatrix} A^\top A & A^\top \vec{b} \end{pmatrix}$.

Example (orthogonal columns). If the columns of A are orthogonal, the normal equations decouple. Take

$$A = \begin{pmatrix} 1 & -6 \\ 1 & -2 \\ 1 & 1 \\ 1 & 7 \end{pmatrix}, \qquad \vec{b} = \begin{pmatrix} -1 \\ 2 \\ 1 \\ 6 \end{pmatrix}.$$

The columns $\vec{a}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$ and $\vec{a}_2 = \begin{pmatrix} -6 \\ -2 \\ 1 \\ 7 \end{pmatrix}$ satisfy $\vec{a}_1 \cdot \vec{a}_2 = -6 - 2 + 1 + 7 = 0$. Therefore

$$\operatorname{proj}_{\operatorname{Col} A}(\vec{b}) = \frac{\vec{b} \cdot \vec{a}_1}{\vec{a}_1 \cdot \vec{a}_1}\vec{a}_1 + \frac{\vec{b} \cdot \vec{a}_2}{\vec{a}_2 \cdot \vec{a}_2}\vec{a}_2,$$

and $\vec{b} \cdot \vec{a}_1 = 8$, $\vec{a}_1 \cdot \vec{a}_1 = 4$, $\vec{b} \cdot \vec{a}_2 = 6 - 4 + 1 + 42 = 45$, $\vec{a}_2 \cdot \vec{a}_2 = 36 + 4 + 1 + 49 = 90$. Hence

$$\vec{s} = \begin{pmatrix} 8/4 \\ 45/90 \end{pmatrix} = \begin{pmatrix} 2 \\ 1/2 \end{pmatrix}, \qquad A\vec{s} = 2\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} -6 \\ -2 \\ 1 \\ 7 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 5/2 \\ 11/2 \end{pmatrix}.$$

Solving a least-squares problem

1. Form $A^\top A$ and $A^\top \vec{b}$ (the normal equations).
2. Row reduce $\begin{pmatrix} A^\top A & A^\top \vec{b} \end{pmatrix}$; the solution set is the set of least-squares solutions.
3. If the columns of A are independent, $A^\top A$ is invertible and the solution is unique.
4. If the columns of A are orthogonal (or orthonormal), skip the normal equations: use $s_i = \dfrac{\vec{b} \cdot \vec{a}_i}{\vec{a}_i \cdot \vec{a}_i}$ directly.

9.9 Lines of Best Fit

A typical least-squares application is fitting a line through data points $(a_1, b_1), \ldots, (a_n, b_n)$. We look for $\beta_0, \beta_1 \in \mathbb{R}$ making $y = \beta_0 + \beta_1 x$ the line of best fit, i.e.

$$b_i \approx \beta_0 + \beta_1 a_i \qquad (i = 1, \ldots, n).$$

If the points were exactly collinear, the vector $\vec{\beta} = \begin{pmatrix}\beta_0\\ \beta_1\end{pmatrix}$ would be an exact solution of $A\vec{\beta} = \vec{b}$ with

$$A = \begin{pmatrix}1 & a_1\\ 1 & a_2\\ \vdots & \vdots\\ 1 & a_n\end{pmatrix}, \qquad \vec{b} = \begin{pmatrix}b_1\\ b_2\\ \vdots\\ b_n\end{pmatrix}.$$

Otherwise we take the least-squares solution, which is the exact solution of $A^\top A\vec{\beta} = A^\top\vec{b}$:

$$A^\top A = \begin{pmatrix}n & \sum a_i\\ \sum a_i & \sum a_i^2\end{pmatrix}, \qquad A^\top\vec{b} = \begin{pmatrix}\sum b_i\\ \sum a_i b_i\end{pmatrix}.$$

Example. For the points $(2,1),(5,2),(7,3),(8,3)$,

$$A = \begin{pmatrix}1 & 2\\ 1 & 5\\ 1 & 7\\ 1 & 8\end{pmatrix}, \qquad \vec{b} = \begin{pmatrix}1\\ 2\\ 3\\ 3\end{pmatrix}, \qquad A^\top A = \begin{pmatrix}4 & 22\\ 22 & 142\end{pmatrix}, \qquad A^\top\vec{b} = \begin{pmatrix}9\\ 57\end{pmatrix}.$$

Here $\sum a_i = 22$, $\sum a_i^2 = 4+25+49+64 = 142$, $\sum b_i = 9$, $\sum a_i b_i = 2+10+21+24 = 57$. Since $\det(A^\top A) = 4\cdot 142 - 22^2 = 568 - 484 = 84 \neq 0$, the matrix is invertible and

$$\vec{\beta} = (A^\top A)^{-1}A^\top\vec{b} = \frac{1}{84}\begin{pmatrix}142 & -22\\ -22 & 4\end{pmatrix}\begin{pmatrix}9\\ 57\end{pmatrix} = \frac{1}{84}\begin{pmatrix}24\\ 30\end{pmatrix} = \begin{pmatrix}2/7\\ 5/14\end{pmatrix}.$$

So the line of best fit is

$$y = \frac{2}{7} + \frac{5}{14}x.$$

Why $A^\top A$ is invertible for a line fit

The columns of A are $(1,\ldots,1)^\top$ and $(a_1,\ldots,a_n)^\top$. They are dependent exactly when some vector $\vec{c} = (c_0, c_1) \neq \vec{0}$ satisfies $c_0 + c_1 a_i = 0$ for all i, i.e. when all a_i are equal (or when there is at most one point). So as soon as the data have at least two distinct x-values, $A^\top A$ is invertible and the line of best fit is unique.

10 Symmetric Matrices and the Singular Value Decomposition

10.1 Symmetric Matrices

Definition. A matrix A is symmetric if $A^\top = A$. This can only happen when A is square, and it means $a_{ij} = a_{ji}$ for all i, j.

Examples. $\begin{pmatrix}1 & 2\\ 2 & 3\end{pmatrix}$, $\begin{pmatrix}1 & 0\\ 0 & -3\end{pmatrix}$, $\begin{pmatrix}0 & -1 & 0\\ -1 & 5 & 8\\ 0 & 8 & -7\end{pmatrix}$ and every diagonal matrix are symmetric; $\begin{pmatrix}1 & 2\\ 3 & 0\end{pmatrix}$ and $\begin{pmatrix}1 & -4 & 0\\ -6 & 1 & -4\\ 6 & -6 & 1\end{pmatrix}$ are not.

Proposition. If A is symmetric and k is a positive integer then A^k is symmetric. If A is symmetric and invertible then A^{-1} is symmetric.

Proof: $(A^k)^\top = (A\cdots A)^\top = A^\top\cdots A^\top = (A^\top)^k = A^k$, and $(A^{-1})^\top = (A^\top)^{-1} =$

A^{-1}. ■

Also, for any matrix A, the product $A^\top A$ is symmetric, since $(A^\top A)^\top = A^\top (A^\top)^\top = A^\top A$. This example will be central in the theory of singular values.

Theorem. Suppose A is symmetric. Then eigenvectors belonging to different eigenvalues are orthogonal: if $A = A^\top$, $A\vec{u} = a\vec{u}$, $A\vec{v} = b\vec{v}$ and $a \neq b$, then $\vec{u} \cdot \vec{v} = 0$.

Proof: Using $A^\top = A$,

$$a(\vec{u} \cdot \vec{v}) = (A\vec{u}) \cdot \vec{v} = (A\vec{u})^\top \vec{v} = \vec{u}^\top A^\top \vec{v} = \vec{u}^\top A\vec{v} = \vec{u} \cdot (A\vec{v}) = \vec{u} \cdot (b\vec{v}) = b(\vec{u} \cdot \vec{v}).$$

Hence $(a - b)(\vec{u} \cdot \vec{v}) = 0$, and since $a \neq b$ we get $\vec{u} \cdot \vec{v} = 0$. ■

Why symmetry matters

The identity $(A\vec{u}) \cdot \vec{v} = \vec{u} \cdot (A\vec{v})$ — which holds exactly for symmetric A — is the whole reason symmetric matrices are special. It forces different eigenspaces to be orthogonal, and it will soon imply that a symmetric matrix can be diagonalized by an orthogonal matrix.

10.2 Orthogonal Diagonalization

Definition. An $n \times n$ matrix A is orthogonally diagonalizable if there is an orthogonal matrix U and a diagonal matrix D such that

$$A = UDU^{-1} = UDU^\top.$$

Then the diagonal entries of D are the eigenvalues of A, and the columns of U are corresponding eigenvectors; because U is orthogonal, these eigenvectors form an orthonormal basis of $\mathbb{R}^n$.

Conversely, by the usual theory of diagonalization, A is orthogonally diagonalizable if and only if $\mathbb{R}^n$ has an orthonormal basis consisting of eigenvectors of A.

Example. Let

$$A = \begin{pmatrix} 6 & -2 & -1 \\ -2 & 6 & -1 \\ -1 & -1 & 5 \end{pmatrix}.$$

A is symmetric. Its characteristic polynomial is $\det(A - xI) = (8 - x)(6 - x)(3 - x)$, so the eigenvalues are $8, 6, 3$. Row reducing the matrices $A - \lambda I$:

$$A - 8I = \begin{pmatrix} -2 & -2 & -1 \\ -2 & -2 & -1 \\ -1 & -1 & -3 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad A - 6I = \begin{pmatrix} 0 & -2 & -1 \\ -2 & 0 & -1 \\ -1 & -1 & -1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1/2 \\ 0 & 1 & 1/2 \\ 0 & 0 & 0 \end{pmatrix},$$

$$A - 3I = \begin{pmatrix} 3 & -2 & -1 \\ -2 & 3 & -1 \\ -1 & -1 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix},$$

so we may take

$$\vec{v}_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \ (8), \qquad \vec{v}_2 = \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix} \ (6), \qquad \vec{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \ (3).$$

These are already pairwise orthogonal, as the theorem predicts:

$$A\vec{v}_1 = \begin{pmatrix} -8 \\ 8 \\ 0 \end{pmatrix} = 8\vec{v}_1, \qquad A\vec{v}_2 = \begin{pmatrix} -6 \\ -6 \\ 12 \end{pmatrix} = 6\vec{v}_2, \qquad A\vec{v}_3 = \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix} = 3\vec{v}_3,$$

and $\vec{v}_1 \cdot \vec{v}_2 = 1-1+0 = 0$, $\vec{v}_1 \cdot \vec{v}_3 = -1+1+0 = 0$, $\vec{v}_2 \cdot \vec{v}_3 = -1-1+2 = 0$. Normalizing,

$$\vec{u}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}-1\\1\\0\end{pmatrix}, \qquad \vec{u}_2 = \frac{1}{\sqrt{6}}\begin{pmatrix}-1\\-1\\2\end{pmatrix}, \qquad \vec{u}_3 = \frac{1}{\sqrt{3}}\begin{pmatrix}1\\1\\1\end{pmatrix},$$

we obtain an orthonormal basis and

$$A = UDU^\top, \qquad U = \begin{pmatrix}\vec{u}_1 & \vec{u}_2 & \vec{u}_3\end{pmatrix}, \qquad D = \begin{pmatrix}8&0&0\\0&6&0\\0&0&3\end{pmatrix}.$$

Theorem. A square matrix is orthogonally diagonalizable if and only if it is symmetric.

We prove this through three lemmas.

Lemma 1. If A is orthogonally diagonalizable then A is symmetric.

Proof: For $n \times n$ matrices, $(XYZ)^\top = Z^\top Y^\top X^\top$. If $A = UDU^\top$ with D diagonal then $D = D^\top$ and

$$A^\top = (UDU^\top)^\top = (U^\top)^\top D^\top U^\top = UDU^\top = A.$$

■

Lemma 2. All complex eigenvalues of a real symmetric matrix are real.

Proof: Let $A = A^\top$ with real entries, and let $\vec{v} \in \mathbb{C}^n$ be an eigenvector with eigenvalue $\lambda \in \mathbb{C}$. The number

$$\overline{\vec{v}}^\top A\vec{v} = \lambda\,\overline{\vec{v}}^\top \vec{v} = \lambda\|\vec{v}\|^2$$

equals its own conjugate, because it is a 1×1 matrix:

$$\overline{\overline{\vec{v}}^\top A\vec{v}} = \vec{v}^\top A^\top \overline{\vec{v}} = \vec{v}^\top A\overline{\vec{v}} = \left(\overline{\vec{v}}^\top A\vec{v}\right)^\top = \overline{\vec{v}}^\top A\vec{v}.$$

Hence $\overline{\vec{v}}^\top A\vec{v} \in \mathbb{R}$. Since $\|\vec{v}\|^2 = \overline{\vec{v}}^\top \vec{v} \neq 0$ is also real, $\lambda = (\overline{\vec{v}}^\top A\vec{v})/\|\vec{v}\|^2 \in \mathbb{R}$. ■

Lemma 3 (Schur factorization). If A is an $n \times n$ matrix all of whose eigenvalues are real, then $A = URU^\top$ for some orthogonal U and some upper-triangular R.

Proof sketch: Choose a unit eigenvector $\vec{u}_1$ for some real eigenvalue λ, and extend it to an orthonormal basis $\vec{u}_1, \ldots, \vec{u}_n$ (Gram–Schmidt). With $U = \begin{pmatrix}\vec{u}_1 & \cdots & \vec{u}_n\end{pmatrix}$, the first column of $U^\top AU$ is $U^\top A\vec{u}_1 = \lambda U^\top \vec{u}_1 = \lambda\vec{e}_1$, so

$$U^\top AU = \begin{pmatrix}\lambda & *\\0 & B\end{pmatrix}$$

for some $(n-1) \times (n-1)$ matrix B. Since $U^\top AU$ is similar to A, its characteristic polynomial equals $\det(A - xI) = (\lambda - x)\det(B - xI)$, so B has all real eigenvalues too. Applying the same construction recursively to B (and embedding the resulting orthogonal matrix as a block) produces an orthogonal X with $X^\top AX$ upper triangular. Taking $R = X^\top AX$ and $U = X$ gives $A = URU^\top$. ■

Proof of the theorem: Lemma 1 gives one direction. Conversely, suppose $A = A^\top$. By Lemma 2 all eigenvalues of A are real, so by Lemma 3 there is a Schur factorization $A = URU^\top$. Then

$$R = U^\top AU = U^\top A^\top U = (U^\top AU)^\top = R^\top.$$

An upper-triangular matrix equal to its transpose must be diagonal, so $A = URU^\top$ is an orthogonal diagonalization. ■

Example. Take the general symmetric 2×2 matrix $A = \begin{pmatrix}a & b\\b & a\end{pmatrix}$. Then

$$\det(A - xI) = (a-x)^2 - b^2 = (a - b - x)(a + b - x),$$

so the eigenvalues are $a-b$ and $a+b$, with eigenvectors

$$\begin{pmatrix}1\\-1\end{pmatrix}\ (a-b), \qquad \begin{pmatrix}1\\1\end{pmatrix}\ (a+b).$$

These are orthogonal, and after normalizing we get

$$U=\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\-1 & 1\end{pmatrix}, \qquad D=\begin{pmatrix}a-b & 0\\0 & a+b\end{pmatrix}, \qquad A=UDU^{\top}.$$

Procedure: orthogonally diagonalizing a symmetric matrix.

1. Factor $\det(A-xI)$ to find the eigenvalues.
2. For each eigenvalue λ, row reduce $A-\lambda I$ to get a basis of $\mathrm{Nul}(A-\lambda I)$.
3. If an eigenspace has dimension >1, apply Gram–Schmidt to its basis so that the basis becomes orthogonal.
4. Normalize all vectors to unit length and combine them into the orthogonal matrix U; the eigenvalues, listed in the same order, fill D.

Eigenspaces for different eigenvalues are automatically orthogonal, so Gram–Schmidt is needed only inside a single eigenspace.

Symmetric = orthogonally diagonalizable

A real square matrix A satisfies $A=UDU^{\top}$ with U orthogonal and D diagonal if and only if $A=A^{\top}$. Moreover, for symmetric A:

- all eigenvalues are real, so D has real entries;
- eigenvectors from different eigenspaces are orthogonal;
- an orthonormal eigenbasis of $\mathbb{R}^n$ always exists.

10.3 The Spectral Decomposition

Corollary. Suppose $A=UDU^{\top}$ where $U=(\vec{u}_1 \ \cdots \ \vec{u}_n)$ has orthonormal columns and $D=\mathrm{diag}(\lambda_1,\dots,\lambda_n)$. Then

$$\boxed{A=\lambda_1\vec{u}_1\vec{u}_1^{\top}+\lambda_2\vec{u}_2\vec{u}_2^{\top}+\cdots+\lambda_n\vec{u}_n\vec{u}_n^{\top}.}$$

Each product $\vec{u}_i\vec{u}_i^{\top}$ is an $n\times n$ matrix of rank 1. This expression is called a spectral decomposition of A.

Why: Multiplying $UDU^{\top}$ out column by column, the (i,j) entry of the product is $\sum_k \lambda_k u_{ik}u_{jk}$, which is exactly the (i,j) entry of $\sum_k \lambda_k\vec{u}_k\vec{u}_k^{\top}$.

Example. Let $A=\begin{pmatrix}7 & 2\\2 & 4\end{pmatrix}$. Its eigenvalues are 8 and 3, with orthonormal eigenvectors

$$\vec{u}_1=\frac{1}{\sqrt{5}}\begin{pmatrix}2\\1\end{pmatrix}, \qquad \vec{u}_2=\frac{1}{\sqrt{5}}\begin{pmatrix}-1\\2\end{pmatrix}.$$

Indeed $A\vec{u}_1=\frac{1}{\sqrt{5}}\begin{pmatrix}16\\8\end{pmatrix}=8\vec{u}_1$ and $A\vec{u}_2=\frac{1}{\sqrt{5}}\begin{pmatrix}-3\\6\end{pmatrix}=3\vec{u}_2$. Hence

$$A=8\vec{u}_1\vec{u}_1^{\top}+3\vec{u}_2\vec{u}_2^{\top}=8\cdot\frac{1}{5}\begin{pmatrix}4 & 2\\2 & 1\end{pmatrix}+3\cdot\frac{1}{5}\begin{pmatrix}1 & -2\\-2 & 4\end{pmatrix}=\begin{pmatrix}32/5 & 16/5\\16/5 & 8/5\end{pmatrix}+\begin{pmatrix}3/5 & -6/5\\-6/5 & 12/5\end{pmatrix},$$

and the two summands add up to $\begin{pmatrix}35/5 & 10/5\\10/5 & 20/5\end{pmatrix}=\begin{pmatrix}7 & 2\\2 & 4\end{pmatrix}$.

The matrices $\vec{u}_i\vec{u}_i^{\top}$ are projections

Put $P_i = \vec{u}_i\vec{u}_i^{\top}$ for an orthonormal eigenbasis of the symmetric matrix A. Then

- $P_i^2 = \vec{u}_i(\vec{u}_i^{\top}\vec{u}_i)\vec{u}_i^{\top} = P_i$, so P_i is the orthogonal projection onto the line $\operatorname{span}\{\vec{u}_i\}$;
- $P_i^{\top} = P_i$, $\operatorname{rank} P_i = 1$, and $P_iP_j = 0$ whenever $i \neq j$;
- $\sum_i P_i = I_n$ (the P_i's split $\mathbb{R}^n$ into orthogonal eigenspaces);
- $A = \sum_i \lambda_i P_i$ and, for any power, $A^k = \sum_i \lambda_i^k P_i$ — powers act on each eigenspace separately.

10.4 Singular Values

For any $m \times n$ matrix A the product $A^{\top}A$ is symmetric, so by the previous subsections it has an orthonormal eigenbasis and all of its eigenvalues are real. In fact more is true.

Lemma. All eigenvalues of $A^{\top}A$ are nonnegative real numbers. More precisely, if λ is an eigenvalue of $A^{\top}A$ and $\vec{v}$ is a unit vector with $A^{\top}A\vec{v} = \lambda\vec{v}$, then $\lambda = \|A\vec{v}\|^2$.

Proof: Suppose $\|\vec{v}\| = 1$ and $A^{\top}A\vec{v} = \lambda\vec{v}$. Then

$$0 \leq \|A\vec{v}\|^2 = (A\vec{v})\cdot(A\vec{v}) = (A\vec{v})^{\top}(A\vec{v}) = \vec{v}^{\top}A^{\top}A\vec{v} = \vec{v}^{\top}(\lambda\vec{v}) = \lambda(\vec{v}^{\top}\vec{v}) = \lambda\|\vec{v}\|^2 = \lambda,$$

so λ is real and equals the nonnegative number $\|A\vec{v}\|^2$. ■

Definition. Let $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0$ be the eigenvalues of $A^{\top}A$ arranged in decreasing order, and define

$$\sigma_i = \sqrt{\lambda_i}, \qquad i = 1, 2, \ldots, n.$$

The numbers $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n \geq 0$ are the singular values of A. In other words, the singular values are the square roots of the (nonnegative) eigenvalues of $A^{\top}A$.

Example. If $A = \operatorname{diag}(d_1, \ldots, d_n)$ is diagonal then $A^{\top}A = \operatorname{diag}(d_1^2, \ldots, d_n^2)$, so the singular values of A are $|d_1|, \ldots, |d_n|$ in decreasing order: the absolute values of the diagonal entries.

Example. Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. Then $A^{\top}A = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$ and

$$\det(A^{\top}A - xI) = (5-x)^2 - 4^2 = (9-x)(1-x),$$

so the singular values of A are $\sigma_1 = 3$ and $\sigma_2 = 1$. (Here A is symmetric, and indeed the singular values are the absolute values of its eigenvalues 3 and 1.)

Theorem. Suppose $\vec{v}_1, \ldots, \vec{v}_n$ is an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of $A^{\top}A$, indexed so that the eigenvalue λ_i of $\vec{v}_i$ satisfies $\lambda_1 \geq \cdots \geq \lambda_n$. If A has r nonzero singular values, then $A\vec{v}_1, \ldots, A\vec{v}_r$ is an orthogonal basis of the column space $\operatorname{Col} A$. Consequently $\operatorname{rank} A = r$.

Proof: Take indices $i \neq j$. Since $\vec{v}_i \cdot \vec{v}_j = 0$,

$$(A\vec{v}_i)^{\top}(A\vec{v}_j) = \vec{v}_i^{\top}A^{\top}A\vec{v}_j = \vec{v}_i^{\top}(\lambda_j\vec{v}_j) = \lambda_j(\vec{v}_i \cdot \vec{v}_j) = 0,$$

so $A\vec{v}_1, \ldots, A\vec{v}_r$ are pairwise orthogonal vectors of $\operatorname{Col} A$. Each of them is nonzero because $\|A\vec{v}_i\| = \sqrt{\lambda_i} > 0$ for $i \leq r$; hence they are linearly independent.

To see that they span $\operatorname{Col} A$, take $\vec{y} \in \operatorname{Col} A$, say $\vec{y} = A\vec{x}$. Writing $\vec{x} = c_1\vec{v}_1 + \cdots + c_n\vec{v}_n$,

$$\vec{y} = A\vec{x} = c_1A\vec{v}_1 + \cdots + c_rA\vec{v}_r + c_{r+1}A\vec{v}_{r+1} + \cdots + c_nA\vec{v}_n = c_1A\vec{v}_1 + \cdots + c_rA\vec{v}_r,$$

because $A\vec{v}_i = \vec{0}$ whenever $i > r$ (then $\lambda_i = 0$, so $\|A\vec{v}_i\|^2 = \lambda_i = 0$). Therefore $A\vec{v}_1, \ldots, A\vec{v}_r$ is a basis of $\operatorname{Col} A$. ■

Corollary. The rank of a matrix equals its number of nonzero singular values.

Example. Let $A = \begin{pmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{pmatrix}$, with columns $\vec{c}_1 = (4, 8)$, $\vec{c}_2 = (11, 7)$, $\vec{c}_3 = (14, -2)$. Then $(A^\top A)_{ij} = \vec{c}_i \cdot \vec{c}_j$, so

$$A^\top A = \begin{pmatrix} 4\cdot 4 + 8\cdot 8 & 4\cdot 11 + 8\cdot 7 & 4\cdot 14 - 8\cdot 2 \\ 11\cdot 4 + 7\cdot 8 & 11\cdot 11 + 7\cdot 7 & 11\cdot 14 - 7\cdot 2 \\ 14\cdot 4 - 2\cdot 8 & 14\cdot 11 - 2\cdot 7 & 14\cdot 14 + 2\cdot 2 \end{pmatrix} = \begin{pmatrix} 80 & 100 & 40 \\ 100 & 170 & 140 \\ 40 & 140 & 200 \end{pmatrix},$$

and $\det(A^\top A - xI) = -x(360 - x)(90 - x)$. Hence the eigenvalues of $A^\top A$ are 360, 90, 0, and the singular values of A are

$$\sigma_1 = \sqrt{360} = 6\sqrt{10}, \qquad \sigma_2 = \sqrt{90} = 3\sqrt{10}, \qquad \sigma_3 = 0.$$

Exactly two singular values are nonzero, so $\operatorname{rank} A = 2$ — as it must be, since the two rows $(4, 11, 14)$ and $(8, 7, -2)$ are plainly independent.

Singular values in practice

- σ_i is always a nonnegative real number, so $\sigma_i = 0$ exactly when $\lambda_i = 0$.
- The number of nonzero singular values is $\operatorname{rank} A$; in particular A has full column rank $\iff$ all n singular values are positive $\iff$ $A^\top A$ is invertible.
- Unlike eigenvalues, singular values are defined for every matrix, square or not, and they always exist and are real.

10.5 The Singular Value Decomposition

The main theorem of this chapter says that every matrix — not just symmetric ones — is a product of two orthogonal matrices with a diagonal matrix in the middle.

Theorem (existence of SVDs). Let A be an $m \times n$ matrix of rank r, with nonzero singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$. Then

$$\boxed{A = U\Sigma V^\top,}$$

where U is an $m \times m$ orthogonal matrix, V is an $n \times n$ orthogonal matrix, and

$$\boxed{\Sigma = \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix}, \qquad D = \begin{pmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_r \end{pmatrix},}$$

is the $m \times n$ matrix obtained by placing D in the upper-left corner and filling every other entry with 0. In the block form the three zeros stand for blocks of sizes $r \times (n - r)$, $(m - r) \times r$ and $(m - r) \times (n - r)$.

Definition. A factorization $A = U\Sigma V^\top$ as above is a singular value decomposition of A, abbreviated SVD. The columns of U are the left singular vectors and the columns of V the right singular vectors of A. The matrices U and V are not uniquely determined by A, but the middle matrix Σ is. Note also that Σ has the same size as A; it is square only when A is.

Proof (existence): Let $\lambda_1 \geq \cdots \geq \lambda_n$ be the eigenvalues of $A^\top A$ and let $\vec{v}_1, \ldots, \vec{v}_n$ be a corresponding orthonormal list of eigenvectors, so that $A^\top A\vec{v}_i = \lambda_i \vec{v}_i$ and $\sigma_i = \sqrt{\lambda_i}$. By the theorem of the previous subsection, $\lambda_{r+1} = \cdots = \lambda_n = 0$ and

$A\vec{v}_1, \dots, A\vec{v}_r$ is an orthogonal basis of $\operatorname{Col} A$. For $1 \le i \le r$ put

$$\vec{u}_i = \frac{1}{\|A\vec{v}_i\|} A\vec{v}_i = \frac{1}{\sqrt{\lambda_i}} A\vec{v}_i = \frac{1}{\sigma_i} A\vec{v}_i,$$

so that $\vec{u}_1, \dots, \vec{u}_r$ is an orthonormal basis of $\operatorname{Col} A$. Extend this list to an orthonormal basis $\vec{u}_1, \dots, \vec{u}_m$ of $\mathbb{R}^m$ (Gram–Schmidt applied to the list $\vec{u}_1, \dots, \vec{u}_r, \vec{e}_1, \dots, \vec{e}_m$, keeping the pivot columns, does the job), and set

$$U = (\vec{u}_1 \ \cdots \ \vec{u}_m), \qquad V = (\vec{v}_1 \ \cdots \ \vec{v}_n).$$

Both are orthogonal by construction, and

$$AV = (A\vec{v}_1 \ \cdots \ A\vec{v}_n) = \big(\sigma_1\vec{u}_1 \ \cdots \ \sigma_r\vec{u}_r \ \ \vec{0} \ \cdots \ \vec{0}\big) = U\Sigma.$$

Multiplying on the right by $V^\top$ gives $A = AVV^\top = U\Sigma V^\top$, since $VV^\top = I_n$. ■

Procedure: computing an SVD of an $m \times n$ matrix A.

1. Find the eigenvalues $\lambda_1 \ge \cdots \ge \lambda_n \ge 0$ of $A^\top A$. For each eigenspace take a basis, apply Gram–Schmidt to make it orthogonal, normalize, and combine the lists into an orthonormal eigenbasis $\vec{v}_1, \dots, \vec{v}_n$, ordered so that $\vec{v}_i$ belongs to λ_i.
2. Set $V = (\vec{v}_1 \ \cdots \ \vec{v}_n)$ and form the $m \times n$ matrix Σ whose entry in position (i,i) is $\sigma_i = \sqrt{\lambda_i}$ for $i = 1, \dots, \min(m,n)$, all other entries being 0.
3. Let $r = \operatorname{rank} A$, the largest index with $\sigma_r > 0$, and put $\vec{u}_i = \dfrac{1}{\sigma_i} A\vec{v}_i$ for $i = 1, \dots, r$. Extend $\vec{u}_1, \dots, \vec{u}_r$ to an orthonormal basis $\vec{u}_1, \dots, \vec{u}_m$ of $\mathbb{R}^m$ and set $U = (\vec{u}_1 \ \cdots \ \vec{u}_m)$.

Example. We compute an SVD of $A = \begin{pmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{pmatrix}$ from the previous example, in three steps.

Step 1: diagonalize $A^\top A$ orthogonally. Row reducing $A^\top A - \lambda I$ for $\lambda = 360, 90, 0$ produces the orthonormal eigenbasis

$$\vec{v}_1 = \frac{1}{3}\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \ (360), \qquad \vec{v}_2 = \frac{1}{3}\begin{pmatrix} -2 \\ -1 \\ 2 \end{pmatrix} \ (90), \qquad \vec{v}_3 = \frac{1}{3}\begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} \ (0).$$

As a check,

$$A^\top A\vec{v}_1 = \frac{1}{3}\begin{pmatrix} 80+200+80 \\ 100+340+280 \\ 40+280+400 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 360 \\ 720 \\ 720 \end{pmatrix} = 360\vec{v}_1,$$

$$A^\top A\vec{v}_2 = \frac{1}{3}\begin{pmatrix} -160-100+80 \\ -200-170+280 \\ -80-140+400 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} -180 \\ -90 \\ 180 \end{pmatrix} = 90\vec{v}_2,$$

$$A^\top A\vec{v}_3 = \frac{1}{3}\begin{pmatrix} 160-200+40 \\ 200-340+140 \\ 80-280+200 \end{pmatrix} = \vec{0},$$

and the three vectors are mutually orthogonal, e.g. $\vec{v}_1 \cdot \vec{v}_2 = \dfrac{1}{9}(-2-2+4) = 0$ and $\vec{v}_1 \cdot \vec{v}_3 = \dfrac{1}{9}(2-4+2) = 0$.

Step 2: form V and Σ.

$$V = (\vec{v}_1 \ \ \vec{v}_2 \ \ \vec{v}_3) = \frac{1}{3}\begin{pmatrix} 1 & -2 & 2 \\ 2 & -1 & -2 \\ 2 & 2 & 1 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} \sqrt{360} & 0 & 0 \\ 0 & \sqrt{90} & 0 \end{pmatrix} = \begin{pmatrix} 6\sqrt{10} & 0 & 0 \\ 0 & 3\sqrt{10} & 0 \end{pmatrix}.$$

Step 3: construct U. Since $r = 2$,

$$A\vec{v}_1 = \frac{1}{3}\begin{pmatrix} 4+22+28 \\ 8+14-4 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 54 \\ 18 \end{pmatrix} = \begin{pmatrix} 18 \\ 6 \end{pmatrix},$$

$$A\vec{v}_2 = \frac{1}{3}\begin{pmatrix} -8-11+28 \\ -16-7-4 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 9 \\ -27 \end{pmatrix} = \begin{pmatrix} 3 \\ -9 \end{pmatrix},$$

whence

$$\vec{u}_1 = \frac{1}{6\sqrt{10}}\begin{pmatrix} 18 \\ 6 \end{pmatrix} = \frac{1}{\sqrt{10}}\begin{pmatrix} 3 \\ 1 \end{pmatrix}, \qquad \vec{u}_2 = \frac{1}{3\sqrt{10}}\begin{pmatrix} 3 \\ -9 \end{pmatrix} = \frac{1}{\sqrt{10}}\begin{pmatrix} 1 \\ -3 \end{pmatrix}.$$

Here $m = 2$, so $\vec{u}_1, \vec{u}_2$ already form an orthonormal basis of $\mathbb{R}^2$ and $U = (\vec{u}_1 \quad \vec{u}_2)$. Putting everything together,

$$A = U\Sigma V^\top = \frac{1}{\sqrt{10}}\begin{pmatrix} 3 & 1 \\ 1 & -3 \end{pmatrix}\begin{pmatrix} \sqrt{360} & 0 & 0 \\ 0 & \sqrt{90} & 0 \end{pmatrix}\frac{1}{3}\begin{pmatrix} 1 & 2 & 2 \\ -2 & -1 & 2 \\ 2 & -2 & 1 \end{pmatrix}.$$

Be careful: the third factor is $V^\top$, not V.

Anatomy of an SVD

For an $m \times n$ matrix A of rank r:

- U is $m \times m$ orthogonal, Σ is $m \times n$ (same shape as A), and the last factor is $V^\top$ with V an $n \times n$ orthogonal matrix;
- the diagonal of Σ lists $\sigma_1 \geq \cdots \geq \sigma_r > 0$ followed by $n - r$ zeros, and below the r-th row Σ is entirely zero;
- $\operatorname{Col} A$ has the orthonormal basis $\vec{u}_1, \ldots, \vec{u}_r$, and $\operatorname{Nul} A$ has the orthonormal basis $\vec{v}_{r+1}, \ldots, \vec{v}_n$; the singular vectors decode both spaces at once;
- the i-th right singular vector is mapped to a multiple of the i-th left singular vector: $A\vec{v}_i = \sigma_i \vec{u}_i$.

10.6 Pseudo-inverses

An SVD produces a substitute for the inverse of a matrix that need not be square or invertible.

Definition. A pseudo-inverse of an $m \times n$ matrix A is an $n \times m$ matrix A^+ such that

$$AA^+A = A \qquad \text{and} \qquad A^+AA^+ = A^+.$$

Example. If A is square and invertible then $A^+ = A^{-1}$: indeed $AA^{-1}A = A$ and $A^{-1}AA^{-1} = A^{-1}$.

Theorem. Every matrix A has a pseudo-inverse. Concretely, if $A = U\Sigma V^\top$ is an SVD and Σ^+ denotes the $n \times m$ matrix obtained from Σ by transposing it and then replacing every nonzero entry by its reciprocal, then

$$\boxed{A^+ = V\Sigma^+U^\top}$$

is a pseudo-inverse of A.

Proof: Since $\Sigma = \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix}$ with $D = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$, we have $\Sigma^+ = \begin{pmatrix} D^{-1} & 0 \\ 0 & 0 \end{pmatrix}$ with $D^{-1} = \operatorname{diag}(1/\sigma_1, \ldots, 1/\sigma_r)$. Hence

$$AA^+A = (U\Sigma V^\top)(V\Sigma^+U^\top)(U\Sigma V^\top) = U(\Sigma\Sigma^+\Sigma)V^\top,$$

$$A^+AA^+ = V(\Sigma^+\Sigma\Sigma^+)U^\top,$$

so it suffices to verify $\Sigma\Sigma^+\Sigma = \Sigma$ and $\Sigma^+\Sigma\Sigma^+ = \Sigma^+$. These hold because

$$\Sigma\Sigma^+ = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}_{m\times m}, \qquad \Sigma^+\Sigma = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}_{n\times n}.$$

For a concrete illustration take

$$\Sigma = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \end{pmatrix}, \qquad \Sigma^+ = \begin{pmatrix} 1/a & 0 \\ 0 & 1/b \\ 0 & 0 \end{pmatrix} \qquad (a, b \neq 0).$$

Then

$$\Sigma\Sigma^+ = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \Sigma^+\Sigma = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

whence $\Sigma\Sigma^+\Sigma = \Sigma$ and $\Sigma^+\Sigma\Sigma^+ = \Sigma^+$ as claimed. ■

One can show that the pseudo-inverse is unique; we will not prove this here.

Example. For the diagonal matrix $A = \operatorname{diag}(1, 2, 0)$ we may read off an SVD as $U = V = I_3$, $\Sigma = A$, so

$$A^+ = \Sigma^+ = \operatorname{diag}(1, 1/2, 0).$$

$$AA^+A = \operatorname{diag}(1, 1, 0)\operatorname{diag}(1, 2, 0) = A, \qquad A^+AA^+ = \operatorname{diag}(1, 1, 0)\operatorname{diag}(1, 1/2, 0) = A^+.$$

Example. For $A = \begin{pmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{pmatrix}$ with the SVD computed above,

$$A^+ = V\Sigma^+U^\top = \frac{1}{3}\begin{pmatrix} 1 & -2 & 2 \\ 2 & -1 & -2 \\ 2 & 2 & 1 \end{pmatrix}\begin{pmatrix} 1/\sqrt{360} & 0 \\ 0 & 1/\sqrt{90} \\ 0 & 0 \end{pmatrix}\frac{1}{\sqrt{10}}\begin{pmatrix} 3 & 1 \\ 1 & -3 \end{pmatrix}$$

is a 3×2 pseudo-inverse of the 2×3 matrix A.

Why pseudo-inverses matter

- Σ^+ is Σ "turned on its side with reciprocal nonzero entries"; the zeros stay zeros. A zero singular value cannot be inverted, which is exactly why A^+ is only a pseudo-inverse.
- If A has independent columns, then $A^+ = (A^\top A)^{-1}A^\top$, so $A^+\vec{b}$ is precisely the least-squares solution of $A\vec{x} = \vec{b}$ studied earlier.
- For invertible square A the pseudo-inverse reduces to the ordinary inverse, so A^+ genuinely generalizes A^{-1}.

10.7 SVDs for Symmetric Matrices

An SVD is intended as an analogue of orthogonal diagonalization, which exists only for symmetric matrices. The next result makes the analogy explicit.

Proposition. Suppose $A = A^\top = UDU^\top$ with U orthogonal and D diagonal, say $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. Order the eigenvalues so that

$$\boxed{|\lambda_1| \geq |\lambda_2| \geq \cdots \geq |\lambda_n| \geq 0.}$$

Put $\epsilon_i = 1$ if $\lambda_i \geq 0$ and $\epsilon_i = -1$ if $\lambda_i < 0$, and let $E = \operatorname{diag}(\epsilon_1, \ldots, \epsilon_n)$. Then

$$\boxed{A = U\Sigma V^\top, \qquad \Sigma = DE, \qquad V = UE}$$

is a singular value decomposition of A, and the singular values of A are $|\lambda_1| \geq |\lambda_2| \geq \cdots \geq |\lambda_n| \geq 0$. In particular, the singular values of a symmetric matrix are the absolute values of its eigenvalues.

Proof: Since each ϵ_i equals ± 1, we have $E = E^\top = E^{-1}$. Also U and E are orthogonal, so $V = UE$ is orthogonal, and

$$U\Sigma V^\top = UDE(UE)^\top = UDE\,E^\top U^\top = UDU^\top = A.$$

So $A = U\Sigma V^\top$ is a factorization with orthogonal outer factors and a diagonal middle factor. It remains to check that the diagonal entries of $\Sigma = DE$ are the singular values in decreasing order. Since A is symmetric,

$$A^\top A = A^2 = (UDU^\top)^2 = (UDU^{-1})^2 = UD^2U^{-1},$$

so $A^\top A$ is similar to D^2 and therefore has eigenvalues $\lambda_1^2 \geq \lambda_2^2 \geq \cdots \geq \lambda_n^2$ (similar matrices share eigenvalues, and the eigenvalues of the diagonal matrix D^2 are its diagonal entries). Taking square roots gives singular values $|\lambda_1| \geq \cdots \geq |\lambda_n|$, which are exactly the diagonal entries of $\Sigma = DE$, because $|d| = d \cdot \operatorname{sign}(d)$. ■

Example. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ the eigenvalues are 3 and 1 with orthonormal eigenvectors $\vec{u}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $\vec{u}_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$. Ordering by absolute value gives $\lambda_1 = 3$, $\lambda_2 = 1$, so $\epsilon_1 = \epsilon_2 = 1$, $E = I_2$, $\Sigma = D = \operatorname{diag}(3, 1)$ and $V = U$. The SVD therefore coincides with the orthogonal diagonalization:

$$A = U\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}U^\top = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Example. Let $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Its eigenvalues are 1 and -1, with orthonormal eigenvectors $\vec{u}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\vec{u}_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$. Ordering by absolute value, both eigenvalues have modulus 1, so we may take $\lambda_1 = 1$, $\lambda_2 = -1$. Then

$$D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \epsilon_1 = 1,\ \epsilon_2 = -1, \qquad E = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and consequently

$$\Sigma = DE = I_2, \qquad V = UE = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

Thus $A = U\Sigma V^\top = UV^\top$, and indeed

$$\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1-1 & 1+1 \\ 1+1 & -1+1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Here all singular values equal 1, so $\Sigma = I$ and the SVD merely exhibits A as a product of two orthogonal matrices — which is consistent with A itself being orthogonal.

Symmetric matrices: diagonalization versus SVD

For a symmetric matrix $A = A^\top$:

- orthogonal diagonalization $A = UDU^\top$ keeps the eigenvalues (possibly negative) on the diagonal;
- an SVD $A = U\Sigma V^\top$ replaces each λ_i by $|\lambda_i|$ and absorbs the signs into E, so $\Sigma = DE$ and $V = UE$;
- consequently $\sigma_i = |\lambda_i|$, and if all eigenvalues are nonnegative then $E = I$, $V = U$ and the SVD is just the orthogonal diagonalization.

10.8 SVDs and the Geometry of Ellipses

For 2×2 matrices an SVD has a very concrete meaning: it records how the unit disc is deformed. We first describe all orthogonal 2×2 matrices.

Proposition. Every orthogonal 2×2 matrix is a rotation matrix times a permutation matrix. Specifically, every 2×2 orthogonal matrix has one of the forms

$$\boxed{R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \qquad \text{or} \qquad \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix} = R_{\theta-\pi/2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}$$

for some angle $0 \le \theta < 2\pi$.

Proof: Both displayed matrices are orthogonal because $(\cos\theta)^2 + (\sin\theta)^2 = 1$ and their columns are orthogonal. Conversely, let $U = \begin{pmatrix} u_1 & v_1 \\ u_2 & v_2 \end{pmatrix}$ be orthogonal, so that $u_1^2 + u_2^2 = 1$: the point (u_1, u_2) lies on the unit circle and hence equals $(\cos\theta, \sin\theta)$ for some angle θ. Orthogonality of the columns forces (v_1, v_2) to be a scalar multiple of $(-\sin\theta, \cos\theta)$, and being a unit vector leaves only the two choices

$$\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix} \qquad \text{or} \qquad \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} \sin\theta \\ -\cos\theta \end{pmatrix}.$$

These give exactly the two forms above. ■

In the first case $\det U = 1$ and U rotates every vector counterclockwise by the angle θ. In the second case $\det U = -1$ and U reflects a vector across the line $y = x$ and then rotates it counterclockwise by some angle. In both cases the angle depends on U but not on the vector.

Let

$$\mathcal{D} = \{\vec{v} \in \mathbb{R}^2 : \vec{v} \cdot \vec{v} \le 1\}$$

be the unit disc. For fixed real numbers $r_1, r_2 \ge 0$, let $\mathcal{E}$ be the set of vectors $\begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ with

$$\left(\frac{v_1}{r_1}\right)^2 + \left(\frac{v_2}{r_2}\right)^2 \le 1,$$

where for $r_i = 0$ we interpret $(v_i/r_i)^2$ as 0 when $v_i = 0$ and as $+\infty$ when $v_i \neq 0$. We call $\mathcal{E}$ a (solid) standard ellipse.

Proposition. $\mathcal{E} = \left\{ \begin{pmatrix} r_1 & 0 \\ 0 & r_2 \end{pmatrix} \vec{v} : \vec{v} \in \mathcal{D} \right\}$.

Proof: For $\vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ we have $\begin{pmatrix} r_1 & 0 \\ 0 & r_2 \end{pmatrix} \vec{v} = \begin{pmatrix} r_1 v_1 \\ r_2 v_2 \end{pmatrix}$, which lies in $\mathcal{E}$ if and only if

$$\left(\frac{r_1 v_1}{r_1}\right)^2 + \left(\frac{r_2 v_2}{r_2}\right)^2 = v_1^2 + v_2^2 \le 1,$$

i.e. if and only if $\vec{v} \in \mathcal{D}$. ■

The radii of $\mathcal{E}$ are the vectors $\pm\begin{pmatrix} r_1 \\ 0 \end{pmatrix}$ and $\pm\begin{pmatrix} 0 \\ r_2 \end{pmatrix}$; they are allowed to be $\vec{0}$. For each radius there are two choices of direction, but any two orthogonal radii determine the ellipse completely. More generally, any rotation of a standard ellipse is called a (solid) ellipse; rotating $\mathcal{E}$ counterclockwise by an angle θ gives radii

$$\pm r_1 \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix} \qquad \text{and} \qquad \pm\, r_2 \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}.$$

Proposition. Suppose U is an orthogonal 2×2 matrix and $\Sigma = \begin{pmatrix} r_1 & 0 \\ 0 & r_2 \end{pmatrix}$ with $r_1, r_2 \ge 0$. Then the set

$$\boxed{\{U\Sigma\vec{v} \in \mathbb{R}^2 : \vec{v} \in \mathcal{D}\}}$$

is an ellipse whose radii have lengths r_1 and r_2, and every such ellipse arises this way.

Proof: By the previous proposition $\{\Sigma\vec{v} : \vec{v} \in \mathcal{D}\} = \mathcal{E}$ is a standard ellipse. Reflecting a standard ellipse across $y = x$ swaps its two radii and produces another

standard ellipse. By the first proposition of this subsection, U is a rotation matrix times a permutation matrix, so $U\mathcal{E}$ is obtained from $\mathcal{E}$ (or from its reflection across $y = x$) by a rotation; hence it is an ellipse with radii of lengths r_1, r_2. Conversely every ellipse is a rotation of some standard ellipse, so every ellipse occurs. ■

Proposition. Let A be a 2×2 matrix. Then the region $\{A\vec{v} : \vec{v} \in \mathcal{D}\}$ is an ellipse, and the lengths of its radii are the singular values $\sigma_1 \geq \sigma_2 \geq 0$ of A.

Proof: Let $A = U\Sigma V^\top$ be an SVD. Multiplication by the orthogonal matrix $V^\top$ preserves lengths, so $\{V^\top\vec{v} : \vec{v} \in \mathcal{D}\} = \mathcal{D}$. Therefore

$$\{A\vec{v} : \vec{v} \in \mathcal{D}\} = \{U\Sigma V^\top\vec{v} : \vec{v} \in \mathcal{D}\} = \{U\Sigma\vec{w} : \vec{w} \in \mathcal{D}\},$$

which is an ellipse with radii σ_1, σ_2 by the previous proposition. ■

Physical reading of an SVD. Suppose the ellipse $\mathcal{E}_A = \{A\vec{v} : \vec{v} \in \mathcal{D}\}$ has radii of lengths $\sigma_1 \geq \sigma_2 \geq 0$, so that $\Sigma = \begin{pmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{pmatrix}$. The columns $\vec{v}_1, \vec{v}_2$ of V are two orthogonal radii of the unit disc $\mathcal{D}$, and each of them is mapped to a radius of $\mathcal{E}_A$:

$$A\vec{v}_1 = U\Sigma V^\top\vec{v}_1 = U\Sigma\vec{e}_1 = U\begin{pmatrix}\sigma_1\\0\end{pmatrix} = \sigma_1\vec{u}_1, \qquad A\vec{v}_2 = U\Sigma\vec{e}_2 = \sigma_2\vec{u}_2.$$

So the SVD singles out which orthogonal radii of the disc are stretched to which orthogonal radii of the ellipse. The inverse $U^{-1} = U^\top$ transforms the ellipse back into a standard ellipse whose radii lie on the coordinate axes. Moreover $\det A = \det U \cdot \sigma_1\sigma_2 \cdot \det V$, and $\sigma_1\sigma_2 > 0$ when A is invertible, so if $\det A$ and $\det V$ have the same sign then $\det U = 1$ and U is a rotation matrix; otherwise $\det U = -1$ and U is a rotation with its two columns interchanged (a reflection followed by a rotation).

A 2×2 matrix A describes a linear map $\mathbb{R}^2 \to \mathbb{R}^2$ by telling us the images of the standard basis vectors, which are the columns of A. The SVD describes the same map differently: it tells us which orthogonal radii of the unit disc (the columns of V) are mapped to which orthogonal radii of the image ellipse (the columns of $U\Sigma$).

This picture extends to all dimensions. Let

$$\mathcal{D}^n = \{\vec{v} \in \mathbb{R}^n : \vec{v} \cdot \vec{v} \leq 1\}$$

and define an m-dimensional ellipse to be any set of the form $\{U\Sigma\vec{v} : \vec{v} \in \mathcal{D}^n\}$, where U is an orthogonal $m \times m$ matrix and Σ is an $m \times n$ matrix whose only nonzero entries lie on the main diagonal. If A is $m \times n$ with SVD $A = U\Sigma V^\top$ and $r = \operatorname{rank} A$, then the first r columns of V are orthogonal vectors of $\mathcal{D}^n$ that are transformed into orthogonal radii of the m-dimensional ellipse $\{A\vec{v} : \vec{v} \in \mathcal{D}^n\}$ (which has $m - r$ radii of length zero), while the last $n - r$ columns of V form an orthogonal basis of $\operatorname{Nul} A$.

SVD = geometry of the unit ball

For a matrix A, the SVD $A = U\Sigma V^\top$ says:
- V chooses an orthonormal set of directions in the domain — the right singular vectors;
- Σ stretches those directions by the factors $\sigma_1, \ldots, \sigma_r$ (and kills the rest);
- U rotates the stretched result into place — the left singular vectors are the directions of the radii of the image ellipse.

Hence the singular values are exactly the lengths of the semi-axes of the ellipsoid $A(\mathcal{D}^n)$, and $\operatorname{rank} A$ is the number of axes of positive length.